

# Power Management

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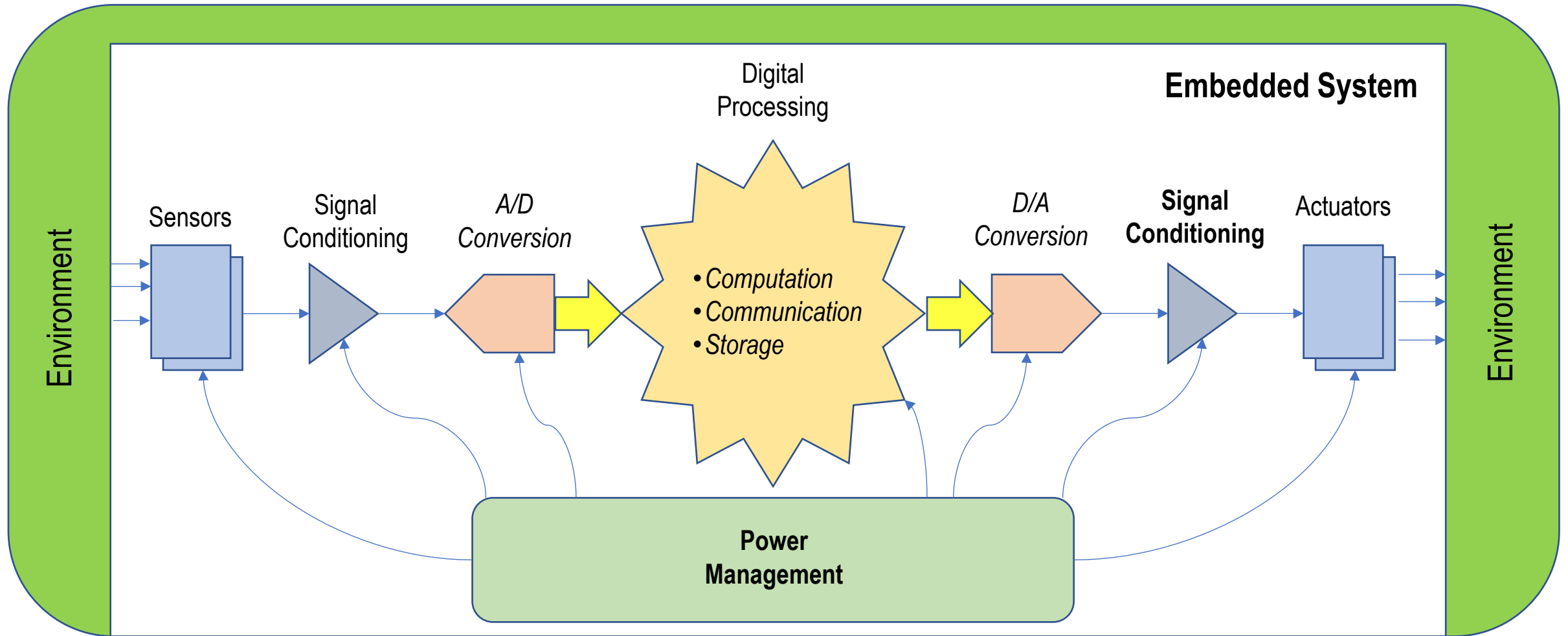
(Electronics for Embedded Systems)

# Outline

- Introduction
- Power Devices
  - Diodes
  - Transistors
- Actuator Driving
  - Low side
  - High side
  - H-Bridge
- Power supply
  - Linear power supply
  - Switching power supply

# Introduction

- Typical architecture of an Embedded System



# Introduction

- Low power applications
  - Logic circuit voltages
    - Around 3.3 V to 5 V
    - Can be as low as 1.2 V in low power processors and memories
  - Logic circuit currents
    - Very small inside integrated circuits
    - Around 20 mA on the outputs
  - Maximum power
    - A logic circuit can drive a maximum power
      - $P_{MAX} = V_{MAX} \cdot I_{MAX} = 5 \cdot 20 \cdot 10^{-3} = 100 \text{ mW}$
    - Example
      - Driving an LED
  - Need special circuits for higher power requirements

# Introduction

- Power Management
  - Provide power to all circuits in an embedded system
    - Power supply
  - Manage high power loads
    - Actuator driving
  - Efficiency
    - Linear circuits provide
      - Very good noise behavior
      - Low efficiency (25 – 50%)
    - Switching circuits provide
      - Very high efficiency (90%)
      - High frequency noise

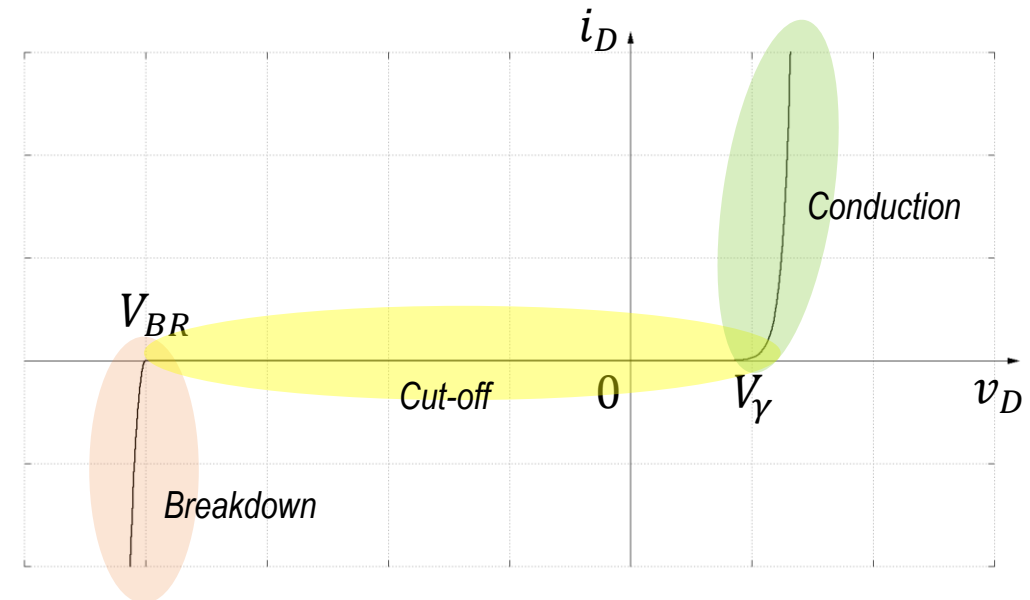
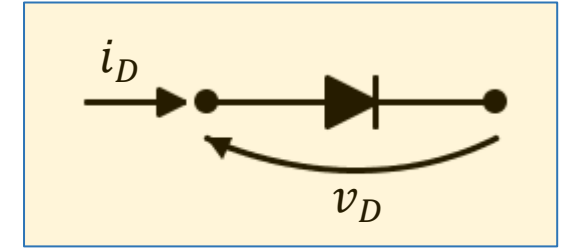
# Introduction

- Power Management
  - Switching amplifier
    - Active devices used as switches
    - Very low power consumption
      - When ON
        - Voltage nearly 0
        - Current set by the rest of the circuit
      - When OFF
        - Current nearly 0
        - Voltage set by the rest of the circuit
      - Static power consumption is 0
    - Dynamic power consumption
      - During transients the device goes through the linear region
      - Fast switching decreases power consumption

# Power Devices

## • Diodes

- Junction of two semiconductors of opposite type
- Two terminal device
  - Anode ( $p$  side)
  - Cathode ( $n$  side)
- When forward biased
  - Conduction
    - Modeled as a short circuit in first approximation
- When reverse biased
  - Cut-off
    - Modeled as an open circuit in first approximation
  - Breakdown
    - Large current when reverse voltage exceeds a certain threshold

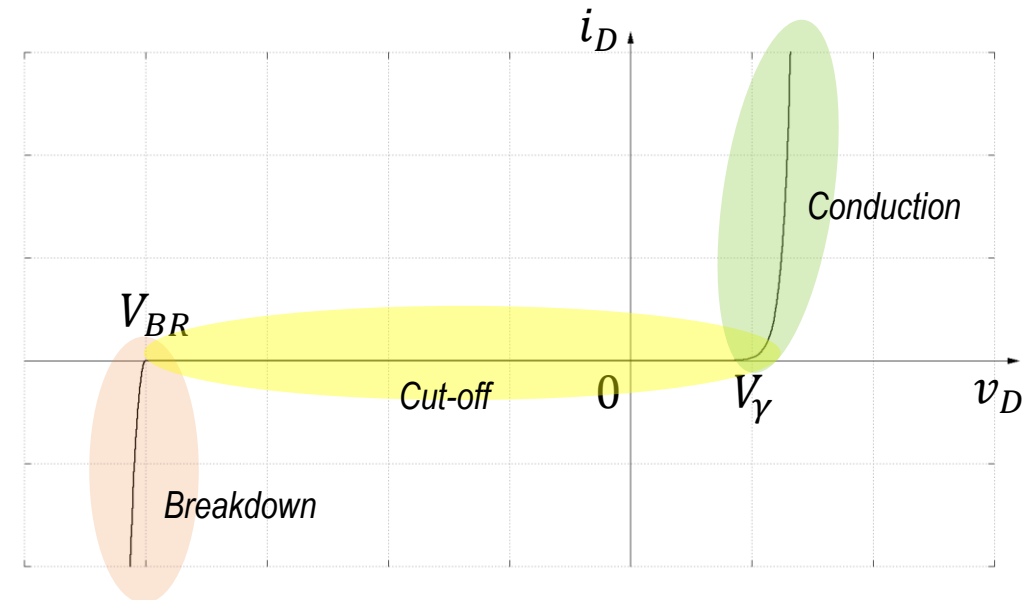
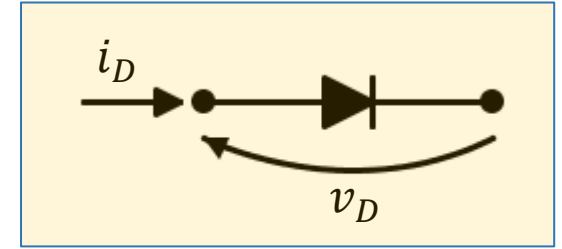


$$i_D(t) = I_S \left( e^{\frac{v_D(t)}{\eta V_T}} - 1 \right)$$

# Power Devices

- Diodes Parameters

- Reverse leakage current  $I_S$ 
  - Small current when diode is reverse biased
  - A few nA in signal devices
  - A few mA for large power devices
  - It varies exponentially with temperature
- Forward voltage  $V_\gamma$ 
  - Approximate value when the diode is switched ON
  - Around 0.6 – 0.7 V in signal diodes
  - Higher for power devices
- Forward ON resistance  $R_{ON}$ 
  - Differential resistance in conduction (slope of characteristic)
  - $R_{ON} = \frac{\Delta v_D}{\Delta i_D}$

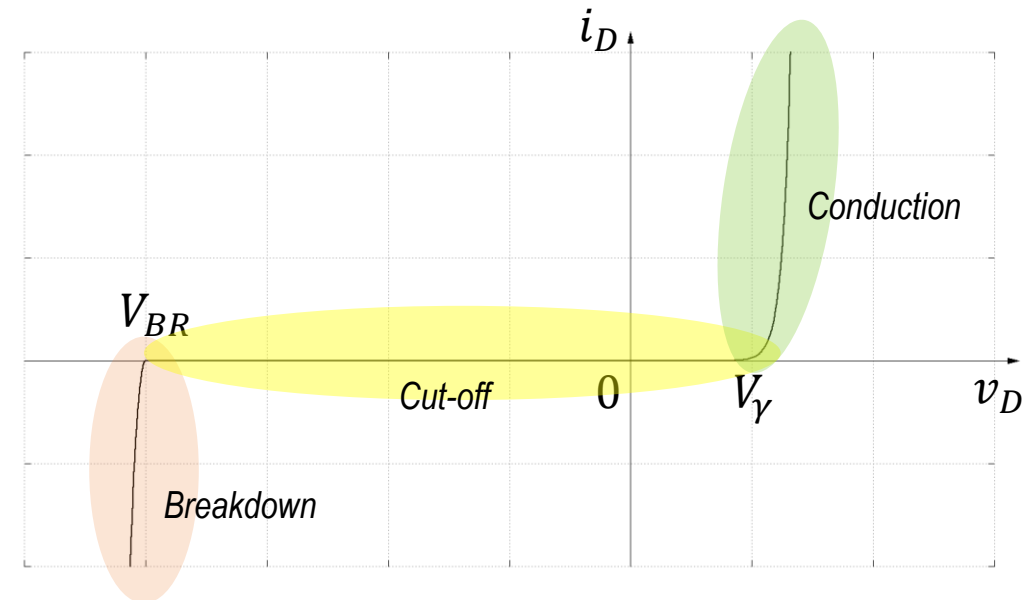
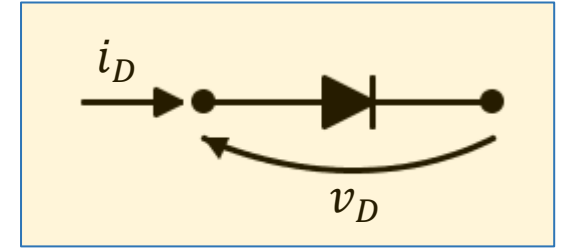


$$i_D(t) = I_S \left( e^{\frac{v_D(t)}{\eta V_T}} - 1 \right)$$

# Power Devices

- Diodes Parameters

- Breakdown voltage  $V_{BR}$ 
  - Exact value not usually known precisely
  - Avalanche or tunneling effect
  - High reverse current
  - Undesirable effect when diode is used as a switch
  - May lead to damages
    - High current, high power consumption
    - High temperature
- It should be higher (in absolute value) than the largest reverse voltage that may occur
  - Typically in the order of tens of Volts
  - Hundreds of Volts for power devices
- Sometimes it is useful
  - Zener diodes

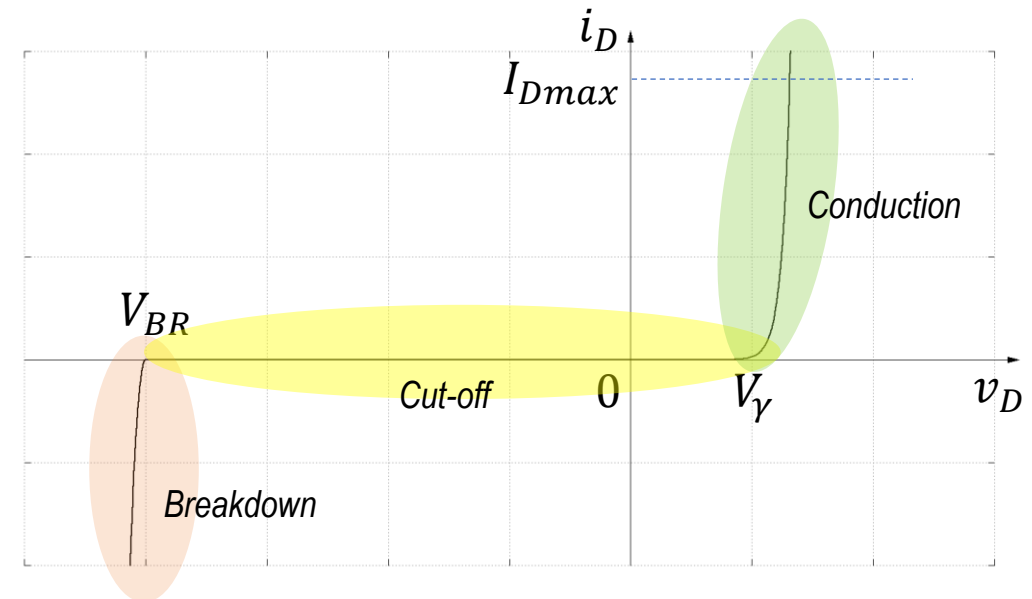
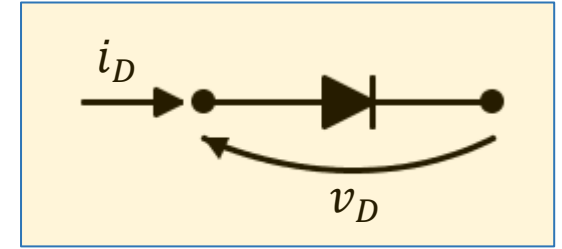


$$i_D(t) = I_S \left( e^{\frac{v_D(t)}{\eta V_T}} - 1 \right)$$

# Power Devices

- Diodes Parameters

- Maximum forward current  $I_{Dmax}$ 
  - Maximum value of the forward current
  - Damages if this value is exceeded
    - High temperatures
- Two specifications
  - Constant maximum current
  - Pulsated maximum current
    - Higher value
- Hot spots
  - Current is higher, temperature is higher
  - $I_S$  grows with temperature
  - Positive feedback, thermal runaway
    - Use multiple parallel devices with smaller current



$$i_D(t) = I_S \left( e^{\frac{v_D(t)}{\eta V_T}} - 1 \right)$$

# Power Devices

- Diodes Parameters

- Reverse recovery time  $t_{rr}$

- When a diode is forward biased

- Large number of carriers injected in the opposite material
      - They are minority carriers (electrons in the  $p$ -side, holes in the  $n$ -side)

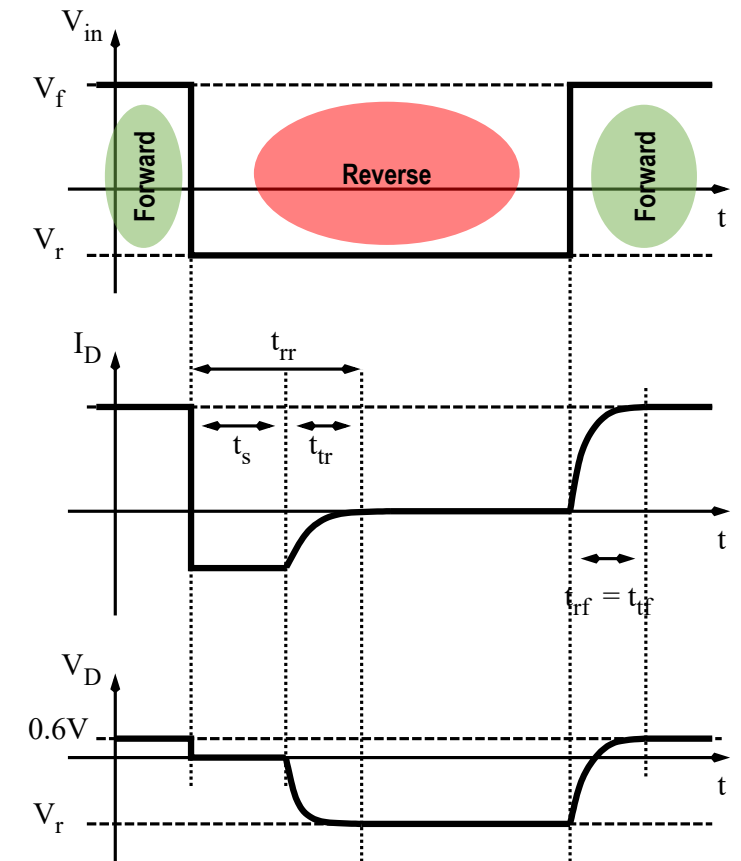
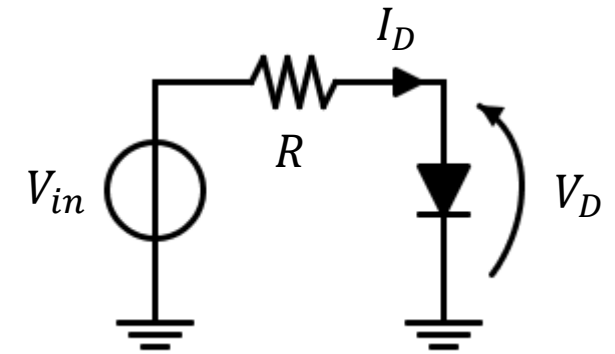
- When switching OFF

- Carriers must be removed
        - Constant reverse current
        - Storage time  $t_s$
      - Electron and holes diffuse and recombine
        - Current becomes  $I_S$
        - Reverse transition time  $t_{tr}$

- $t_{rr} = t_s + t_{tr}$

- Forward recovery time  $t_{rf}$

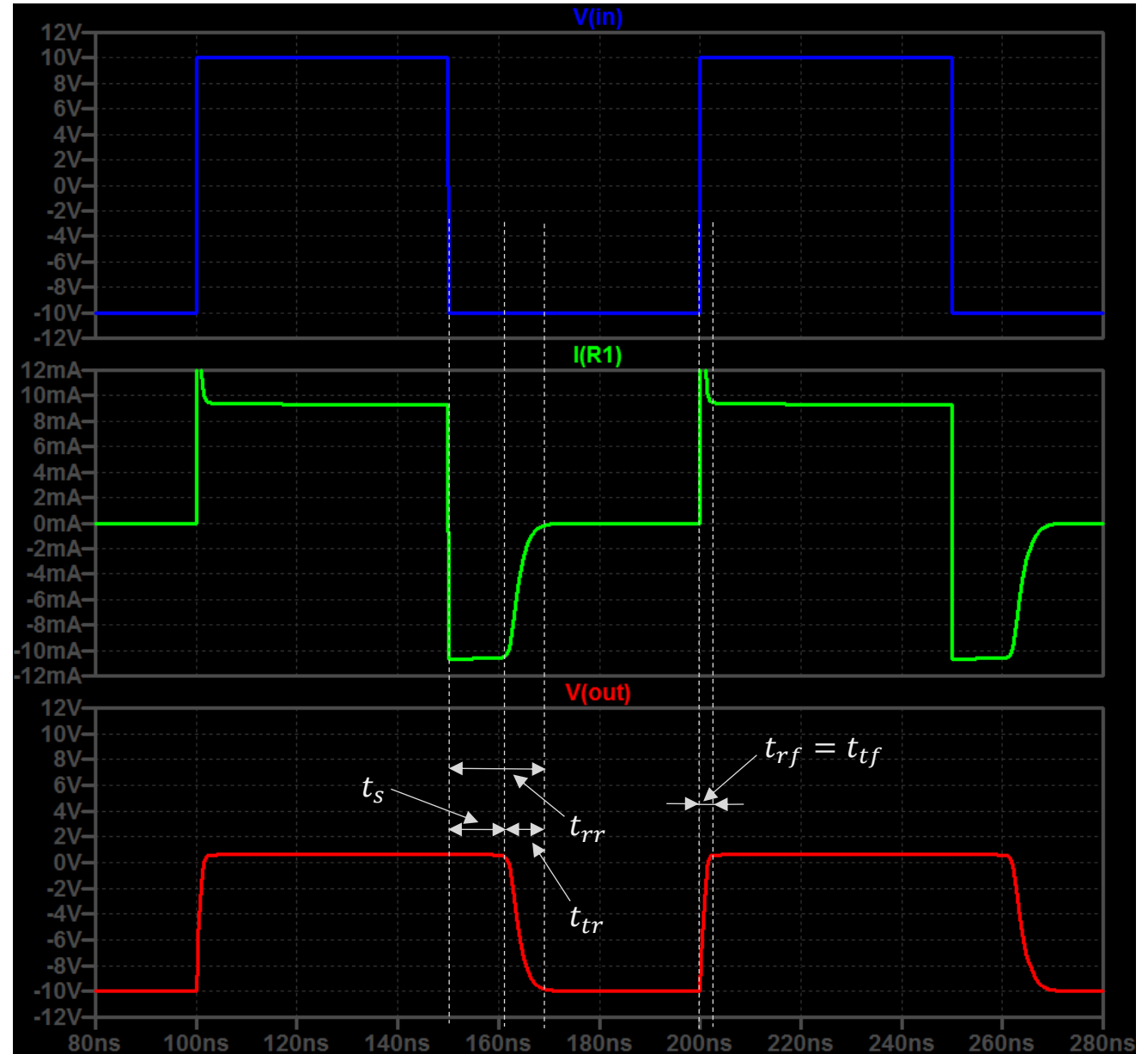
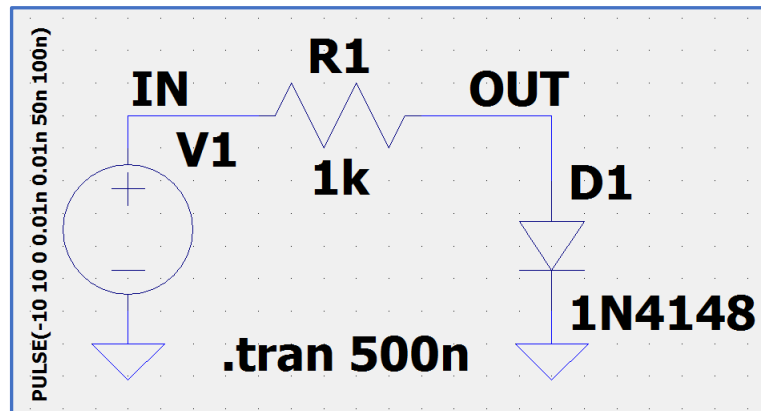
- No charges to remove
    - Forward transition time  $t_{tf}$
    - $t_{rf} = t_{tf}$



# Power Devices

- Diodes Parameters

- Reverse recovery time  $t_{rr}$ 
  - Storage time  $t_s$
  - Reverse transition time  $t_{tr}$
  - $t_{rr} = t_s + t_{tr}$
- Forward recovery time  $t_{rf}$ 
  - Forward transition time  $t_{tf}$
  - $t_{rf} = t_{tf}$



# Power Devices

- P.I.N. Diode
  - Intrinsic region between  $p$  and  $n$  regions
    - Breakdown voltage  $V_{BR}$  greatly increased
      - Advantage in power applications involving large voltages
      - In the order of hundreds of Volts
    - ON resistance  $R_{ON}$  and forward voltage  $V_f$  increased
      - Disadvantage when used as a switch
      - Increased power dissipation
  - Also used as attenuator and RF switch



# Power Devices

- Schottky Diode

- Metal – silicon junction

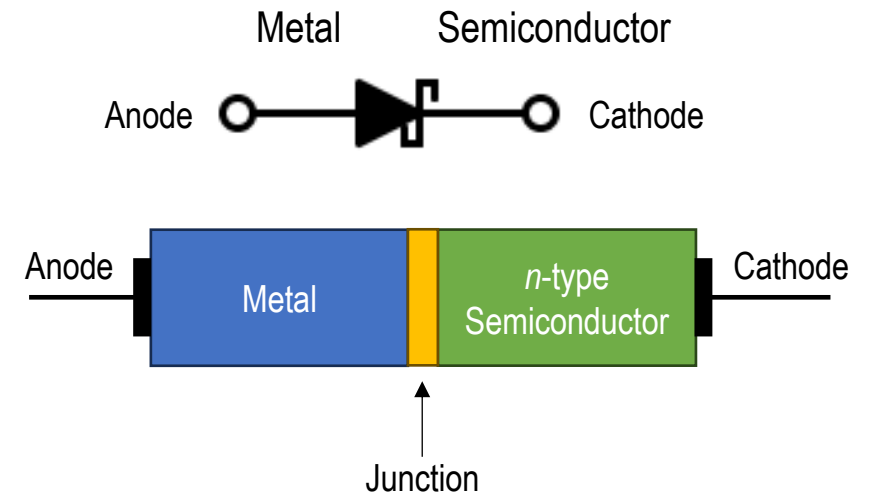
- Conduction through majority carriers only
    - Silicon region lightly doped
      - If highly doped it becomes an ohmic contact

- Advantages

- Fast switching in both transitions
      - In the order of hundreds of ps
      - No charges stored to be removed
    - Lower ON resistance  $R_{ON}$
    - Lower forward voltage  $V_F$ 
      - In the order of 0.2 – 0.3 V

- Disadvantages

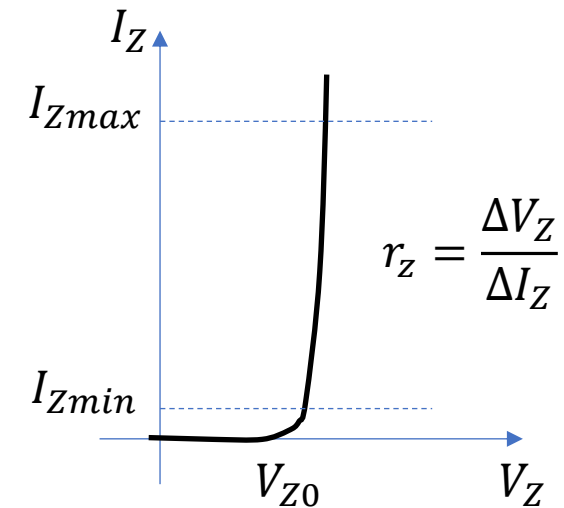
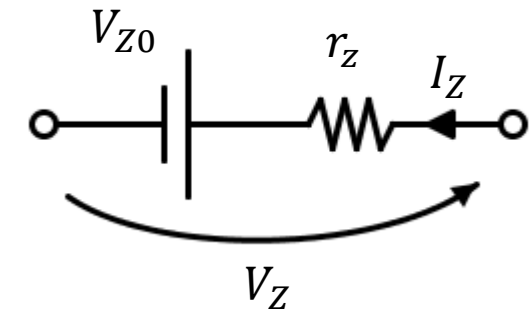
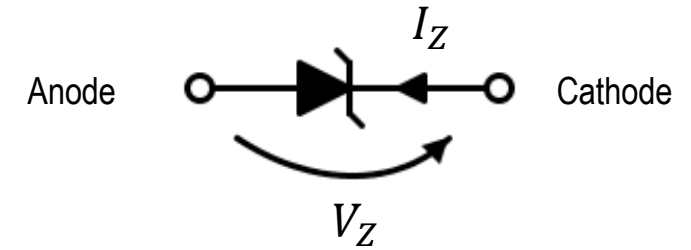
- High capacitance in the OFF state
    - Higher reverse current  $I_S$
    - Lower breakdown voltage  $V_{BR}$  in absolute value



- Electrons are majority carriers in the  $n$ -side
- When forward biased, electrons go towards the metal side
- They don't become minority carriers in the metal side
- Since there is no  $p$ -side, there are no holes that become minority carriers in the  $p$ -side
- Therefore, there are no charges to be removed when switching OFF → no storage time!

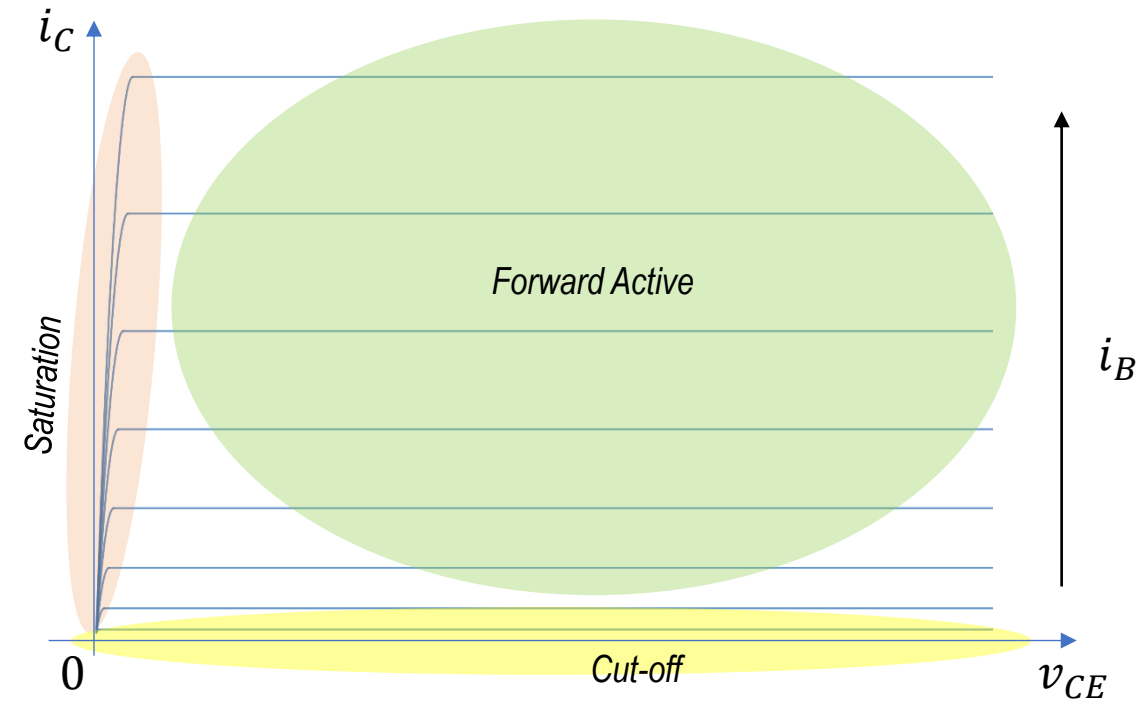
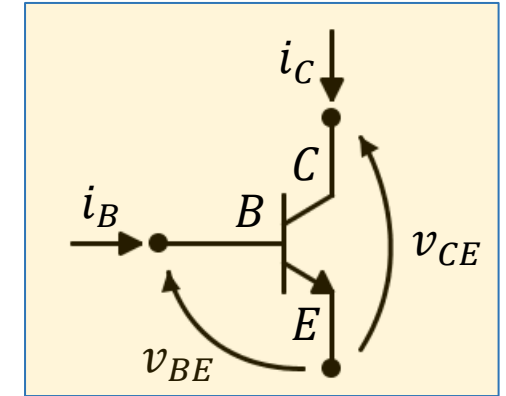
# Power Devices

- Zener Diode
  - Breakdown voltage  $V_{BR}$  accurately determined
    - Diode works in breakdown without degradation
    - When forward biased it works like a standard diode
    - When reverse biased
      - A significant reverse current flows
      - Current should be at least  $I_{Zmin}$
      - Current should not exceed  $I_{Zmax}$
      - Diode modeled as
        - A constant voltage source of value  $V_{Z0}$
        - A differential resistance of value  $r_z$ 
          - Small, a few  $\Omega$
        - $V_Z = V_{Z0} + I_Z r_z$
  - Used as a voltage reference
    - Breakdown voltage can be as low as a few Volts



# Power Devices

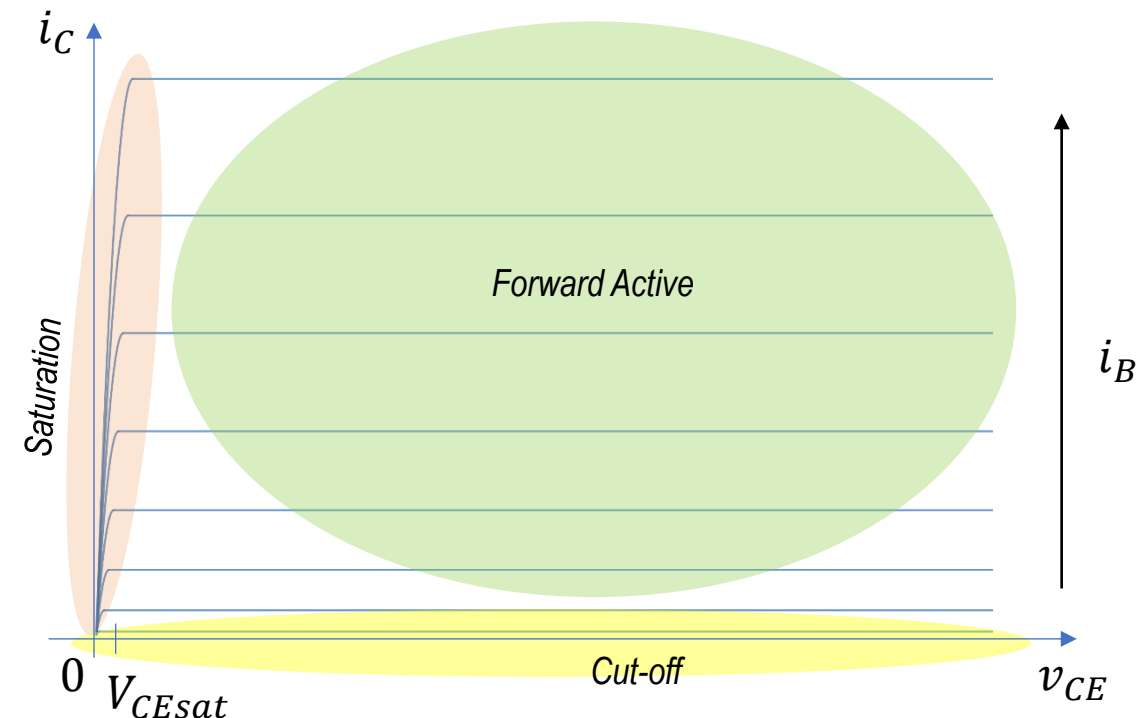
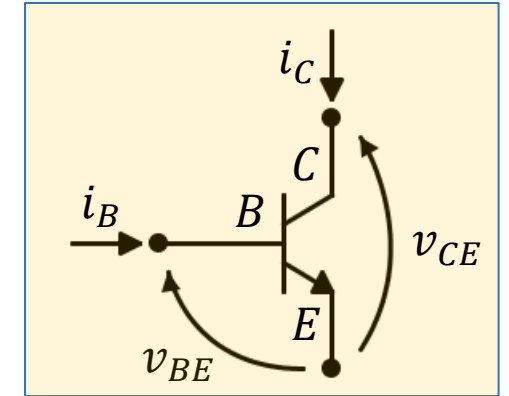
- Bipolar Junction Transistor (BJT)
  - Three terminal device
  - Used as an amplifier or as a switch
  - Cut-off region
    - BE and BC junctions reverse biased
    - Open switch,  $i_C = 0$
  - Forward active region
    - BE junction forward, BC junction reverse
    - Linear amplifier,  $i_C = \beta i_B$
  - Saturation region
    - BE and BC junctions forward biased
    - Closed switch,  $v_{CE} \simeq 0$



# Power Devices

- Bipolar Junction Transistor (BJT) Parameters

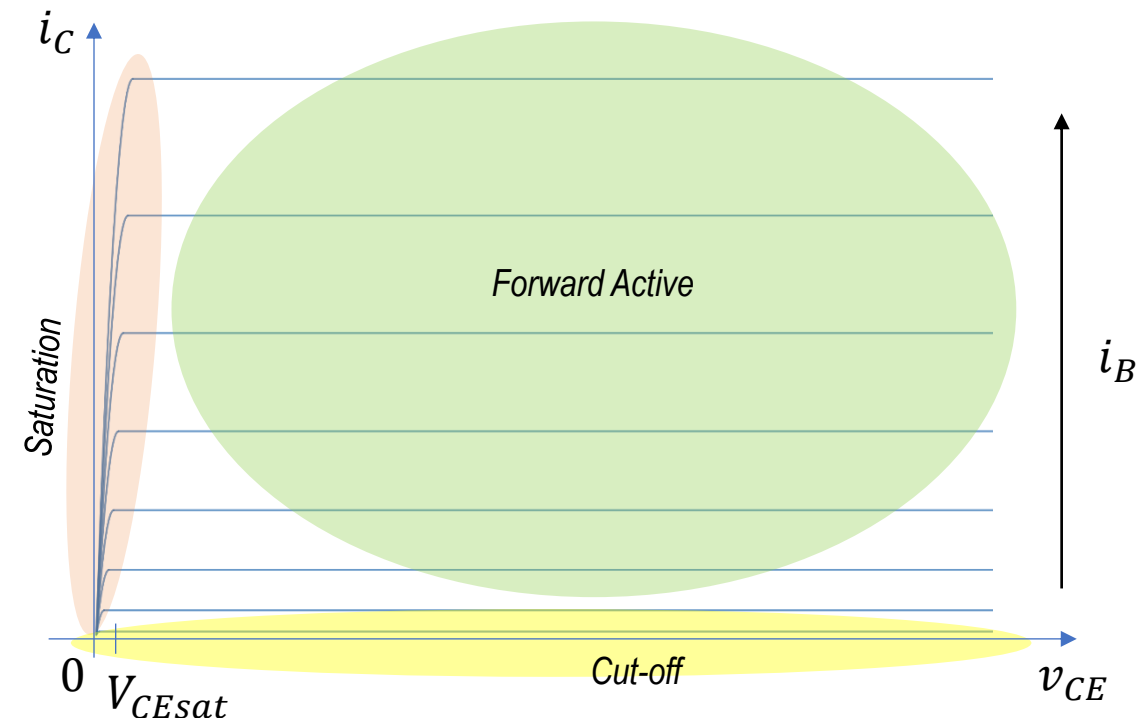
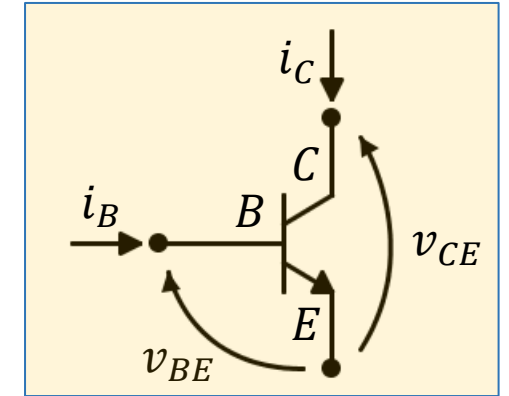
- Current gain  $\beta$ 
  - In forward active region,  $i_C = \beta i_B$
  - Not known accurately
- Minimum BE voltage  $V_{BEon}$ 
  - Minimum voltage to turn on the transistor
  - Collector current is very small at this voltage
  - Around 0.4 V
- CE saturation voltage  $V_{CEsat}$ 
  - CE voltage when the transistor is in saturation
  - Around 0.2 – 0.3 V
- BE saturation voltage  $V_{BEsat}$ 
  - BE voltage when the transistor is in saturation
  - Around 1 V or more



# Power Devices

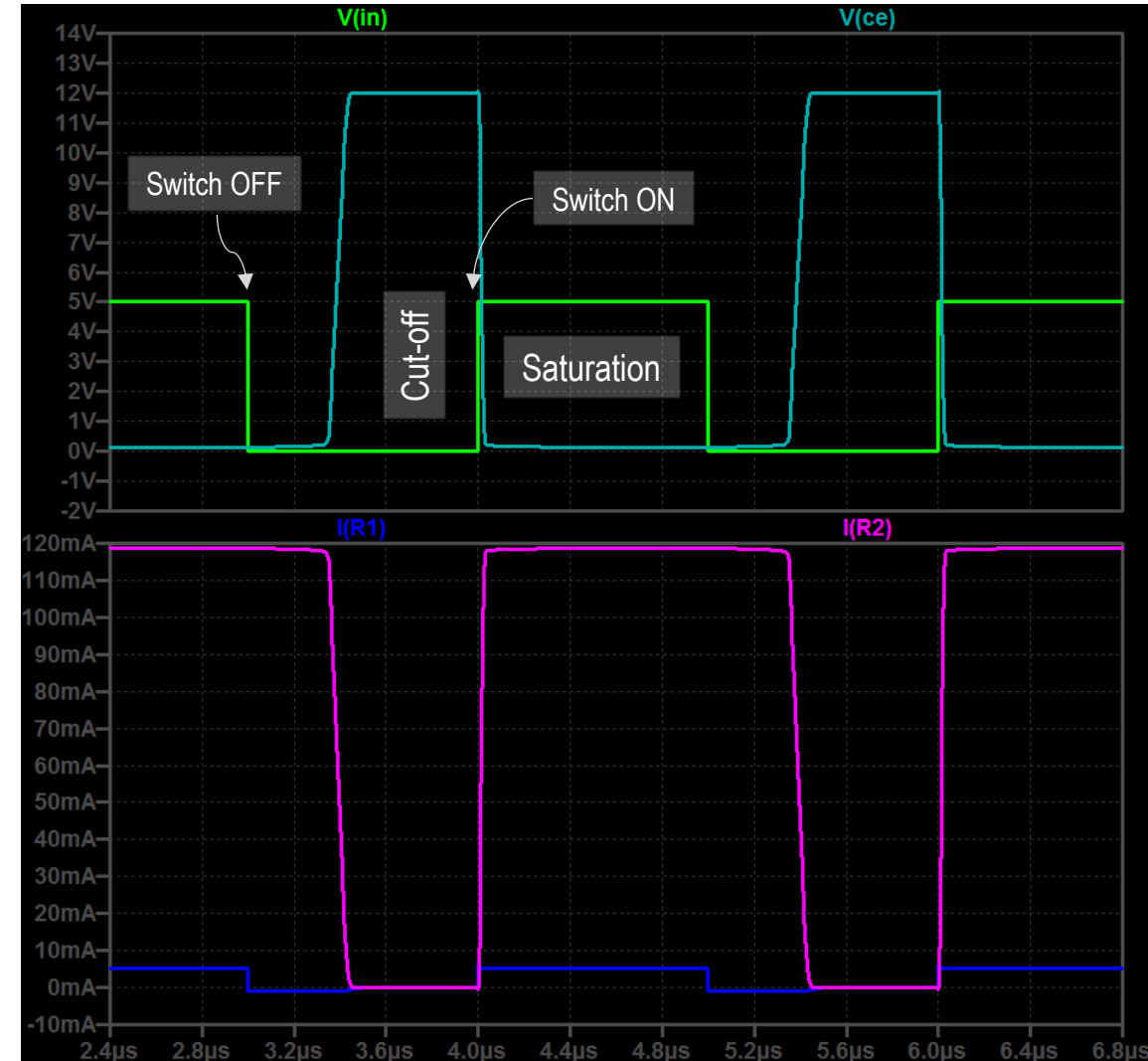
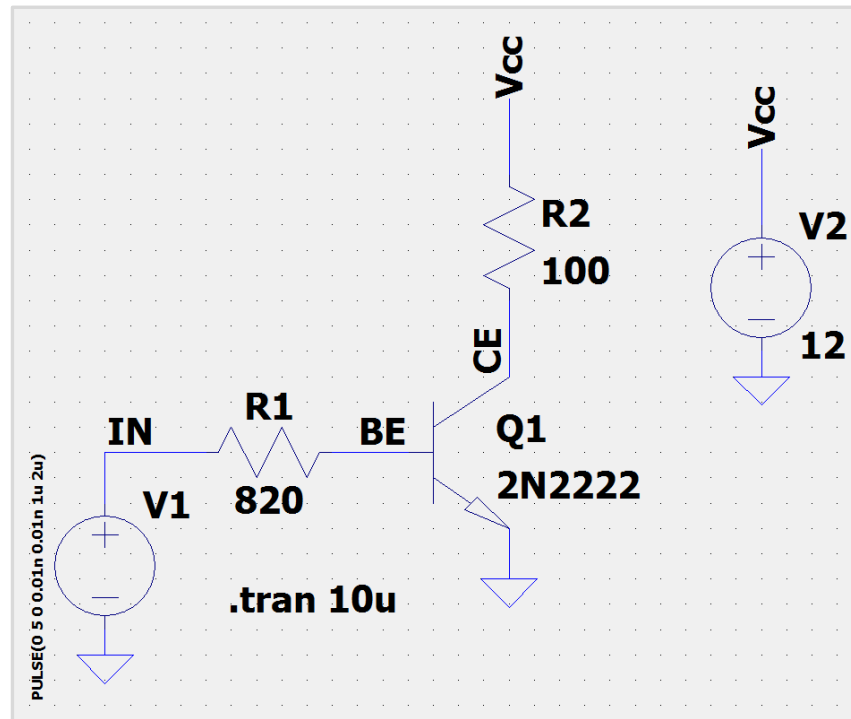
- Bipolar Junction Transistor (BJT) Parameters

- Reverse recovery time  $t_{rr}$ 
  - Similar to the one already seen for diodes
    - Applies to the base-collector junction
  - When transistor is in saturation, BC is forward biased
    - Large number of stored charges
  - When switching OFF the transistor, charges need to be removed
    - Higher switching time
  - The opposite transition does not show the same problem
    - Asymmetrical switching time



# Power Devices

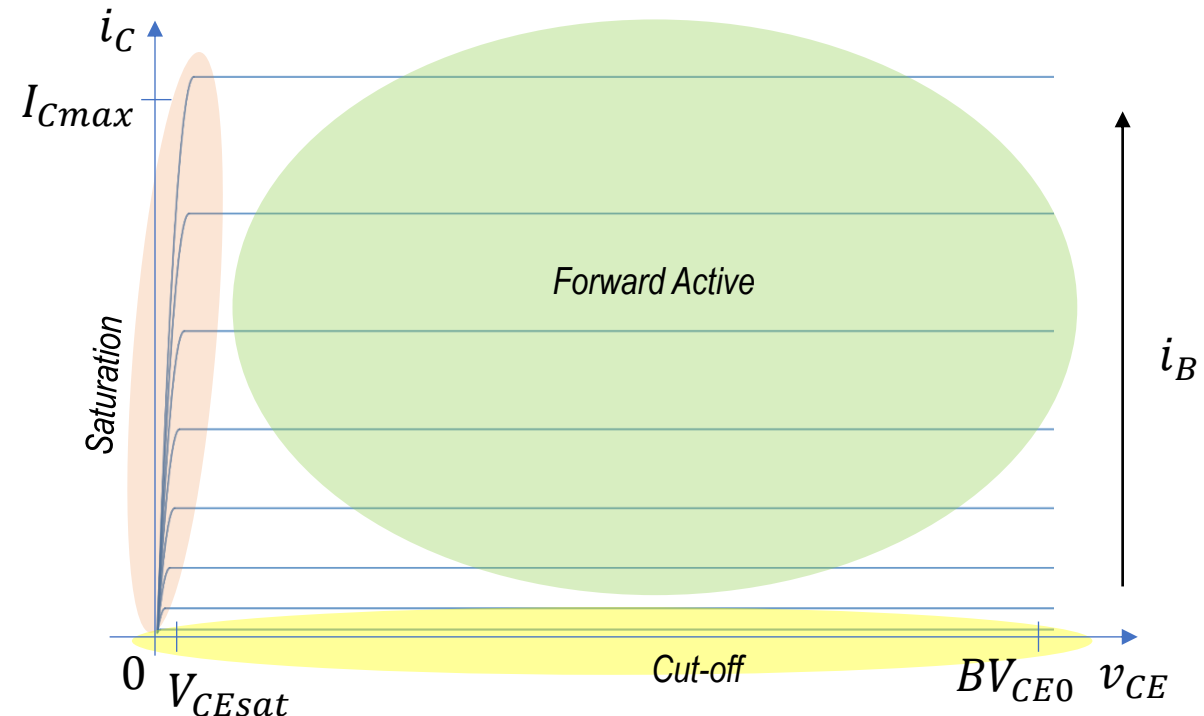
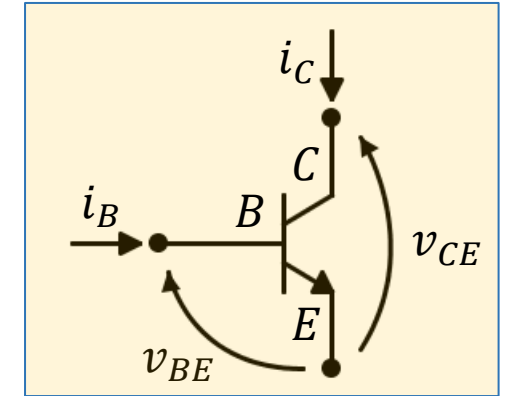
- Bipolar Junction Transistor (BJT) Parameters
  - Reverse recovery time  $t_{rr}$ 
    - When switching OFF the transistor
      - Higher switching time



# Power Devices

- Bipolar Junction Transistor (BJT) Parameters

- Maximum CE voltage  $BV_{CE0}$  (or  $V_{CEbr}$ )
  - Maximum rating for CE voltage
  - Breakdown for higher values
- Maximum collector current  $I_{Cmax}$ 
  - Maximum rating for collector current
  - Exceeding this value leads to damages
- Maximum power dissipation  $P_{Dmax}$ 
  - Power dissipation is  $P_D = I_C \cdot V_{CE}$
  - Exceeding the maximum value leads to damages of the device
  - Parameter given in different conditions
    - Pulsating current
    - Presence of heatsink and temperature



# Power Devices

- Bipolar Junction Transistor (BJT) Parameters

- Safe Operating Area (SOA)

- Defined by the interaction of

- $I_{Cmax}$
      - $BV_{CE0}$
      - $P_{Dmax}$

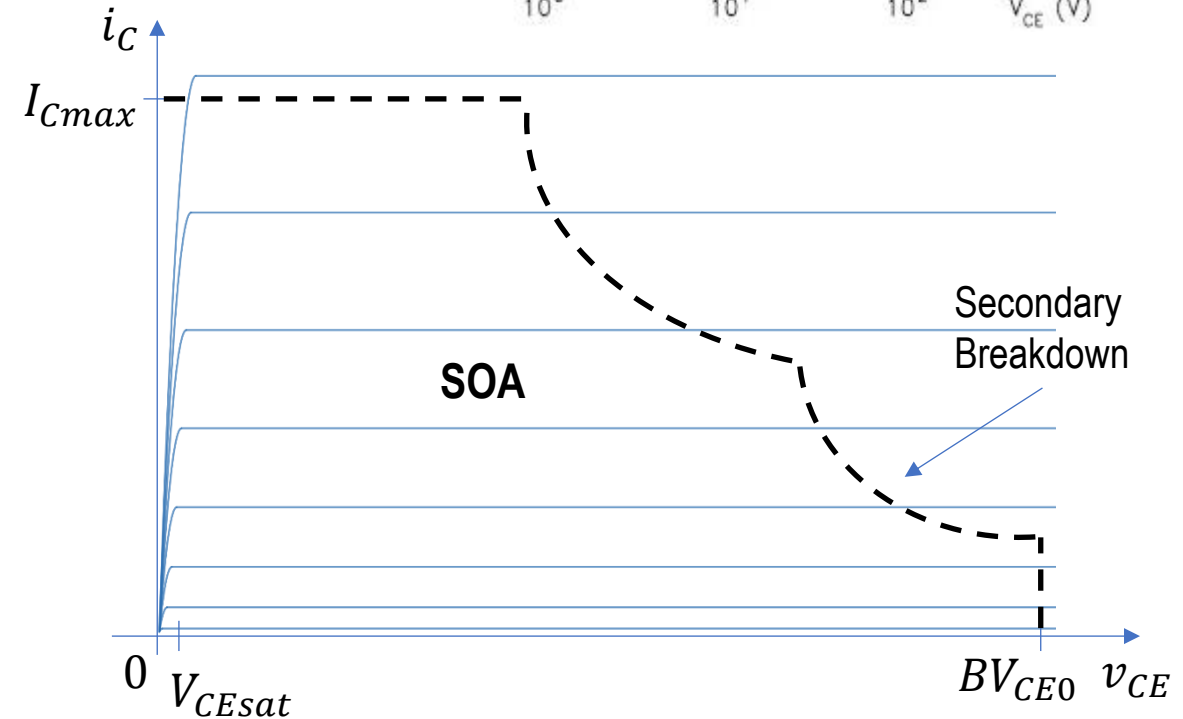
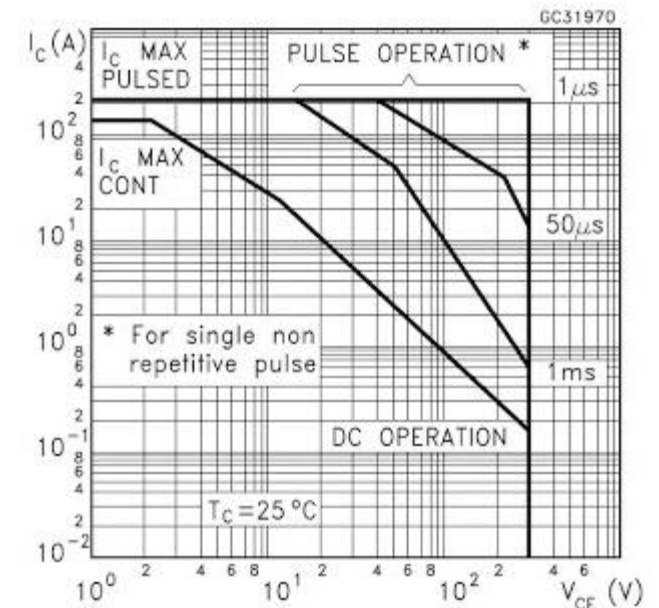
- Secondary breakdown due to hotspots

- Use multiple parallel transistors to distribute the current

- SOA often illustrated using logarithmic scales

- SOA often illustrated for different pulsating currents

- Higher allowed power dissipation if pulses are short



# Power Devices

- Metal-Oxide-Semiconductor (MOS) Transistor

- Three terminal device
- Used as an amplifier or as a switch
- Threshold voltage  $v_{th}$
- Cut-off region

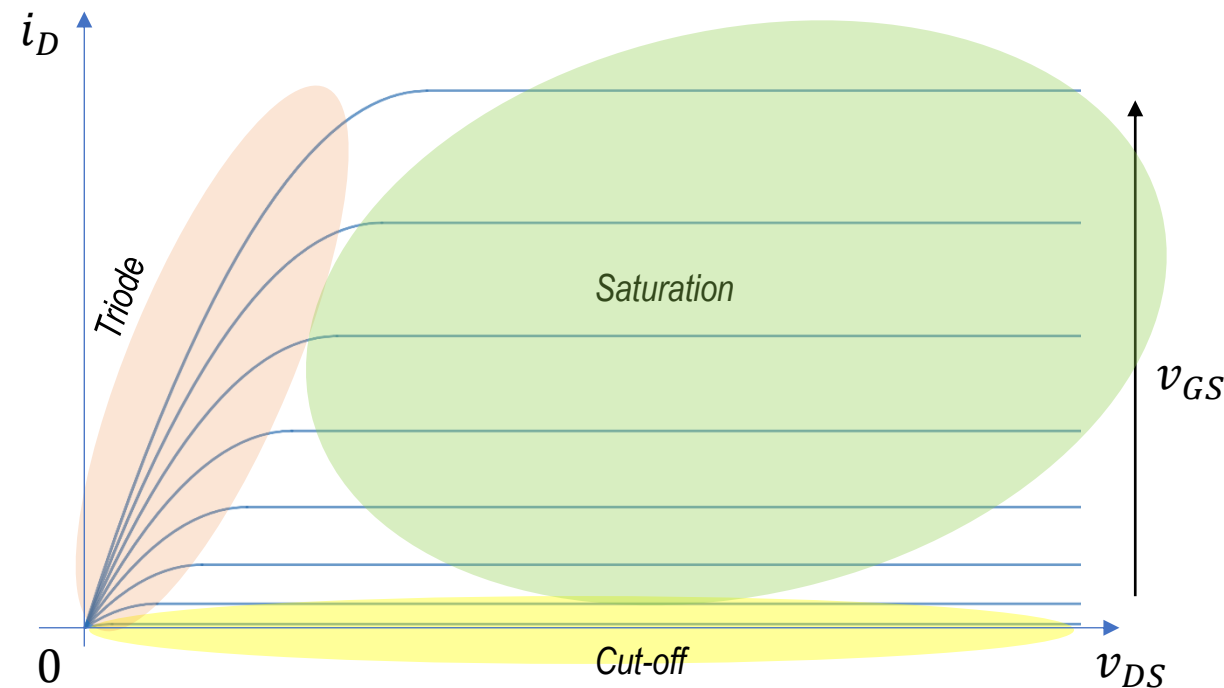
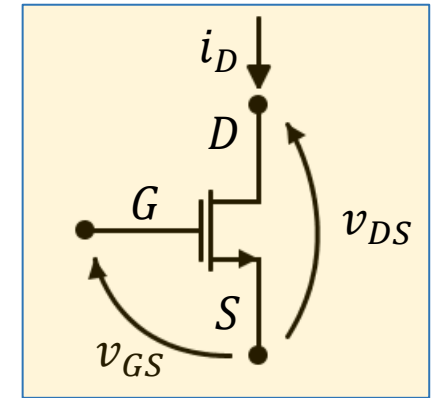
- $v_{GS} < v_{th}$ ,  $i_D = 0$
- Open switch

- Triode region

- $v_{GS} > v_{th}$ ,  $v_{DS} < v_{GS} - v_{th}$
- $i_D = \mu C_{ox} \frac{W}{L} \left[ (v_{GS} - v_{th}) v_{DS} - \frac{v_{DS}^2}{2} \right]$
- Closed switch,  $v_{DS} \simeq 0$

- Saturation region

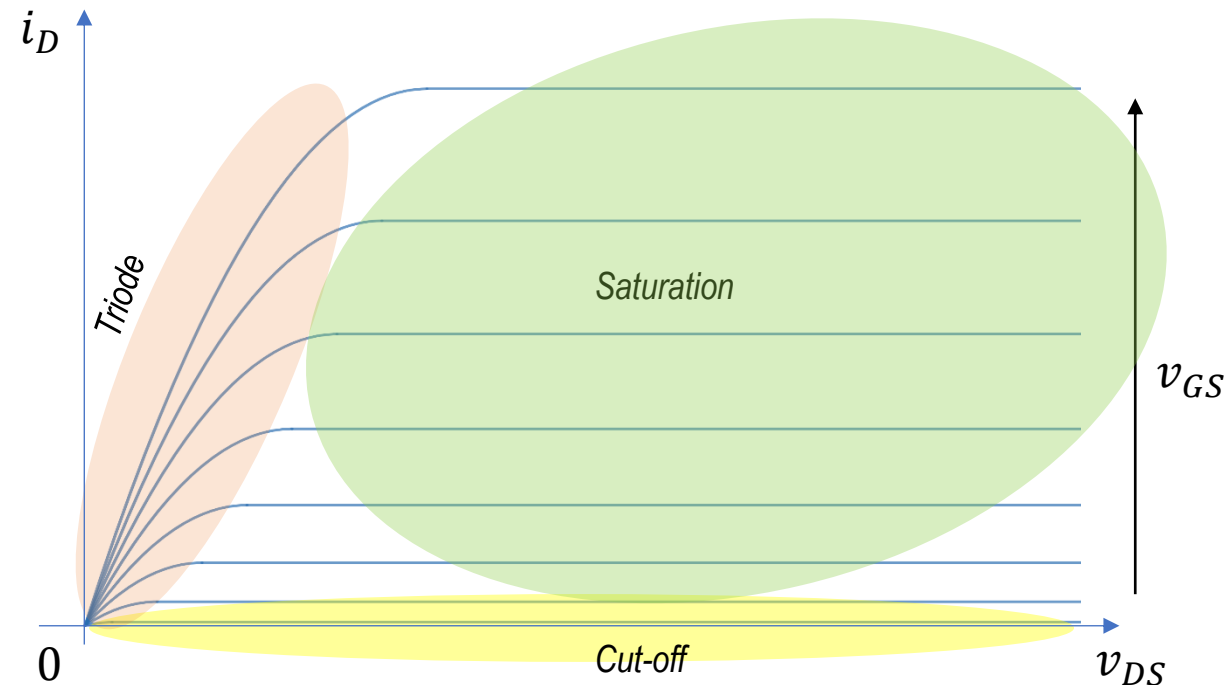
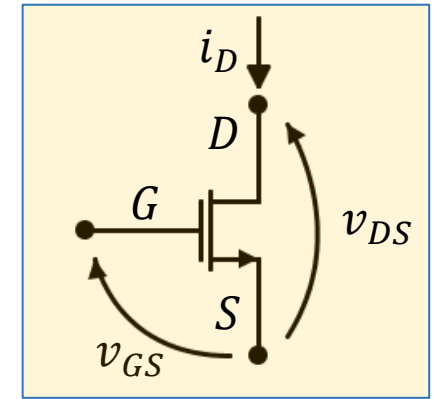
- $v_{GS} > v_{th}$ ,  $v_{DS} > v_{GS} - v_{th}$
- Linear amplifier,  $i_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (v_{GS} - v_{th})^2$



# Power Devices

- Metal-Oxide-Semiconductor (MOS) Transistor Parameters

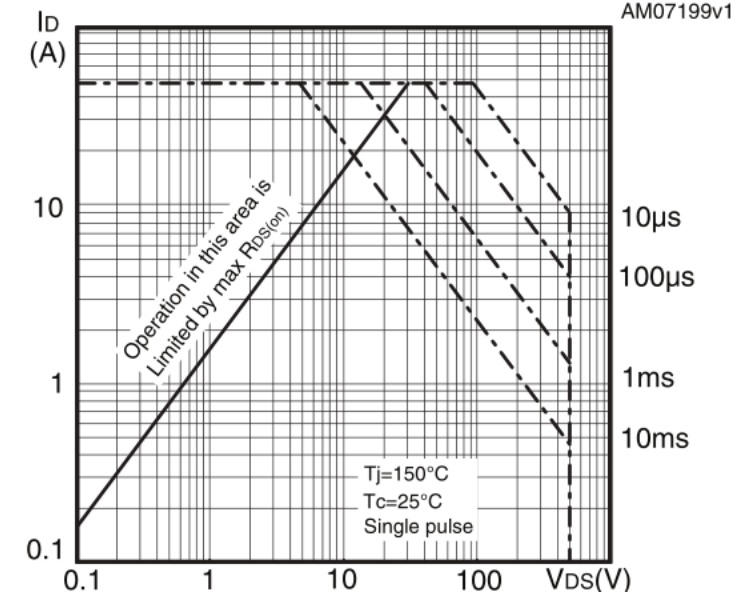
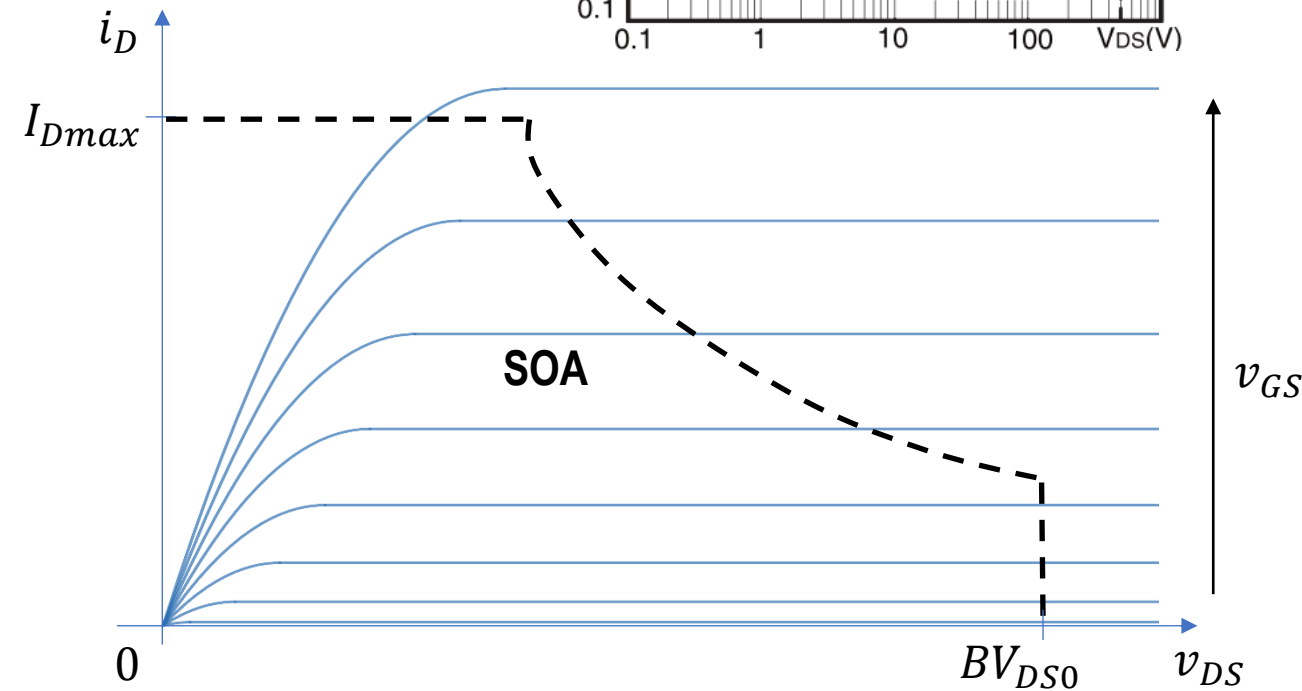
- Threshold voltage  $v_{th}$ 
  - Minimum  $v_{GS}$  to turn on the transistor
  - Drain current is very small at this voltage
  - Around 1 – 1.5 V
  - Changes with temperature
- ON resistance  $R_{DSon}$ 
  - Slope of characteristic in linear region
    - Triode region for  $v_{DS} \approx 0$
  - Equivalent resistance when transistor is used as a closed switch
    - $R_{DSon} = \frac{v_{DS}}{i_D} = \frac{1}{\mu C_{ox} \frac{W}{L} (v_{GS} - v_{th})}$
    - A few Ohms



# Power Devices

- Metal-Oxide-Semiconductor (MOS) Transistor Parameters

- Maximum DS voltage  $BV_{DS0}$ 
  - Maximum rating for DS voltage
  - Breakdown for higher values
- Maximum drain current  $I_{Dmax}$ 
  - Maximum rating for drain current
  - Exceeding this value leads to damages
- Maximum power dissipation  $P_{Dmax}$ 
  - Power dissipation is  $P_D = I_D \cdot V_{DS}$
  - Exceeding the maximum value leads to damages of the device
  - Parameter given in different conditions
    - Pulsating current
    - Presence of heatsink and temperature



# Power Devices

## • Metal-Oxide-Semiconductor (MOS) Transistor Parameters

### • Dynamic switching from OFF to ON

#### • Initial phase

- At  $t = 0$  the input makes an ideal low to high transition
- Transistor is in cut-off,  $V_{GS}$  grows with time constant  $\tau_1$

$$\tau_1 = R_G \cdot C_{GS}$$

#### • Middle phase

- When  $V_{GS} = V_{th}$  and  $V_{DS}$  is high the transistor is in saturation (linear amplifier with gain  $k$ )
- $V_{GS}$  then grows with time constant  $\tau_2$

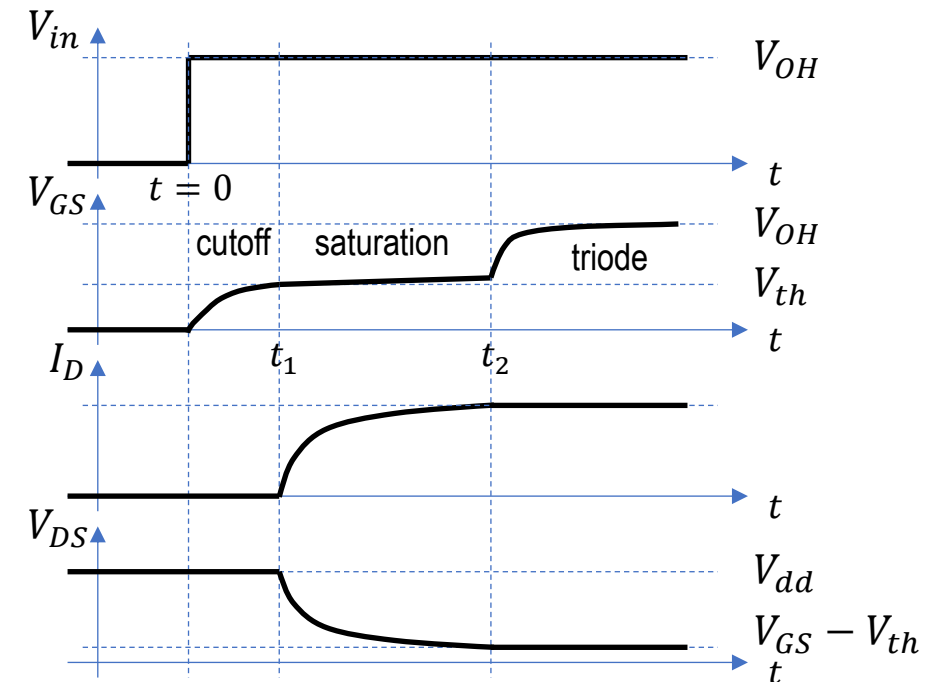
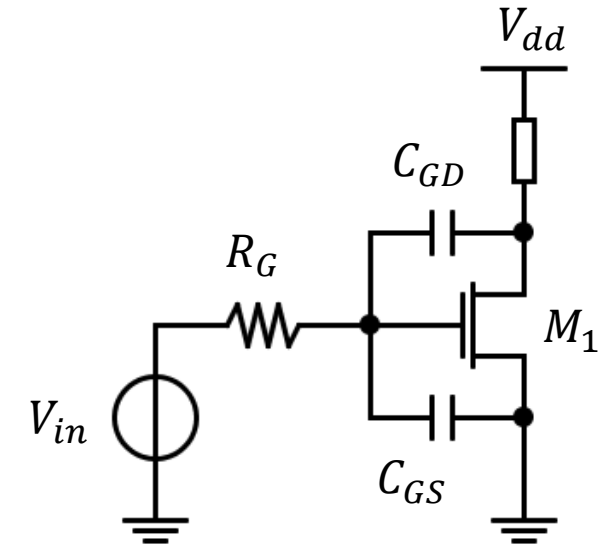
$$\tau_2 = R_G \cdot (C_{GS} + kC_{GD})$$

- Miller effect

#### • Final phase

- When  $V_{DS} = V_{GS} - V_{th}$  the transistor goes in triode region
- $V_{GS}$  then grows with time constant  $\tau_3$

$$\tau_3 = R_G \cdot (C_{GS} + C_{GD})$$

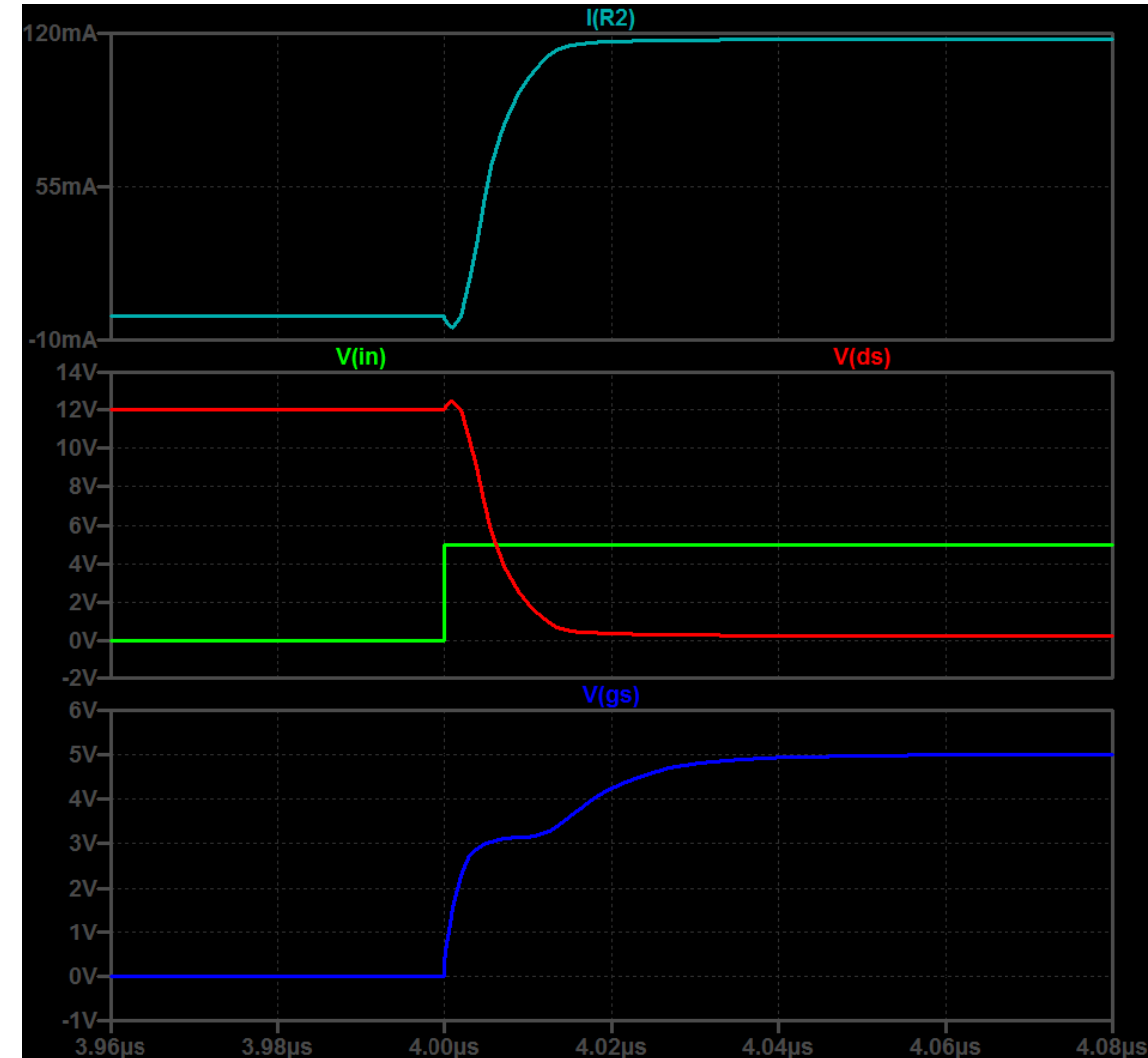
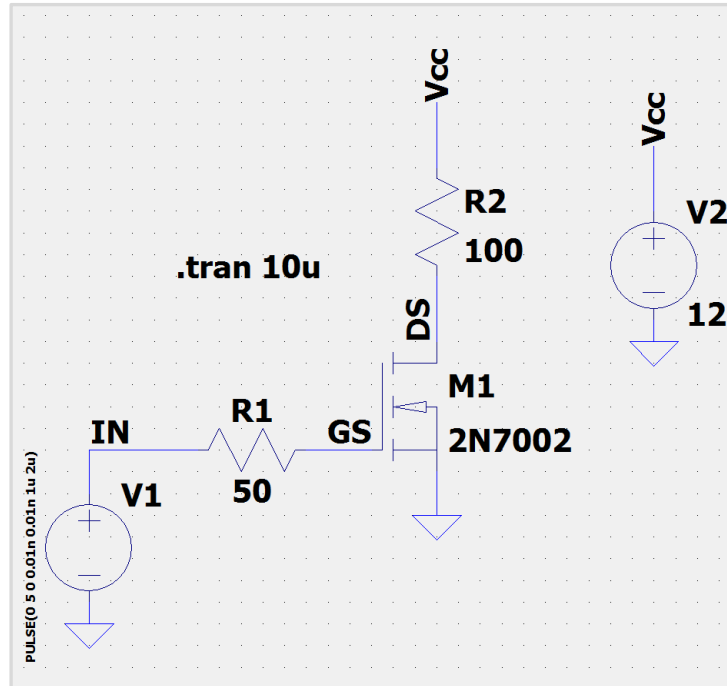


# Power Devices

- Metal-Oxide-Semiconductor (MOS) Transistor Parameters

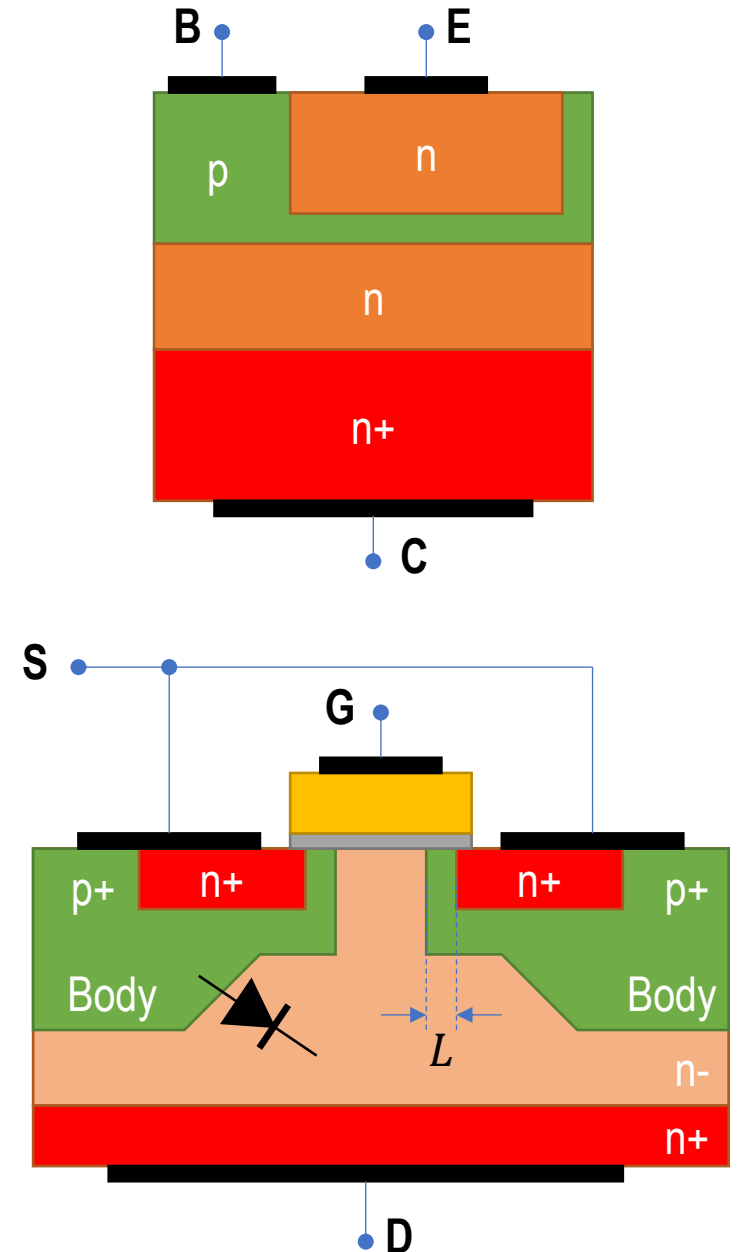
- Dynamic switching

|               | Time constant                           |
|---------------|-----------------------------------------|
| Initial phase | $\tau_1 = R_G \cdot C_{GS}$             |
| Middle phase  | $\tau_2 = R_G \cdot (C_{GS} + kC_{GD})$ |
| Final phase   | $\tau_3 = R_G \cdot (C_{GS} + C_{GD})$  |



# Power Devices

- Technologies for power transistors
  - Planar process not suitable for high currents
  - Vertical structure
  - BJT
    - Lightly doped base and collector regions
    - High breakdown voltage  $BV_{CE0}$
    - Low current gain  $\beta$
  - MOS (DMOS, Double Diffused)
    - Drain lightly doped
    - High breakdown voltage  $BV_{DS0}$
    - Small  $L$ , large  $W$ , large current  $I_D$
    - Source and Body connected together
    - Parasitic diode between Source and Drain
      - Current can flow from S to D



# Power Devices

- Bipolar Junction Transistor (BJT)
  - TIP31 Datasheet

| MAXIMUM RATINGS: ( $T_C=25^\circ\text{C}$ unless otherwise noted) |                |       |             |        |        |                    | ELECTRICAL CHARACTERISTICS: ( $T_C=25^\circ\text{C}$ unless otherwise noted) |                                                            |     |     |     |               |
|-------------------------------------------------------------------|----------------|-------|-------------|--------|--------|--------------------|------------------------------------------------------------------------------|------------------------------------------------------------|-----|-----|-----|---------------|
|                                                                   | SYMBOL         | TIP31 | TIP31A      | TIP31B | TIP31C | UNITS              | SYMBOL                                                                       | TEST CONDITIONS                                            | MIN | TYP | MAX | UNITS         |
| Collector-Base Voltage                                            | $V_{CBO}$      | 40    | 60          | 80     | 100    | V                  | $I_{CEO}$                                                                    | $V_{CE}=30\text{V}$ (TIP31, TIP31A)                        |     |     | 0.3 | mA            |
| Collector-Emitter Voltage                                         | $V_{CEO}$      | 40    | 60          | 80     | 100    | V                  | $I_{CEO}$                                                                    | $V_{CE}=60\text{V}$ (TIP31B, TIP31C)                       |     |     | 0.3 | mA            |
| Emitter-Base Voltage                                              | $V_{EBO}$      |       |             | 5.0    |        | V                  | $I_{CES}$                                                                    | $V_{CE}=\text{Rated } V_{CEO}$                             |     |     | 0.2 | mA            |
| Continuous Collector Current                                      | $I_C$          |       |             | 3.0    |        | A                  | $I_{EBO}$                                                                    | $V_{EB}=5.0\text{V}$                                       |     |     | 1.0 | mA            |
| Peak Collector Current                                            | $I_{CM}$       |       |             | 5.0    |        | A                  | $BV_{CEO}$                                                                   | $I_C=30\text{mA}$ (TIP31)                                  | 40  |     |     | V             |
| Continuous Base Current                                           | $I_B$          |       |             | 1.0    |        | A                  | $BV_{CEO}$                                                                   | $I_C=30\text{mA}$ (TIP31A)                                 | 60  |     |     | V             |
| Power Dissipation                                                 | $P_D$          |       |             | 40     |        | W                  | $BV_{CEO}$                                                                   | $I_C=30\text{mA}$ (TIP31B)                                 | 80  |     |     | V             |
| Power Dissipation ( $T_A=25^\circ\text{C}$ )                      | $P_D$          |       |             | 2.0    |        | W                  | $BV_{CEO}$                                                                   | $I_C=30\text{mA}$ (TIP31C)                                 | 100 |     |     | V             |
| Operating and Storage Junction Temperature                        | $T_J, T_{stg}$ |       | -65 to +150 |        |        | $^\circ\text{C}$   | $V_{CE(SAT)}$                                                                | $I_C=3.0\text{A}, I_B=375\text{mA}$                        |     |     | 1.2 | V             |
| Thermal Resistance                                                | $\theta_{JA}$  |       | 62.5        |        |        | $^\circ\text{C/W}$ | $V_{BE(ON)}$                                                                 | $V_{CE}=4.0\text{V}, I_C=3.0\text{A}$                      |     |     | 1.8 | V             |
| Thermal Resistance                                                | $\theta_{JC}$  |       | 3.13        |        |        | $^\circ\text{C/W}$ | $h_{FE}$                                                                     | $V_{CE}=4.0\text{V}, I_C=1.0\text{A}$                      | 25  |     |     |               |
|                                                                   |                |       |             |        |        |                    | $h_{FE}$                                                                     | $V_{CE}=4.0\text{V}, I_C=3.0\text{A}$                      | 10  |     | 50  |               |
|                                                                   |                |       |             |        |        |                    | $h_{fe}$                                                                     | $V_{CE}=10\text{V}, I_C=0.5\text{A}, f=1.0\text{kHz}$      | 20  |     |     |               |
|                                                                   |                |       |             |        |        |                    | $f_T$                                                                        | $V_{CE}=10\text{V}, I_C=0.5\text{A}, f=1.0\text{MHz}$      | 3.0 |     |     | MHz           |
|                                                                   |                |       |             |        |        |                    | $t_{on}$                                                                     | $I_C=1.0\text{A}, I_{B1}=I_{B2}=0.1\text{A}, R_L=30\Omega$ |     | 0.3 |     | $\mu\text{s}$ |
|                                                                   |                |       |             |        |        |                    | $t_{off}$                                                                    | $I_C=1.0\text{A}, I_{B1}=I_{B2}=0.1\text{A}, R_L=30\Omega$ |     | 1.0 |     | $\mu\text{s}$ |

# Power Devices

- MOS Transistor
  - IRF640 Datasheet

## Absolute Maximum Ratings

|                                   | Parameter                                           | Max.                  | Units |
|-----------------------------------|-----------------------------------------------------|-----------------------|-------|
| $I_D$ @ $T_C = 25^\circ\text{C}$  | Continuous Drain Current, $V_{GS}$ @ 10 V           | 18                    | A     |
| $I_D$ @ $T_C = 100^\circ\text{C}$ | Continuous Drain Current, $V_{GS}$ @ 10 V           | 11                    |       |
| $I_{DM}$                          | Pulsed Drain Current ①                              | 72                    |       |
| $P_D$ @ $T_C = 25^\circ\text{C}$  | Power Dissipation                                   | 125                   | W     |
|                                   | Linear Derating Factor                              | 1.0                   | W/°C  |
| $V_{GS}$                          | Gate-to-Source Voltage                              | $\pm 20$              | V     |
| $E_{AS}$                          | Single Pulse Avalanche Energy ②                     | 580                   | mJ    |
| $I_{AR}$                          | Avalanche Current ①                                 | 18                    | A     |
| $E_{AR}$                          | Repetitive Avalanche Energy ①                       | 13                    | mJ    |
| $dv/dt$                           | Peak Diode Recovery $dv/dt$ ③                       | 5.0                   | V/ns  |
| $T_J$<br>$T_{STG}$                | Operating Junction and<br>Storage Temperature Range | -55 to +150           | °C    |
|                                   | Soldering Temperature, for 10 seconds               | 300 (1.6mm from case) |       |
|                                   | Mounting Torque, 6-32 or M3 screw                   | 10 lbf•in (1.1 N•m)   |       |

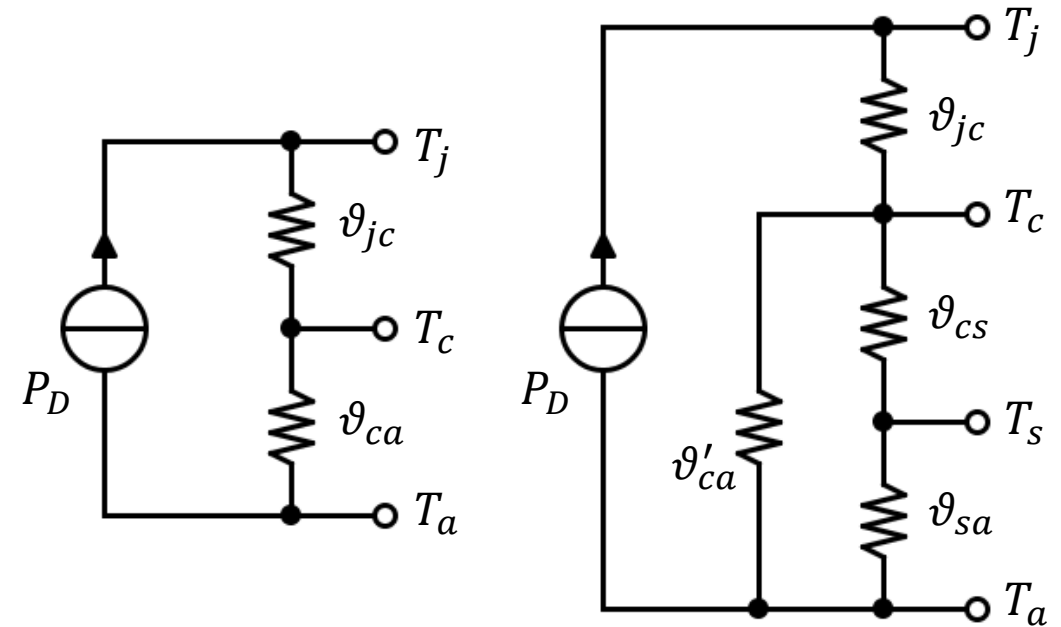
## Thermal Resistance

|                 | Parameter                           | Min. | Typ. | Max. | Units |
|-----------------|-------------------------------------|------|------|------|-------|
| $R_{\theta JC}$ | Junction-to-Case                    | —    | —    | 1.0  | °C/W  |
| $R_{\theta CS}$ | Case-to-Sink, Flat, Greased Surface | —    | 0.50 | —    |       |
| $R_{\theta JA}$ | Junction-to-Ambient                 | —    | —    | 62   |       |

# Power Devices

## • Thermal analysis

- Power dissipation produces heat
  - Heat causes a temperature increase
  - Over a certain temperature the device is damaged
- Electrical model of thermal behavior
  - Power dissipation  $\leftrightarrow$  Current source
  - Temperature difference  $\leftrightarrow$  Voltage drop
  - Thermal resistance  $\leftrightarrow$  Electrical resistance
    - $\vartheta_{jc}$ : between junction and case
    - $\vartheta_{ca}$ : between case and ambient without an heat sink
    - $\vartheta'_{ca}$ : between case and ambient with an heat sink
      - Higher than  $\vartheta_{ca}$  because it considers only the case surface not covered by the heat sink
    - $\vartheta_{cs}$ : between case and heat sink
    - $\vartheta_{sa}$ : between heat sink and ambient



# Power Devices

- Thermal analysis

- Datasheet

- $T_{jmax} = 150\text{ }^{\circ}\text{C}$
    - $P_{D\infty} = 40\text{ W @ } T_a = 25\text{ }^{\circ}\text{C}$ 
      - With ideal heat sink ( $\vartheta_{cs} = \vartheta_{sa} = 0\text{ }^{\circ}\text{C/W}$ )
      - $T_c = T_a = 25\text{ }^{\circ}\text{C}$
    - $P_{D0} = 2\text{ W @ } T_a = 25\text{ }^{\circ}\text{C}$ 
      - Without an heat sink

- Then

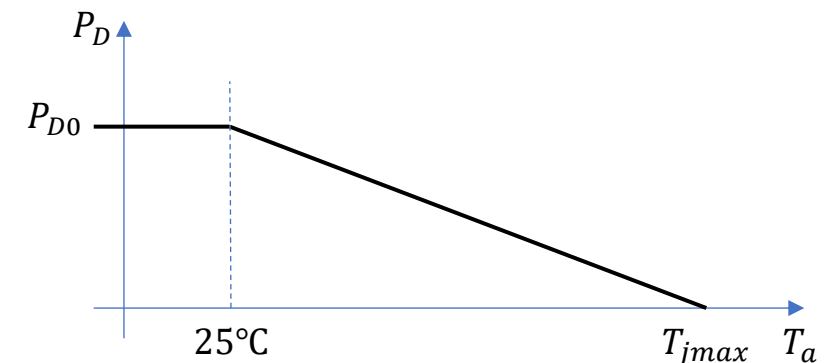
- $\vartheta_{jc} = \frac{T_{jmax} - T_c}{P_{D\infty}} = \frac{T_{jmax} - T_a}{P_{D\infty}} = \frac{150 - 25}{40} = 3.125\text{ }^{\circ}\text{C/W}$
    - $\vartheta_{ca} = \frac{T_{jmax} - T_a}{P_{D0}} - \vartheta_{jc} = \frac{150 - 25}{2} - 3.125 = 59.375\text{ }^{\circ}\text{C/W}$
    - $\vartheta_{ja} = \vartheta_{jc} + \vartheta_{ca} = 3.125 + 59.375 = 62.5\text{ }^{\circ}\text{C/W}$

- Power dissipation derating

|                                                                                              |       |              |                                      |
|----------------------------------------------------------------------------------------------|-------|--------------|--------------------------------------|
| Total Power Dissipation<br>@ $T_C = 25^{\circ}\text{C}$<br>Derate above $25^{\circ}\text{C}$ | $P_D$ | 40<br>0.32   | Watts<br>$\text{W}/^{\circ}\text{C}$ |
| Total Power Dissipation<br>@ $T_A = 25^{\circ}\text{C}$<br>Derate above $25^{\circ}\text{C}$ | $P_D$ | 2.0<br>0.016 | Watts<br>$\text{W}/^{\circ}\text{C}$ |

## THERMAL CHARACTERISTICS

| Characteristic                          | Symbol          | Max   | Unit                 |
|-----------------------------------------|-----------------|-------|----------------------|
| Thermal Resistance, Junction to Ambient | $R_{\theta JA}$ | 62.5  | $^{\circ}\text{C/W}$ |
| Thermal Resistance, Junction to Case    | $R_{\theta JC}$ | 3.125 | $^{\circ}\text{C/W}$ |



# Power Devices

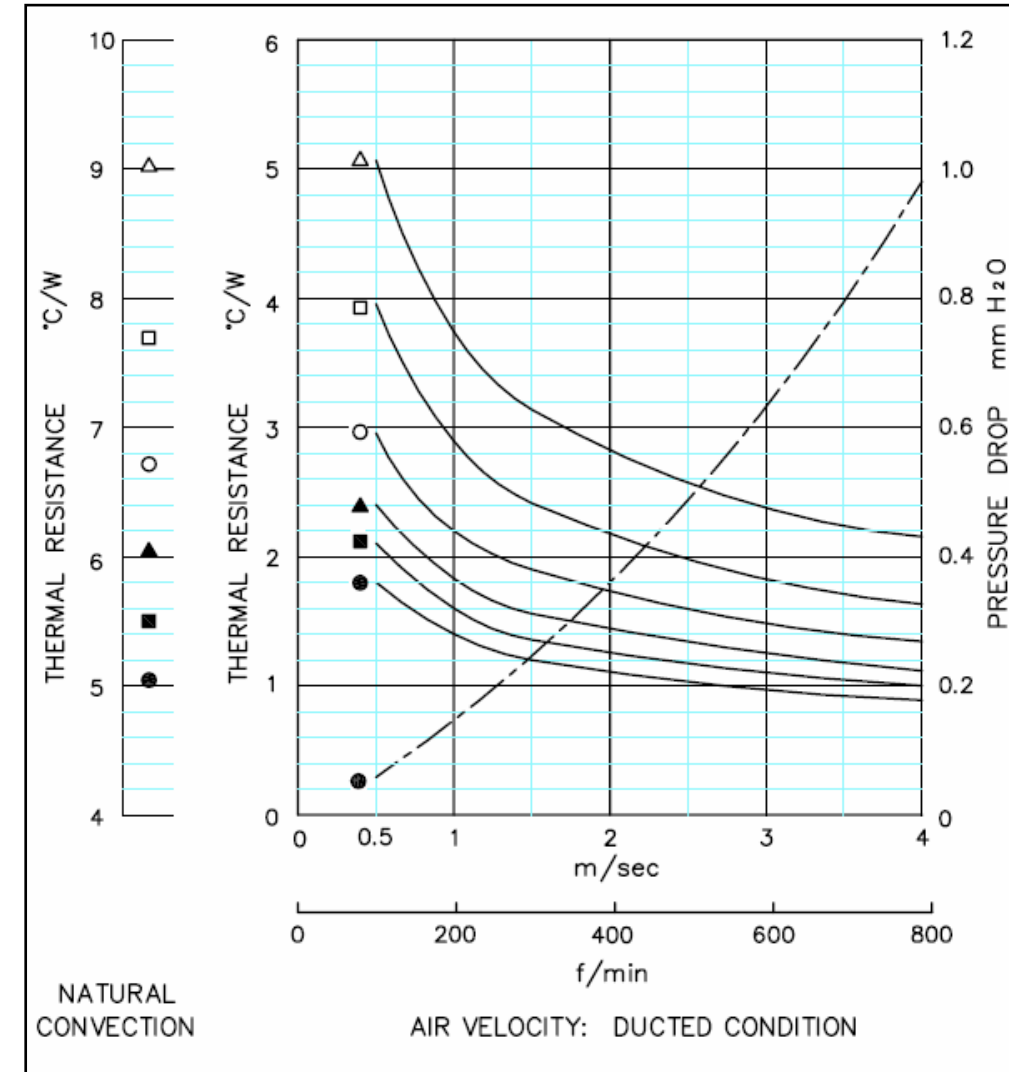
- Thermal analysis

- Design

- Evaluate the maximum power  $P_D$  to be dissipated
  - Depends on voltages and currents in the application
- Evaluate the maximum ambient temperature  $T_{amax}$
- If  $P_D < \frac{T_{jmax}-T_{amax}}{\vartheta_{ja}}$  and  $P_D < P_{D0}$ 
  - There is no need of an heat sink
- Else, neglecting  $\vartheta'_{ca}$ 
  - $\frac{T_{jmax}-T_{amax}}{\vartheta_{jc}+\vartheta_{cs}+\vartheta_{sa}} > P_D \Rightarrow \vartheta_{cs} + \vartheta_{sa} < \frac{T_{jmax}-T_{amax}}{P_D} - \vartheta_{jc}$
  - Find heat sink with proper specifications
  - Possibly using forced ventilation

- Example

- $P_D = 18 \text{ W}$ ,  $T_{amax} = 60 \text{ }^\circ\text{C}$ ,  $\vartheta_{jc} = 3.125 \text{ }^\circ\text{C/W}$
- $\frac{T_{jmax}-T_{amax}}{P_D} - \vartheta_{jc} = 1.875 \text{ }^\circ\text{C/W}$



LPD45 series Heatsink

# Power Devices

- Thermal analysis

- Example

- Power transistor with  $T_{jmax} = 200\text{ °C}$  and  $\vartheta_{jc} = 1.17\text{ °C/W}$
    - It dissipates  $P_D = 20\text{ W}$
    - Ambient temperature is between  $15\text{ °C}$  and  $40\text{ °C}$
    - Heatsink compound with  $\vartheta_{cs} = 0.36\text{ °C/W}$
    - Determine heat sink thermal resistance such that
      - Junction temperature is at least  $\Delta T_{margin} = 60\text{ °C}$  below the maximum
      - The heat sink temperature is at maximum  $T_{smax} = 80\text{ °C}$

- Solution

- $P_D < \frac{(T_{jmax} - \Delta T_{margin}) - T_{amax}}{\vartheta_{ja}} \Rightarrow \vartheta_{ja} < \frac{(T_{jmax} - \Delta T_{margin}) - T_{amax}}{P_D} = 5\text{ °C/W}$ 
      - $\vartheta_{sa} = \vartheta_{ja} - \vartheta_{jc} - \vartheta_{cs} = 3.47\text{ °C/W}$  (it should be less than this)
    - $T_{smax} = T_{amax} + P_D \cdot \vartheta_{sa} < 80\text{ °C} \Rightarrow \vartheta_{sa} < \frac{T_{smax} - T_{amax}}{P_D} = 2\text{ °C/W}$

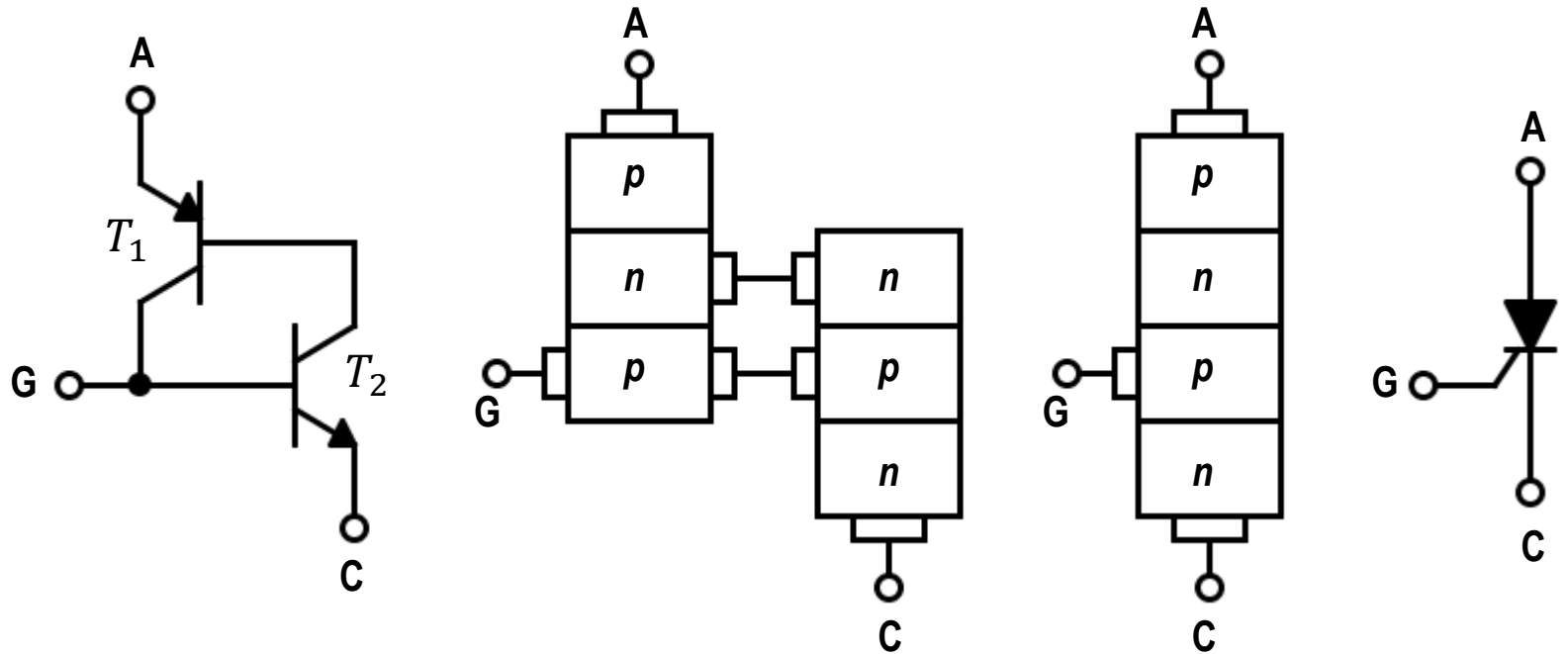
# Power Devices

- Silicon Controlled Rectifier (SCR)

- Two interconnected BJTs
- Three terminals
  - Gate
  - Anode
  - Cathode

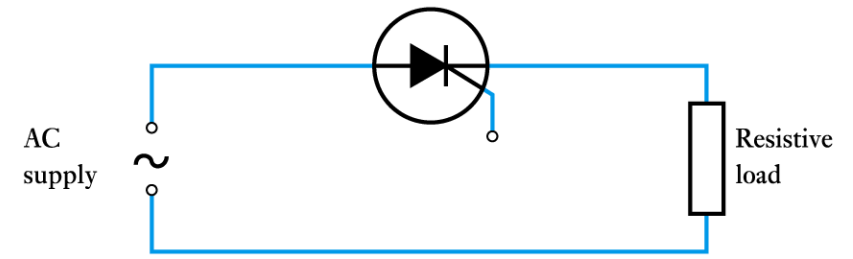
- Behavior

- Drive the gate
- Turn on  $T_2$ 
  - Provides base current to  $T_1$
- $T_1$ , turned on, provides base current to  $T_2$
- Both transistors remain ON, regardless of what the gate does after the initial trigger
- When the current goes to 0 because of what happens in the rest of the circuit
  - Both transistors turn OFF

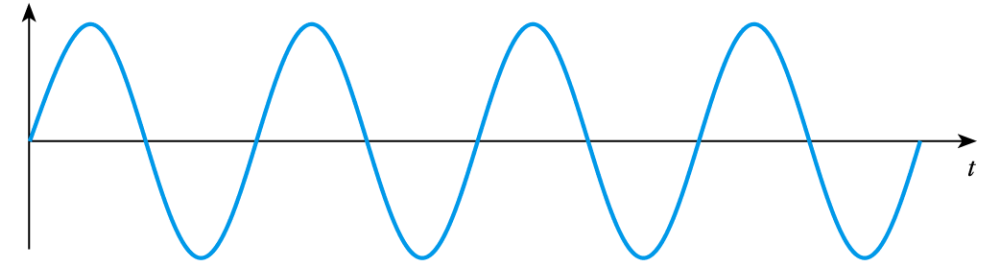


# Power Devices

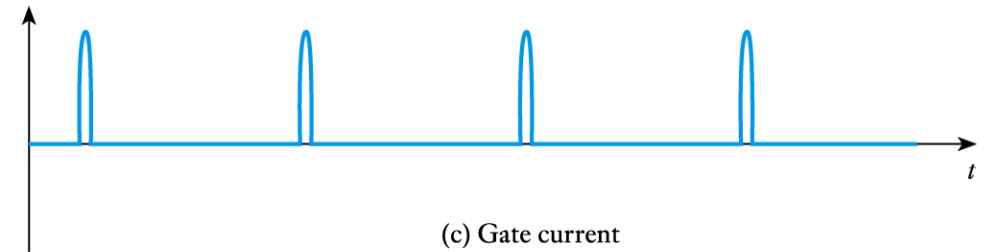
- Silicon Controlled Rectifier (SCR)
  - Also known as Thyristor
  - Power flows from Anode to Cathode
  - Gate is the control input
  - Used to control power on a load
    - Depending on firing instant
    - From 0 to 50%
  - Single direction



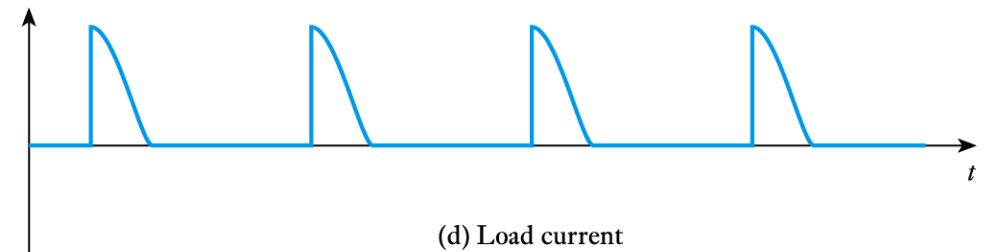
(a) Circuit



(b) AC supply



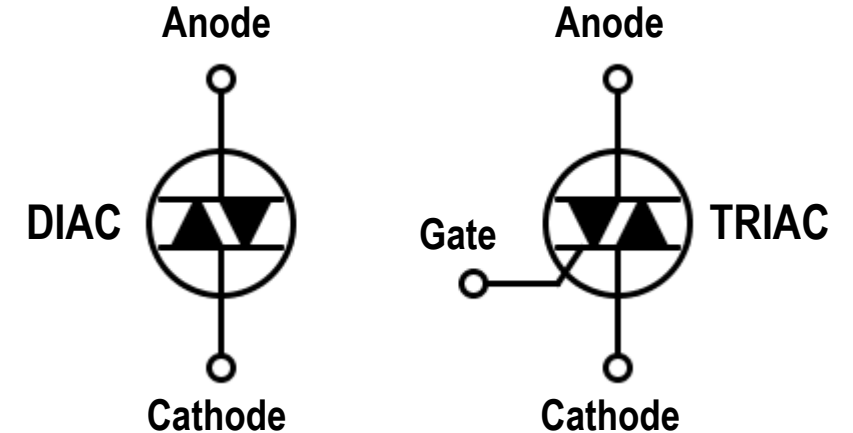
(c) Gate current



(d) Load current

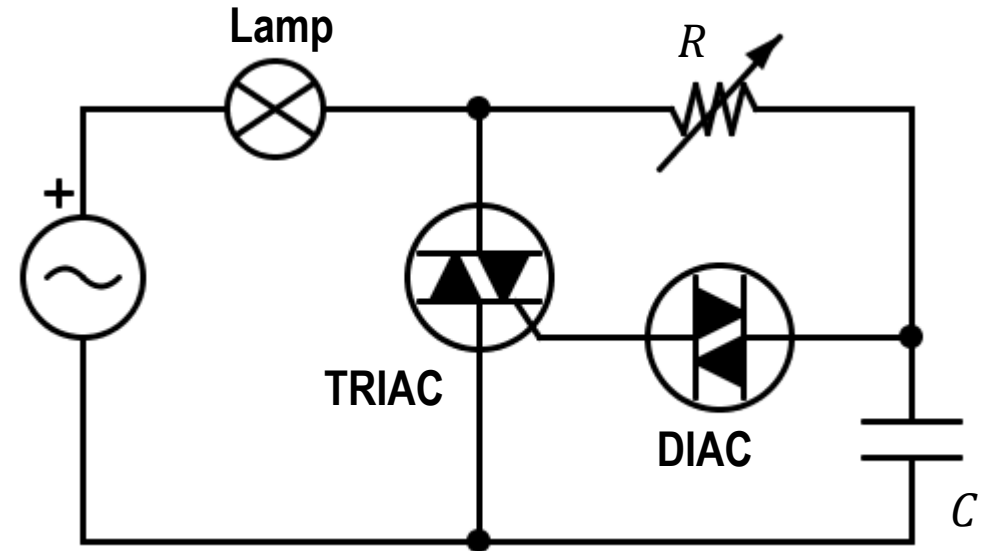
# Power Devices

- DIAC
  - Bidirectional
  - When voltage is higher than a threshold
    - It starts conducting
  - It remains in conduction until the voltage drops way below the threshold
- TRIAC
  - Bidirectional SCR
  - A solid state AC switch
  - Often triggered by a DIAC



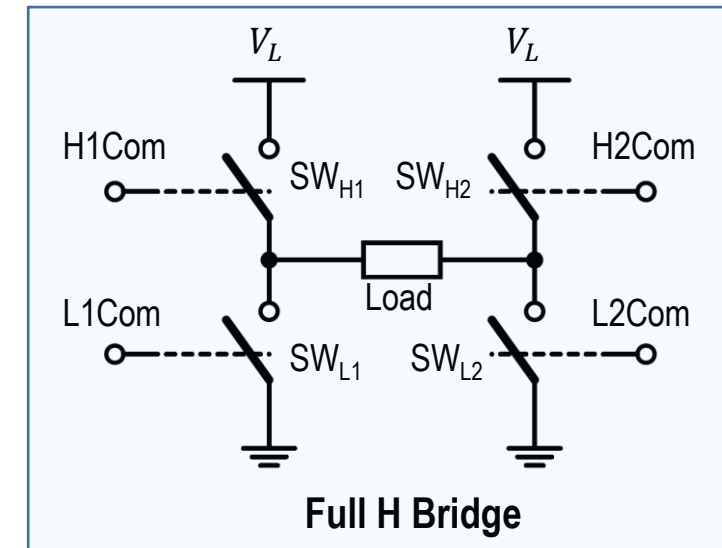
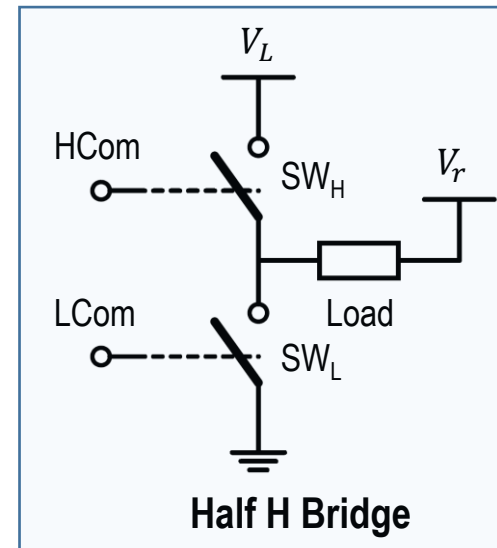
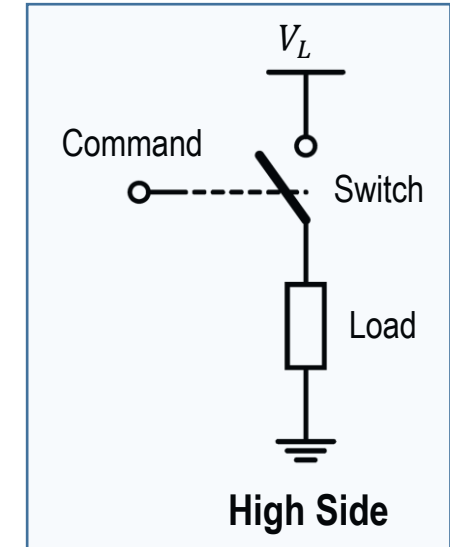
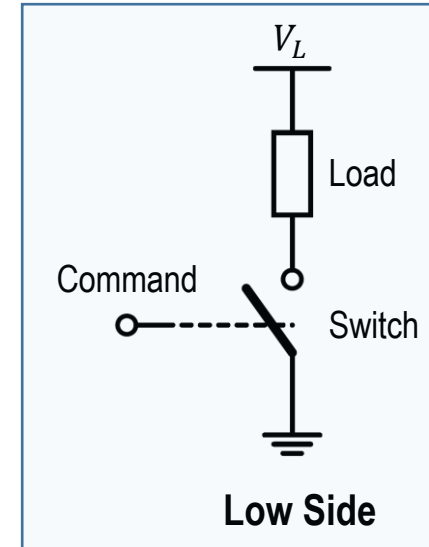
# Power Devices

- TRIAC example
  - Lamp dimmer
  - Variable resistor and capacitor are a phase shift network
    - They provide the firing voltage to the DIAC
    - In turn, the DIAC triggers the TRIAC
      - It remains ON until the voltage drops to 0
    - Varying the phase shift changes the amount of time in a period that the lamp stays on
      - From full brightness
      - To total darkness
    - Little power consumption
      - Using a voltage divider would be highly inefficient



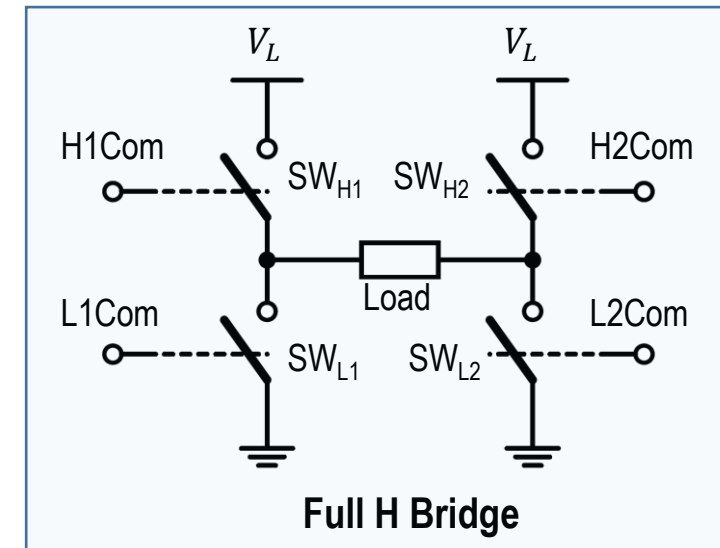
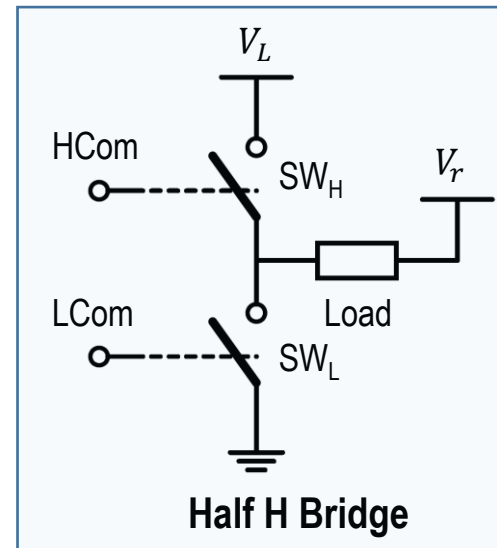
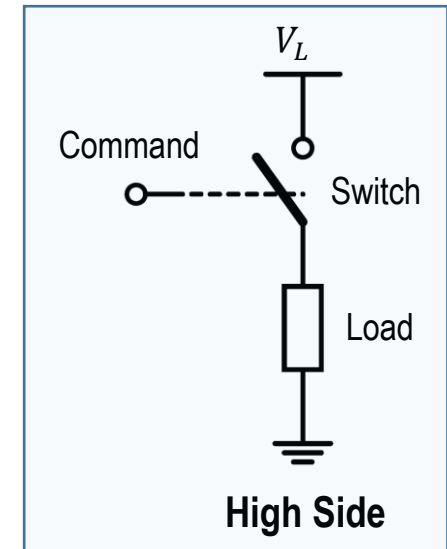
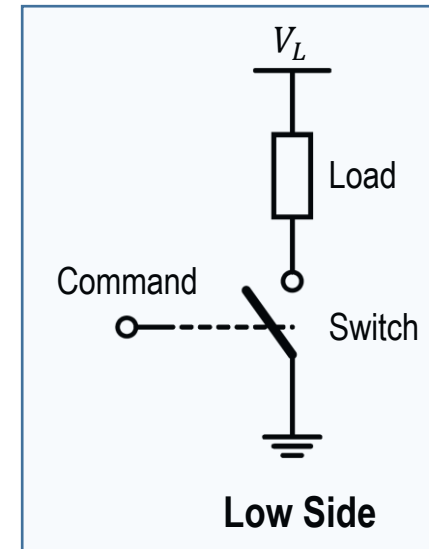
# Actuator driving

- Why do we want actuator driving?
  - Load requires high power
    - High current  $I_L$
    - High voltage  $V_L$
  - Digital circuits cannot provide the high power
    - Low voltages and currents
    - They even break at high power
  - Voltage and current provided to the load by a separate voltage supply
    - Power modulated using switches
      - For instance using PWM
      - Switches do not consume static power
    - Switches driven by low power commands
      - Commands given by digital circuits



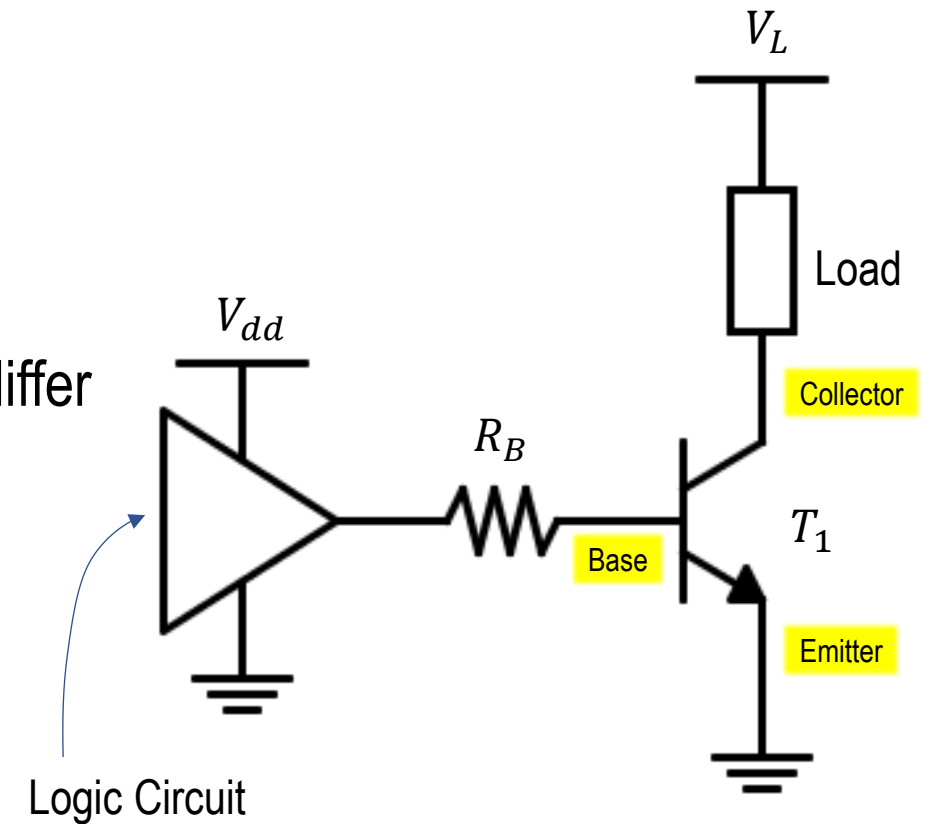
# Actuator driving

- Low side
  - Load connected to  $V_{dd}$ , switch to GND
  - Not all loads can be driven
  - Current always in the same direction
- High side
  - Load connected to GND, switch to  $V_{dd}$
  - Current always in the same direction
  - Problems controlling the switch
- H-Bridge
  - Current in both directions
  - Load floating or connected to a reference voltage (half bridge)



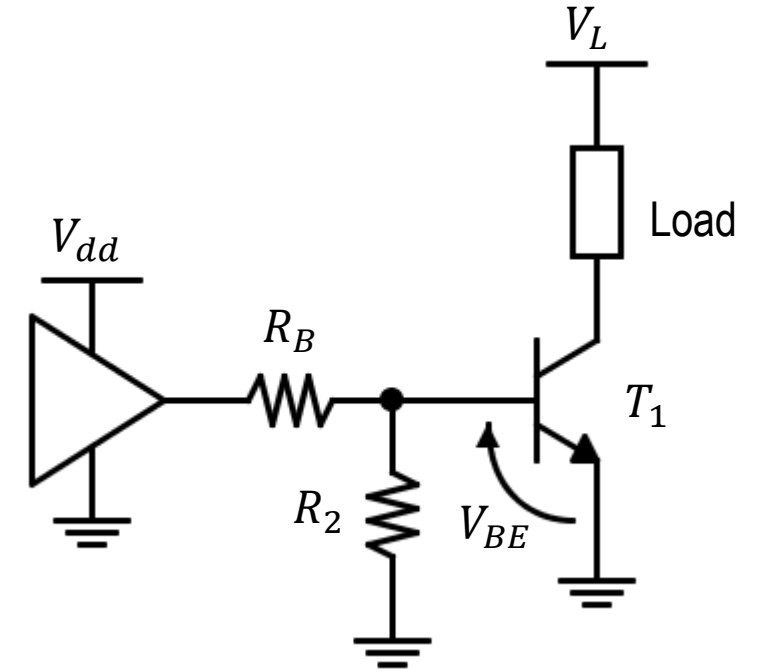
# Actuator driving

- Low side driving, npn BJT
  - Common emitter configuration
    - Load on collector
    - Base current to control the switch
  - Voltage supplies for the load and the digital circuit may differ
    - $V_L$  often higher than  $V_{dd}$
  - Transistor must withstand load voltages and currents
    - See power devices characteristics
  - Transistor must be driven in cut-off or saturation
    - Cut-off is an open switch
    - Saturation is a closed switch



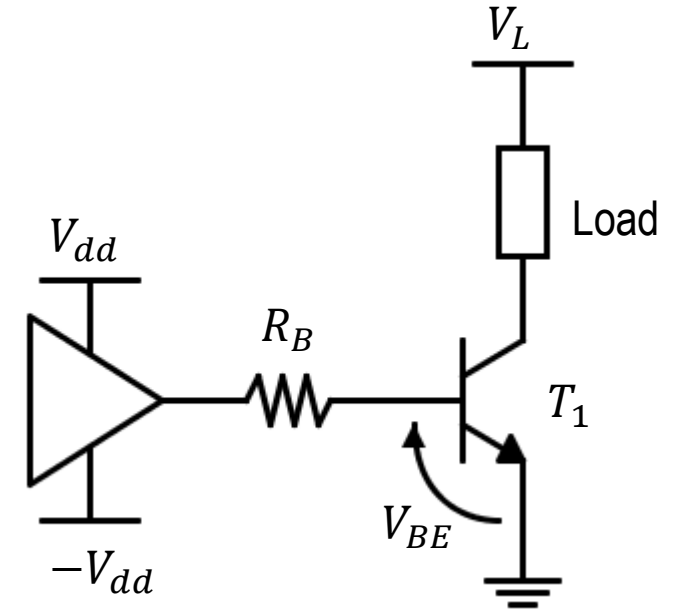
# Actuator driving

- Low side driving, npn BJT
  - Transistor OFF
    - Force  $V_{BE} < V_{BEon}$
  - Common values
    - $V_{BEon} = 0.4\text{ V}$
    - Power transistors have higher  $V_{BEon}$
  - Many logic families have  $V_{OL} < V_{BEon}$ 
    - A low logic state at the output of the digital circuit drives the transistor in cut-off region
    - When digital output is tri-stated or not yet initialized
      - Connect base to GND using optional resistor  $R_2$
      - The resistor value is not critical, but should be at least 10 times higher than  $R_B$ 
        - Can be neglected in the following computations
          - $I_{R_2} \ll I_{R_B}$
      - $R_2$  also useful to steer  $I_{cb0}$  away from the base and avoid thermal runaway



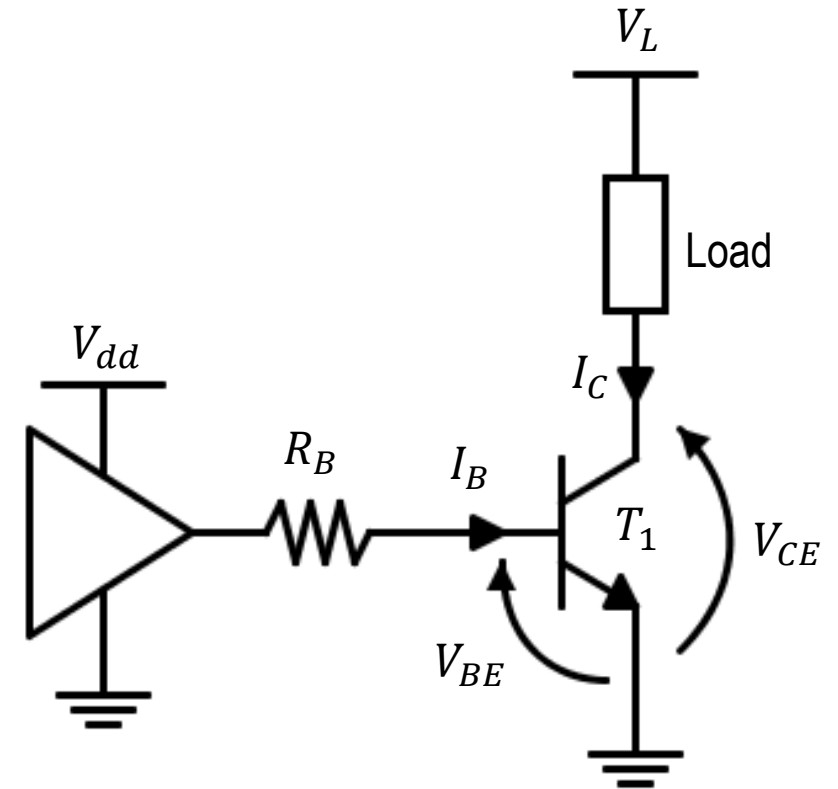
# Actuator driving

- Low side driving, npn BJT
  - Transistor OFF
    - May also reverse bias the BE junction
      - $V_B < 0$
    - Need logic circuits with negative power supply
      - As well as positive
    - Decrease leakage currents
    - Be careful not to breakdown the BE junction



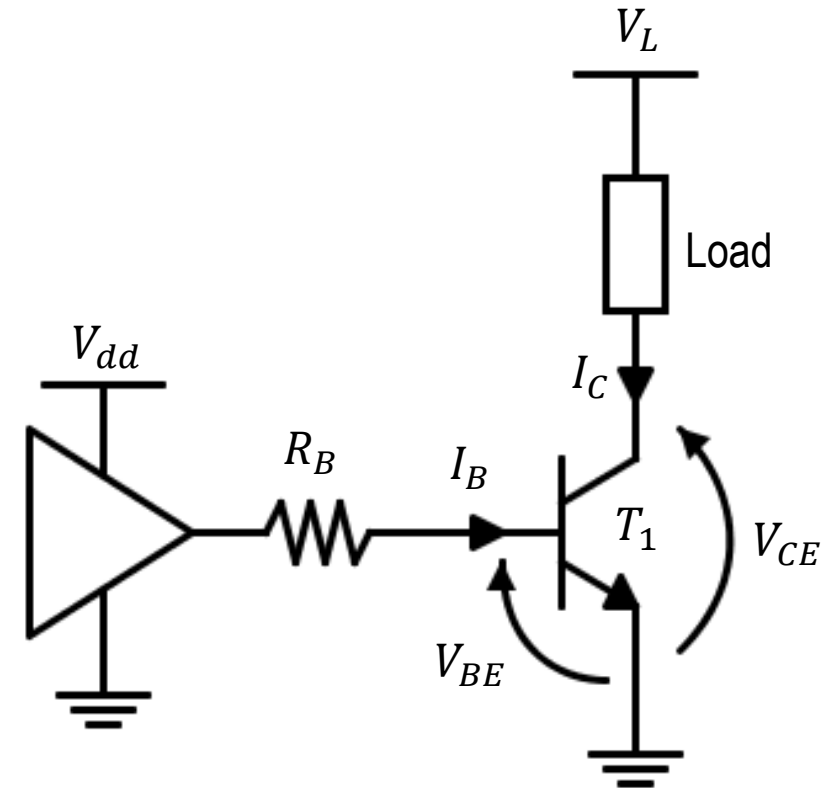
# Actuator driving

- Low side driving, npn BJT
  - Transistor ON
    - Main relationship in linear region
      - $I_C = \beta I_B$
    - To force saturation
      - $I_B \gg \frac{I_C}{\beta}$
      - $\beta_f = \frac{I_C}{I_B}$  (forced  $\beta$ )
    - When in saturation
      - $V_{CE} = V_{CEsat} = 0.2 \div 0.3 \text{ V}$
      - $V_L$  is almost completely applied to the load
      - The load determines  $I_C$ , not the transistor



# Actuator driving

- Low side driving, npn BJT
  - Transistor ON
    - When in saturation
      - $V_{BE} = V_{BEsat} = 0.8 \div 1 \text{ V}$
      - $I_B$  depends on (ignoring  $R_2$ , not shown)
        - $V_{OH}$  of the digital circuit
        - Base resistance  $R_B$
        - $I_B = \frac{V_{OH} - V_{BEsat}}{R_B}$
    - To force saturation
      - $\frac{V_{OH} - V_{BEsat}}{R_B} \gg \frac{I_C}{\beta} \Rightarrow R_B \ll \frac{\beta}{I_C} (V_{OH} - V_{BEsat})$
      - Upper bound for the base resistance  $R_B$



# Actuator driving

- Low side driving, npn BJT

- Transistor ON

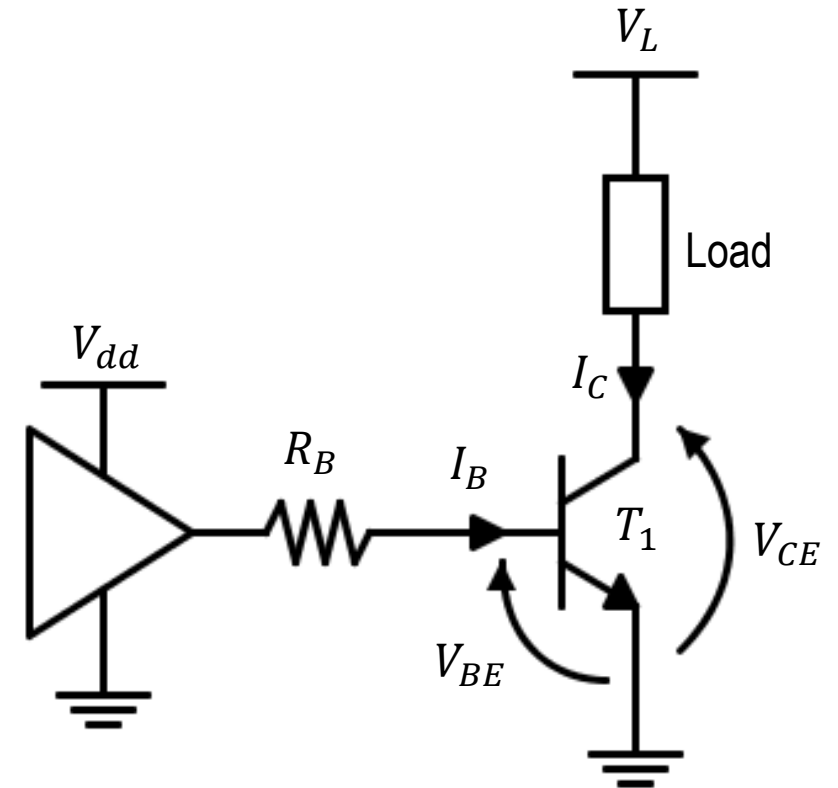
- $I_B$  should not exceed the maximum output current  $I_{OH}$  of the digital circuit

- $I_B < I_{OH} \Rightarrow \frac{V_{OH} - V_{BEsat}}{R_B} < I_{OH} \Rightarrow R_B > \frac{V_{OH} - V_{BEsat}}{I_{OH}}$

- Lower bound for the base resistance  $R_B$

- If a valid interval for  $R_B$  exists

- The circuit works!
    - Higher  $R_B$  (towards the upper bound)
      - Lower power consumption in the base mesh
      - Weaker saturation
    - Lower  $R_B$  (towards the lower bound)
      - Higher power consumption in the base mesh
      - Stronger saturation



# Actuator driving

- Low side driving, npn BJT

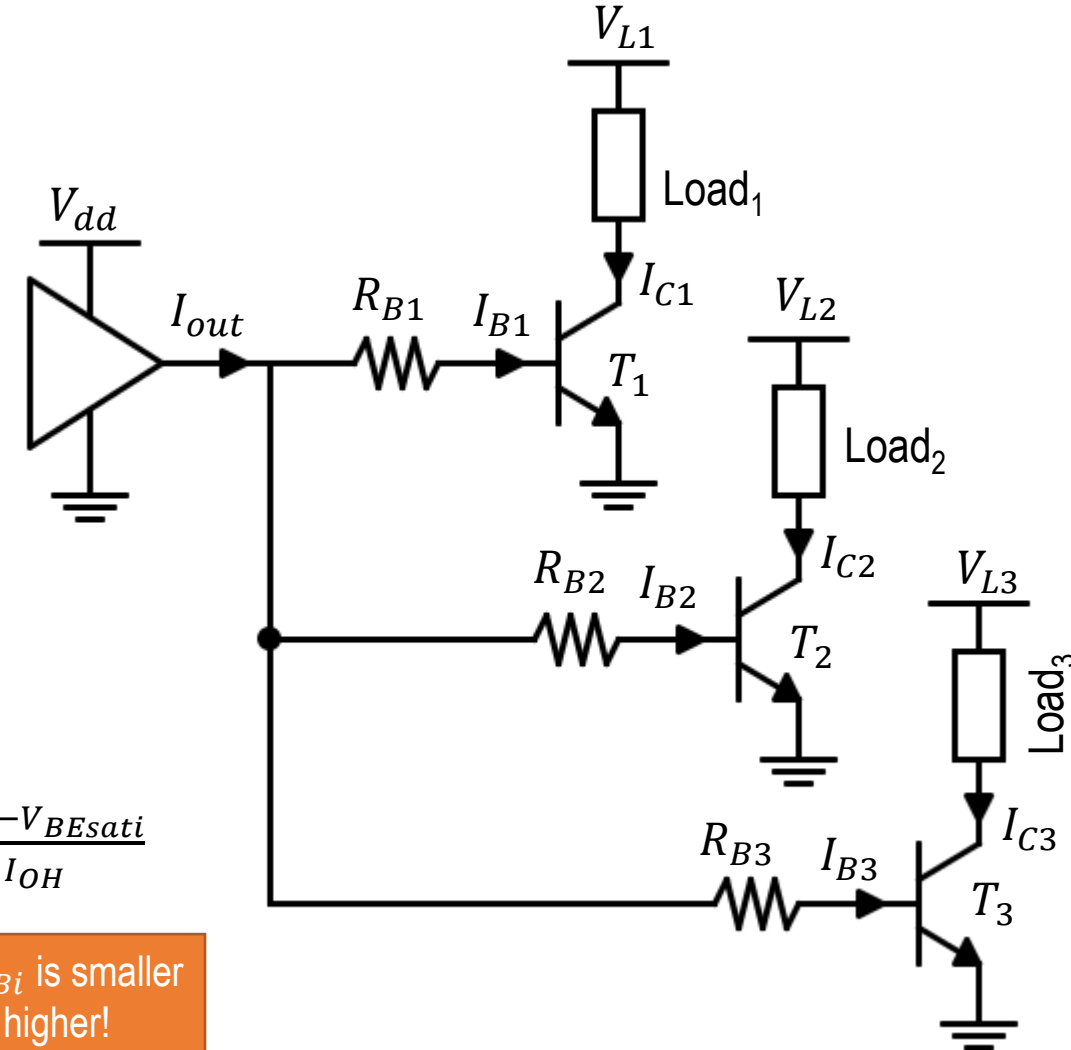
- Transistor ON

- If the digital circuit controls in parallel more than one low side transistor
  - The output current  $I_{out}$  of the digital circuit is divided among the multiple low side transistors
- The sum of all the base currents  $I_{Bi}$  should not exceed the maximum output current  $I_{OH}$  of the digital circuit
  - If  $N$  low side transistors are controlled by the digital circuit
  - If the current  $I_{out}$  is evenly divided among all low side transistors

- $I_{Bi} < \frac{I_{OH}}{N} \Rightarrow \frac{V_{OH} - V_{BEsati}}{R_{Bi}} < \frac{I_{OH}}{N} \Rightarrow R_{Bi} > N \frac{V_{OH} - V_{BEsati}}{I_{OH}}$

- $\Rightarrow \sum_{i=1}^N I_{Bi} < I_{OH}$

The valid range for  $R_{Bi}$  is smaller as the lower bound is higher!



# Actuator driving

- Low side driving, npn BJT
  - Example
    - Lamp:  $V_L = 12\text{ V}$ ,  $P_L = 2\text{ W}$
    - Transistor:  $\beta = 100$ ,  $V_{CEsat} = 0.3\text{ V}$ ,  $V_{BEsat} = 0.8\text{ V}$
    - Logic:  $V_{dd} = 5\text{ V}$ ,  $V_{OH} = 4.7\text{ V}$ ,  $I_{OH} = 10\text{ mA}$
    - Equivalent lamp resistance
      - $R_L = \frac{V_L}{I_L} = \frac{V_L}{P_L/V_L} = \frac{12}{2/12} = 72\ \Omega$
    - Current at full power
      - $I_{Lmax} = \frac{P_L}{V_L} = \frac{2}{12} = 167\text{ mA}$
    - Current when transistor is in saturation
      - $I_C = \frac{V_L - V_{CEsat}}{R_L} = \frac{12 - 0.3}{72} = 162.5\text{ mA}$ ,  $I_C \simeq I_{Lmax}$
      - Using  $I_{Lmax}$  instead of  $I_C$  in computing the  $R_B$  upper bound is conservative
        - It makes the upper bound slightly lower

# Actuator driving

- Low side driving, npn BJT

- Example

- Lamp:  $V_L = 12\text{ V}$ ,  $P_L = 2\text{ W}$ ,  $I_{Lmax} = 167\text{ mA}$
- Transistor:  $\beta = 100$ ,  $V_{CEsat} = 0.3\text{ V}$ ,  $V_{BEsat} = 0.8\text{ V}$
- Logic:  $V_{dd} = 5\text{ V}$ ,  $V_{OH} = 4.7\text{ V}$ ,  $I_{OH} = 10\text{ mA}$

- Range of  $R_B$

- $R_B \ll \frac{\beta}{I_C} (V_{OH} - V_{BEsat}) = \frac{100}{167 \cdot 10^{-3}} (4.7 - 0.8) = 2335\ \Omega$
- $R_B > \frac{V_{OH} - V_{BEsat}}{I_{OH}} = \frac{5 - 0.8}{10 \cdot 10^{-3}} = 420\ \Omega$ 
  - Using  $V_{dd}$  instead of  $V_{OH}$  in computing the lower bound is conservative
    - It makes the lower bound slightly higher

- Value of  $R_B$

- $R_B = 470\ \Omega$ ,  $I_B = \frac{V_{OH} - V_{BEsat}}{R_B} = \frac{5 - 0.8}{470} \simeq 9\text{ mA}$ ,  $\beta_f = \frac{I_C}{I_B} = \frac{167 \cdot 10^{-3}}{9 \cdot 10^{-3}} = 18.5$
- $R_B = 1\text{ k}\Omega$ ,  $I_B = \frac{V_{OH} - V_{BEsat}}{R_B} = \frac{5 - 0.8}{1000} \simeq 4.2\text{ mA}$ ,  $\beta_f = \frac{I_C}{I_B} = \frac{167 \cdot 10^{-3}}{4.2 \cdot 10^{-3}} = 40$

# Actuator driving

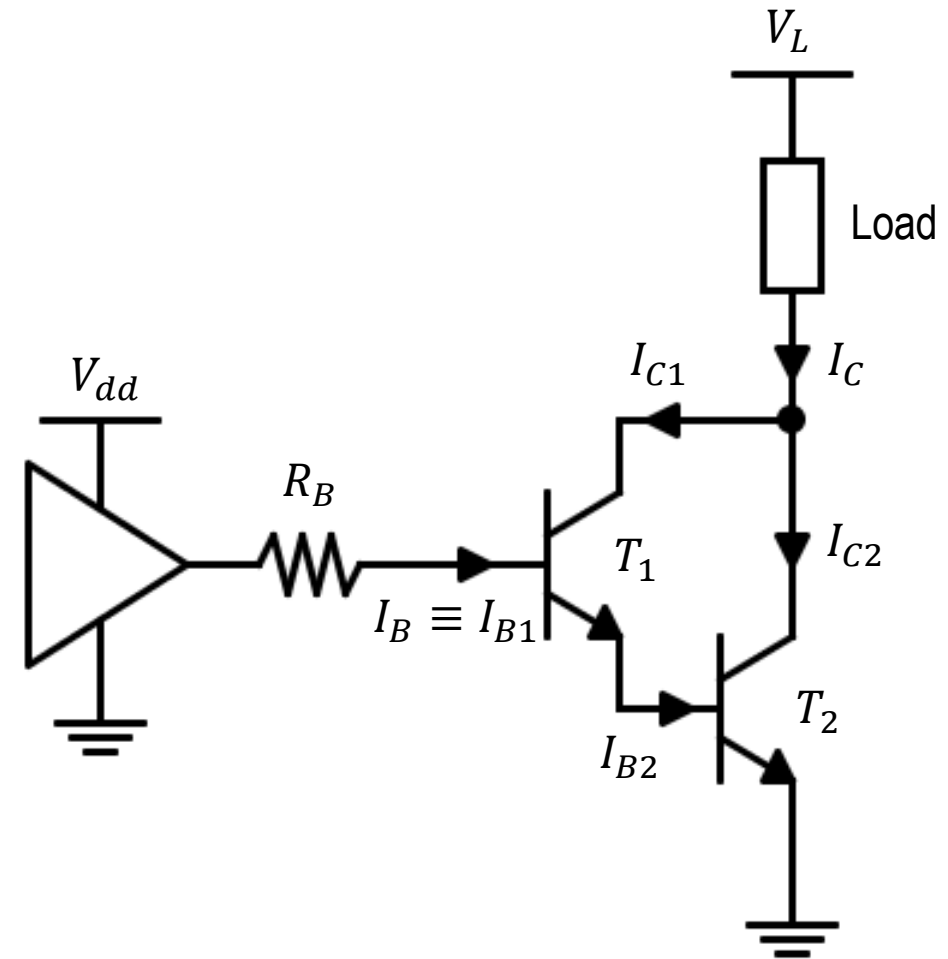
- Low side driving, npn BJT

- Example

- Lamp:  $V_L = 12\text{ V}$ ,  $P_L = 6\text{ W}$ ,  $R_L = 24\ \Omega$ ,  $I_{Lmax} = 500\text{ mA}$
- Transistor:  $\beta = 100$ ,  $V_{CEsat} = 0.3\text{ V}$ ,  $V_{BEsat} = 0.8\text{ V}$
- Logic:  $V_{dd} = 5\text{ V}$ ,  $V_{OH} = 4.7\text{ V}$ ,  $I_{OH} = 10\text{ mA}$
- Range of  $R_B$ 
  - $R_B \ll \frac{\beta}{I_C} (V_{OH} - V_{BEsat}) = \frac{100}{500 \cdot 10^{-3}} (4.7 - 0.8) = 780\ \Omega$
  - $R_B > \frac{V_{OH} - V_{BEsat}}{I_{OH}} = \frac{5 - 0.8}{10 \cdot 10^{-3}} = 420\ \Omega$
  - Range is much smaller
    - The transistor never really reaches saturation
    - If  $P_L = 12\text{ W}$  there is no valid range
- Solutions if the range is not valid (or too small)
  - Increase  $\beta$
  - Increase  $I_{OH}$

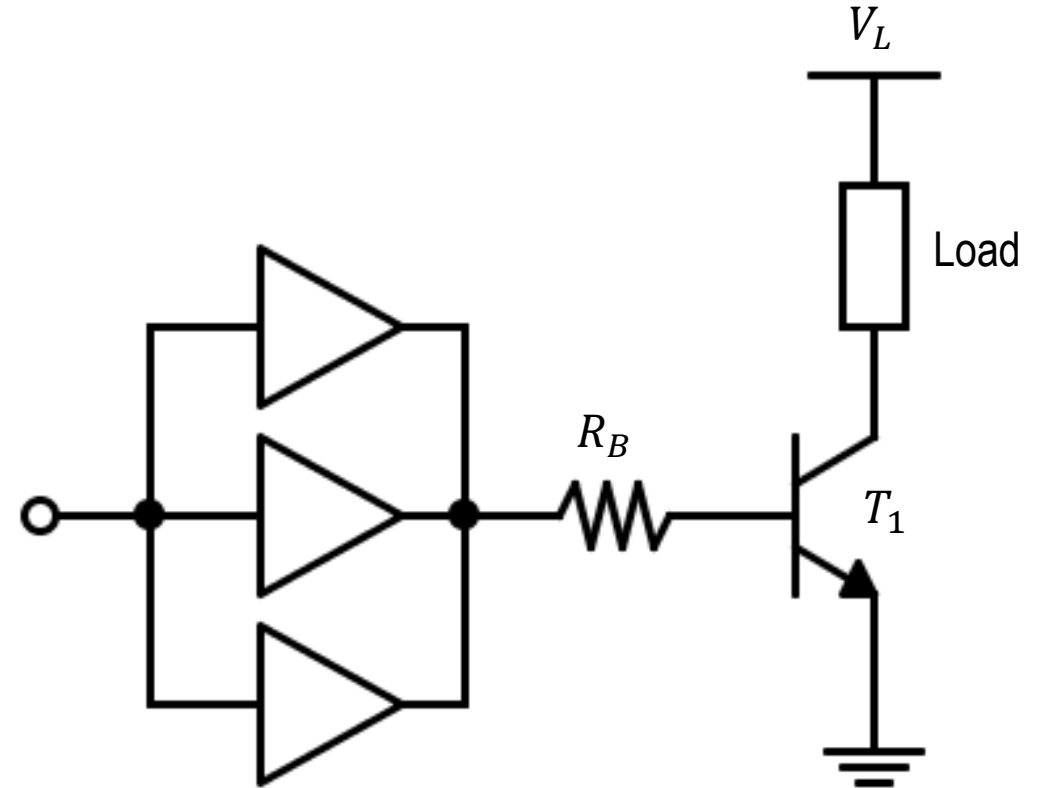
# Actuator driving

- Low side driving, npn BJT
  - Darlington transistor
    - Increase  $\beta$ 
      - $I_{B2} = \beta_1 I_{B1}$
      - $I_{C2} = \beta_2 I_{B2} = \beta_2 \beta_1 I_{B1}$
      - $I_C \simeq I_{C2}$
      - $\beta_{equiv} = \frac{I_C}{I_B} = \frac{I_{C2}}{I_{B1}} = \beta_2 \beta_1$
    - Available as three terminal transistor
      - In a single package
    - Higher  $V_{CEsat}$
    - Higher  $V_{BEsat}$
    - Higher costs



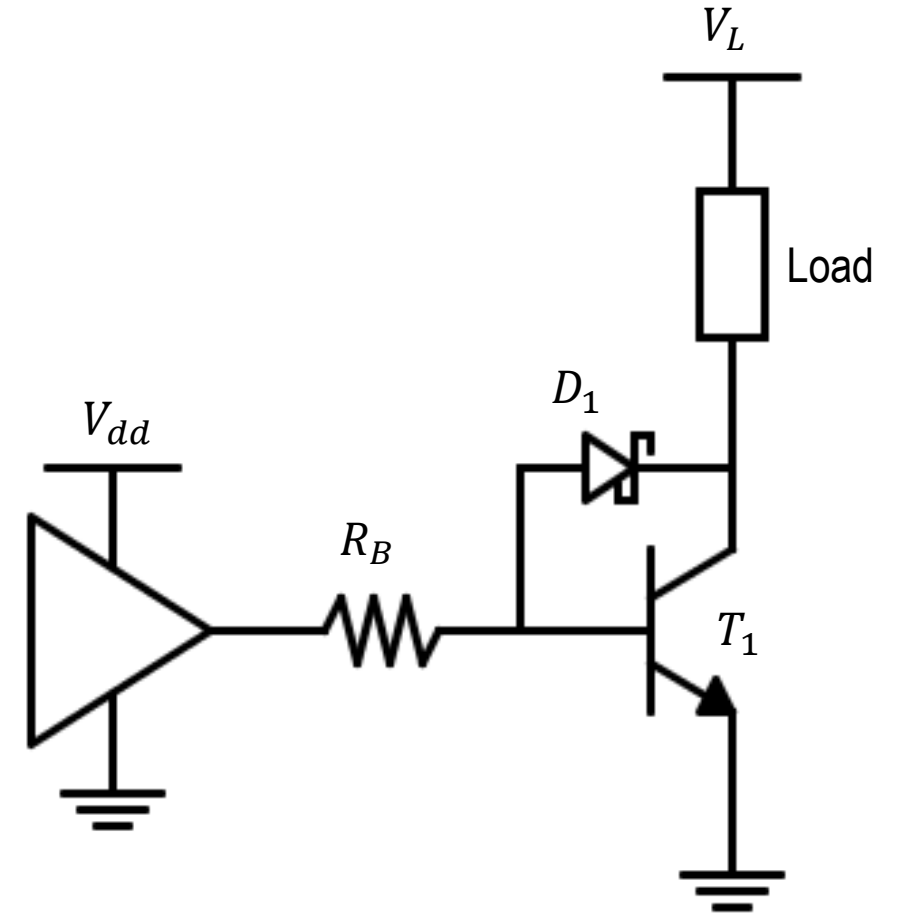
# Actuator driving

- Low side driving, npn BJT
  - Multiple driving logic gates
    - Increase  $I_{OH}$
    - Parallel outputs provide additional current
      - $I_{OHequiv} = n \cdot I_{OH}$
  - Higher costs
    - More complex to design
    - Avoid conflicts due to different delays



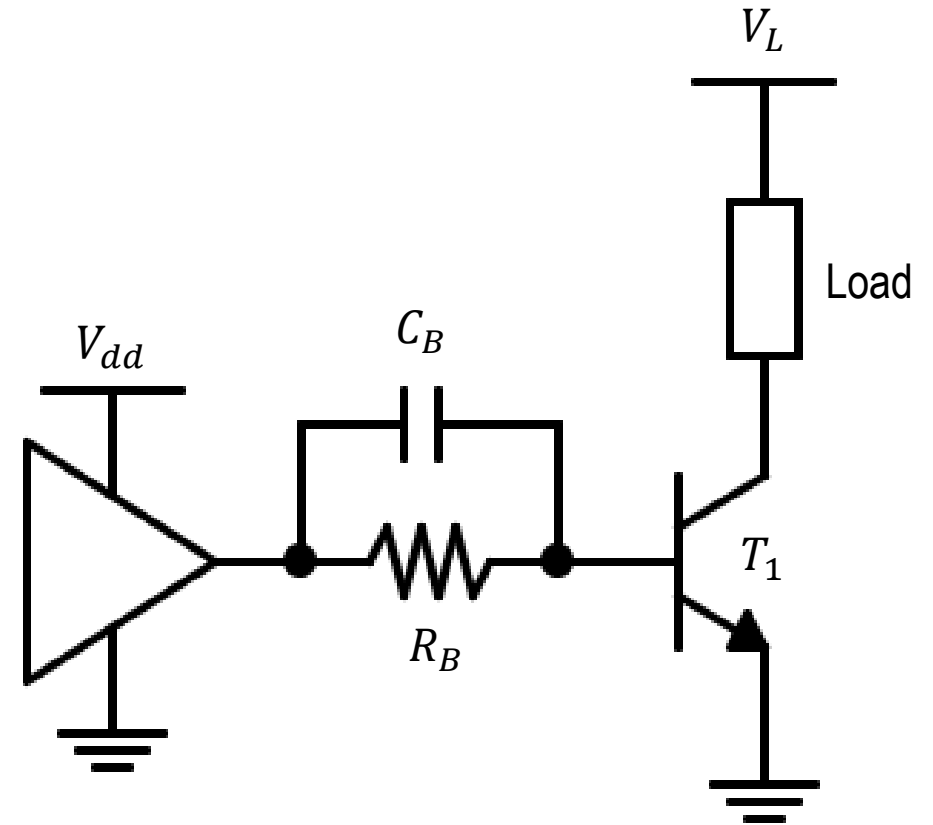
# Actuator driving

- Low side driving, npn BJT
  - Dynamic behavior
    - Switching OFF a transistor from saturation can take a long time
    - Reverse recovery time = storage time + transfer time
      - $t_{rr} = t_s + t_{rt}$
  - Solution
    - Use a Schottky diode between base and collector to avoid saturating the BC junction
    - No charges to remove
  - Schottky transistor
    - Transistor plus diode in a single package



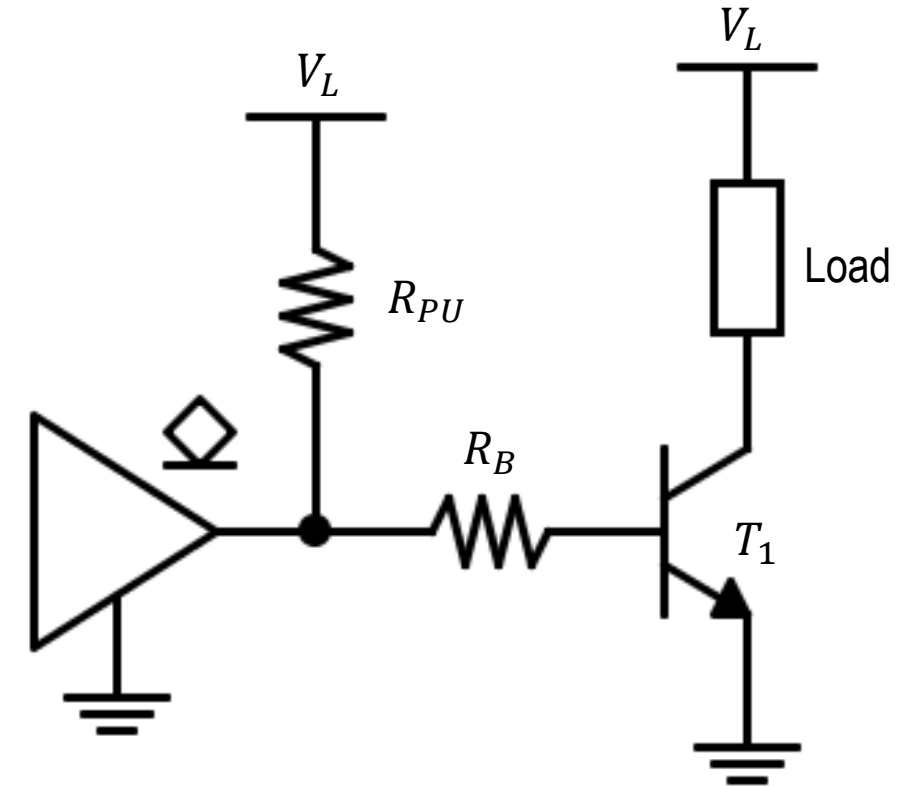
# Actuator driving

- Low side driving, npn BJT
  - Dynamic behavior
    - Switching OFF a transistor from saturation can take a long time
    - Increase base current during transient
      - Reduce base resistor using a parallel capacitor
        - During transient, it acts as a short circuit
        - Afterwards, it becomes an open circuit
      - Current may be limited by the logic gate maximum output current



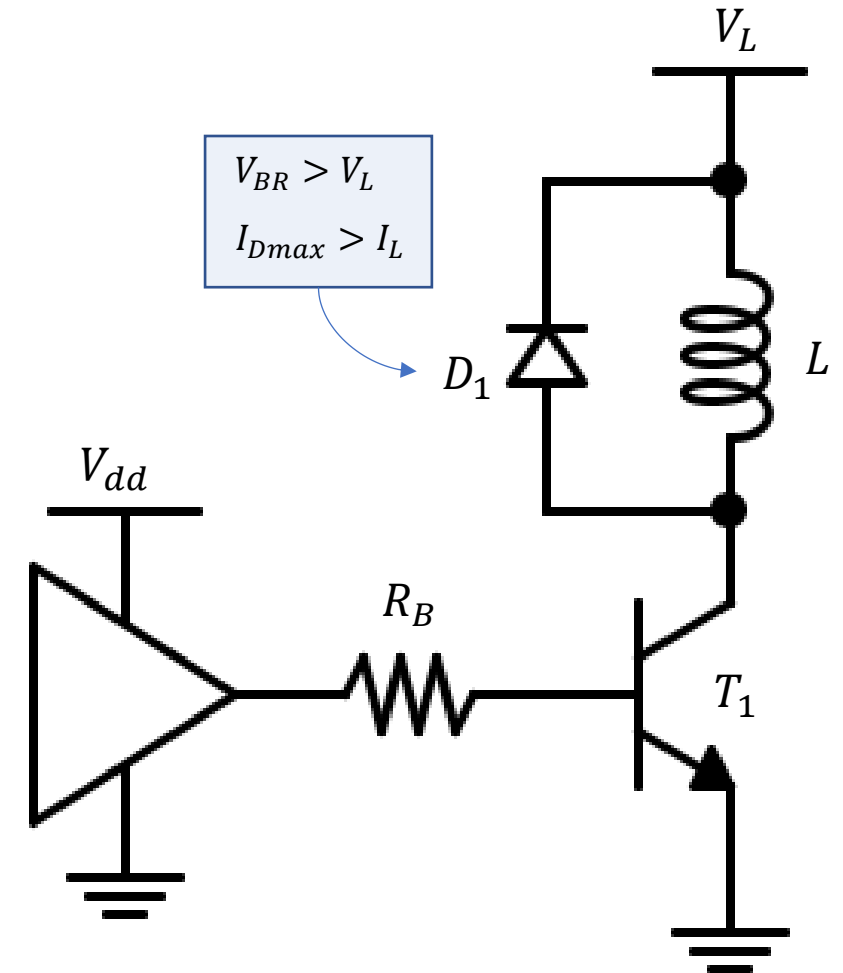
# Actuator driving

- Low side driving, npn BJT
  - Digital circuit with open collector output
    - Use a pull-up resistor to set the logic high voltage
    - $R_{PU}$  should be high to limit the current when digital output is low
      - $\frac{V_L}{R_{PU}} < I_{OL} \Rightarrow R_{PU} > \frac{V_L}{I_{OL}}$
    - When the digital output is high
      - The digital circuit is in high impedance
      - $R_{PU}$  is in series with  $R_B$
      - $R_{PU}$  should be low to force saturation
        - $R_{PU} + R_B \ll \frac{\beta}{I_C} (V_L - V_{BEsat})$
    - Two competing requirements, especially with low beta
    - Easier if  $R_{PU}$  is connected to  $V_L$  rather than  $V_{dd}$ 
      - When  $V_L$  is higher than  $V_{dd}$
    - Check that the logic circuit tolerates high voltages



# Actuator driving

- Low side driving, npn BJT
  - Inductive load
    - Motors, relays and loudspeakers are inductive
    - Voltage is the derivative of current
      - $v(t) = L \frac{di(t)}{dt}$
  - When load is switched OFF
    - A sudden drop in current causes a high voltage spike
    - Might damage the transistor (exceeds  $BV_{CE0}$ )
    - Use a diode to recirculate and dampen the current
      - Diode is OFF (reverse biased) when load is ON
      - Optional resistor is series with diode
      - The diode is also known as:
        - Flywheel diode
        - Freewheeling diode

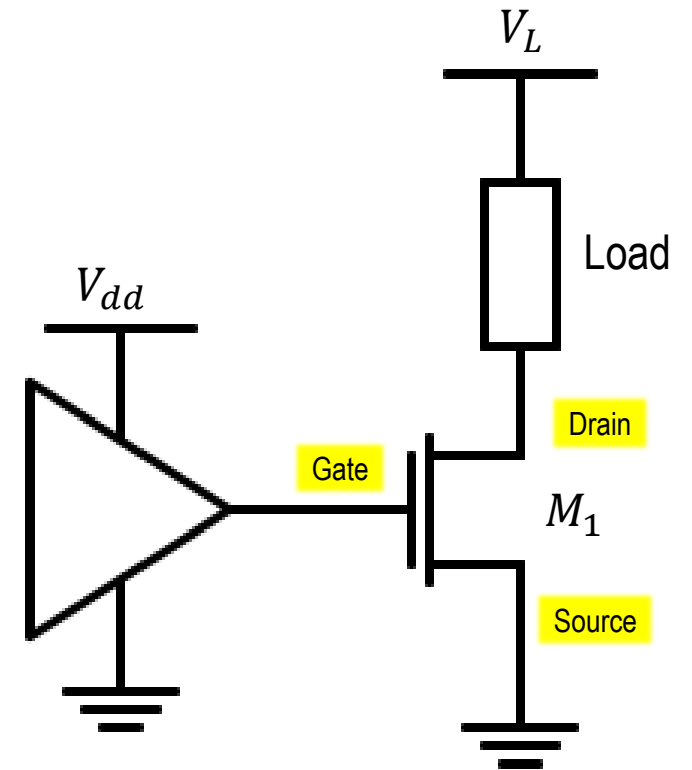


# Actuator driving

- Low side driving, npn BJT
  - Summary
    - Choose transistor based on
      - Maximum  $V_{CE}$  voltage ( $BV_{CE0}$ ) higher than load supply voltage
      - Maximum  $I_C$  current higher than maximum load current
      - Current gain  $\beta$  high enough to drive the transistor using digital circuits
        - Compute range for  $R_B$  and choose value based on other requirements (power, speed, ...)
    - To increase breakdown voltages
      - Low doping, larger base
      - Power transistors have low beta
      - Difficult to reach saturation with small base currents provided by digital circuits
    - Transistor dissipates power during transient
      - Check that it remains in the SOA

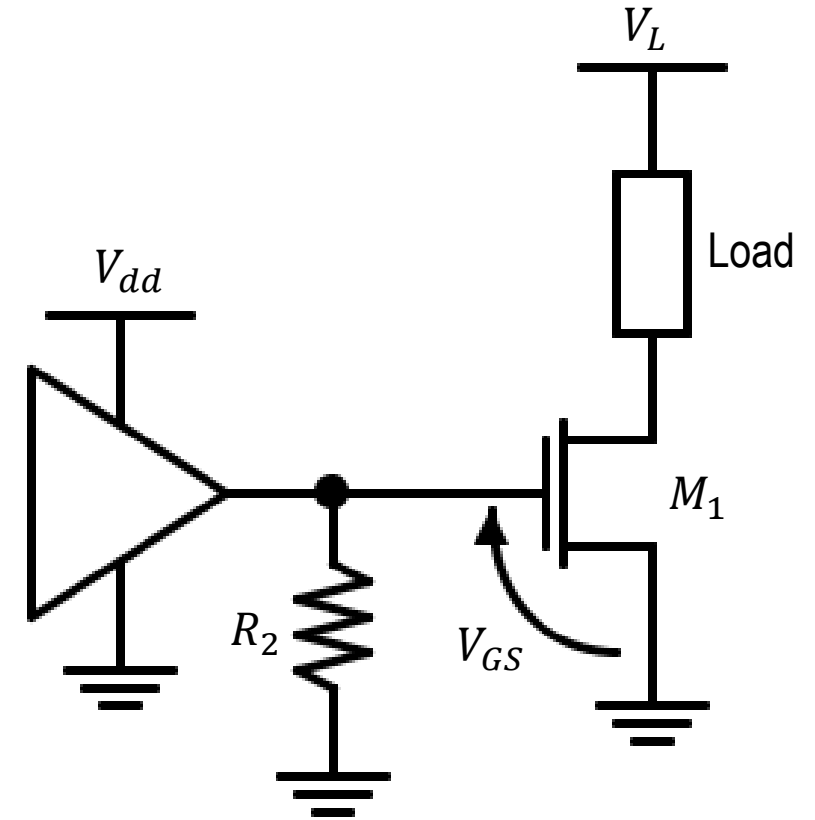
# Actuator driving

- Low side driving, nMOS
  - Same configuration (common Source) as in BJT
  - Driven using the gate voltage, rather than current
  - Transistor alternatively in cut-off and triode (linear) regions
    - Saturation in MOS corresponds to the Forward Active Region in BJT
    - Triode in MOS corresponds to saturation in BJT
    - When in triode region, the MOS transistor behaves as a (low) variable resistor controlled by  $V_{GS}$ 
      - $R_{DSon}$  given in the datasheet



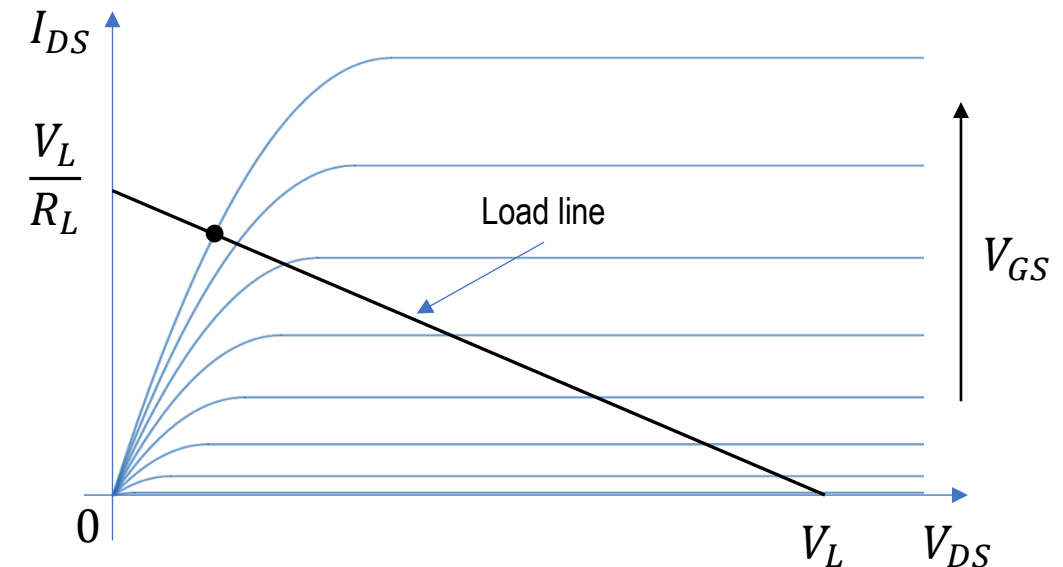
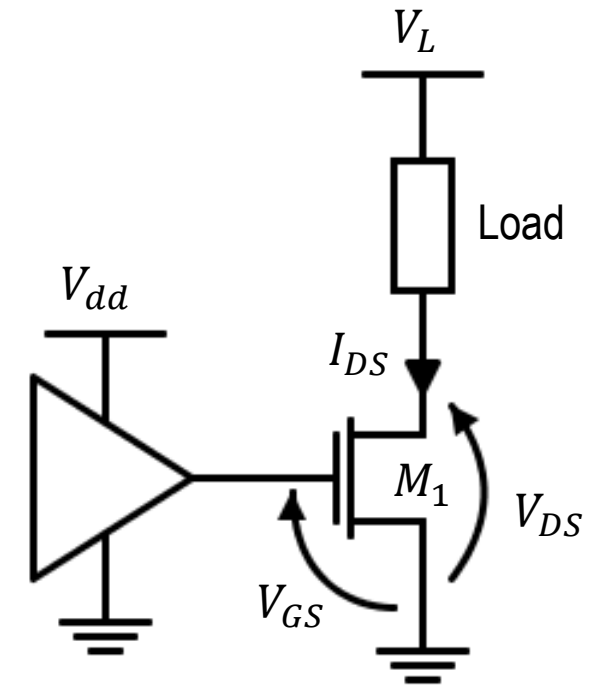
# Actuator driving

- Low side driving, nMOS
  - Transistor OFF
    - Set gate voltage  $V_{GS} < V_{th}$
    - Most power MOS transistors have  $V_{th} > 1\text{ V}$ 
      - Any digital circuit has  $V_{OL}$  low enough to switch it off
    - Insert a resistor from the gate to GND to keep the transistor off when the digital output is tri-stated or not yet initialized
      - Resistor value can be high



# Actuator driving

- Low side driving, nMOS
  - Transistor ON
    - Rise the gate voltage  $V_{GS}$  high enough to bring the transistor in triode region
      - Exceed threshold voltage
        - $V_{GS} \gg V_{th}$
  - Intersect load line with output transfer characteristic
    - The higher  $V_{OV} = V_{GS} - V_{th}$  the more the MOS transistor is in triode region
    - The higher  $V_{OV}$ , the lower  $V_{DS}$



# Actuator driving

- Low side driving, nMOS

- Transistor ON

- nMOS transfer characteristics in linear, triode and saturation regions

- In all cases  $V_{GS} > V_{th}$

- Linear region

- $I_{DS} = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th}) V_{DS}$

$$V_{DS} \approx 0$$

- $R_{DSon} = \frac{V_{DS}}{I_{DS}} = \frac{1}{\mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})}$

- Triode region

- $I_{DS} = \mu C_{ox} \frac{W}{L} \left[ (V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right]$

$$V_{DS} < V_{GS} - V_{th}$$

- Saturation region

- $I_{DS} = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$

$$V_{DS} \geq V_{GS} - V_{th}$$

# Actuator driving

- Low side driving, nMOS

- Transistor ON

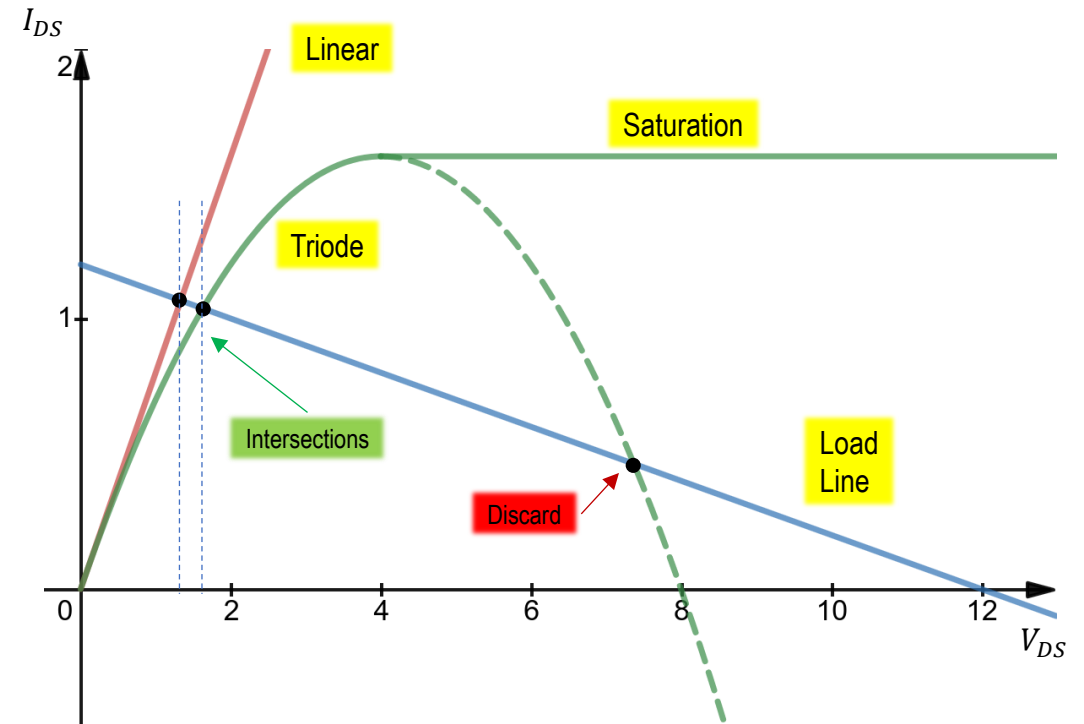
- Assume the nMOS is in linear (or triode) region
    - Assume a value for  $V_{GS}$ 
      - It should be  $V_{GS} > V_{th}$
      - Typically  $V_{GS} = V_{OH}$  of the digital circuit
    - Intersect load line and nMOS characteristic
      - Check that  $V_{DS} < V_{GS} - V_{th}$
      - If linear region was assumed, check that  $V_{DS}$  is small
      - For a linear (resistor) load  $R_L$ , the currents are

- $I_{DS} = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th}) V_{DS}$  or  $I_{DS} = \mu C_{ox} \frac{W}{L} \left[ (V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right]$

- $I_{DS} = \frac{V_L - V_{DS}}{R_L}$

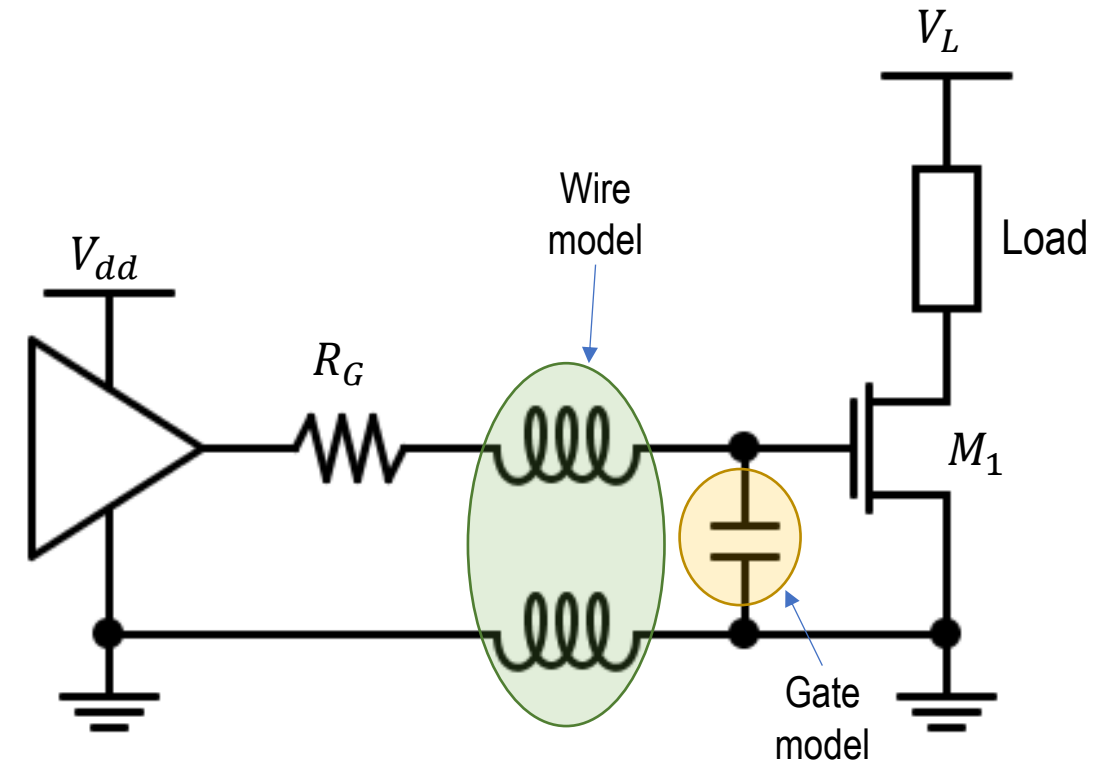
- The only unknown is  $V_{DS}$

- If using the triode region expression, you get two values for  $V_{DS}$  and one is to be discarded



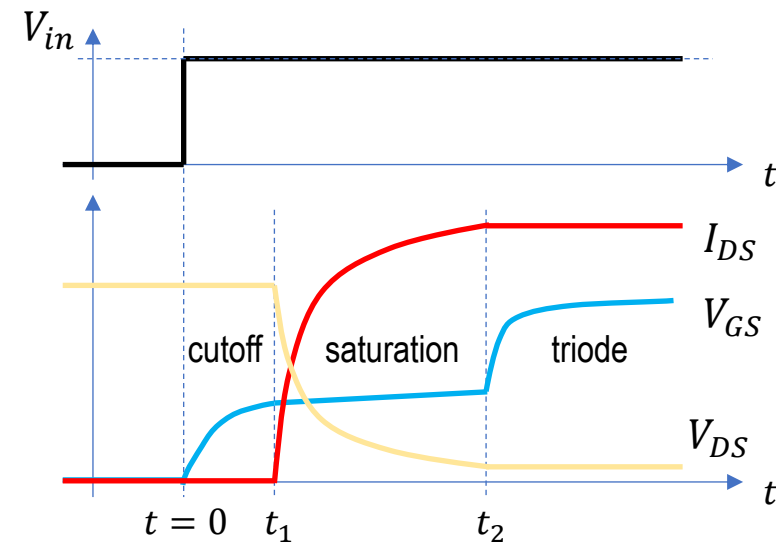
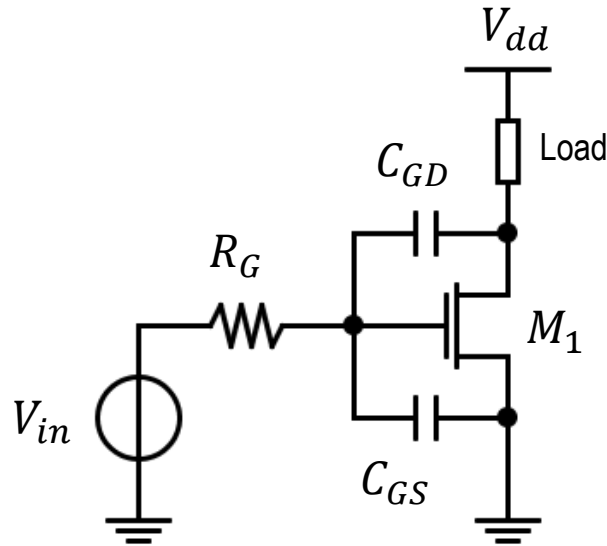
# Actuator driving

- Low side driving, nMOS
  - Dynamic behavior
    - At steady state
      - No current on gate, low power consumption
    - During transitions
      - Current needed to charge and discharge the gate capacitance
      - Wire inductance, together with gate capacitance, make a resonant circuit
        - Risk of oscillations and high overvoltage
        - Use a damping resistor  $R_G$  on the gate mesh



# Actuator driving

- Low side driving, nMOS
  - Dynamic behavior
    - Switching performance
    - Parasitic capacitance  $C_{GD}$  amplified by Miller effect during transient when MOS transistor is in saturation
      - MOS acts as a  $k$  time amplifier
    - Drive the MOS gate with low output resistance (high current) to increase switching performance

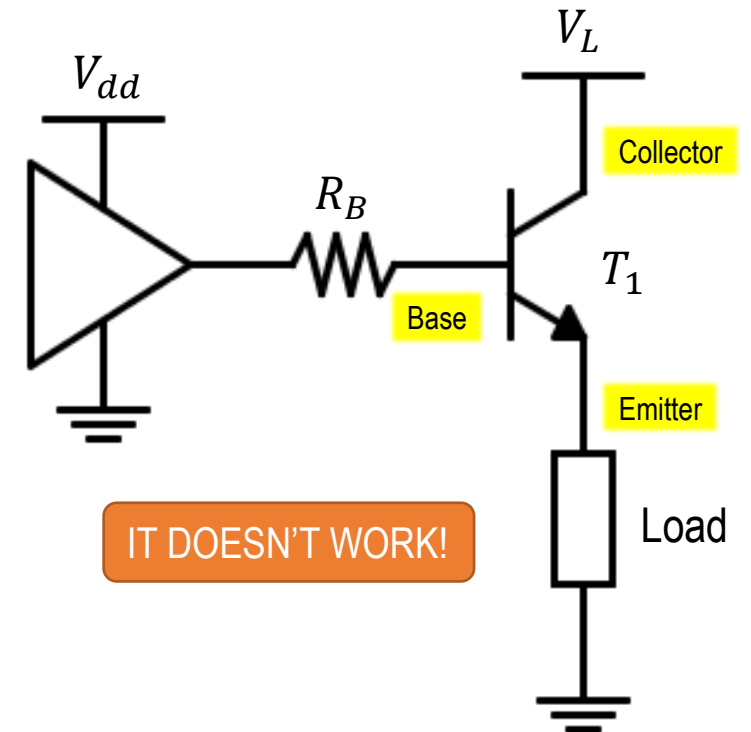


# Actuator driving

- Low side driving, nMOS
  - Summary
    - Choose transistor based on:
      - Breakdown voltage  $BV_{DS0}$
      - Maximum  $I_{DS}$  current
      - Threshold voltage  $V_{th}$
    - Advantages with respect to BJT
      - No gate current, lower power consumption
        - Power dissipation only during transients
      - No gate current, easier driving
        - No need of Darlington configuration
      - Higher efficiency (lower  $R_{DSon}$ ) and higher currents
    - Disadvantages with respect to BJT
      - Possibly slower due to high gate capacitance in fast switching applications

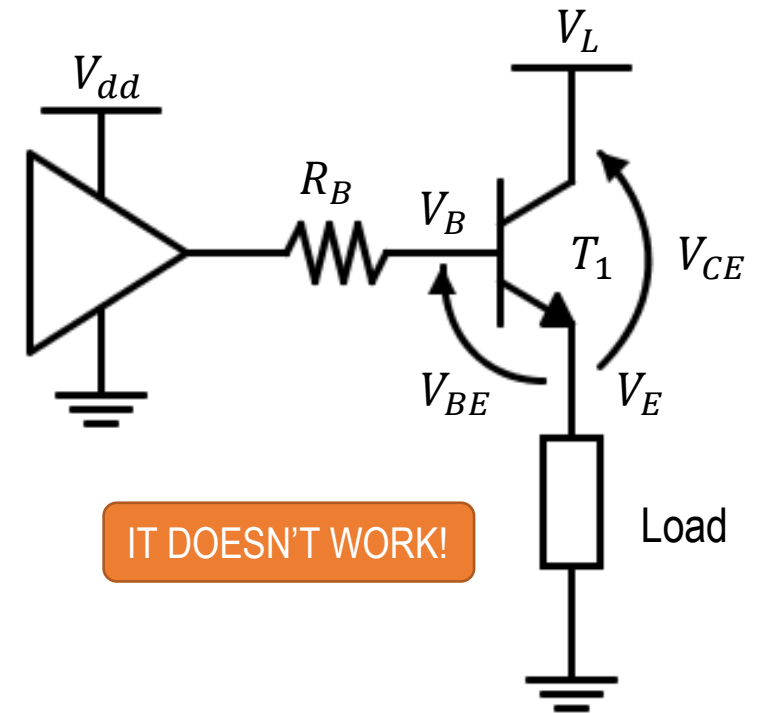
# Actuator driving

- High side driving, npn BJT
  - npn BJT transistors have better characteristics than pnp BJT
    - Electrons have better mobility than holes
      - In intrinsic silicon
        - $\mu_n \simeq 1500 \text{ cm}^2/(\text{Vs})$
        - $\mu_p \simeq 450 \text{ cm}^2/(\text{Vs})$
  - Naive approach
    - Move npn BJT to high side
      - Collector connected to  $V_L$
    - Move load between emitter and GND
    - IT DOESN'T WORK
      - Transistor easy to switch OFF
      - Transistor cannot be switched ON



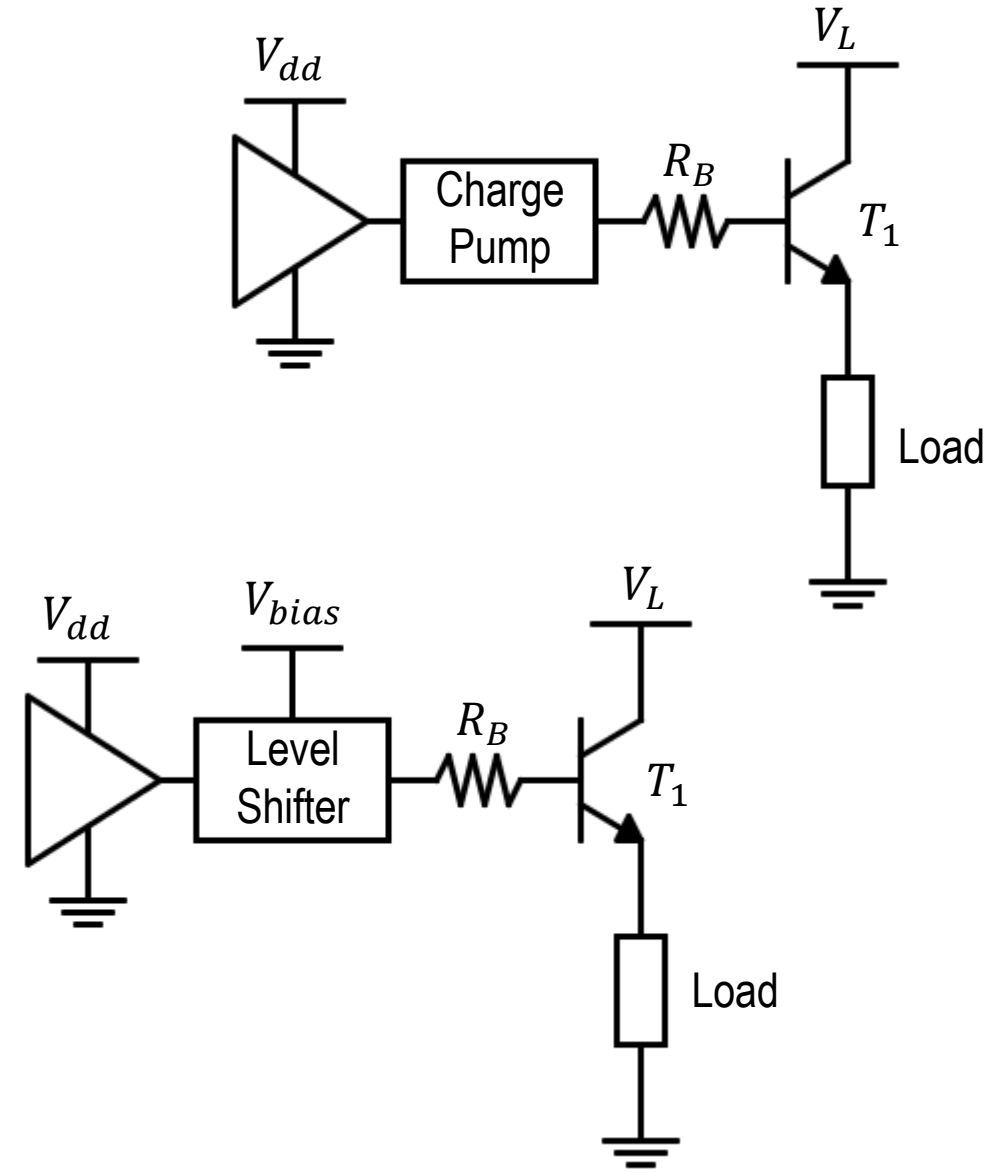
# Actuator driving

- High side driving, npn BJT
  - npn BJT cannot be switched ON
    - When in saturation
      - $V_{CE} = 0.2 \div 0.3 \text{ V}$
      - Hence  $V_E$  is almost equal to  $V_L$
    - To switch on the BJT
      - We need  $V_{BE} > V_{BEon}$ 
        - We actually want  $V_{BE} > V_{BEsat}$
      - We need  $V_B > V_{BEon} + V_E = V_{BEon} + V_L$
    - But  $V_B \leq V_{dd}$ 
      - If  $V_{dd} \leq V_L$  (as it almost always occurs), the requirement cannot be satisfied
      - $V_{BE}$  is often negative
        - BE junction reversed, transistor in cut-off region



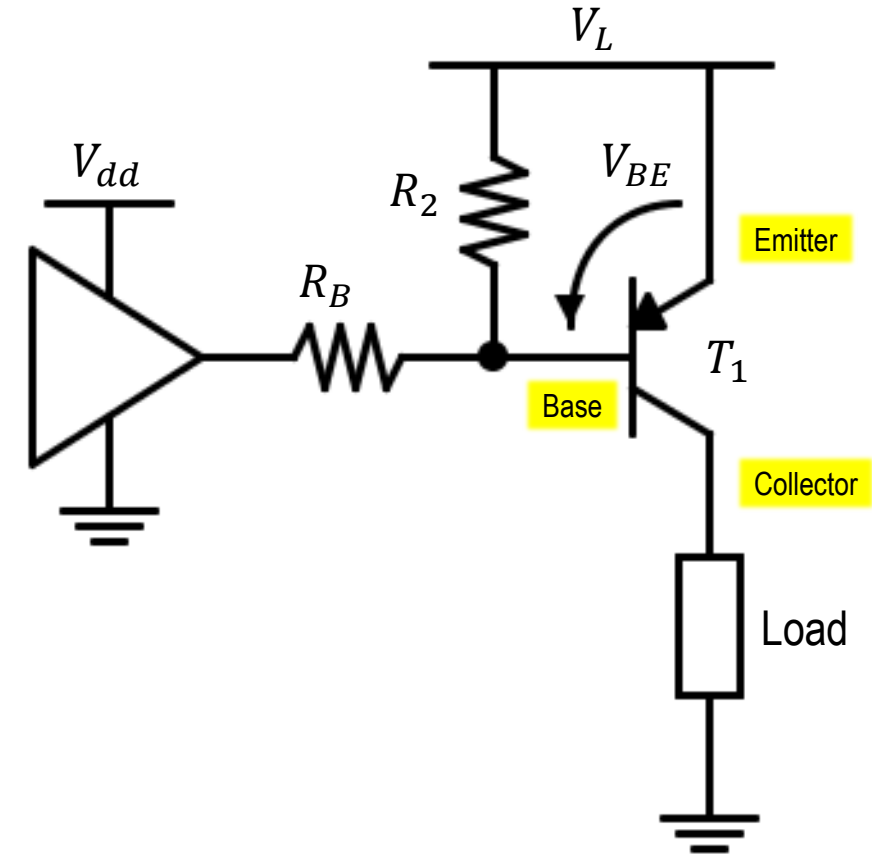
# Actuator driving

- High side driving, npn BJT
  - npn BJT cannot be switched ON
    - Solutions
      - Charge pump: a switching circuit employing a capacitor is used to momentarily rise the output voltage of the digital circuit to a value higher than the supply voltages
      - Level shifter: the digital supply voltage for the driver of the npn transistor is increased with respect to the rest of the digital circuit using a dedicated  $V_{bias}$  supply rail
        - $V_{bias} > V_L$
  - More complex circuits than low side driving
    - More expensive



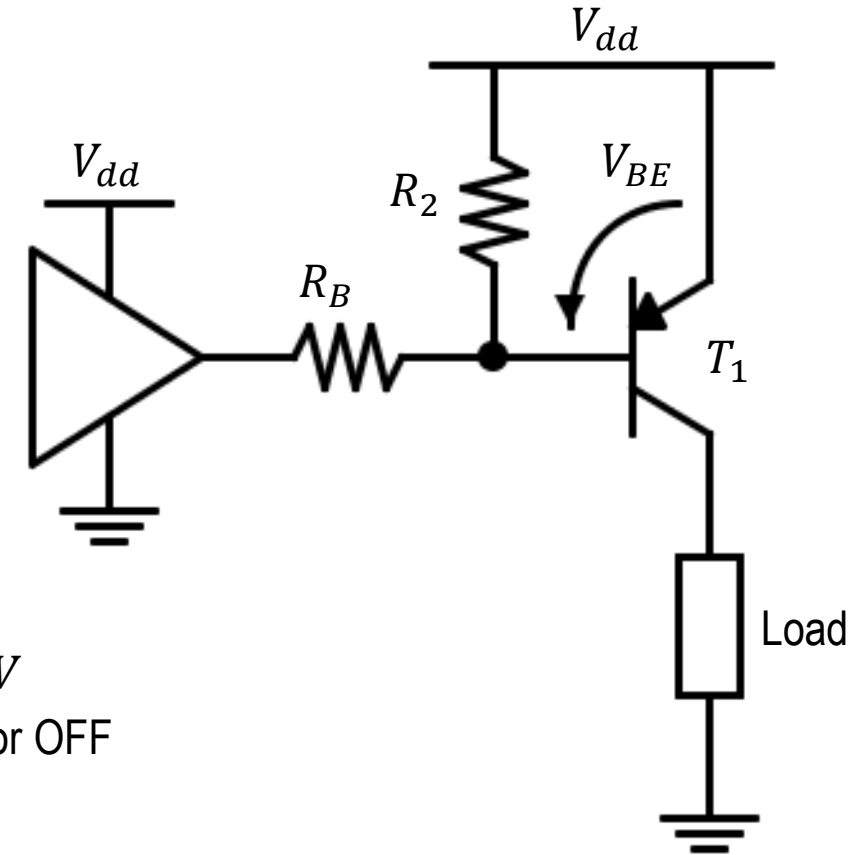
# Actuator driving

- High side driving, pnp BJT
  - Common emitter configuration
    - Load on collector
    - Base current to control the switch
  - Complementary to low side driving with npn BJT
    - High logic output switches transistor OFF
    - Low logic output switches transistor ON
    - All voltages and currents have opposite signs in pnp
    - But pnp BJTs often have a lower  $\beta$
  - Two cases
    - $V_{dd} = V_L$ 
      - Same considerations as low side driving
    - $V_{dd} < V_L$ 
      - Difficult to switch the transistor OFF



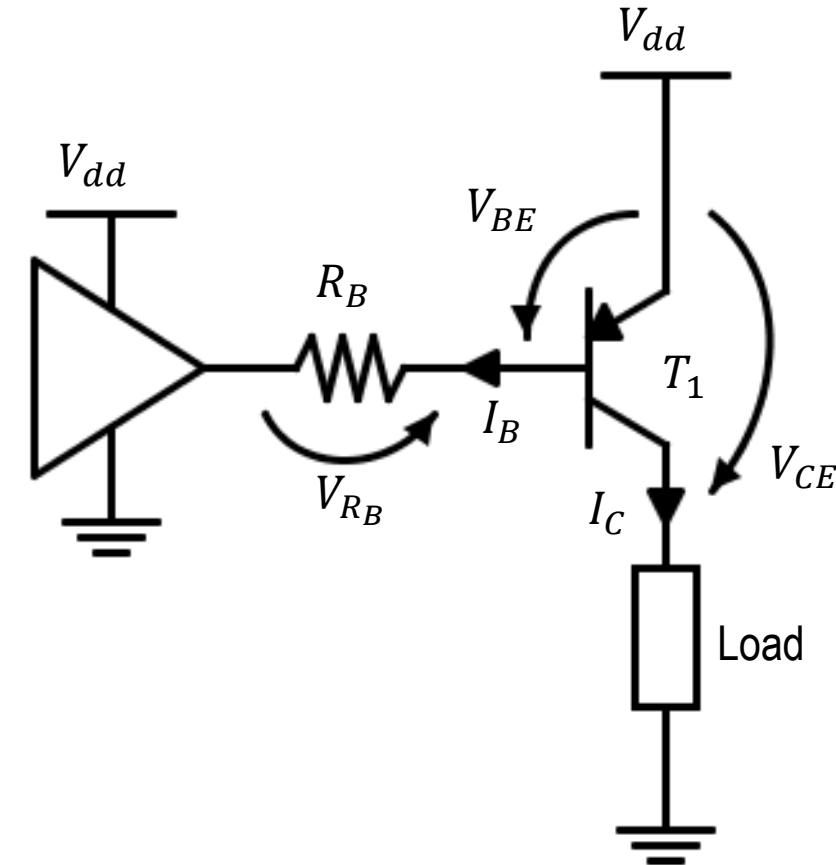
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} = V_L$ 
    - Transistor OFF
      - The output of the digital circuit is high
      - Force  $V_{BE} > V_{BEon}$ 
        - They are negative
        - $|V_{BE}| < |V_{BEon}|$
      - Assuming the transistor is in cut-off region
        - The base current  $I_B = 0$
        - There is no voltage drop across resistance  $R_B$
        - $V_{BE} = V_{OH} - V_{dd} \simeq V_{dd} - V_{dd} = 0$
        - It is negative, but very close to zero  $\rightarrow V_{BE} > V_{BEon} \simeq -0.4\text{ V}$
      - $R_2$  used when the digital output is in high impedance to keep transistor OFF
        - $R_2$  at least ten times higher than  $R_B$



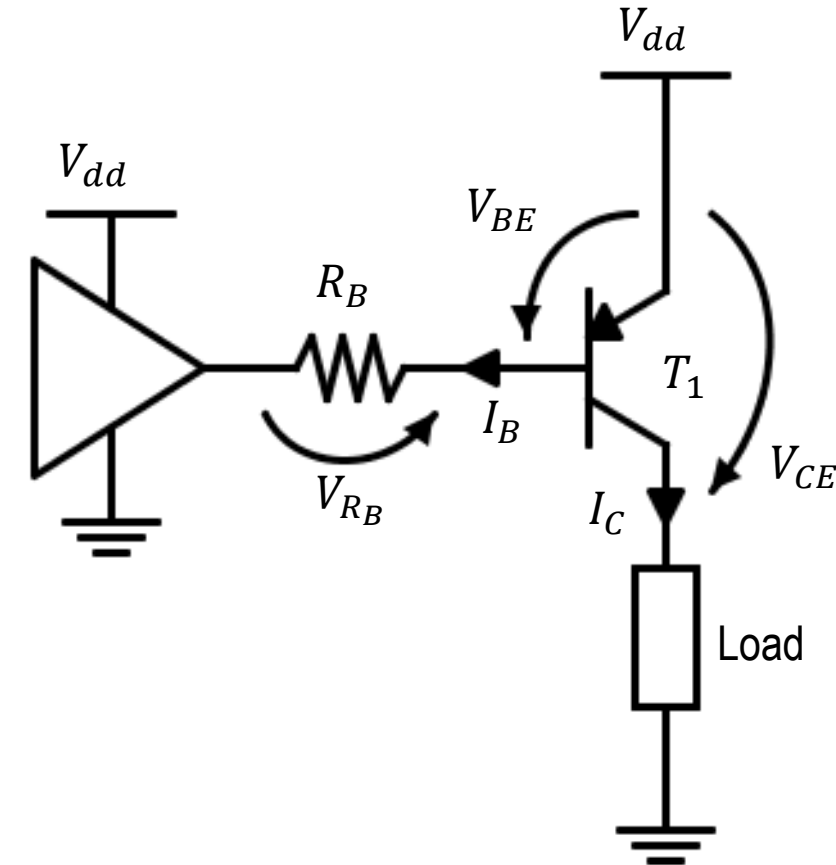
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} = V_L$ 
    - Transistor ON
      - The output of the digital circuit is low
      - $V_{BE}$ ,  $V_{BEsat}$ ,  $V_{CE}$  and  $V_{CEsat}$  are negative in the picture
        - $V_{CEsat} \approx -0.3\text{ V}$ ,  $V_{BEsat} \approx -0.9\text{ V}$
      - $I_C$ ,  $I_B$  and  $V_{RB}$  are positive in the picture
      - The load determines  $I_C$ , not the transistor
      - Force saturation
        - $I_B \gg \frac{I_C}{\beta}$
        - $V_{RB} = V_{dd} + V_{BEsat} - V_{OL}$
        - $I_B = \frac{V_{dd} + V_{BEsat} - V_{OL}}{R_B}$
        - $R_B \ll \frac{\beta}{I_C} (V_{dd} + V_{BEsat} - V_{OL})$
        - Upper bound for the base resistance  $R_B$



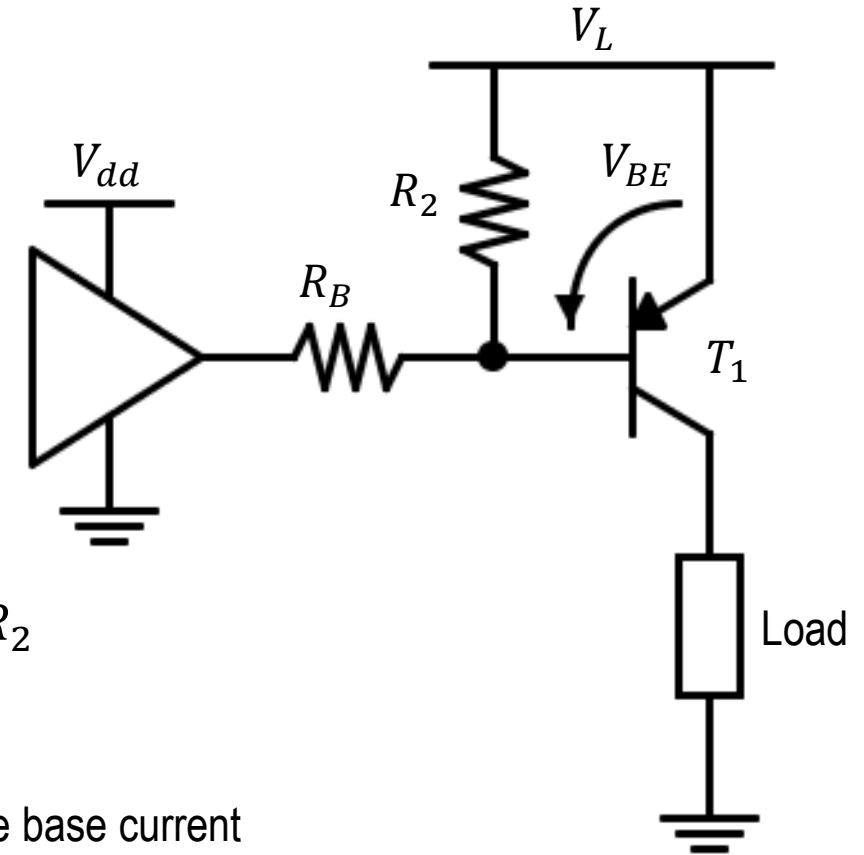
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} = V_L$ 
    - Transistor ON
      - $I_B$  should not exceed the maximum output current  $I_{OL}$  of the digital circuit
      - $I_B < I_{OL} \Rightarrow \frac{V_{dd} + V_{BEsat} - V_{OL}}{R_B} < I_{OL}$
      - $\Rightarrow R_B > \frac{V_{dd} + V_{BEsat} - V_{OL}}{I_{OL}}$
      - Lower bound for the base resistance  $R_B$
      - Assuming  $V_{OL} = 0$  is a conservative choice
        - It makes the lower bound higher
    - If a valid interval for  $R_B$  exists
      - The circuit works!



# Actuator driving

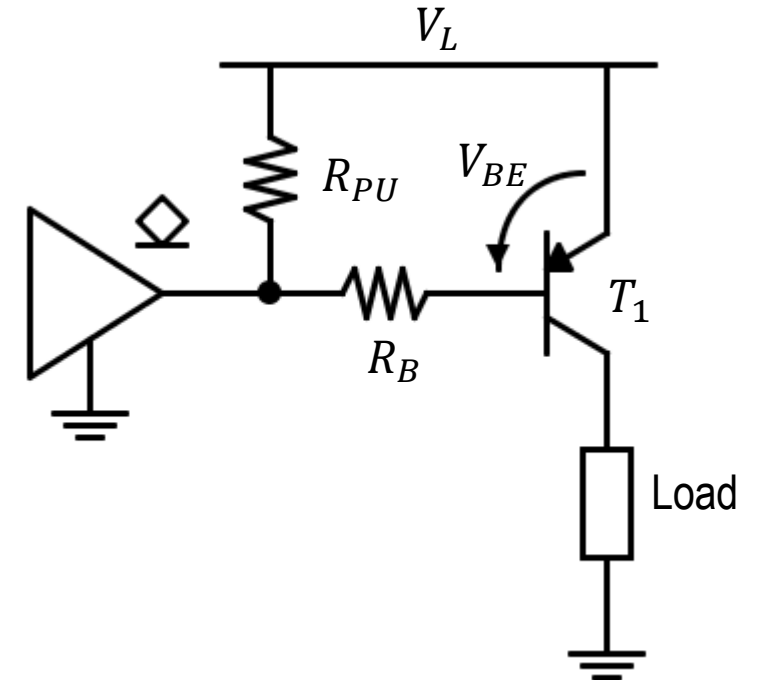
- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Low logic output switches transistor ON
    - Transistor cannot be switched OFF!
    - High logic output, assume transistor is OFF
      - $V_{BE} = V_{dd} - V_L$
      - Greater than  $V_{BEon}$  in absolute value
        - $V_{BEon}$  is negative
        - $V_{BE}$  is even more negative
        - Often even greater than  $V_{BEsat}$
    - Example, assume transistor is OFF, current in  $R_B$  is 0 and neglect  $R_2$ 
      - $V_{dd} = 5\text{ V}$ ,  $V_L = 12\text{ V}$
      - $V_{BE} = V_{OH} - V_{R_B} - V_L = 5 - 0 - 12 = -7\text{ V}$
      - Impossible! It means the transistor is in saturation,  $V_{BE} = V_{BEsat}$ , the base current is not 0 and there is a voltage drop across resistance  $R_B$



# Actuator driving

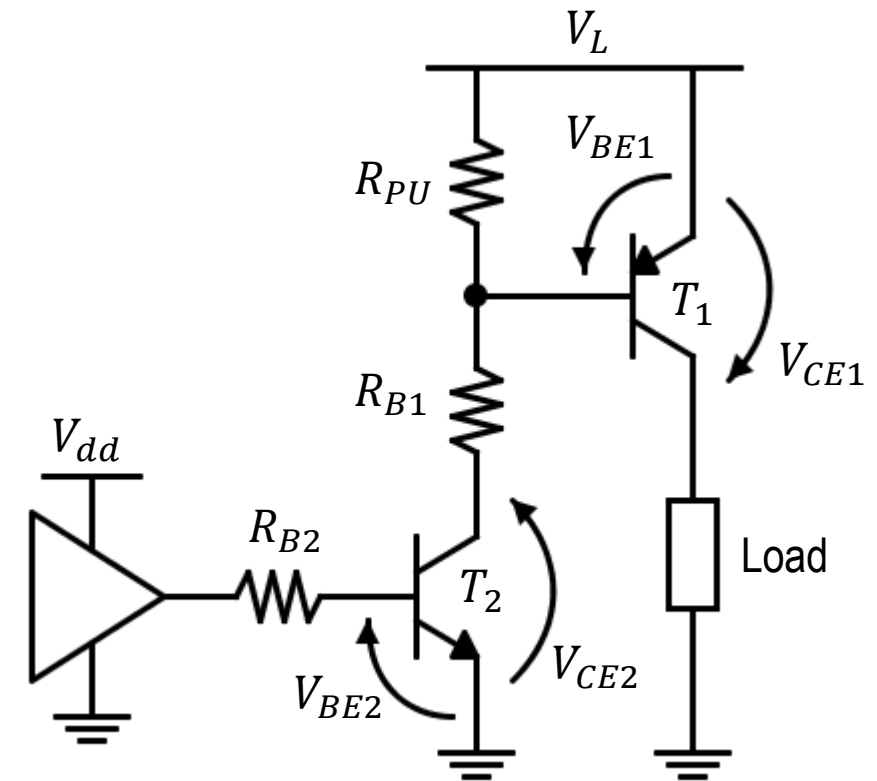
- High side driving, pnp BJT

- $V_{dd} < V_L$ 
  - Use an open collector digital circuit and a pull-up resistor  $R_{PU}$  to  $V_L$ 
    - Cannot be used if  $V_L$  exceeds the maximum voltage sustained by the low side output transistor of the digital circuit
- Value of  $R_{PU}$ 
  - When the digital output is low (transistor ON)
    - Current in  $R_{PU}$  flows in the output transistor of the logic circuit
    - Additional to the base current of the pnp transistor
    - Better a high  $R_{PU}$  value
  - When switching OFF the transistor
    - Need enough current to remove charges and speed-up the transient
    - Better a low  $R_{PU}$  value
- Not practical for low  $\beta$  cases
  - Power pnp transistors often have very low  $\beta$



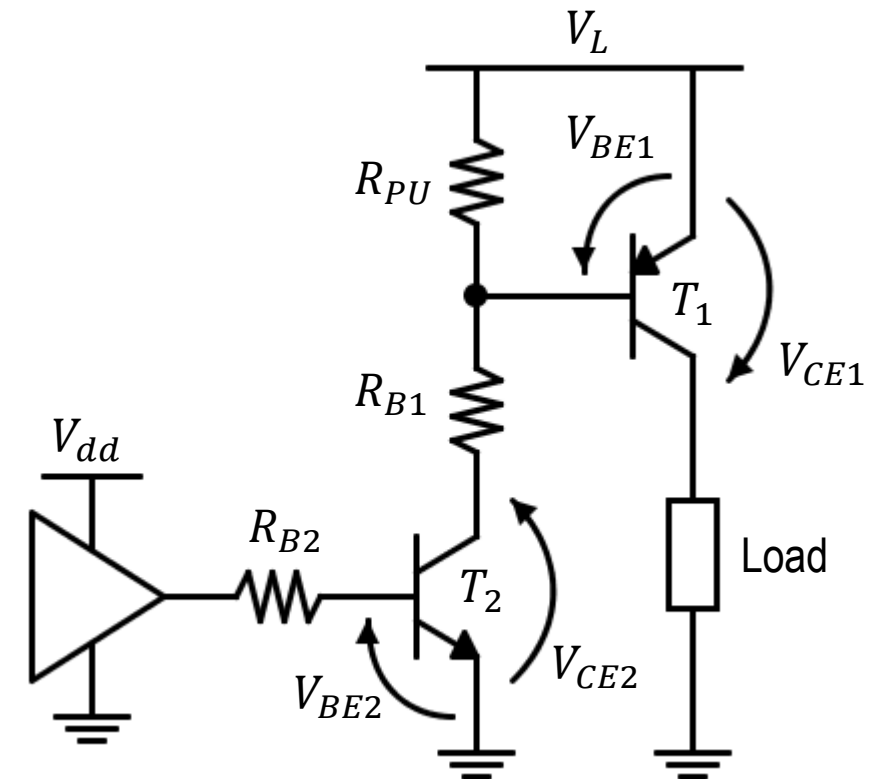
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Use a driving low side npn transistor  $T_2$ 
      - Handles the higher voltage and current required by the switching pnp transistor  $T_1$
      - Similar to the open collector case
        - The low side npn driving transistor is now external
        - It can be chosen with appropriate characteristics
          - Choose a transistor with  $BV_{CE0} > V_L$
      - The npn transistor  $T_2$  should handle the same voltage as the load, but a lower current
      - A resistor from the base of the npn transistor  $T_2$  to GND may be required



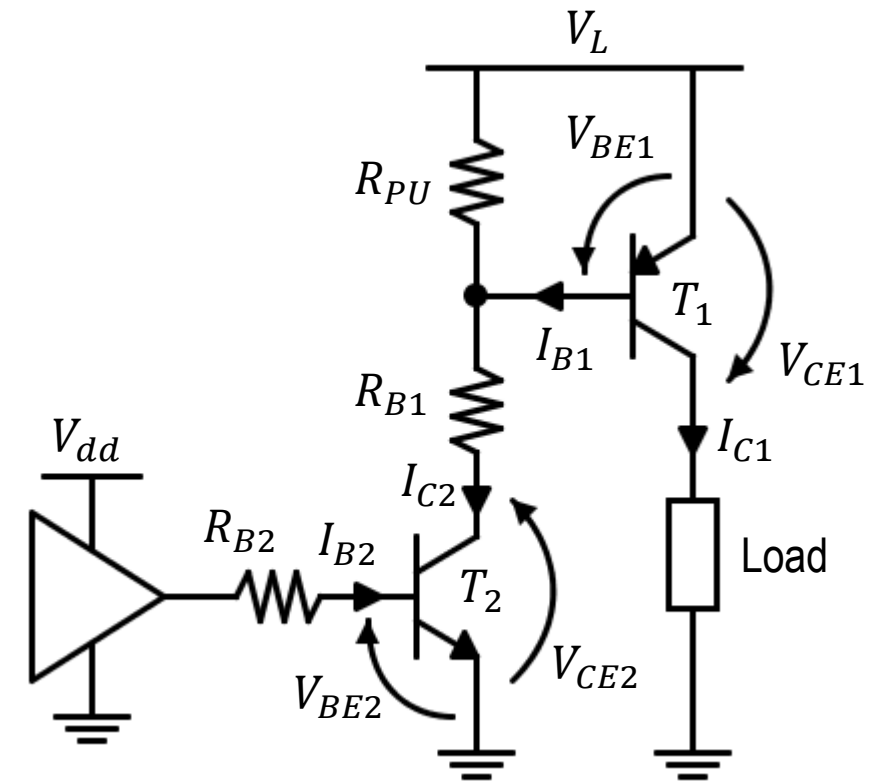
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Use a driving low side npn transistor  $T_2$
    - Command from the digital circuit complemented
      - A low logic output switches OFF the low side npn transistor  $T_2$ 
        - The high side pnp transistor  $T_1$  is also OFF
          - The base of the pnp transistor  $T_1$  is tied to  $V_L$
          - $V_{BE1} = 0$  V
        - A high logic output switches ON (saturates) the low side npn transistor  $T_2$ 
          - If properly designed, current flows in the base of the pnp transistor  $T_1$  and saturates it
          - The high side pnp transistor  $T_1$  is ON



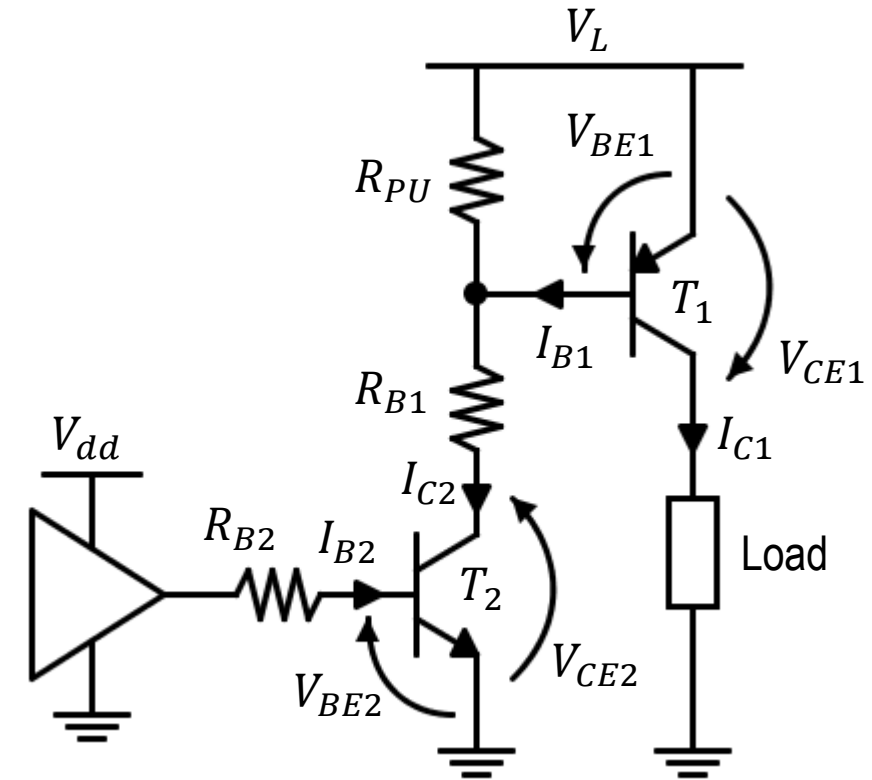
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Use a driving low side npn transistor  $T_2$
    - Design to switch ON transistor  $T_1$ 
      - Choose the high side pnp transistor  $T_1$  based on the load voltage  $V_L$  and maximum current
        - Maximum current depends on the load and corresponds to current  $I_{C1}$  when the pnp transistor  $T_1$  is in saturation
      - Evaluate  $\beta_1$  in the working conditions
        - See the datasheet
      - Compute the needed base current to saturate the pnp transistor  $T_1$ 
        - $I_{B1} \gg \frac{I_{C1}}{\beta_1}$
      - Currents  $I_{B1}$  and  $I_{C1}$  are positive as shown in the picture



# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Use a driving low side npn transistor  $T_2$
    - Design to switch ON transistor  $T_1$ 
      - Choose the low side npn transistor  $T_2$  based on the load voltage  $V_L$  and required pnp base current  $I_{B1}$ 
        - Ignore current in  $R_{PU}$  for the moment
        - Much lower power requirements than pnp transistor  $T_1$
      - Compute pnp transistor  $T_1$  base resistor  $R_{B1}$  upper bound
        - Voltage across  $R_{B1}$ 
          - $V_{R_{B1}} = V_L + V_{BE1sat} - V_{CE2sat}$ 
            - $V_{BE1sat}$  is negative
          - $R_{B1} = \frac{V_{R_{B1}}}{I_{R_{B1}}} \Rightarrow I_{R_{B1}} = I_{C2} = I_{B1} = \frac{V_{R_{B1}}}{R_{B1}} \gg \frac{I_{C1}}{\beta_1}$
          - $R_{B1} \ll \frac{\beta_1}{I_{C1}} (V_L + V_{BE1sat} - V_{CE2sat})$



# Actuator driving

- High side driving, pnp BJT

- $V_{dd} < V_L$

- Use a driving low side npn transistor  $T_2$

- Design to switch ON transistor  $T_1$

- Lower bound for resistor  $R_{B1}$  to limit the current  $I_{C2}$

- Choose a maximum current  $I_{C2up}$

- It should be  $I_{C2up} < I_{C2max}$  from the datasheet

- Set the condition

- $I_{B1} < I_{C2up} \Rightarrow \frac{V_{RB1}}{R_{B1}} < I_{C2up}$

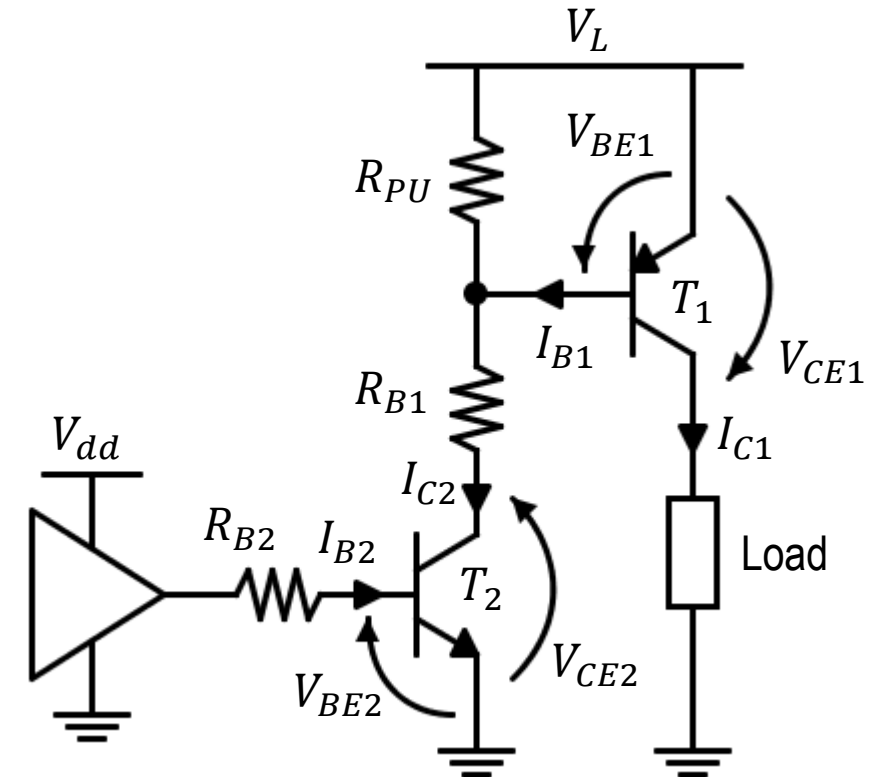
- $R_{B1} > \frac{V_L + V_{BE1sat} - V_{CE2sat}}{I_{C2up}}$

- If a valid range for  $R_{B1}$  exists, choose a value in the range

- You can compute the current  $I_{B1}$  with the chosen resistor

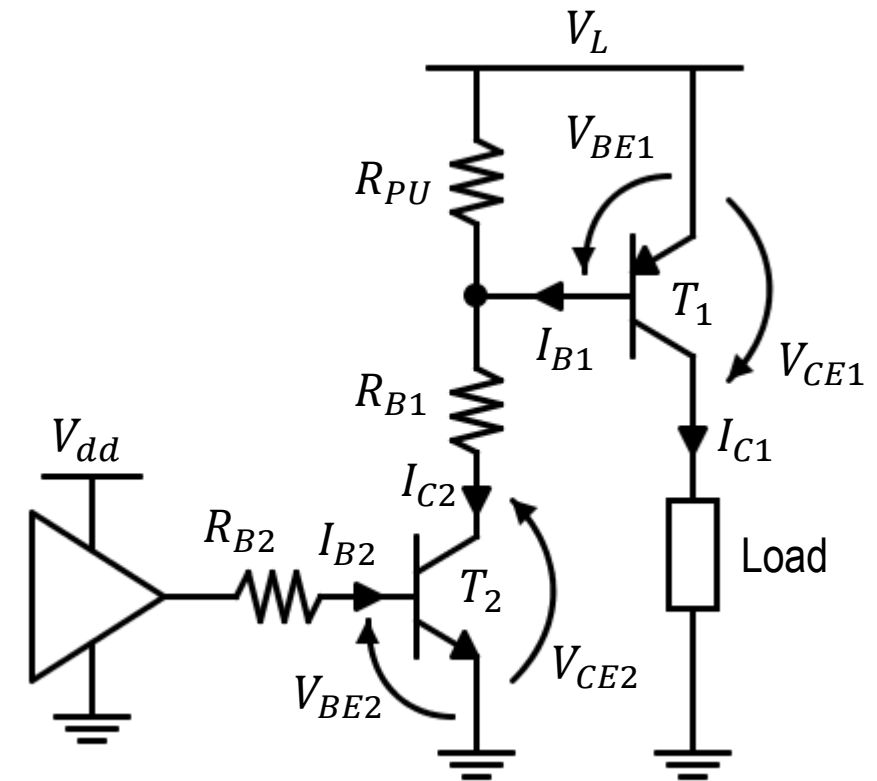
- Evaluate power dissipation on resistor  $R_{B1}$

- $P_{R_{B1}} = V_{RB1} \cdot I_{B1}$



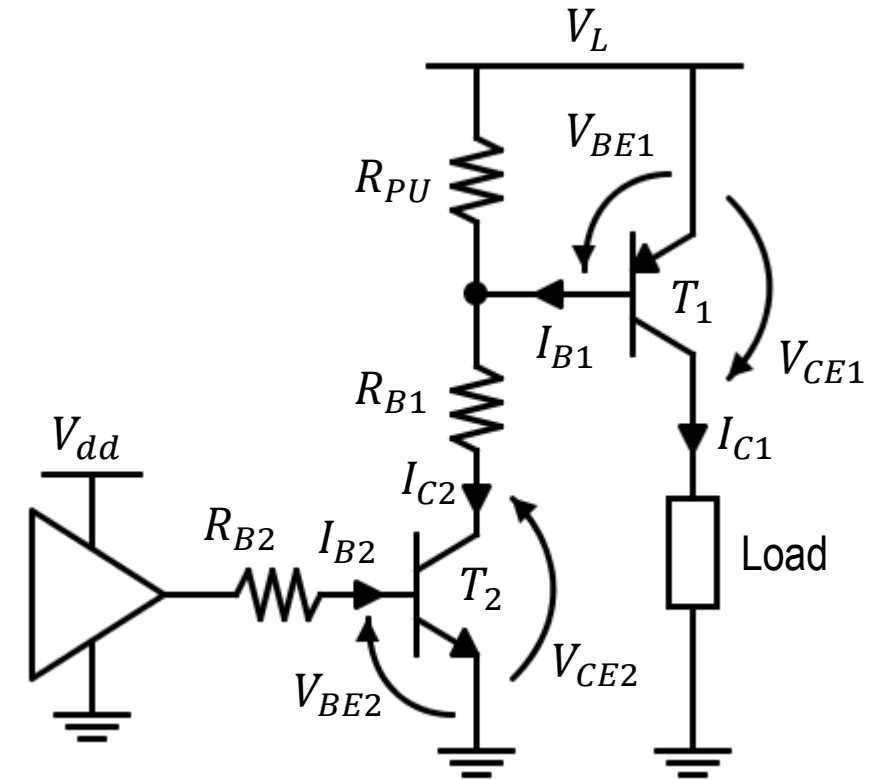
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Use a driving low side npn transistor  $T_2$
    - Design to switch ON transistor  $T_1$ 
      - Choose npn transistor  $T_2$  base resistor
        - Collector current in  $T_2$ 
          - $I_{C2} = I_{B1}$
          - Keep ignoring current in  $R_{PU}$
      - Compute the needed base current to saturate the npn transistor  $T_2$ 
        - $I_{B2} \gg \frac{I_{C2}}{\beta_2} = \frac{I_{B1}}{\beta_2}$
        - It should be small enough to be handled by the digital circuit



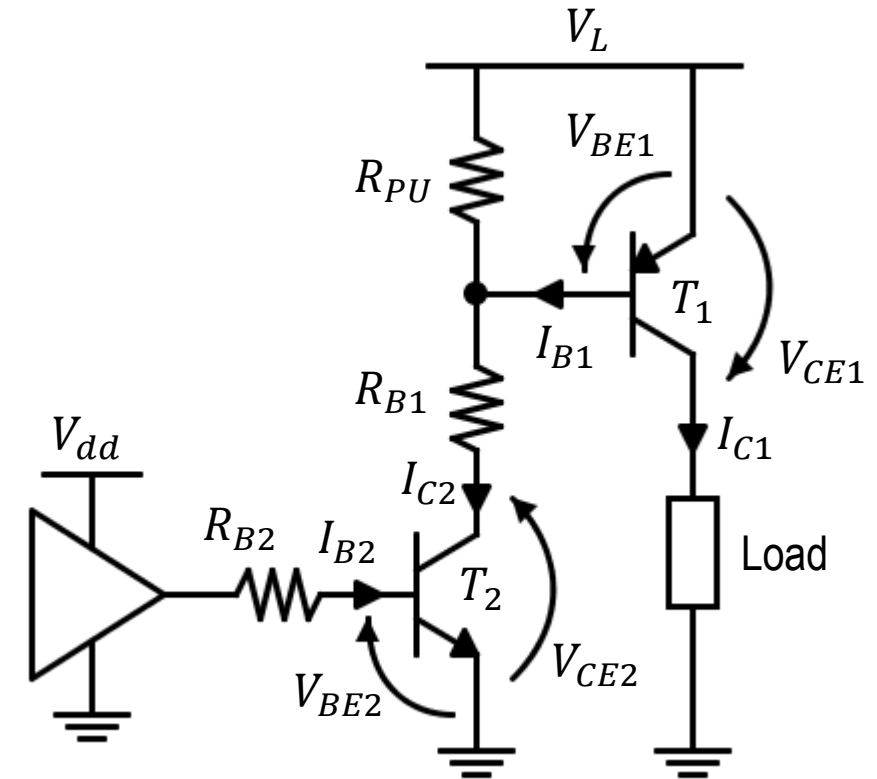
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Use a driving low side npn transistor  $T_2$
    - Design to switch ON transistor  $T_1$ 
      - Compute the upper bound for base resistor  $R_{B2}$ 
        - $V_{R_{B2}} = V_{OH} - V_{BE2sat}$
        - $R_{B2} = \frac{V_{R_{B2}}}{I_{B2}} \Rightarrow I_{B2} = \frac{V_{R_{B2}}}{R_{B2}} \gg \frac{I_{C2}}{\beta_2}$ 
          - $R_{B2} \ll \frac{\beta_2}{I_{C2}} (V_{OH} - V_{BE2sat})$
          - $R_{B2} \ll \frac{\beta_1 \beta_2}{I_{C1}} (V_{OH} - V_{BE2sat})$
      - Compute lower bound for base resistor  $R_{B2}$ 
        - $R_{B2} > \frac{V_{OH} - V_{BE2sat}}{I_{OH}}$  or  $R_{B2} > \frac{V_{dd} - V_{BE2sat}}{I_{OH}}$
      - Power dissipation is less of a concern in this case
        - Much lower current



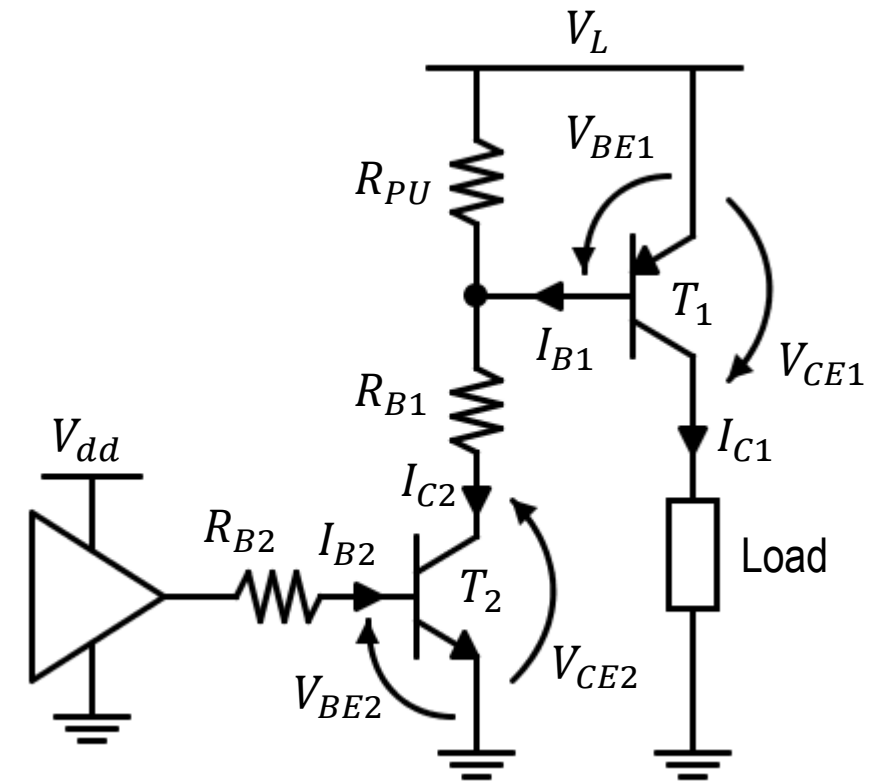
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Use a driving low side npn transistor  $T_2$
    - Design to switch ON transistor  $T_1$ 
      - Compute  $R_{PU}$ 
        - $R_{PU}$  is needed to switch OFF the pnp transistor  $T_1$  when the npn transistor  $T_2$  is also OFF
        - Otherwise we don't want it to affect the circuit when the two transistors are in saturation
      - If  $R_{PU}$  is too small
        - More current flows in  $R_{B1}$  and rises the base voltage of the pnp transistor  $T_1$
        - Lower saturation or no saturation



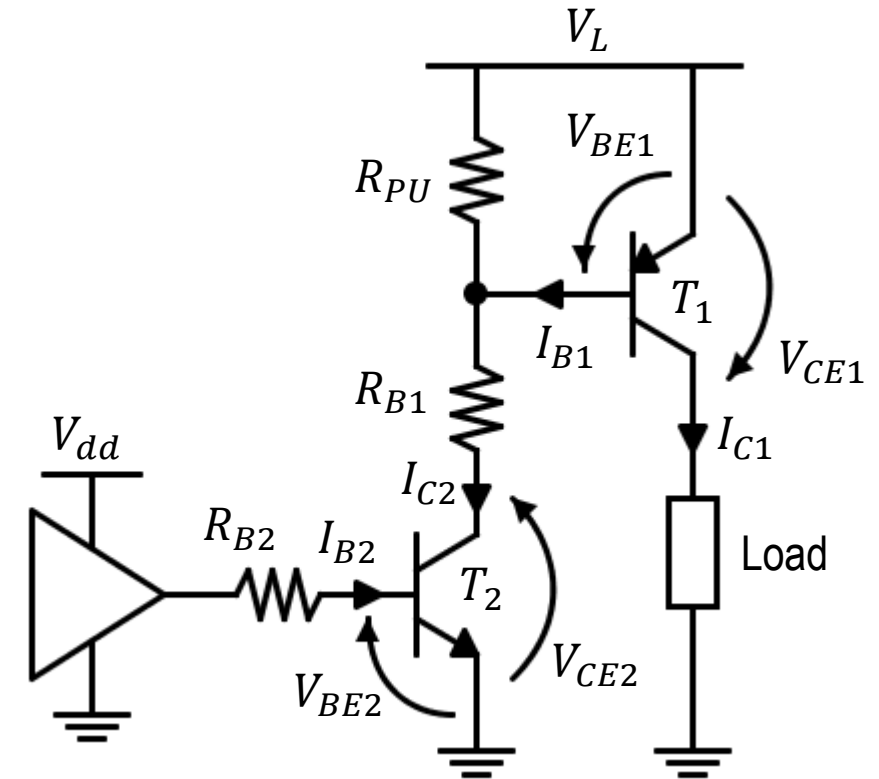
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Use a driving low side npn transistor  $T_2$
    - Design to switch ON transistor  $T_1$ 
      - Compute  $R_{PU}$
      - Make current in  $R_{PU}$  at least ten times smaller than  $I_{B1}$ 
        - Voltage across  $R_{PU}$  is  $V_{BE1sat}$
        - $\frac{|V_{BE1sat}|}{R_{PU}} < \frac{I_{B1}}{10} \Rightarrow R_{PU} > 10 \frac{|V_{BE1sat}|}{I_{B1}}$
        - $R_{PU}$  can be even higher
          - But not too much, otherwise switching OFF the transistor  $T_1$  takes a long time
      - Decrease  $R_{B2}$  a little bit to take into account the additional  $T_2$  collector current
      - Check that the digital circuit and  $T_2$  can sustain the additional currents



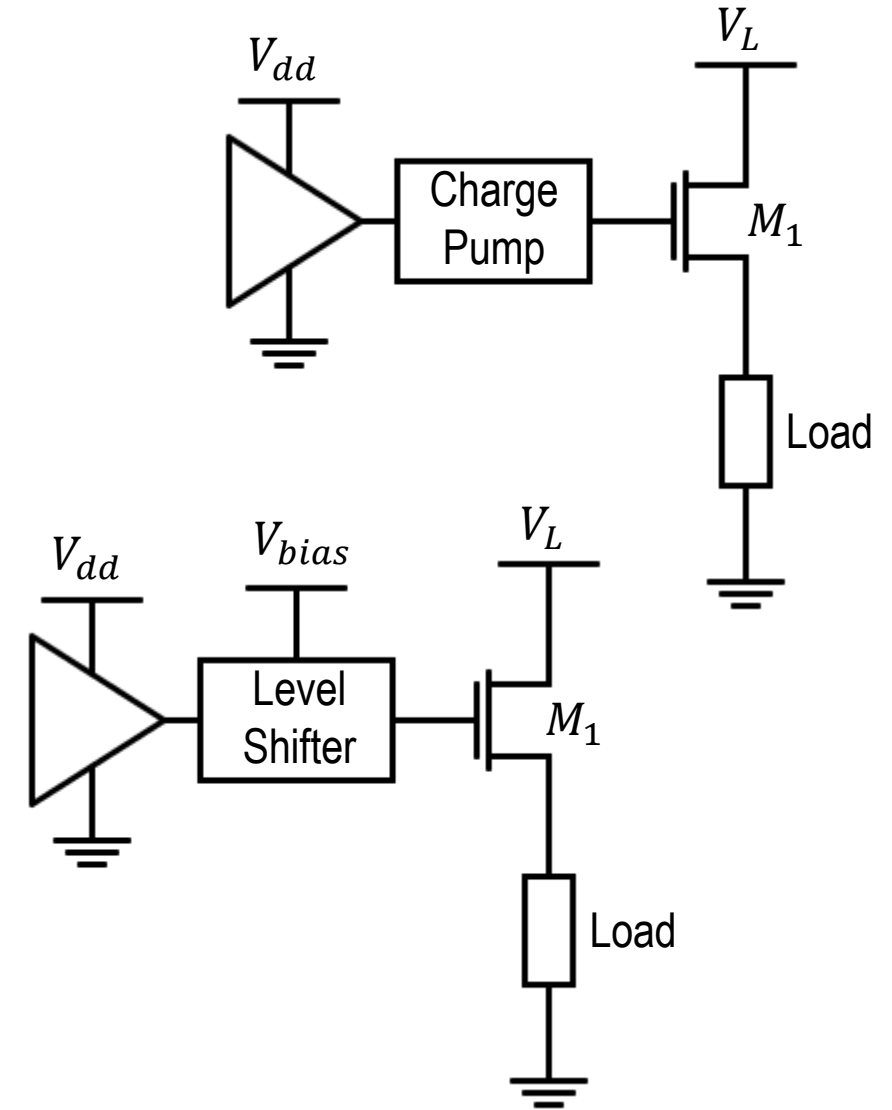
# Actuator driving

- High side driving, pnp BJT
  - $V_{dd} < V_L$ 
    - Is a charge pump or a level shifter useful?
      - Not really!
      - Voltages greater than  $V_L$  are not necessary
        - The pnp transistor  $T_1$  is easy to turn ON
        - It is just difficult to turn it OFF
      - The low side transistor  $T_2$  is enough
        - Cheaper and less noisy than a charge pump
        - Don't need an additional power line



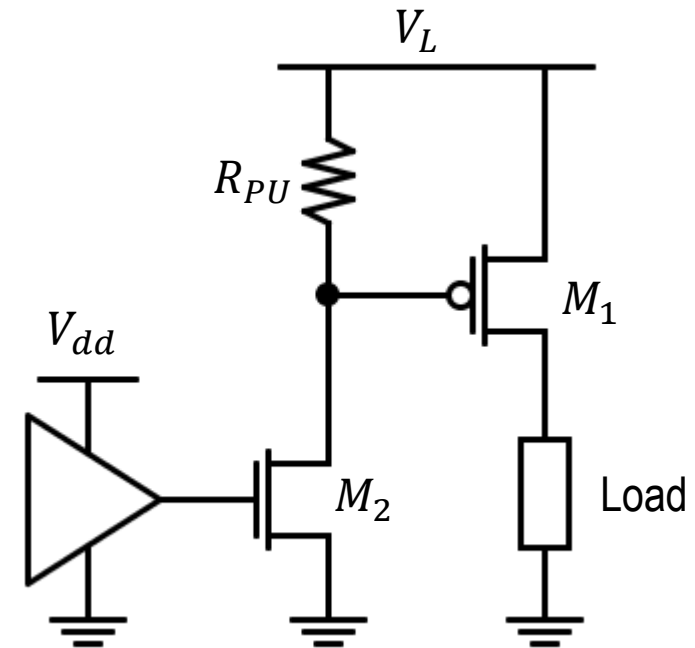
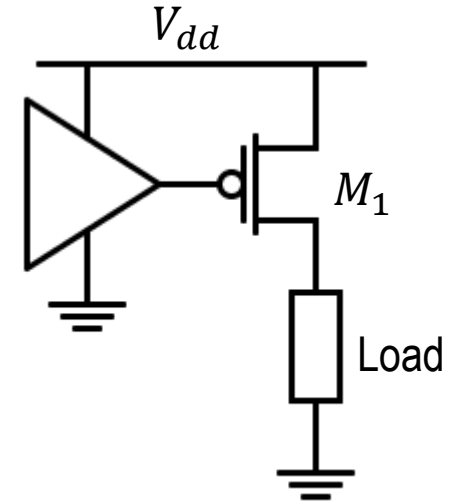
# Actuator driving

- High side driving, nMOS
  - Transistor cannot be easily switched ON
    - Charge pump
    - Level shifter
    - Complexity added to circuit
  - Advantages
    - nMOS transistors have lower  $R_{DSon}$  compared to pMOS
      - Lower power dissipation
      - Higher current loads
      - Lower gate capacitance
      - Faster switching



# Actuator driving

- High side driving, pMOS
  - Complementary to nMOS low side driving
    - When load voltage equal to digital circuit voltage supply
      - Use a complementary circuit compared to low side with a pMOS transistor
    - When load voltage higher than what the digital circuit can sustain
      - Use a low side nMOS driving transistor
      - $M_2$  sustains the same voltage as  $M_1$ , but a much lower current
      - $R_{PU}$  high
        - Used to switch OFF  $M_1$  when  $M_2$  is OFF
        - Current in  $R_{PU}$  when transistors are ON flows in  $M_2$
      - May require a resistor from gate to GND for  $M_2$
      - A low side npn BJT driving transistor can also be used
        - In place of  $M_2$



# Actuator driving

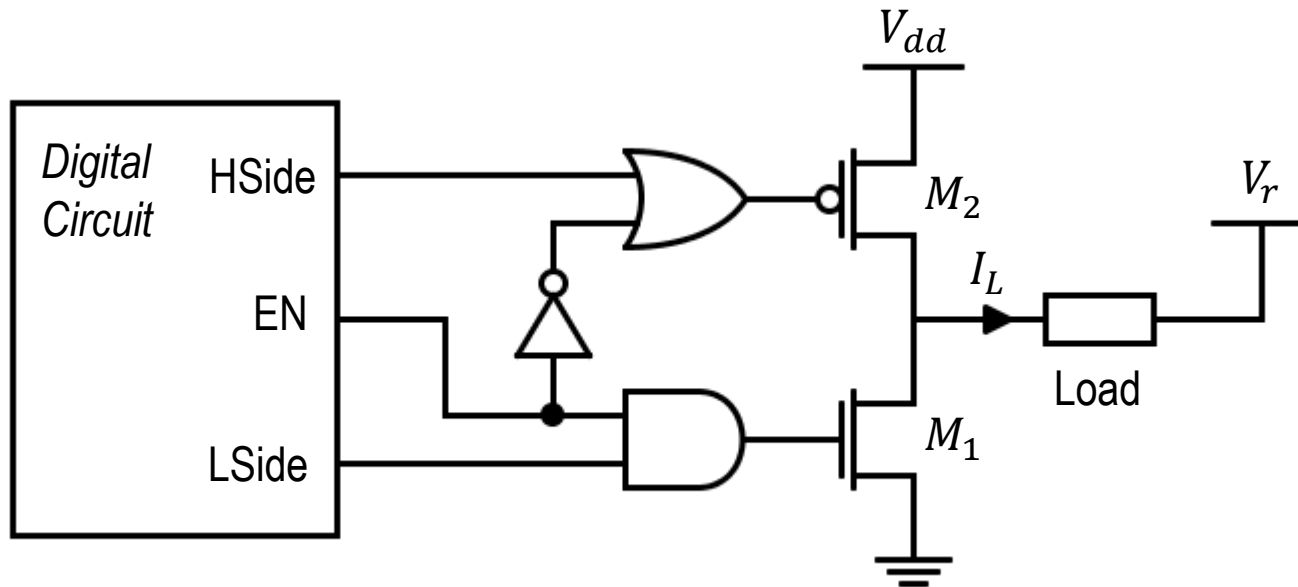
- High side driving
  - Summary
    - High side npn BJT or nMOS transistor
      - Cannot switch the transistor ON
      - Use a charge pump or a level shifter
    - High side pnp BJT or pMOS transistor
      - $V_{dd} = V_L$ 
        - Same as low side driving
        - Compute the valid range for the  $R_B$  resistance in case of the pnp BJT transistor
        - Evaluate the voltage  $V_{DS}$  to bring the transistor in triode region for the pMOS transistor
      - $V_{dd} < V_L$ 
        - Cannot switch the transistor OFF
        - Use a low side npn BJT or nMOS transistor to drive the high side pnp BJT or pMOS transistor
    - Choose transistor based on
      - Breakdown voltage ( $BV_{CE0}$ ,  $BV_{DS0}$ ), maximum current ( $I_{Cmax}$ ,  $I_{DSmax}$ ), SOA

# Actuator driving

- Floating load
  - With low and high side driving, current always flows in the same direction (or does not flow)
  - Some loads need current in both directions
    - Motors that turn in both directions
- Solutions
  - Use both high side and low side switches with load in between
  - Half H bridge
    - When an intermediate voltage reference is available
  - Full H bridge
  - Two commands
    - Enable EN
      - ON (high) or OFF (low)
    - Direction DIR
      - POSITIVE (high) or NEGATIVE (low)

# Actuator driving

- Floating load
  - Half H bridge
    - An intermediate voltage reference  $V_r$  is available
      - $0 < V_r < V_{dd}$
    - With Enable and direct control of low and high side transistors



|    |       | LSide    |          |
|----|-------|----------|----------|
|    |       | 0        | 1        |
| EN | NO 0  | 0<br>OFF | 0<br>OFF |
|    | YES 1 | 0<br>OFF | 1<br>ON  |

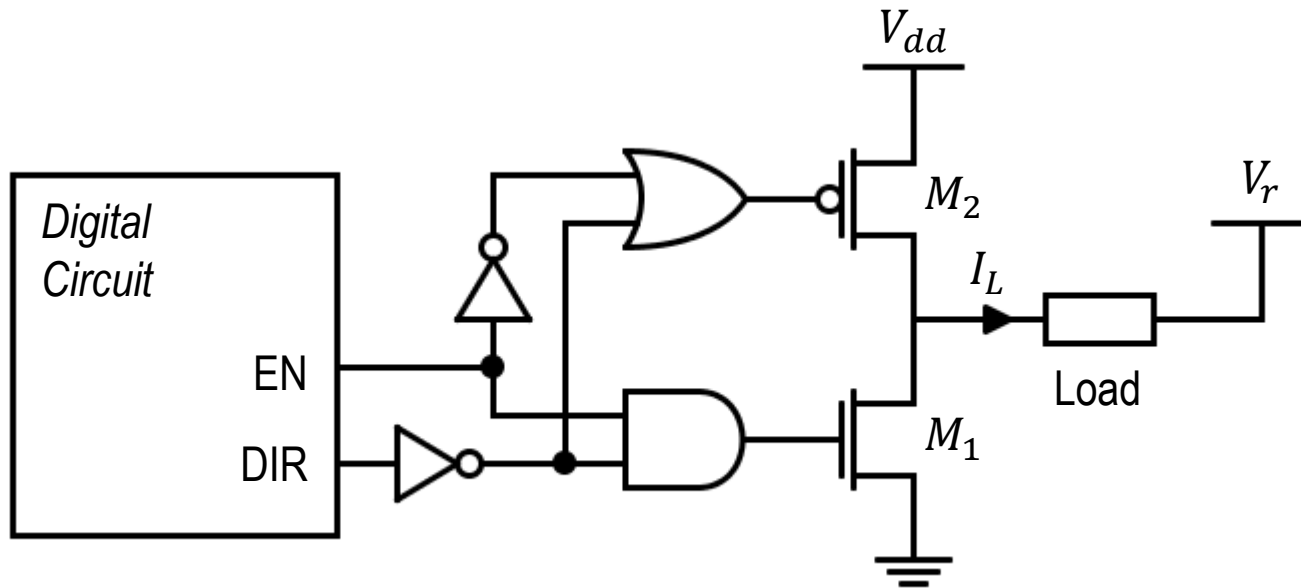
$$Y_{M_1} = \text{EN} \cdot \text{LSide}$$

|    |       | HSide    |          |
|----|-------|----------|----------|
|    |       | 0        | 1        |
| EN | NO 0  | 1<br>OFF | 1<br>OFF |
|    | YES 1 | 0<br>ON  | 1<br>OFF |

$$Y_{M_2} = \overline{\text{EN}} + \text{HSide}$$

# Actuator driving

- Floating load
  - Half H bridge
    - An intermediate voltage reference  $V_r$  is available
      - $0 < V_r < V_{dd}$
    - With Enable and Direction signals



| EN \ DIR |   | Negative | Positive |
|----------|---|----------|----------|
|          |   | 0        | 1        |
| NO       | 0 | 0<br>OFF | 0<br>OFF |
|          | 1 | 1<br>ON  | 0<br>OFF |

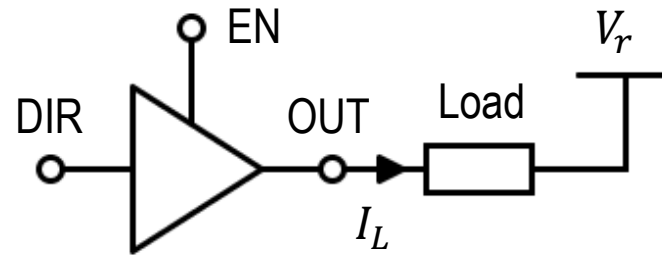
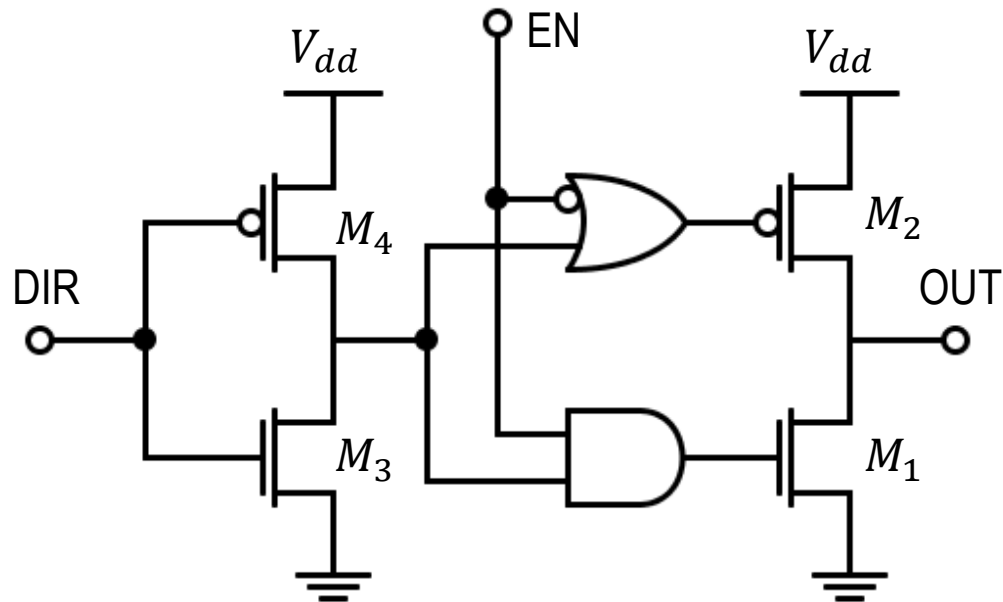
$$Y_{M_1} = EN \cdot \overline{DIR}$$

| EN \ DIR |   | Negative | Positive |
|----------|---|----------|----------|
|          |   | 0        | 1        |
| NO       | 0 | 1<br>OFF | 1<br>OFF |
|          | 1 | 1<br>OFF | 0<br>ON  |

$$Y_{M_2} = \overline{EN} + \overline{DIR}$$

# Actuator driving

- Floating load
  - Half H bridge
    - Similar to a buffer
      - With totem pole output stage
      - With enable signal



| EN | DIR | Negative | Positive |
|----|-----|----------|----------|
|    |     | 0        | 1        |
| NO | 0   | 0<br>OFF | 0<br>OFF |
|    | 1   | 1<br>ON  | 0<br>OFF |

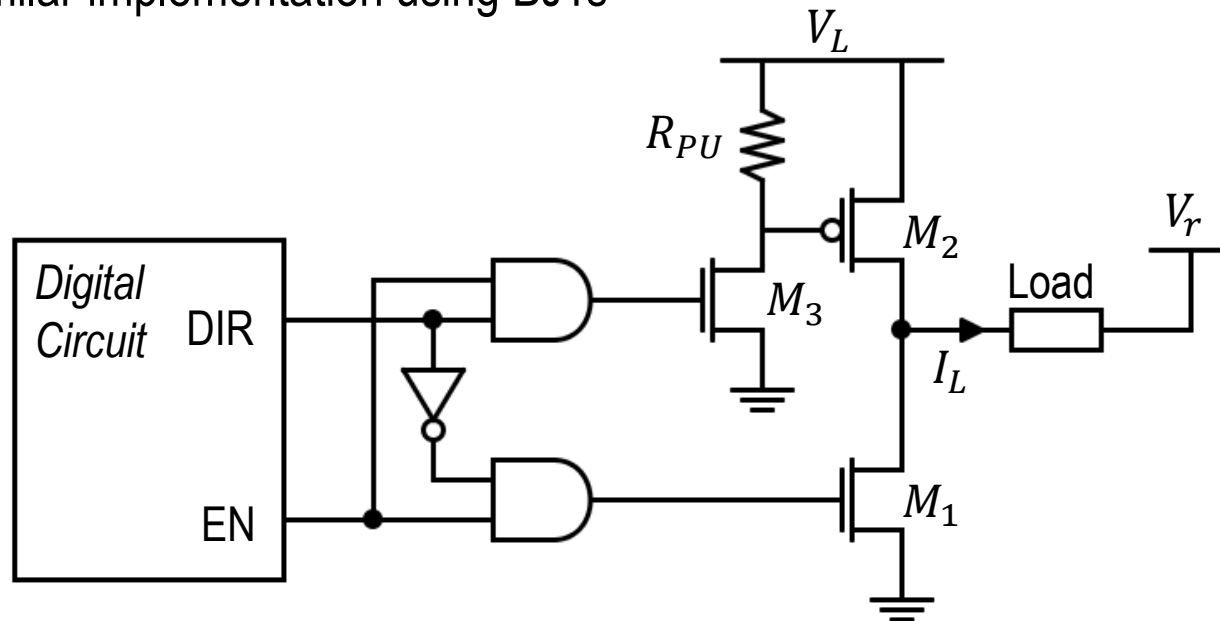
$$Y_{M_1} = EN \cdot \overline{DIR}$$

| EN | DIR | Negative | Positive |
|----|-----|----------|----------|
|    |     | 0        | 1        |
| NO | 0   | 1<br>OFF | 1<br>OFF |
|    | 1   | 1<br>OFF | 0<br>ON  |

$$Y_{M_2} = \overline{EN} + \overline{DIR}$$

# Actuator driving

- Floating load
  - Half H bridge
    - pMOS may require low side driving nMOS
      - If  $V_L$  higher than  $V_{dd}$
    - pMOS may be substituted with nMOS and charge pump or level shifter
    - Similar implementation using BJTs



|       |      | DIR      |          |
|-------|------|----------|----------|
|       |      | Negative | Positive |
| EN    |      | 0        | 1        |
|       | NO 0 | 0<br>OFF | 0<br>OFF |
| YES 1 |      | 1<br>ON  | 0<br>OFF |

$$Y_{M_1} = EN \cdot \overline{DIR}$$

|       |      | DIR      |          |
|-------|------|----------|----------|
|       |      | Negative | Positive |
| EN    |      | 0        | 1        |
|       | NO 0 | 0<br>OFF | 0<br>OFF |
| YES 1 |      | 0<br>OFF | 1<br>ON  |

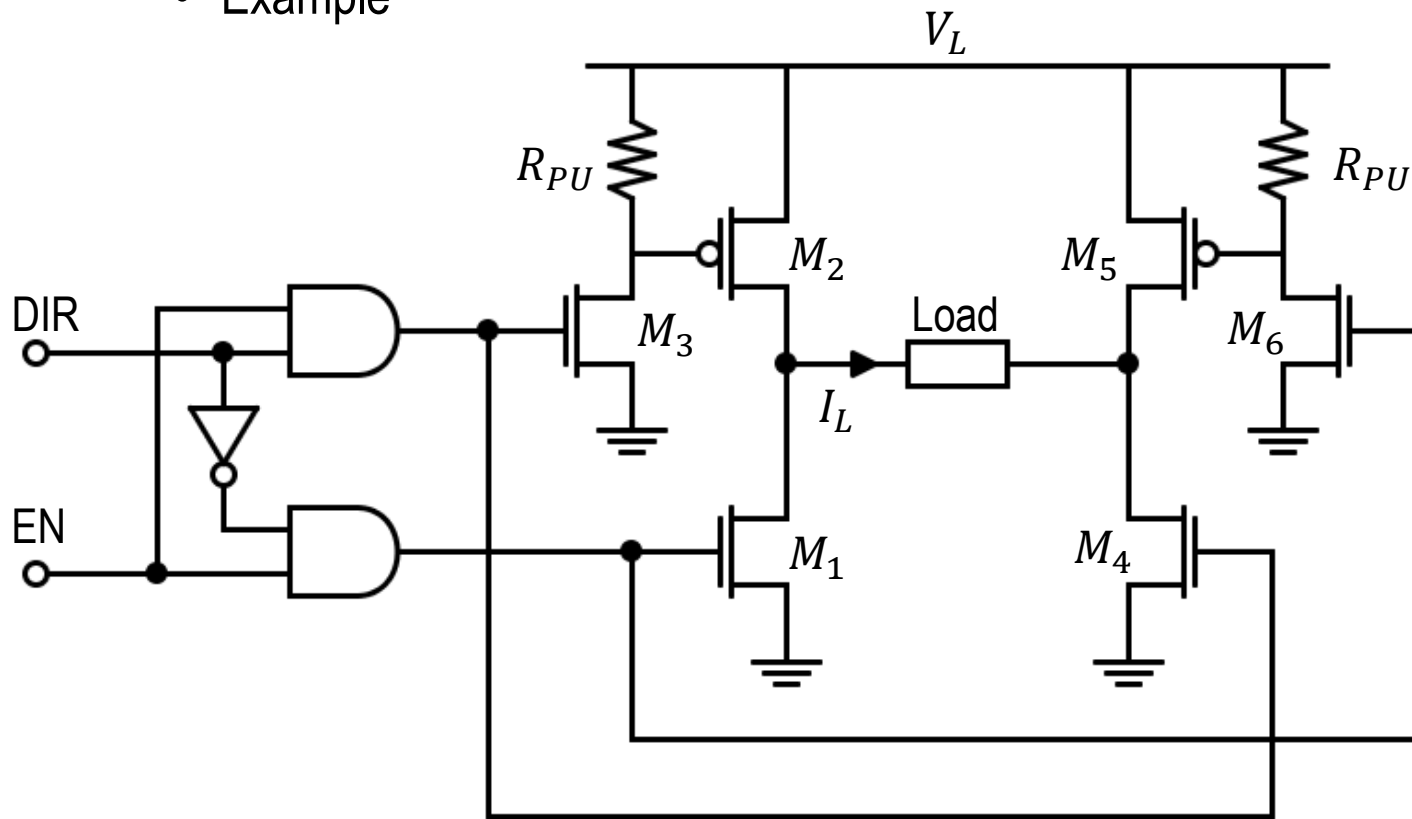
$$Y_{M_3} = EN \cdot DIR$$

# Actuator driving

- Floating load
  - Full H bridge
    - Two Half H Bridges
      - Each Half H Bridge drives one of the load terminals
      - No additional reference voltage
    - Transistors always switched ON in opposite couples to drive current in one of the two directions
      - One low side and the opposite high side
        - Motor turns clockwise or counterclockwise
      - Load can be left floating (all transistor OFF)
        - Motor is free running
      - If the two low (high) side transistors are ON, while the high (low) side are OFF, both load terminals are grounded
        - Motor brakes
  - pMOS may require a low side driving nMOS
  - pMOS may be substituted with nMOS and charge pump or level shifter
  - Similar implementation using BJTs

# Actuator driving

- Floating load
  - Full H bridge
  - Example



|    |       | DIR      |          |
|----|-------|----------|----------|
|    |       | Negative | Positive |
| EN |       | 0        | 1        |
|    | NO 0  | 0<br>OFF | 0<br>OFF |
|    | YES 1 | 1<br>ON  | 0<br>OFF |

$$Y_{M_1} = EN \cdot \overline{DIR}$$

|    |       | DIR      |          |
|----|-------|----------|----------|
|    |       | Negative | Positive |
| EN |       | 0        | 1        |
|    | NO 0  | 0<br>OFF | 0<br>OFF |
|    | YES 1 | 0<br>OFF | 1<br>ON  |

$$Y_{M_4} = EN \cdot DIR$$

|    |       | DIR      |          |
|----|-------|----------|----------|
|    |       | Negative | Positive |
| EN |       | 0        | 1        |
|    | NO 0  | 0<br>OFF | 0<br>OFF |
|    | YES 1 | 0<br>OFF | 1<br>ON  |

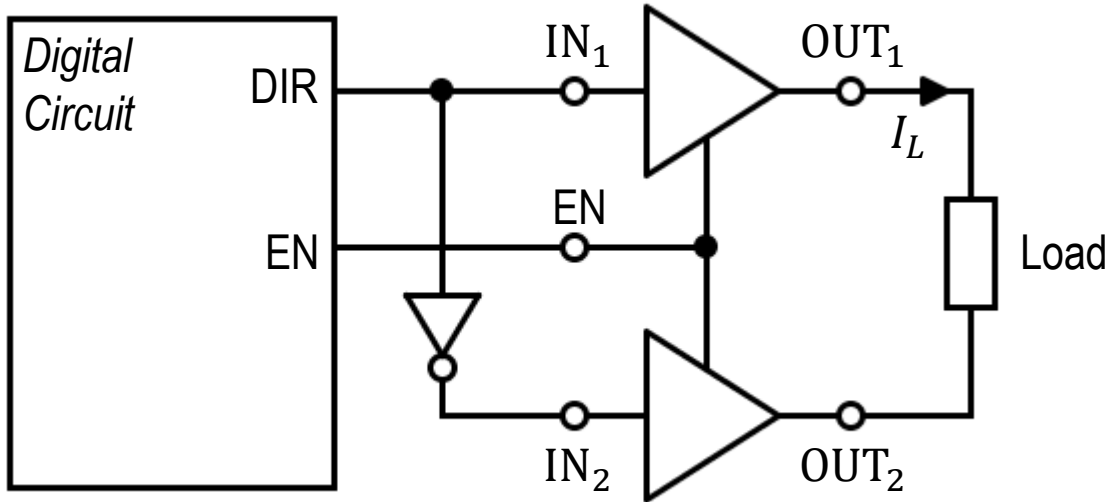
$$Y_{M_3} = EN \cdot \overline{DIR}$$

|    |       | DIR      |          |
|----|-------|----------|----------|
|    |       | Negative | Positive |
| EN |       | 0        | 1        |
|    | NO 0  | 0<br>OFF | 0<br>OFF |
|    | YES 1 | 1<br>ON  | 0<br>OFF |

$$Y_{M_6} = EN \cdot \overline{DIR}$$

# Actuator driving

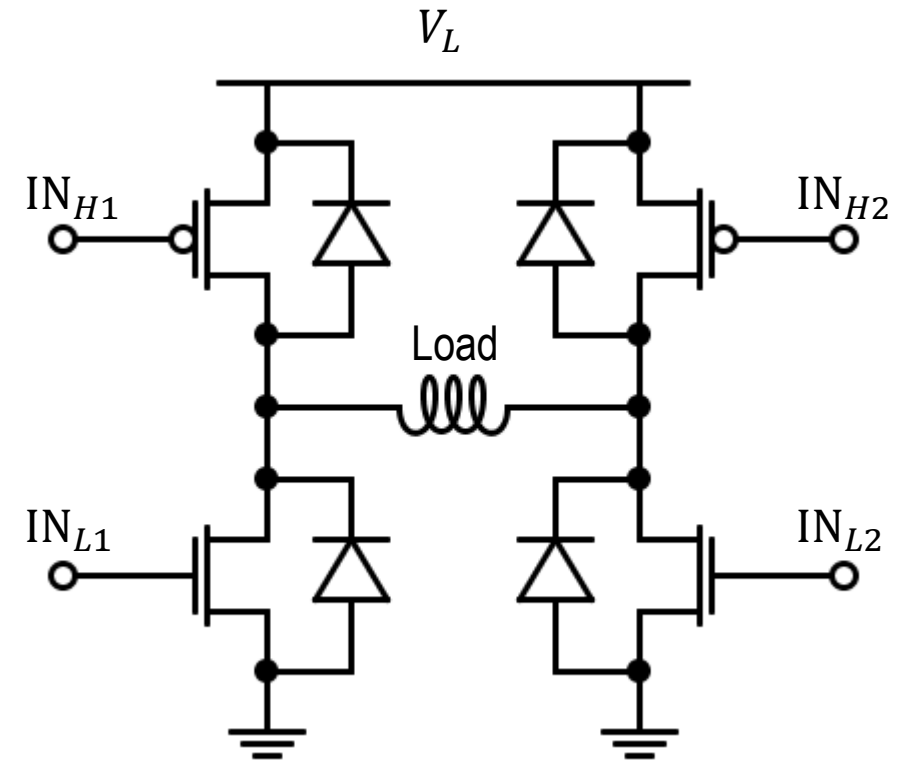
- Floating load
  - Full H bridge
    - Example



| EN | DIR | IN <sub>2</sub> | IN <sub>1</sub> | OUT <sub>2</sub> | OUT <sub>1</sub> |                 |
|----|-----|-----------------|-----------------|------------------|------------------|-----------------|
| 0  | -   | 0               | 0               | Z                | Z                | Load floating   |
| 0  | 1   | 0               | 1               | Z                | Z                | Load floating   |
| 0  | 0   | 1               | 0               | Z                | Z                | Load floating   |
| 0  | -   | 1               | 1               | Z                | Z                | Load floating   |
| 1  | -   | 0               | 0               | 0                | 0                | Load grounded   |
| 1  | 1   | 0               | 1               | 0                | 1                | $I_L$ positive  |
| 1  | 0   | 1               | 0               | 1                | 0                | $I_L$ negative  |
| 1  | -   | 1               | 1               | 1                | 1                | Load "grounded" |

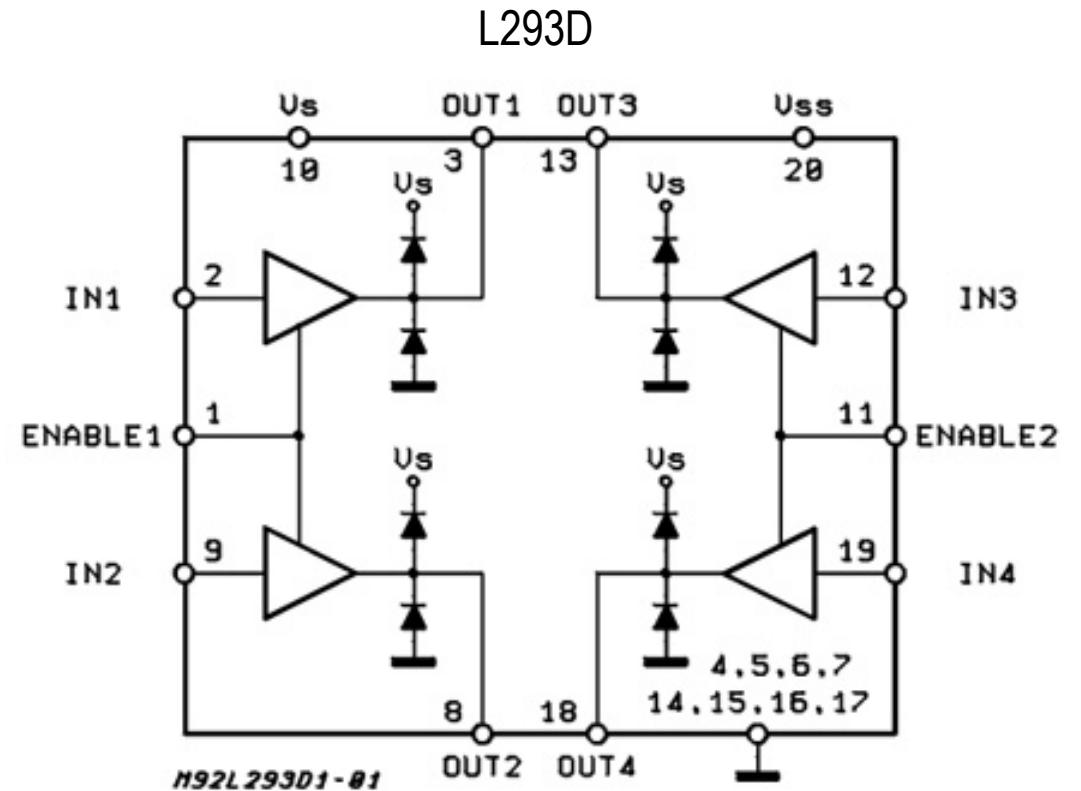
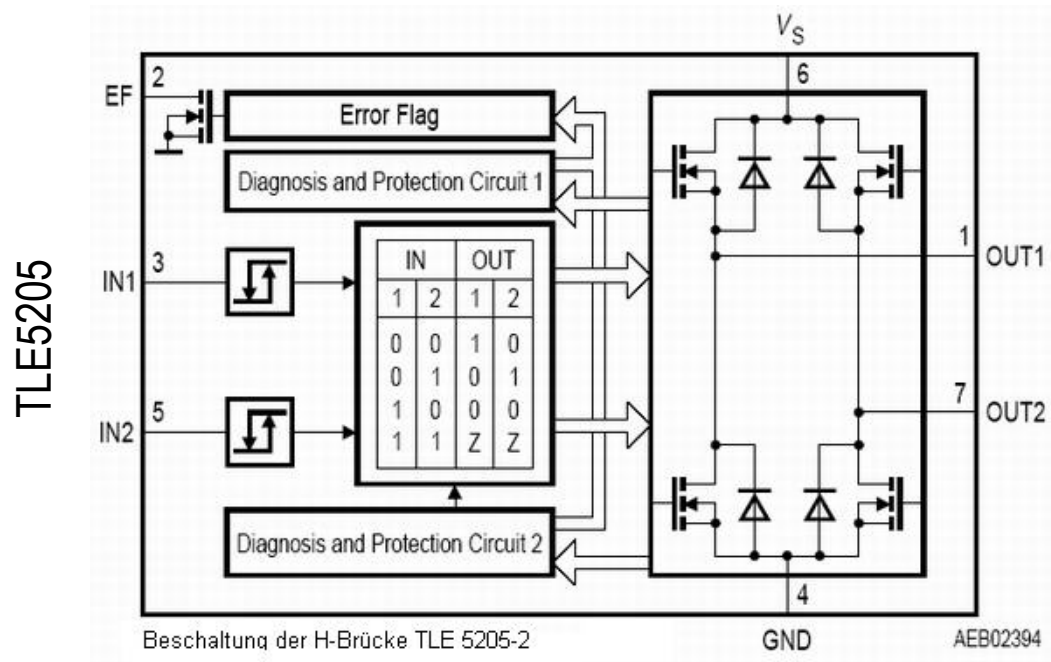
# Actuator driving

- Floating load
  - Full H bridge
    - With inductive loads
      - Need diodes to avoid strong voltage spikes
    - Power MOS transistors already have parasitic diodes between Source and Drain
      - They can be used as protection in certain circumstances
      - Otherwise use external diodes



# Actuator driving

- Floating load
  - Half and Full H bridge
    - They can be found as integrated circuits
    - IC may also contains direction and enable logic
    - Often use npn or nMOS in high side, as well

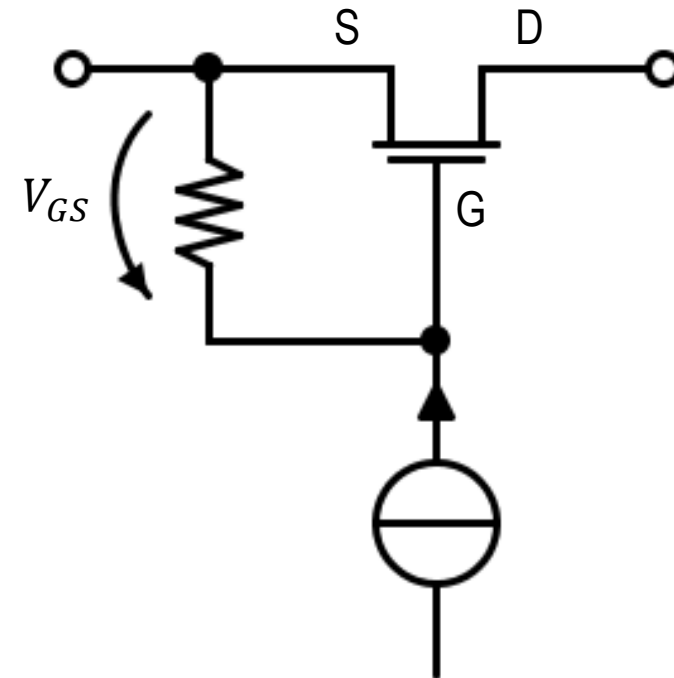


# Actuator driving

- MOS Gate drivers
  - Special integrated circuits used to drive a power MOS (or IGBT) from digital logic
    - Provide high current, fast edges and fast switching capabilities
    - Required in high power applications
      - The digital circuit cannot switch the MOS fast enough (low driving current)
      - Slow switching causes a high power consumption
        - The transistor stays in active region too long
    - May contain a charge pump for high side nMOS
      - $R_{DSon}$  is roughly 1/3 of a similar pMOS

# Actuator driving

- Floating switch nMOS
  - Both switch terminals at any voltage
    - Need a special circuit to control  $V_{GS}$ 
      - Current flows in the resistance only
      - It causes a voltage drop between Gate and Source
        - No current  $\rightarrow$  Switch open
        - Current flowing  $\rightarrow$  Switch closed
  - Used as a pass gate
    - In analog multiplexers
    - In Sample&Hold circuits



# Power supply

- What is a Power Supply Unit (PSU)
  - All electronic circuits require energy to work correctly
    - They need a (constant) supply voltage and withdraw a certain maximum current
  - Energy may come from a number of primary sources
    - National electrical distribution grid
      - Main power outlet, 230  $V_{RMS}$  @ 50 Hz
    - Battery
    - Solar panel
    - Dynamo
    - ...
  - Power Supply Unit (PSU)
    - A circuit that transfers energy from a primary source to the electronic circuit
    - The PSU is itself an electronic circuit that requires energy to work
    - Different circuits have different characteristics

# Power supply

- Power supply parameters
  - Nominal output voltage  $V_{out}$ 
    - Constant voltage
    - Positive or negative, fixed or variable
  - Output voltage tolerance
    - Expressed as a relative error of the nominal output voltage
  - Maximum output current  $I_{outmax}$ 
    - For higher currents the nominal output voltage regulation is not guaranteed
  - Type, nominal value and range of the input signal
    - Alternating
      - Frequency, amplitude, shape
    - Continuous
      - Tolerance

# Power supply

- Power supply parameters
  - Load regulation or sensitivity
    - Change of the output voltage  $V_{out}$  for a change of the output current  $I_{out}$ 
      - The output current  $I_{out}$  depends on the load requirements
      - Depending on the load state, the output current  $I_{out}$  can significantly change
        - It should be  $I_{out} < I_{outmax}$
        - Sometimes  $I_{out} = 0$ 
          - It may have negative effects on the power supply
    - Equivalent to an output resistance
    - $S_I = R_{out} = \frac{\Delta V_{out}}{\Delta I_{out}}$
    - Ideally  $S_I = 0$ 
      - No output voltage change if the load current changes
      - In practice it should be within acceptable specifications

# Power supply

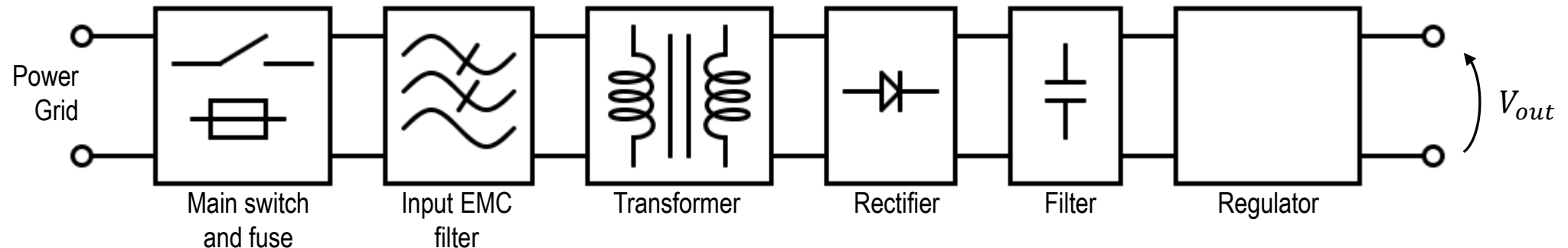
- Power supply parameters
  - Line regulation or sensitivity
    - Change of the output voltage  $V_{out}$  for a change of the input voltage  $V_{in}$ 
      - The input voltage  $V_{in}$  depends on the source
      - A power supply works when  $V_{in}$  remains within the allowed range
    - $S_V = \frac{\Delta V_{out}}{\Delta V_{in}}$
    - Ideally  $S_V = 0$ 
      - No output voltage change if the input voltage changes
      - In practice it should be within acceptable specifications

# Power supply

- Power supply parameters
  - Efficiency  $\eta$ 
    - Ratio between
      - The power  $P_L$  delivered to the load
      - The total power  $P_{tot}$  withdrawn from the primary source
    - $\eta = \frac{P_L}{P_{tot}}$
    - Ideally  $\eta = 1$  (or 100%, as it is often expressed in percentage)
    - In practice it is always lower
      - Linear regulators
        - Low efficiency
        - Excellent sensitivities
      - Switching regulators
        - High efficiency
        - Good sensitivities

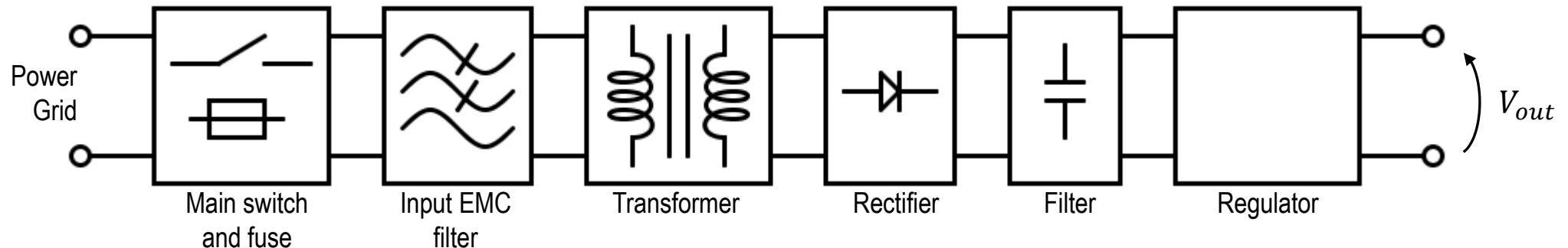
# Power supply

- Power supply block diagram
  - Starting from an alternating source
    - Protection and filtering
    - A pseudo-DC voltage is generated
      - Rectifier and filtering
    - A regulator provides the required output voltage
  - Starting from a pseudo-continuous source
    - A regulator provides the required output voltage



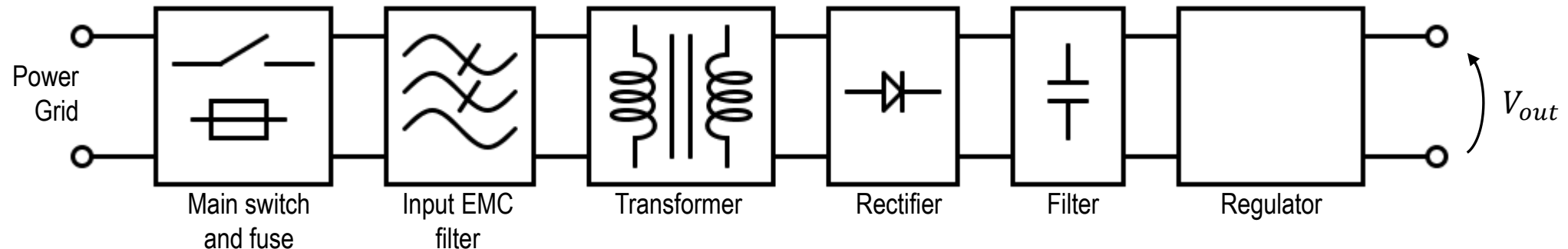
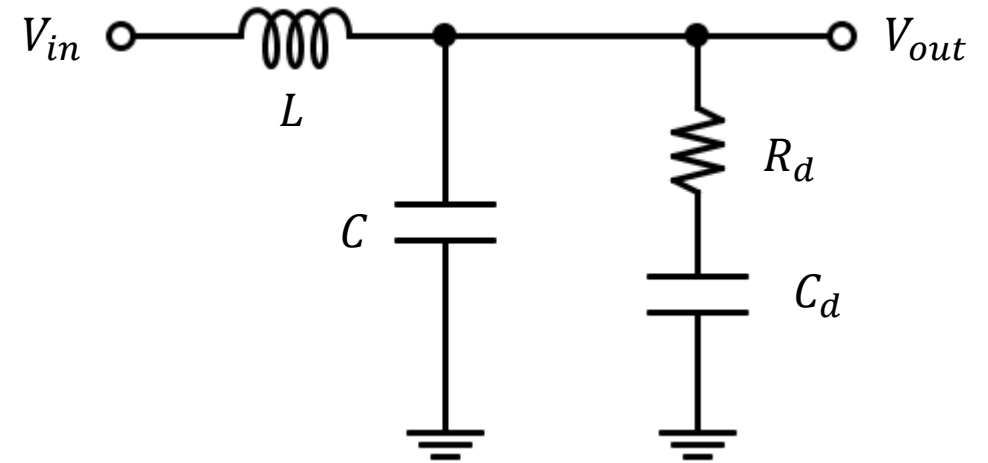
# Power supply

- Power supply block diagram
  - Main switch and fuse
    - Switch used to turn ON and OFF the power supply
      - Provides safety
      - May be absent
    - Fuse protects the power supply and the load from excessive currents
      - When it blows, no more power delivered to the load
      - Needs a replacement



# Power supply

- Power supply block diagram
  - Input EMC filter
    - Avoids noise from the power supply and load to reach the Power Grid
      - Required by national regulations (CE certification)
    - Avoids noise from the Power Grid to reach the power supply and the load

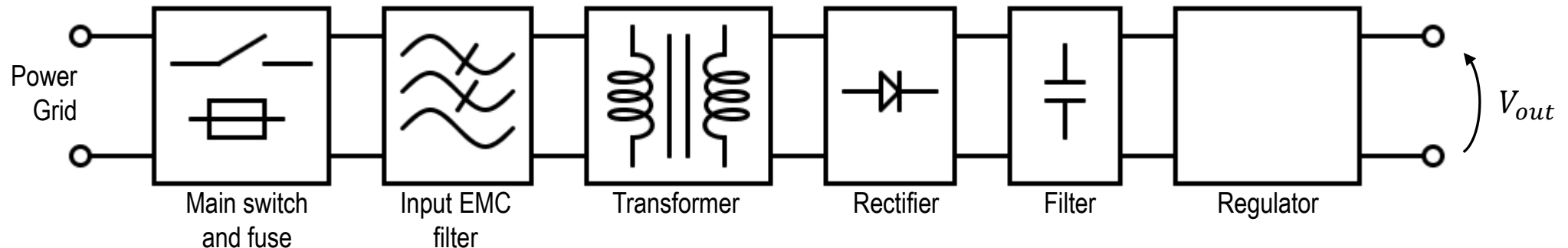


# Power supply

- Power supply block diagram

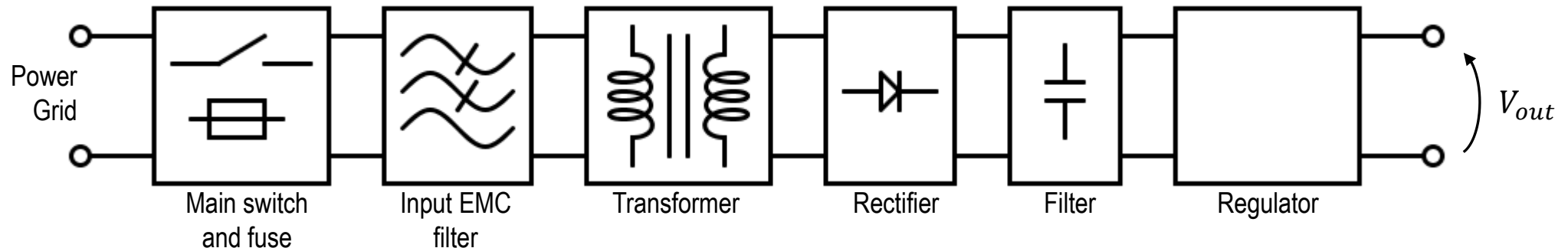
- Transformer

- Adapts the input voltage swing to be closer to the required output
    - Primary and secondary windings
      - $N_P$ : number of turns in the primary
      - $N_S$ : number of turns in the secondary
  - $\frac{V_P}{V_S} = \frac{N_P}{N_S} \Rightarrow V_S = \frac{N_S}{N_P} V_P$
  - No power increase or decrease
    - $V_P I_P = V_S I_S$



# Power supply

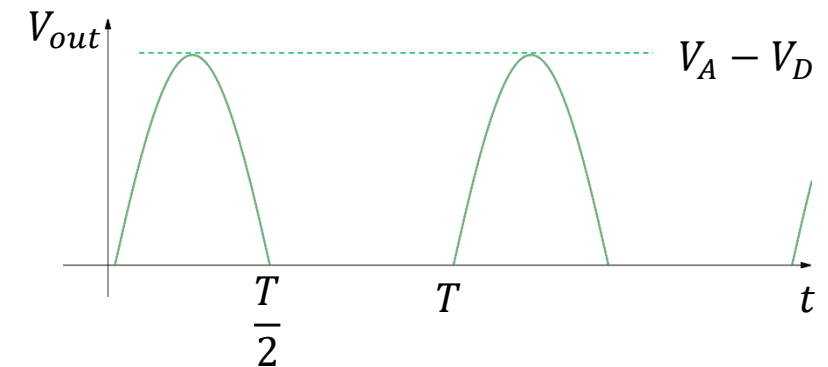
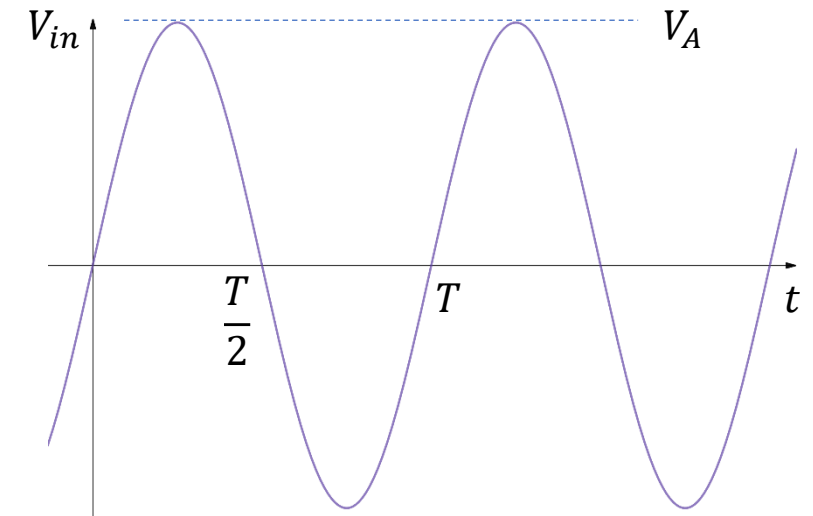
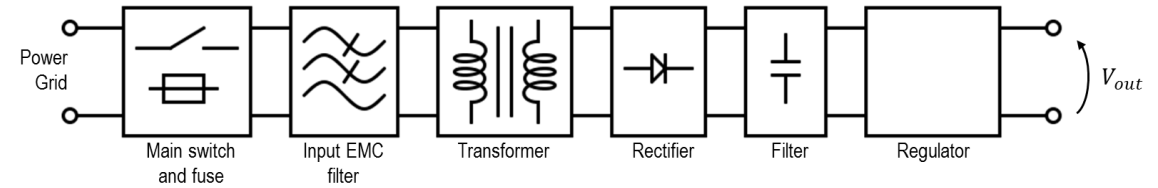
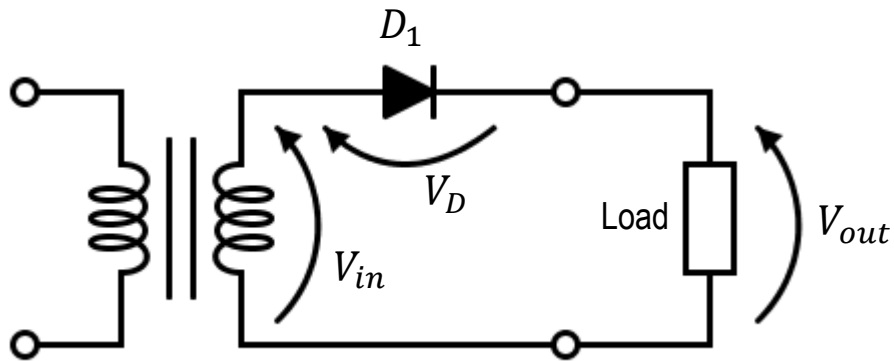
- Power supply block diagram
  - Transformer
    - A transformer requires the input to change
      - It doesn't let the DC go through
      - If the input is sinusoidal, the output is sinusoidal
      - The output of a transformer has both positive and negative portions
      - The subsequent blocks, rectifier and filter, turn this signal into a pseudo-DC
    - A transformer provides galvanic isolation
      - Good for safety



# Power supply

- Rectifier

- Turns an AC voltage into a positive (or negative) waveform
- Half wave rectifier
  - Single diode
  - Very simple and cheap
  - Output is only the positive part of the input waveform
  - Peak value  $V_A$  decreased by the voltage drop  $V_D$  across the diode
    - $V_D \approx 1\text{ V}$

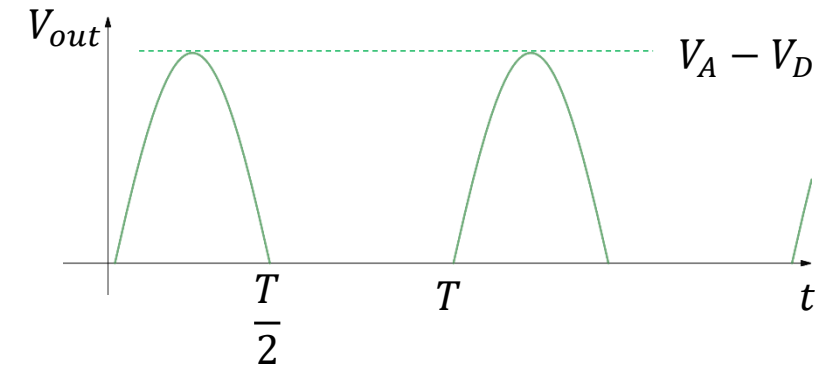
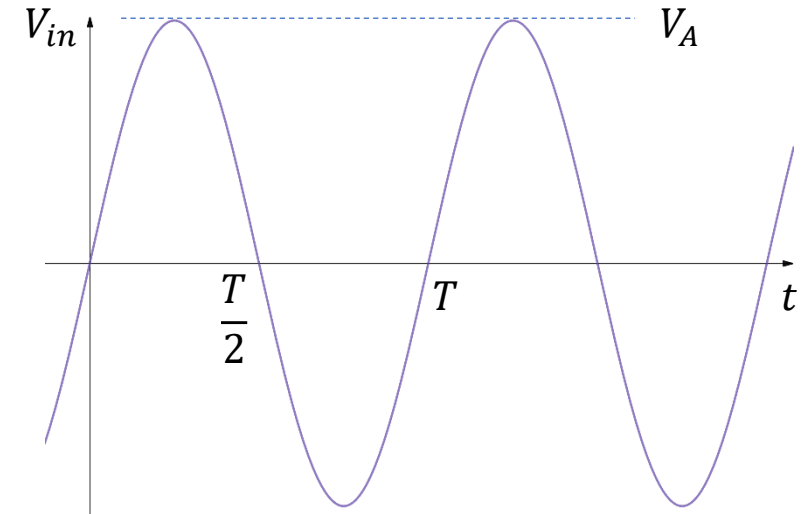
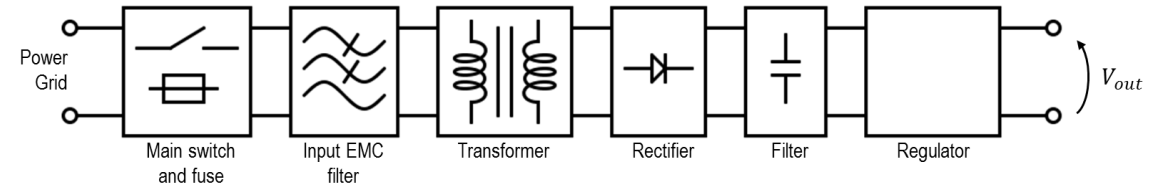
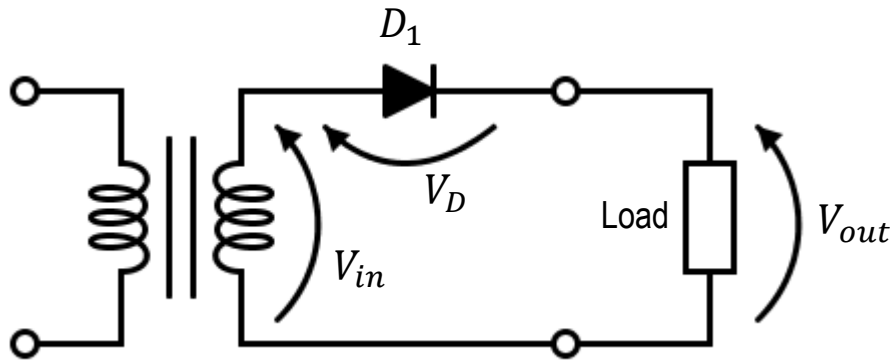


# Power supply

- Rectifier

- Half wave rectifier

- Not the best choice for the subsequent filter
    - Current in the secondary of the transformer has a DC component
      - Bias on the core
      - Core oversized to avoid saturation
      - Increased weight and cost
    - Evaluate breakdown voltage and maximum current after considering the filter

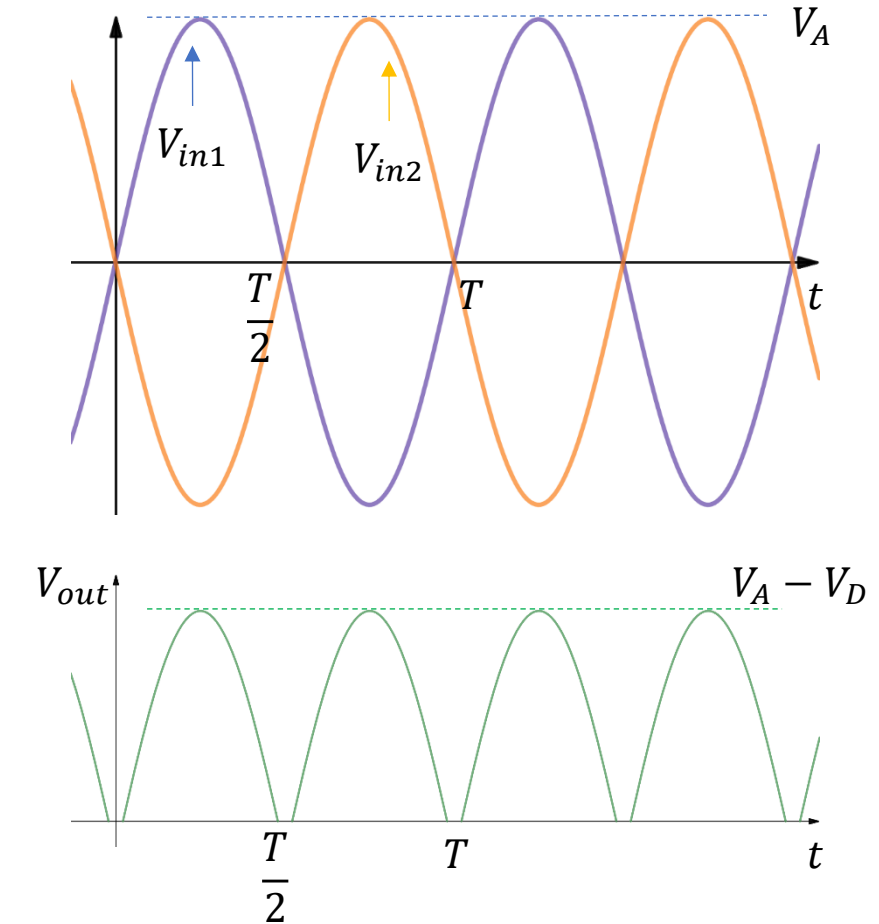
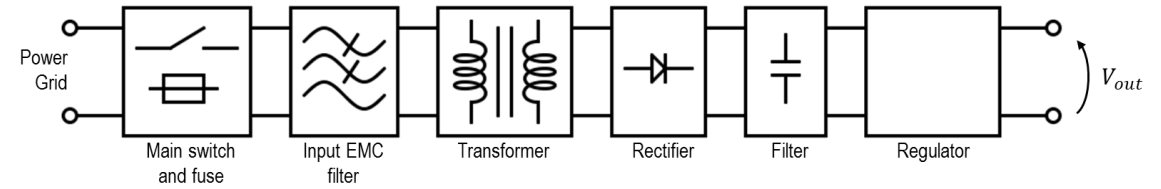
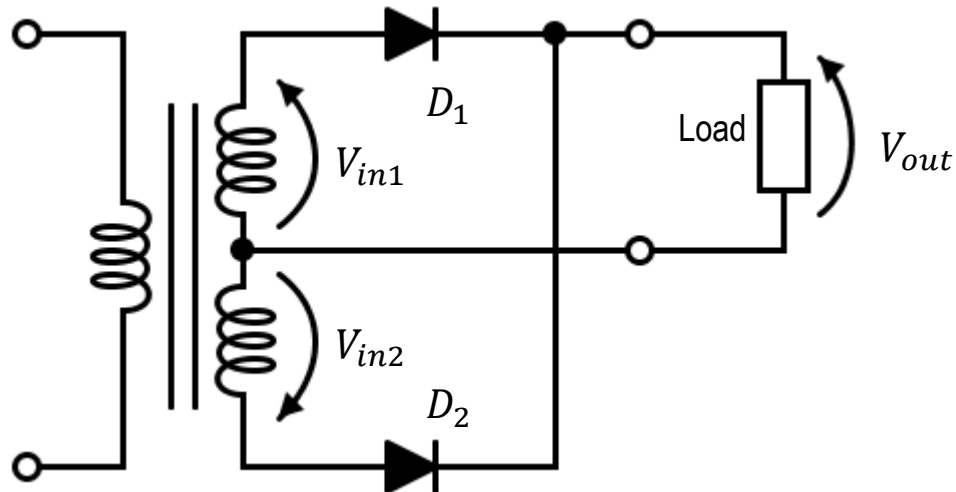


# Power supply

- Rectifier

- Full wave rectifier

- Two diodes
    - Transformer with two secondary windings and a center tap
      - Heavier and bulkier
      - More expensive

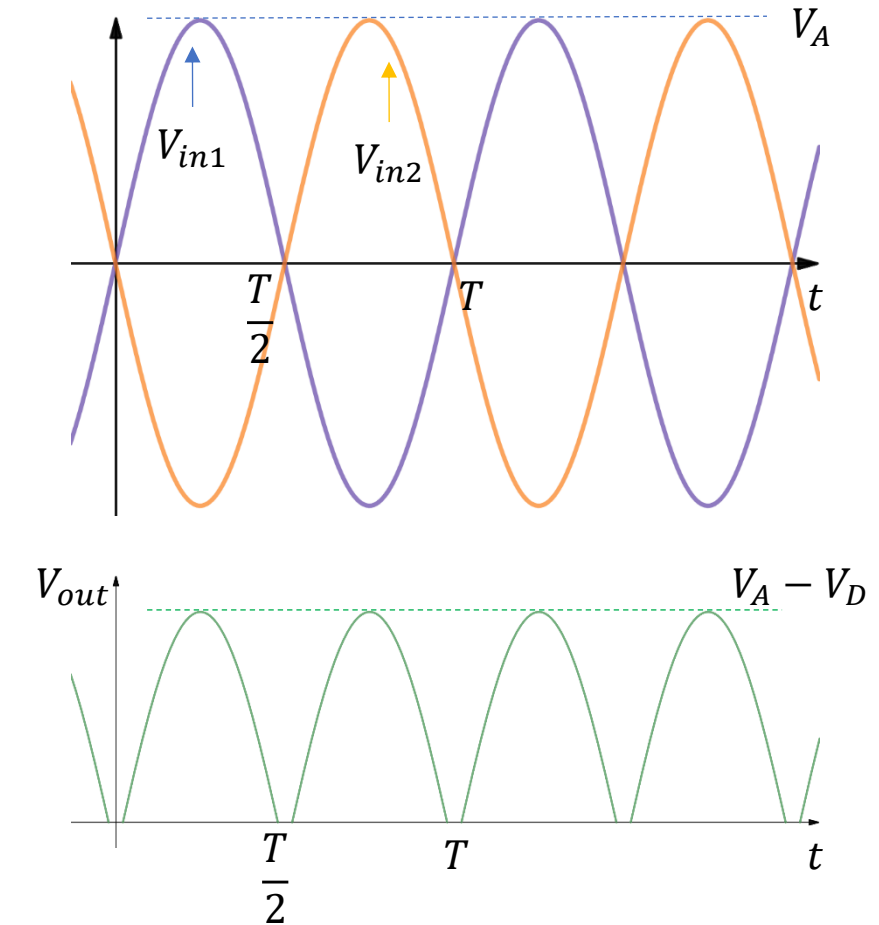
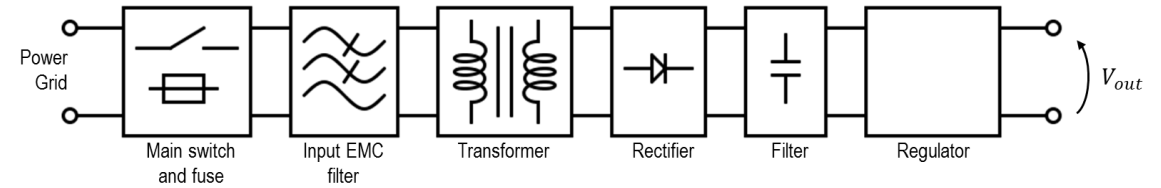
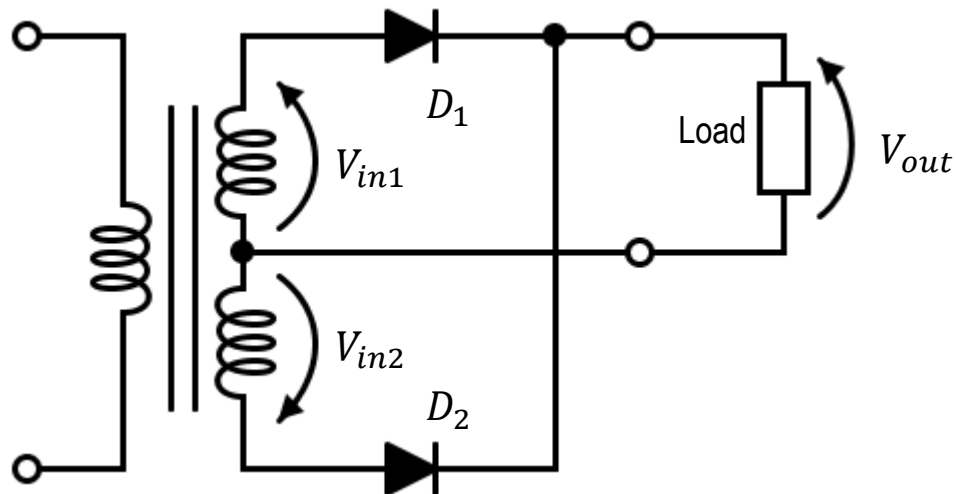


# Power supply

- Rectifier

- Full wave rectifier

- During the positive half of the input waveform
      - $D_1$  conducts and  $D_2$  is in cut-off
    - During the negative half of the input waveform
      - $D_1$  is in cut-off and  $D_2$  conducts

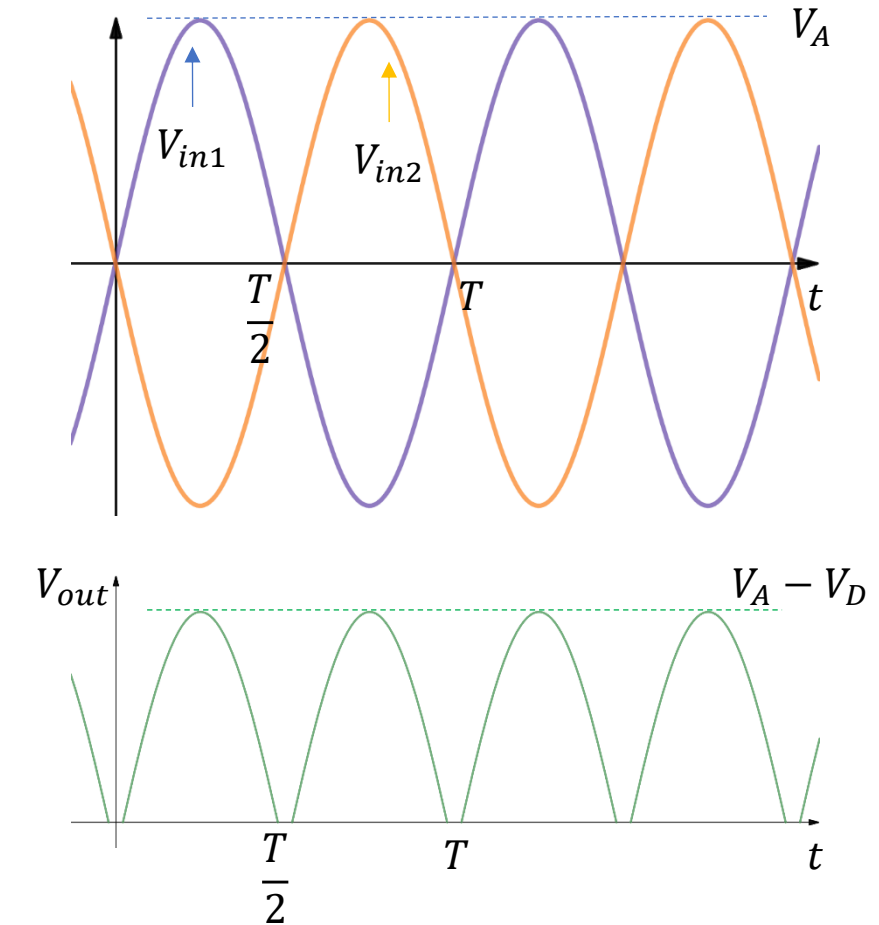
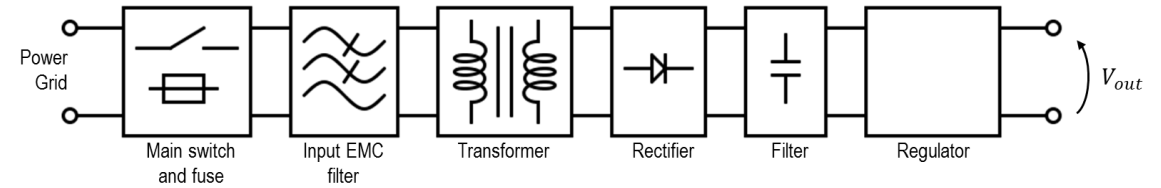
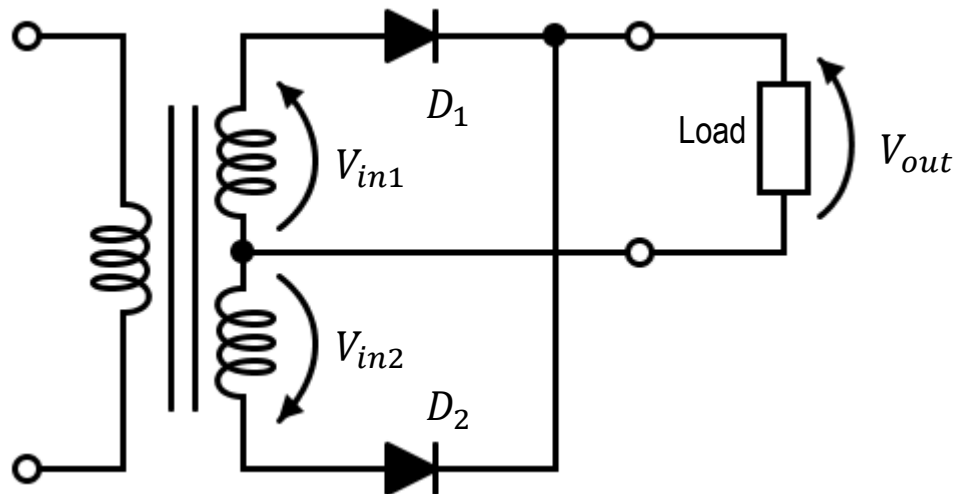


# Power supply

- Rectifier

- Full wave rectifier

- Diodes operate alternatively
      - Still a single  $V_D$  drop on the output voltage
    - Output current divided between the two diodes
      - Lower power dissipation in each diode compared to the half wave rectifier
      - Cheaper components



# Power supply

- Rectifier

- Full wave rectifier

- Four diodes

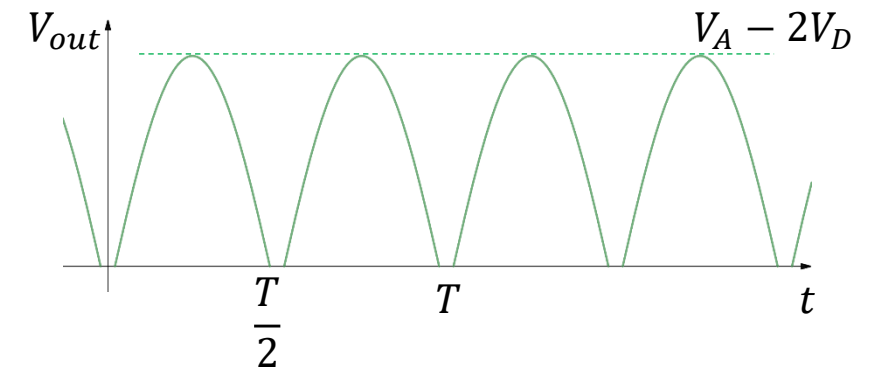
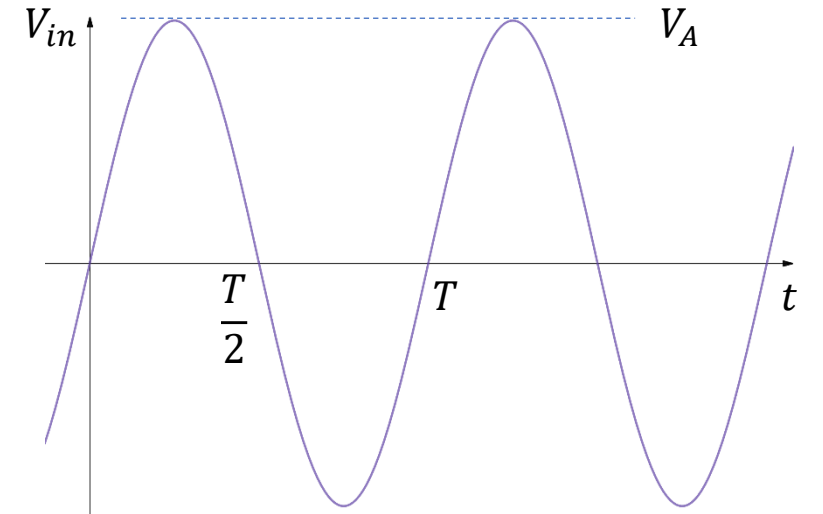
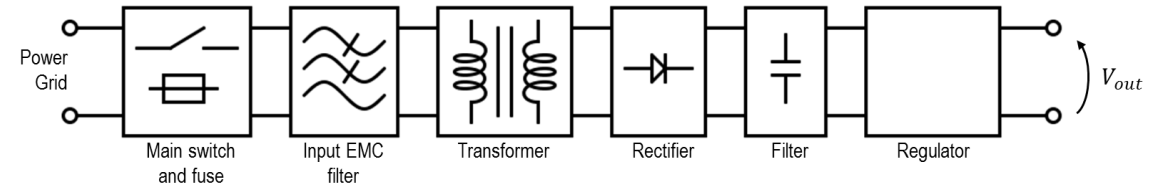
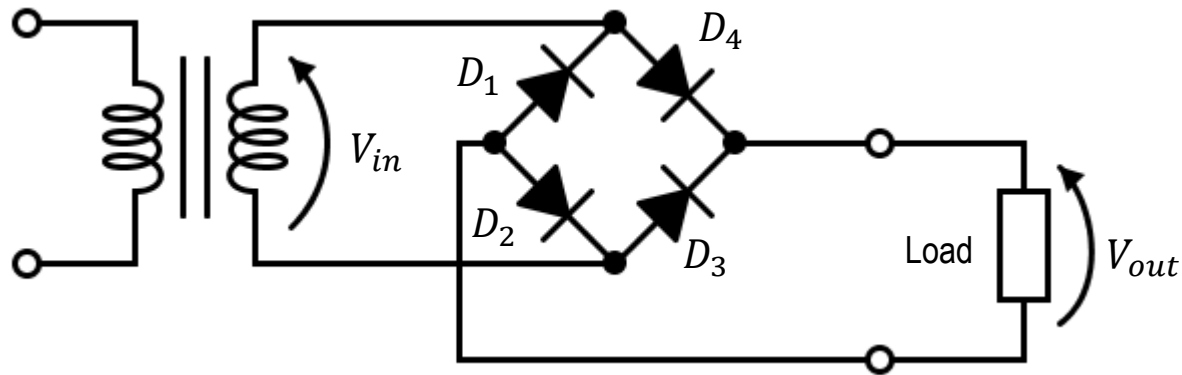
- Graetz Bridge

- Twice the number of diodes

- More expensive

- Normal transformer

- Smaller and cheaper than full wave rectifier with 2 diodes

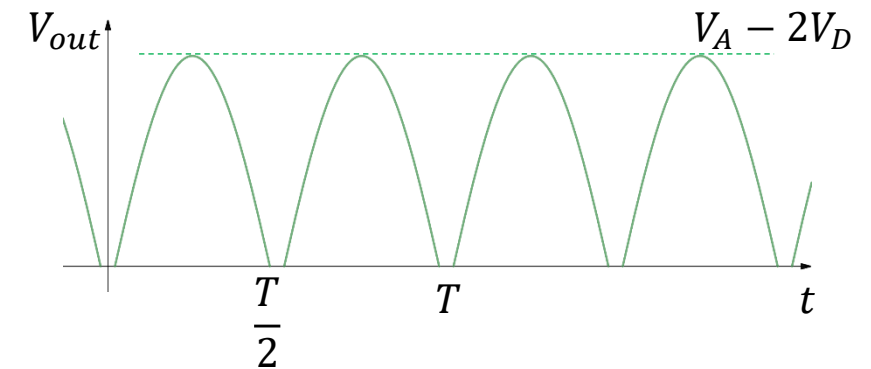
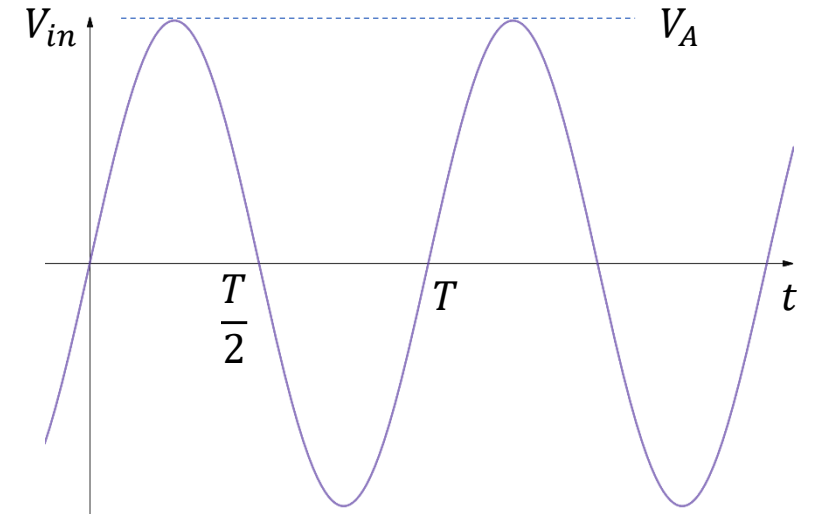
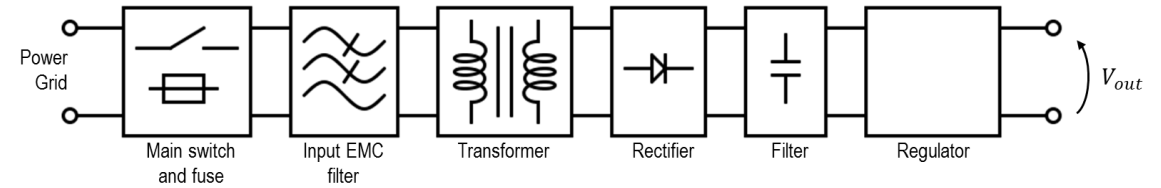
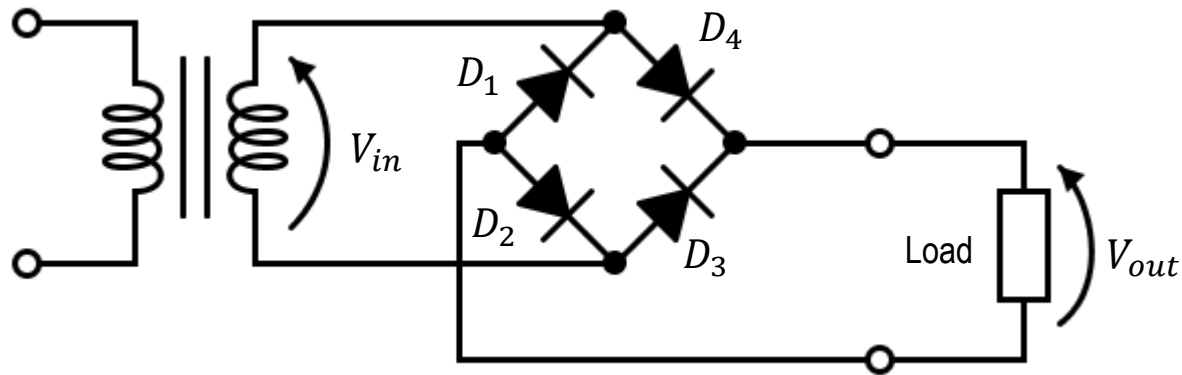


# Power supply

- Rectifier

- Full wave rectifier

- During the positive half of the input waveform
      - $D_4$  and  $D_2$  conduct
      - $D_1$  and  $D_3$  are in cut-off
      - The output voltage is decreased by two  $V_D$  voltage drops

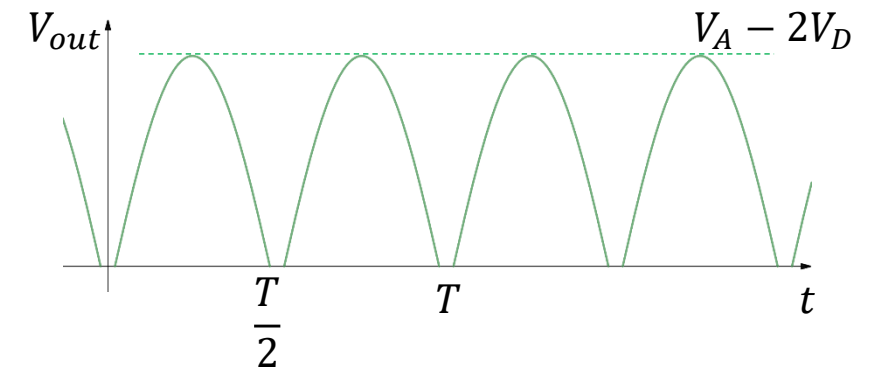
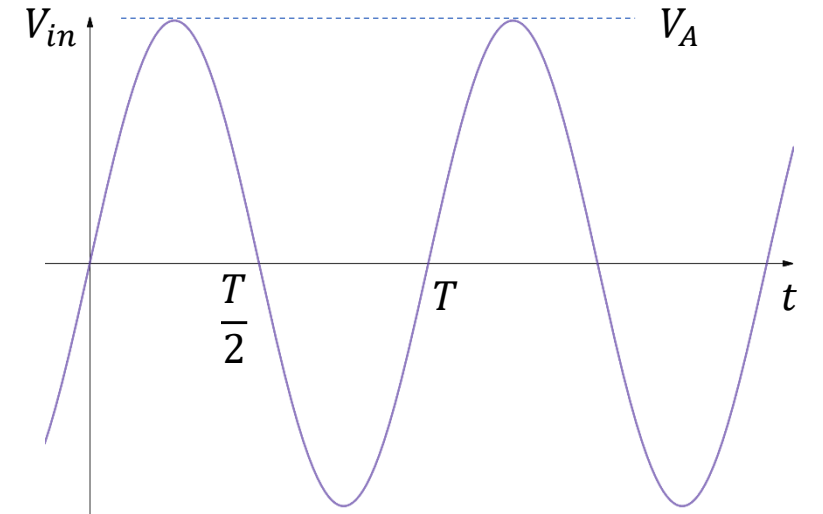
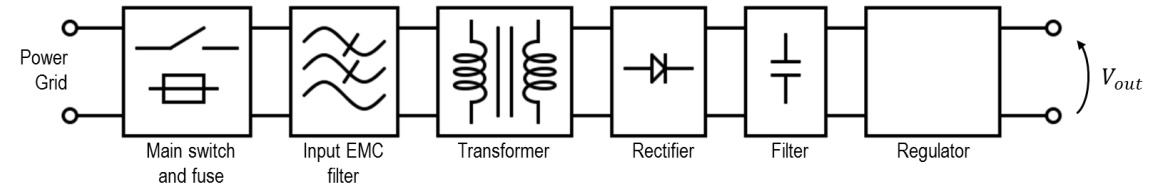
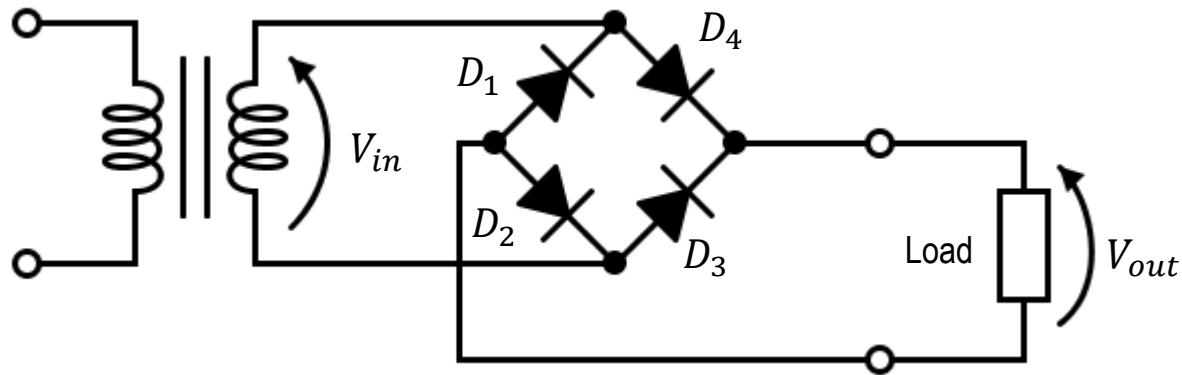


# Power supply

- Rectifier

- Full wave rectifier

- During the negative half of the input waveform
      - $D_4$  and  $D_2$  are in cut-off
      - $D_1$  and  $D_3$  conduct
      - The output voltage is decreased by two  $V_D$  voltage drops



# Power supply

- Rectifier

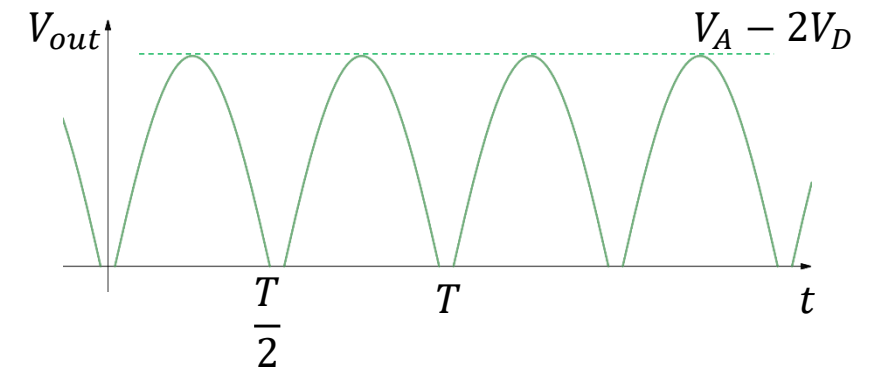
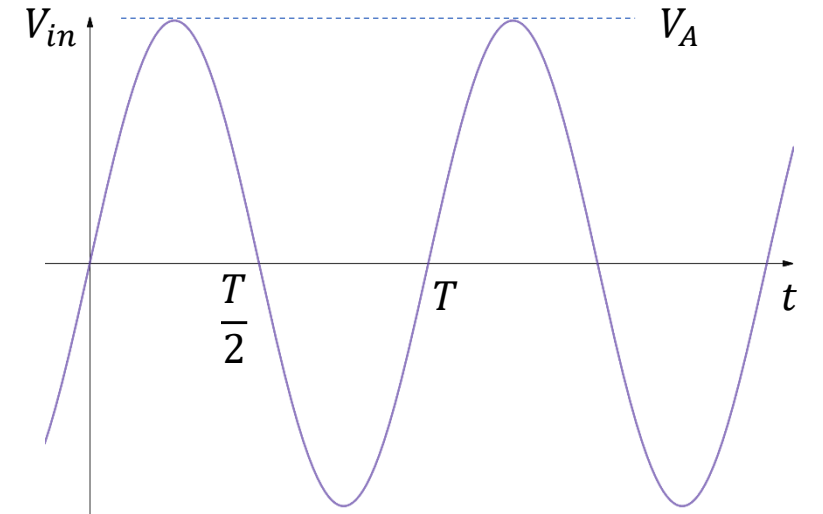
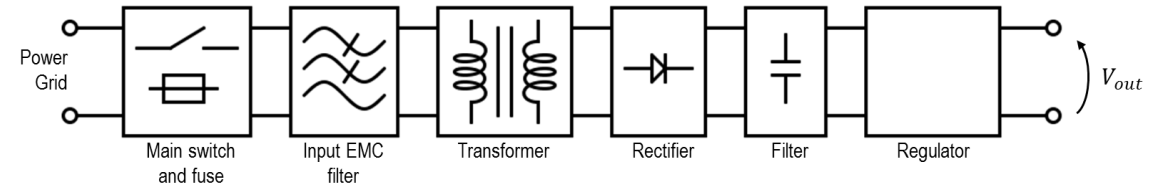
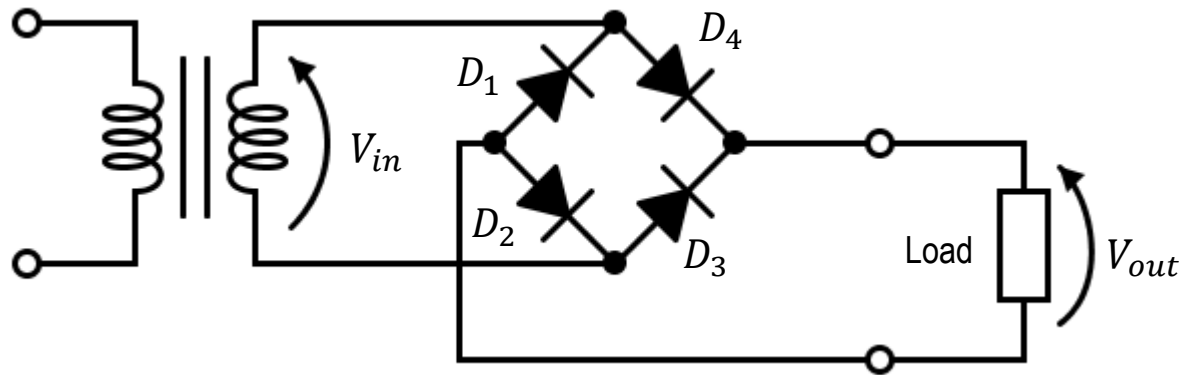
- Full wave rectifier

- Advantage

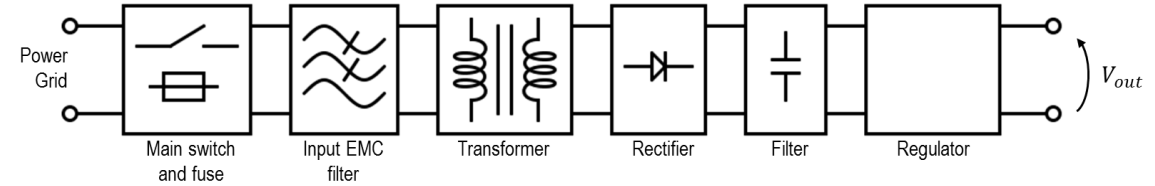
- Simple transformer

- Disadvantages

- Higher power consumption in diodes
      - Four diodes instead of two
      - Higher decrease of the output voltage



# Power supply



- Rectifier

- Full wave is better than half wave
  - It allows producing a better pseudo-DC after the filter
  - When the input waveform has a low amplitude
    - Full wave rectifier with 2 diodes is better, because of the lower voltage drop
    - Otherwise the full wave rectifier with 4 diodes is generally preferable
  - It is more expensive
- Still to evaluate
  - Maximum reverse voltage on the diodes
    - The diode breakdown voltage should be higher in absolute value
  - Maximum current flowing in the diodes
    - The maximum allowed current in the diode should be higher
  - Different values affect the costs of the diodes

# Power supply

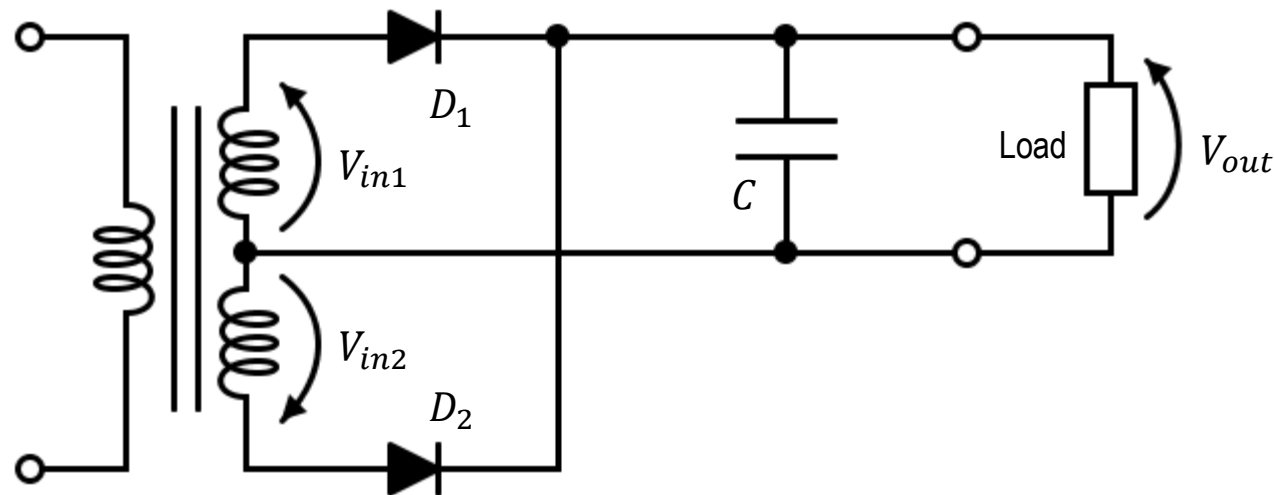
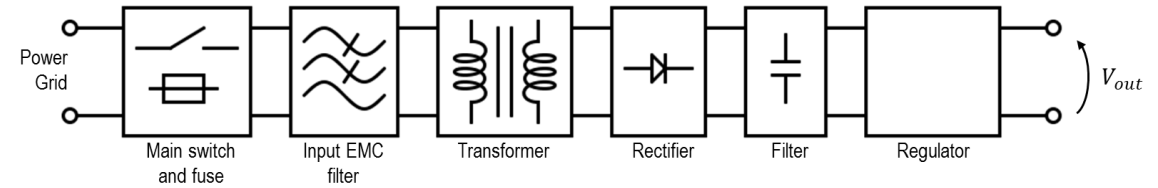
- Filter

- Peak detect and low pass

- Converts the pulsating output of the rectifier into a pseudo-DC
    - Constant value with a residual ripple

- Most simple filter

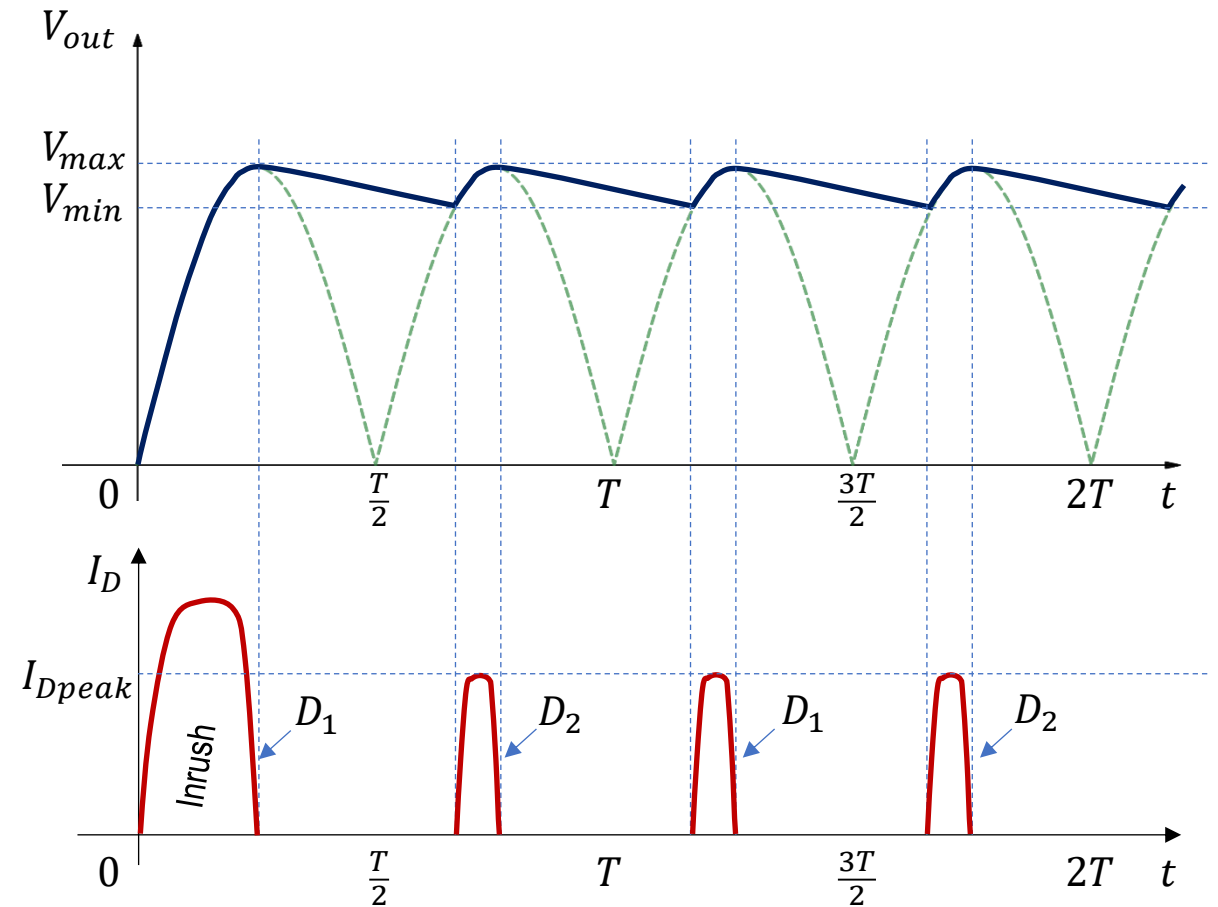
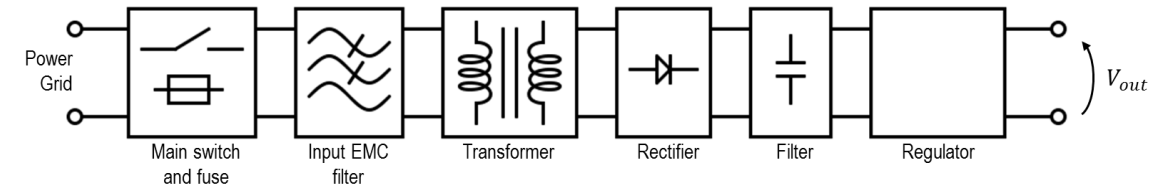
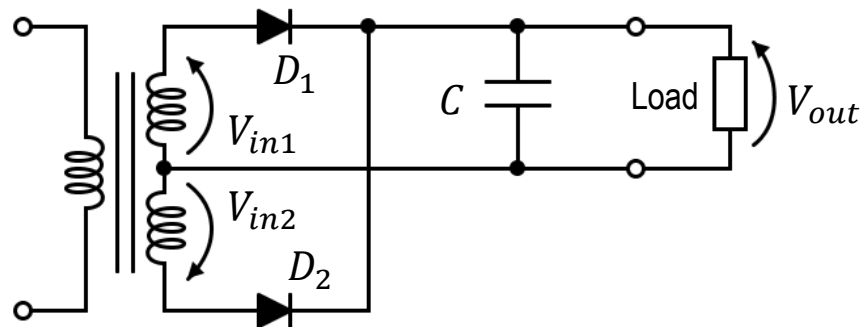
- A single capacitor in parallel with the load
    - It can be made more complex



# Power supply

- Filter

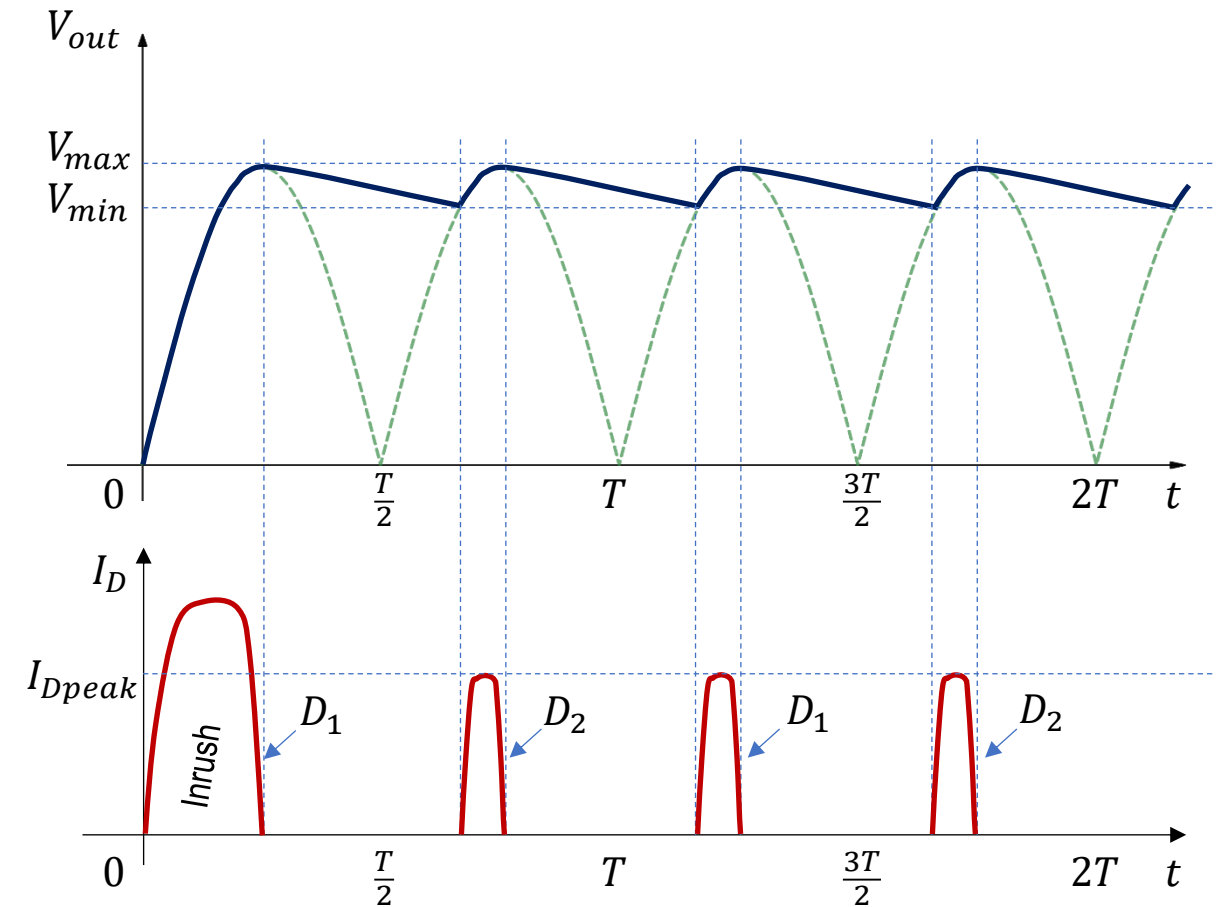
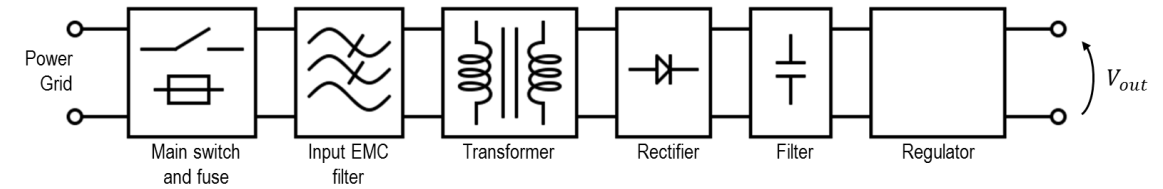
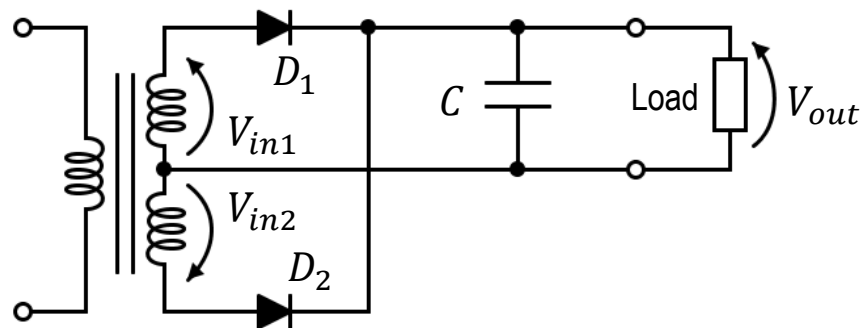
- Voltage on capacitor equal to output voltage
  - $V_C \equiv V_{out}$
- Assume capacitor is initially discharged
  - $V_{out} = 0\text{ V}$
- During the first half period
  - Diode  $D_1$  conducts
  - Diode  $D_2$  is in cut-off



# Power supply

## • Filter

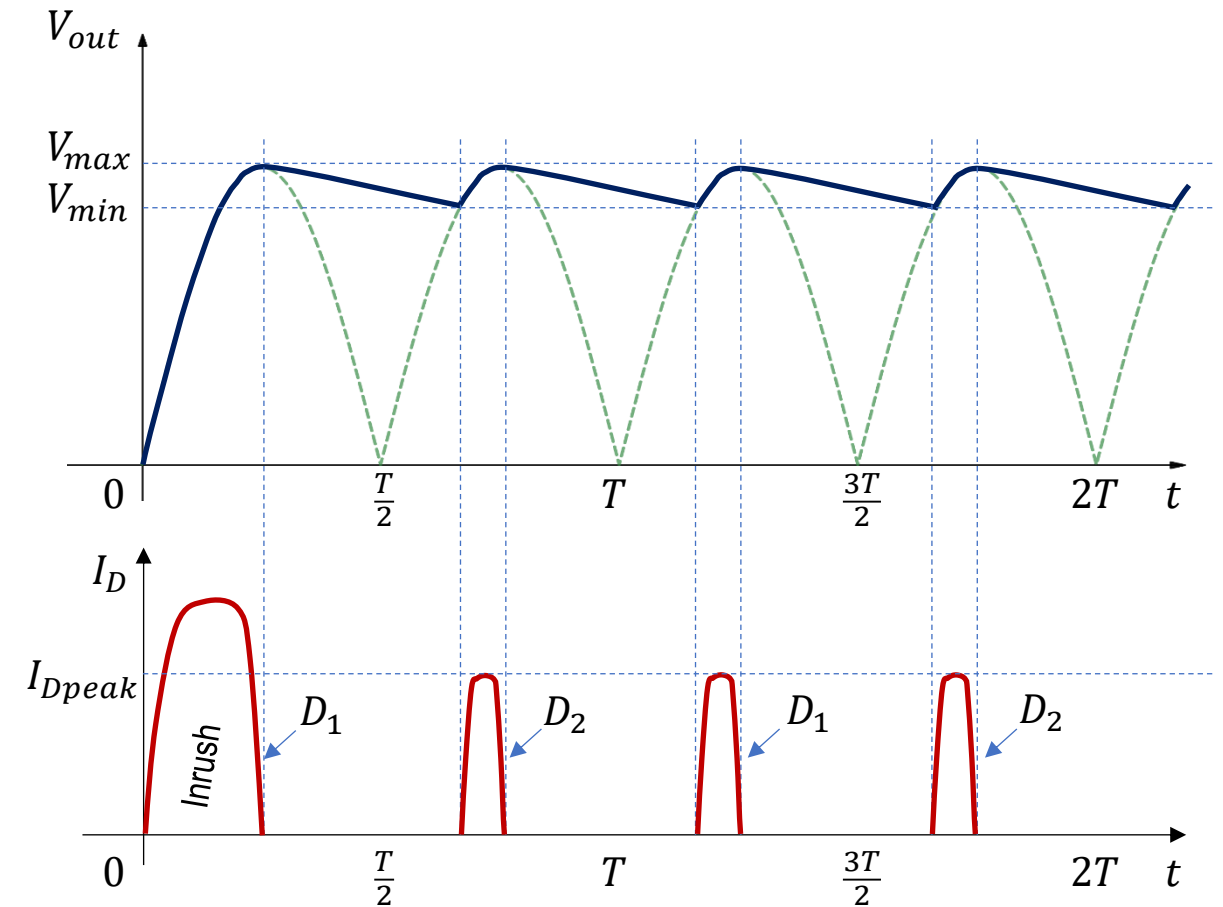
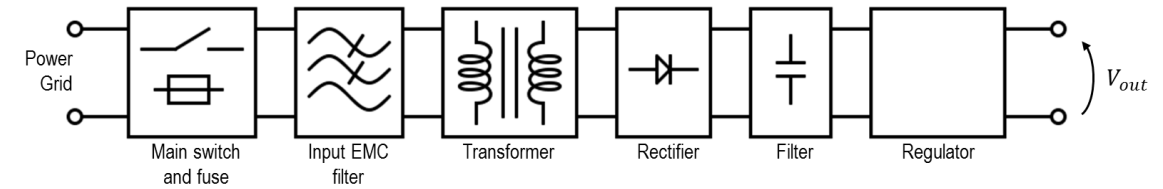
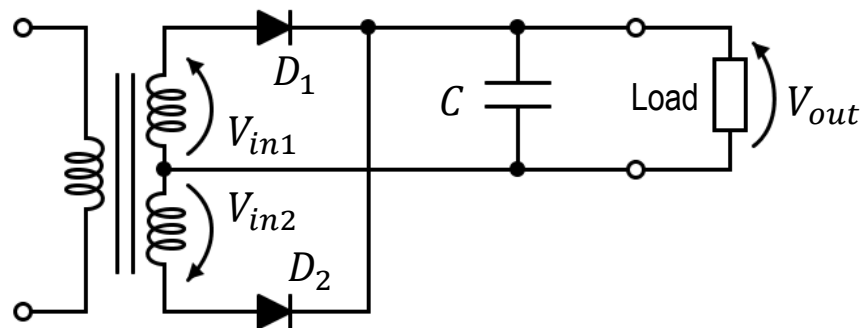
- During the first half period
  - The capacitor is charged to the peak voltage
  - $V_{max} = V_A - V_D$ 
    - With a four diodes rectifier (Graetz bridge)
      - $V_{max} = V_A - 2V_D$
- Inrush current in the diode  $D_1$ 
  - Only at power-up



# Power supply

- Filter

- When input  $V_{in1}$  decreases
  - The capacitor maintains a higher voltage
  - Diode  $D_1$  is reversed biased
    - It goes in cut-off
  - Diode  $D_2$  is still in cut-off
- The output voltage slowly decreases
  - Depends on the current in the load



# Power supply

- Filter

- Resistive load

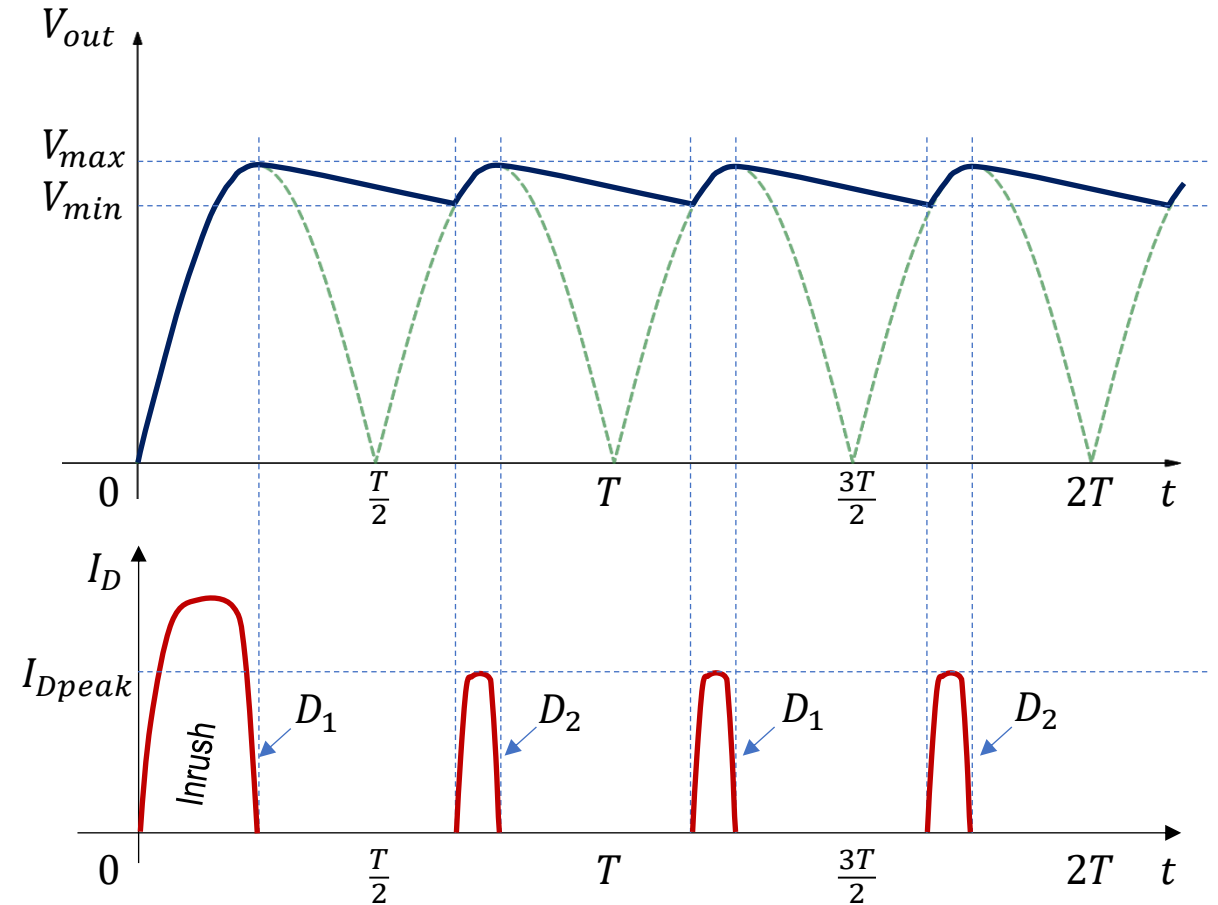
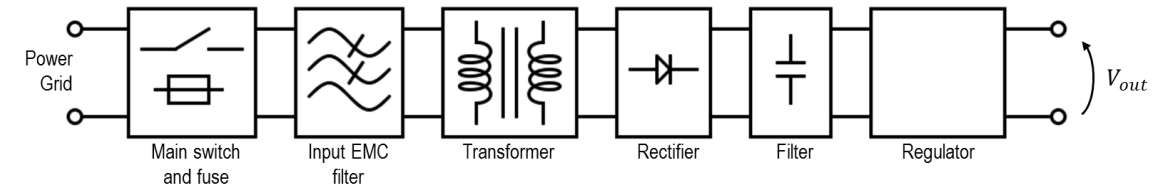
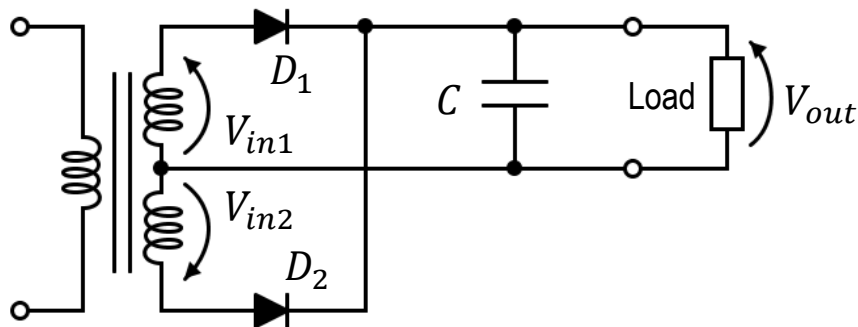
- The output voltage decreases according to an exponential law

- Constant current load

- The output voltage decreases linearly

- $V_{out}(t) = V_{max} - \frac{I_{load} \cdot t}{C}$

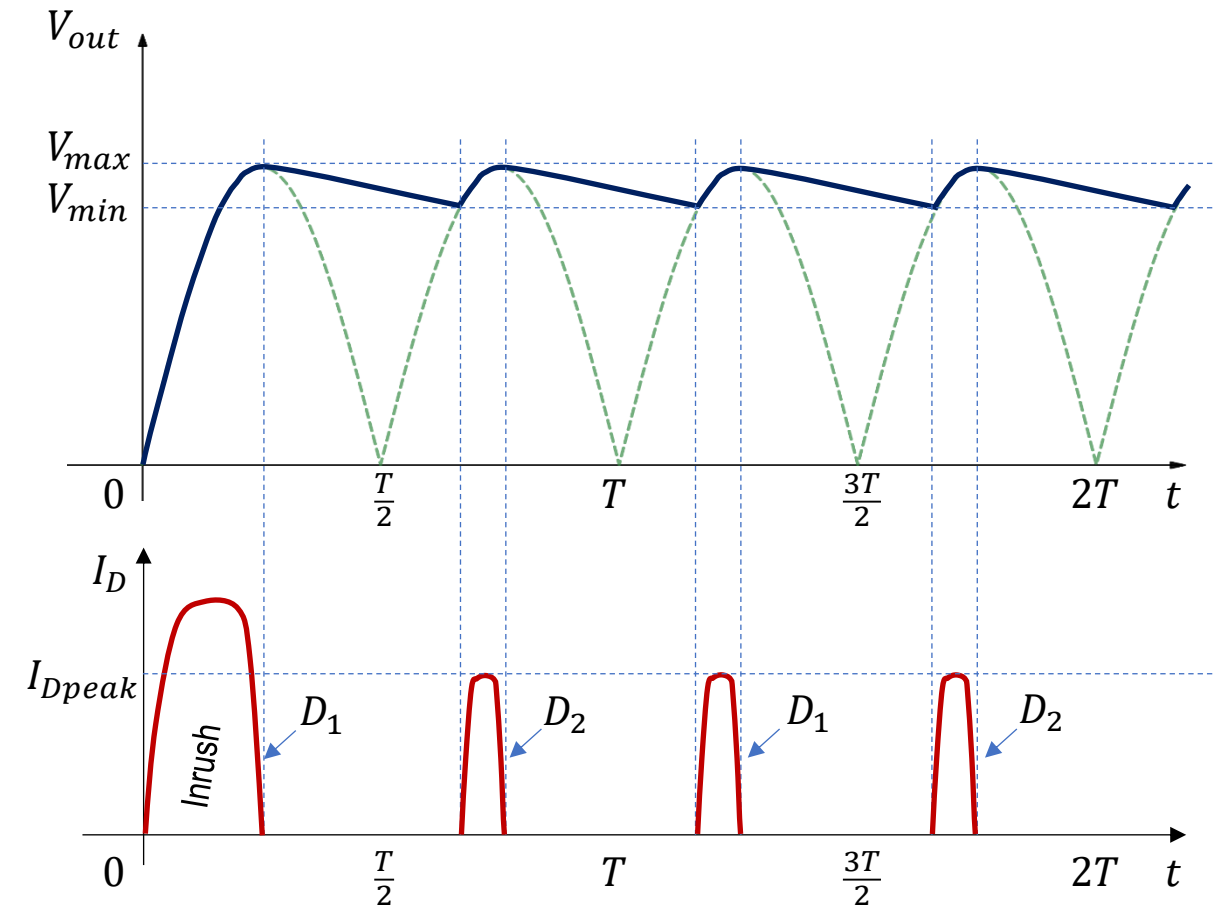
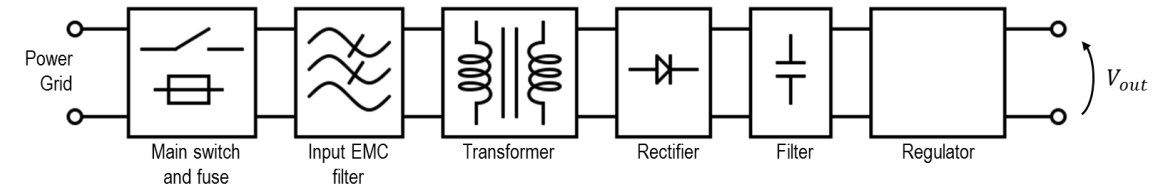
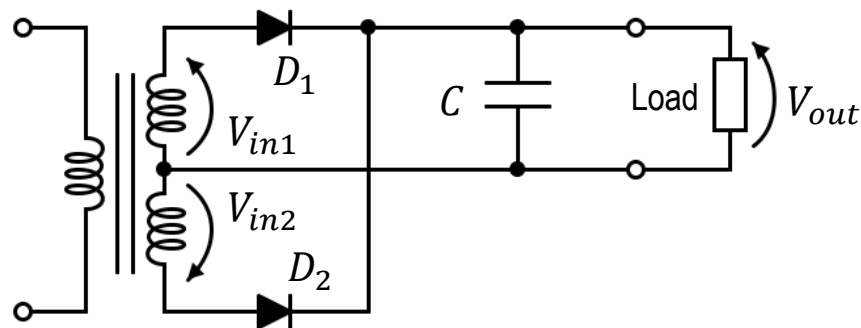
- $\Delta V_{out}(t) = V_{max} - V_{out}(t) = \frac{I_{load} \cdot t}{C}$



# Power supply

## • Filter

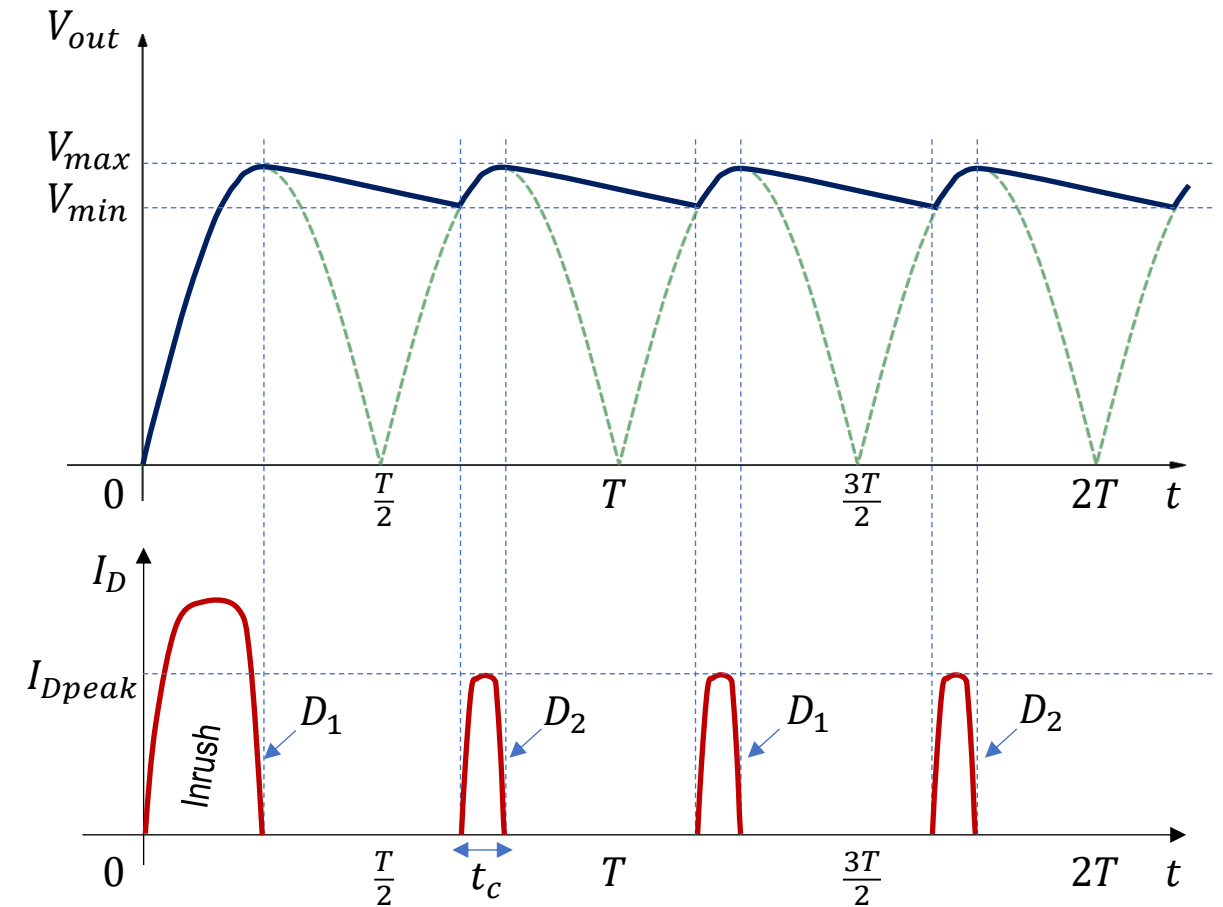
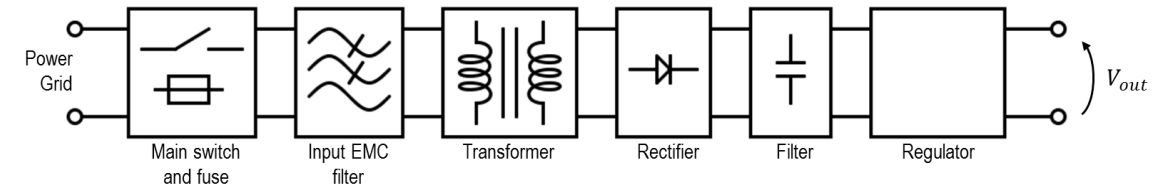
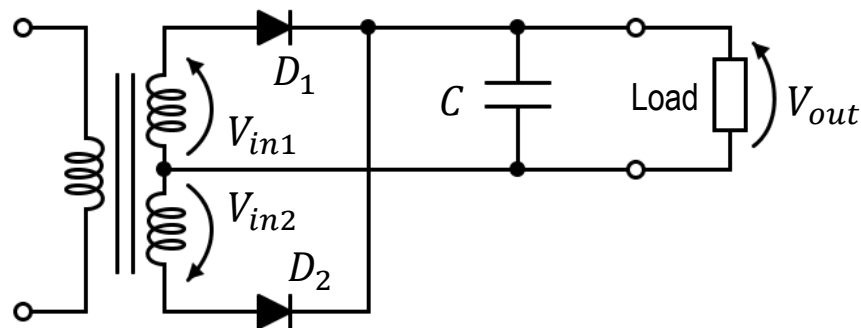
- At the beginning of the second half period
  - Both diodes are still in cut-off
  - The output voltage is still slowly decreasing
- Diode  $D_2$  becomes forward biased when
  - $V_{out} = V_{in2}$



# Power supply

- Filter

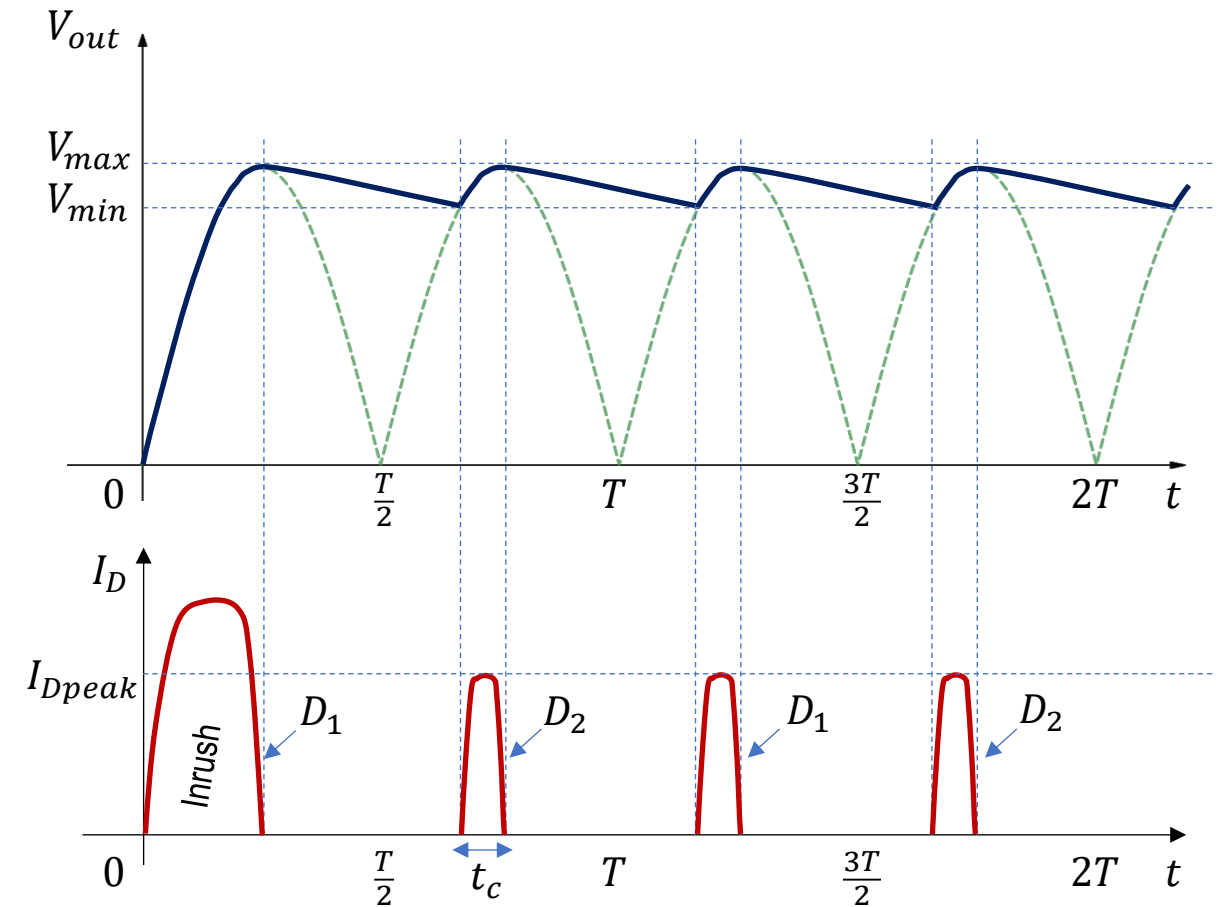
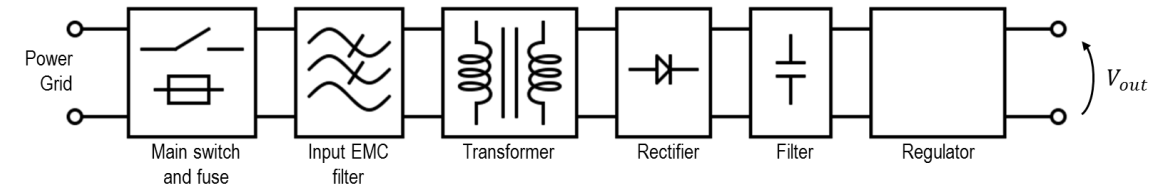
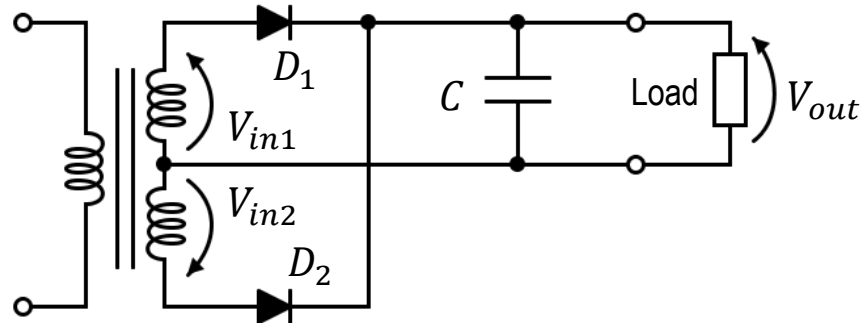
- During the second period
  - The capacitor is charged to the peak voltage
  - $V_{max} = V_A - V_D$
- Current in diode  $D_2$  is impulsive
  - It reaches a peak  $I_{Dpeak}$
  - The diode conducts for a short time  $t_c$ 
    - Computed later



# Power supply

## • Filter

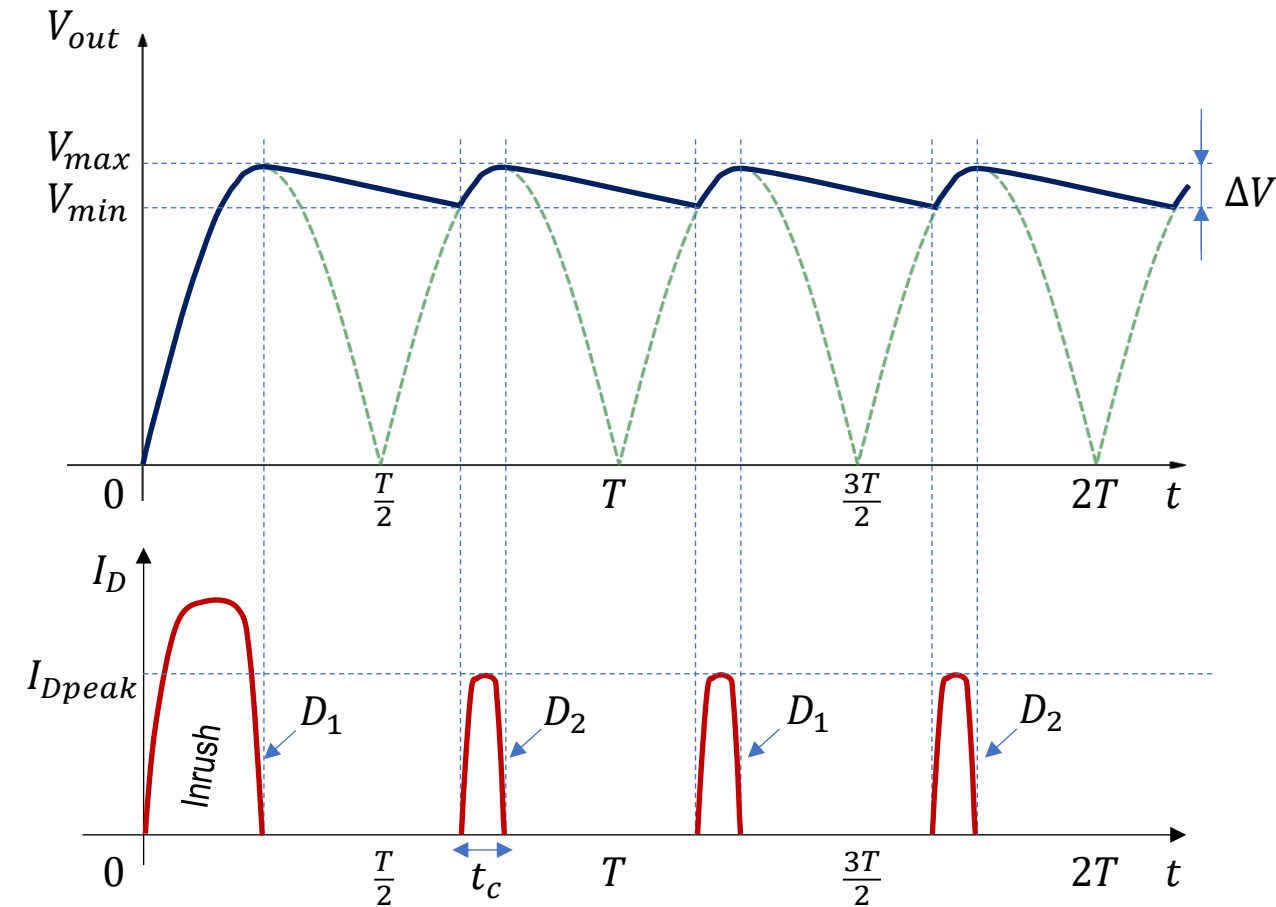
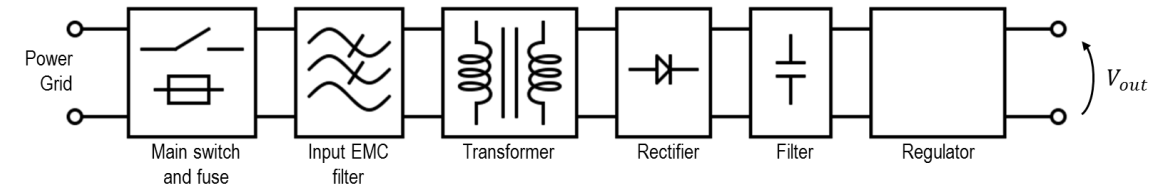
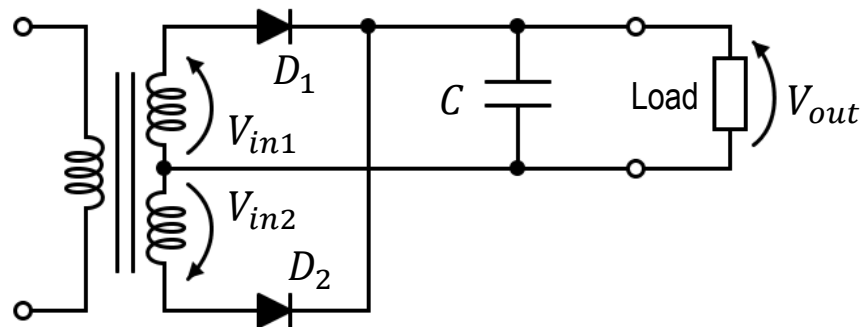
- When input  $V_{in2}$  decreases
  - The capacitor maintains a higher voltage
  - Diode  $D_2$  is reversed biased
    - It goes in cut-off
  - Diode  $D_1$  is still in cut-off
- The output voltage slowly decreases
  - Same laws as before
    - Resistive
    - Constant current



# Power supply

## • Filter

- The cycle repeats infinite times
  - Diodes  $D_1$  and  $D_2$  conduct alternatively
  - The output voltage resembles a sawtooth waveform
  - The maximum  $\Delta V$  is approximately
    - $\Delta V = \frac{I_{load}}{C} \cdot \frac{T}{2}$
    - $V_{min} = V_{max} - \Delta V$
    - $V_{avg} = \frac{V_{max} + V_{min}}{2} = V_{max} - \frac{\Delta V}{2}$



# Power supply

- Filter

- For a half wave rectifier

- The cycle is similar

- Higher maximum  $\Delta V$

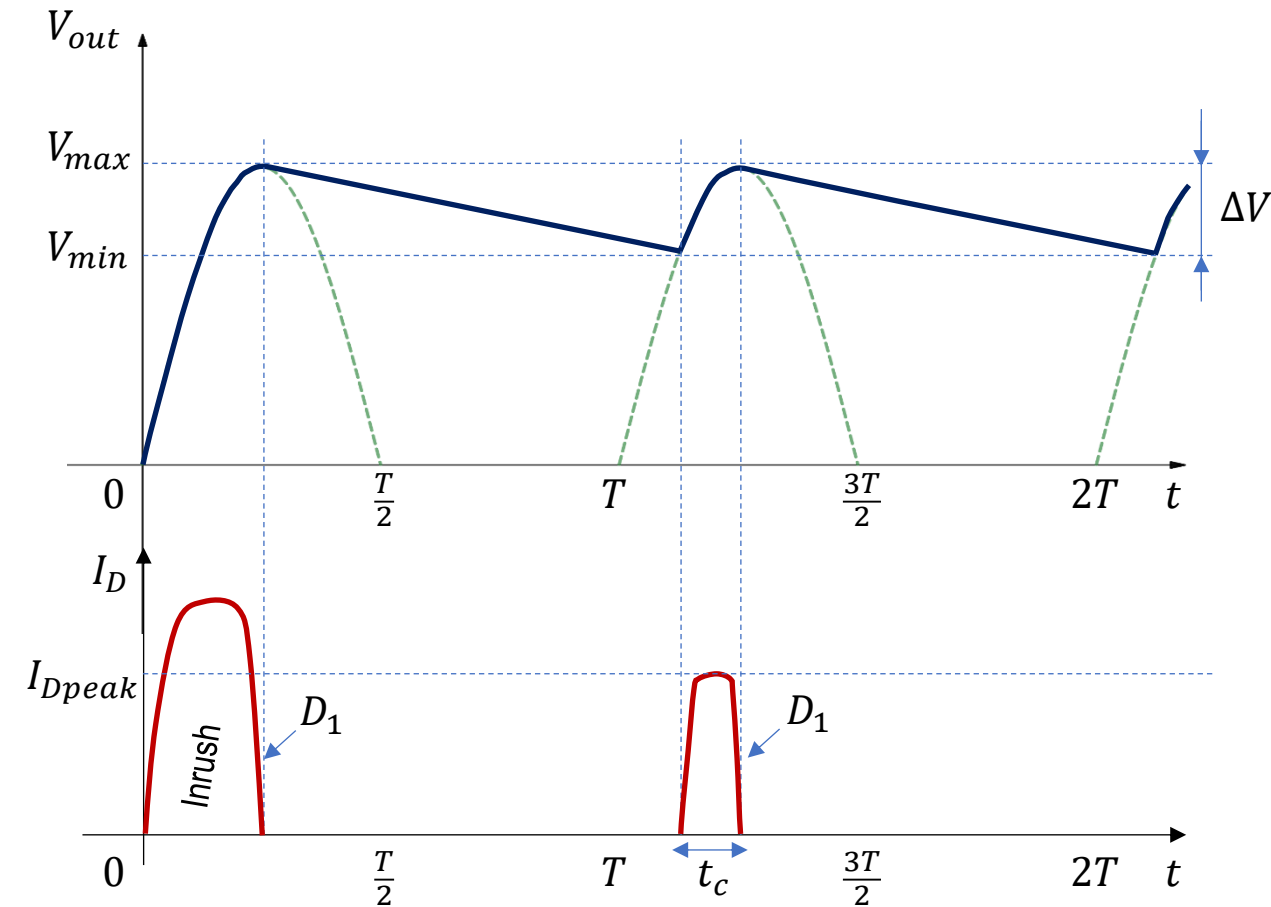
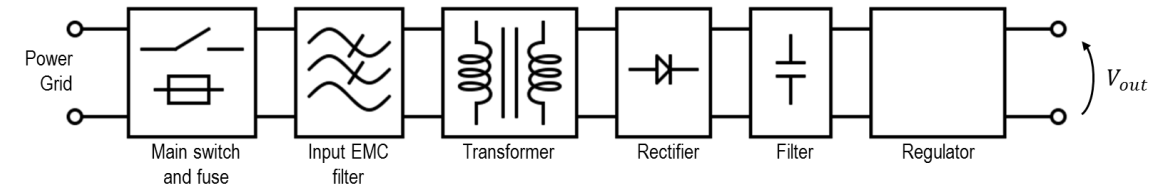
- $\Delta V = \frac{I_{load}}{C} \cdot T$

- $V_{min} = V_{max} - \Delta V$

- $V_{avg} = \frac{V_{max} + V_{min}}{2} = V_{max} - \frac{\Delta V}{2}$

- Longer conduction time

- But higher peak (see later)



# Power supply

- Filter

- Maximum reverse voltage on the diodes

- When a diode is not conducting

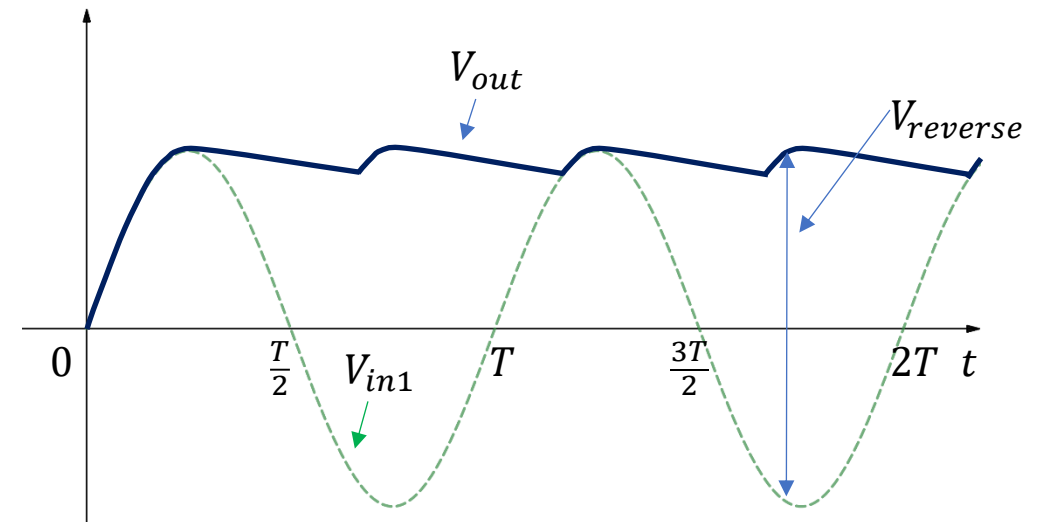
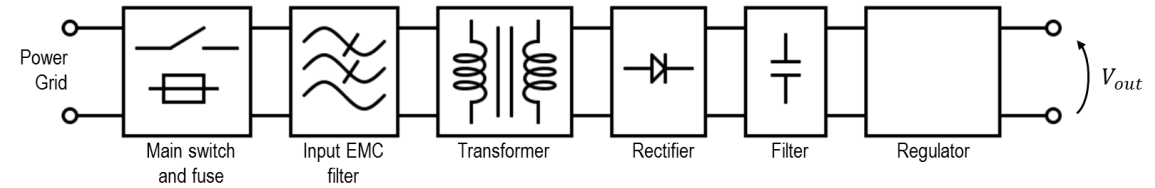
- The output voltage is almost at its peak value  $V_{max} \simeq V_A$
      - The input voltage goes to the minimum voltage  $-V_A$
      - The total reverse voltage is approximately  $2V_A$ 
        - It is the peak-to-peak voltage at the output of the transformer

- For the half wave rectifier and full wave rectifier with two diodes

- Choose diodes such that  $V_{BR} > 2V_A$

- For the full wave rectifier with four diodes

- Two non conducting diodes are always in series
    - The reverse voltage is divided among the two diodes
    - Choose diodes such that  $V_{BR} > V_A$

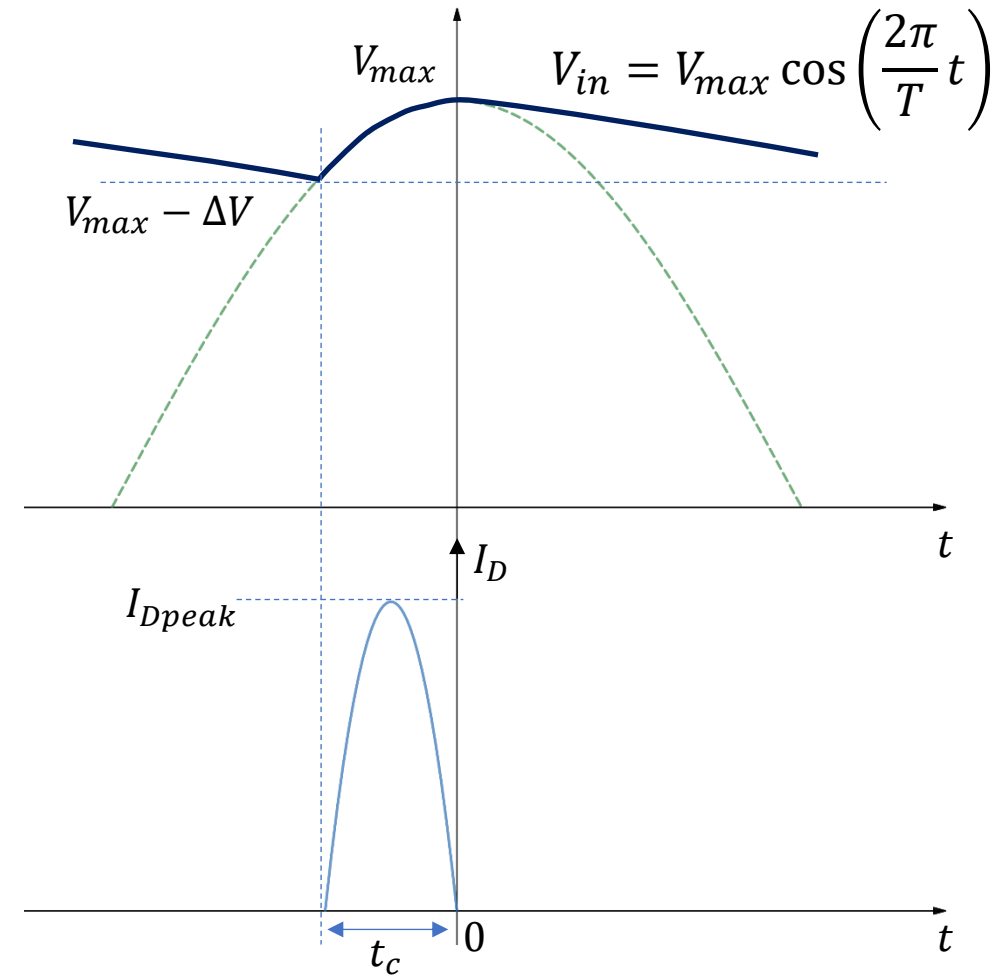
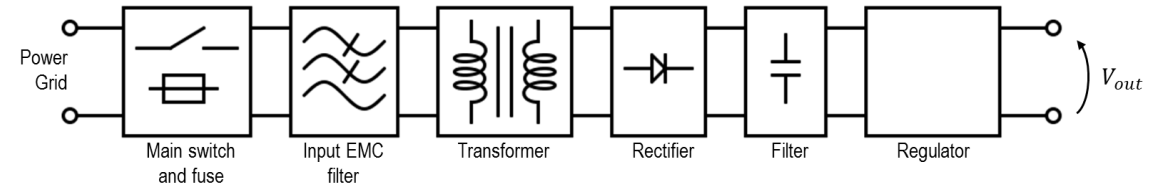


# Power supply

- Filter

- Current peak on the diode

- The load withdraws a constant current  $I_{out}$
    - The integral of the load current over a period should be equal to the integral of the diode currents over the same period
      - Assuming no losses in the transformer
    - Assume diode current has a triangular shape
      - Base  $t_c$ , height  $I_{Dpeak}$
      - $t_c = \frac{T}{2\pi} \cos^{-1} \left( \frac{V_{max} - \Delta V}{V_{max}} \right)$
    - Conduction time  $t_c$  is longer the higher is the residual ripple  $\Delta V$



# Power supply

- Filter

- Current peak on the diode

- Half wave rectifier

- $\frac{t_c I_{Dpeak}}{2} = T I_{out} \Rightarrow I_{Dpeak} = 2 I_{out} \frac{T}{t_c}$

- Full wave rectifier (two or four diodes)

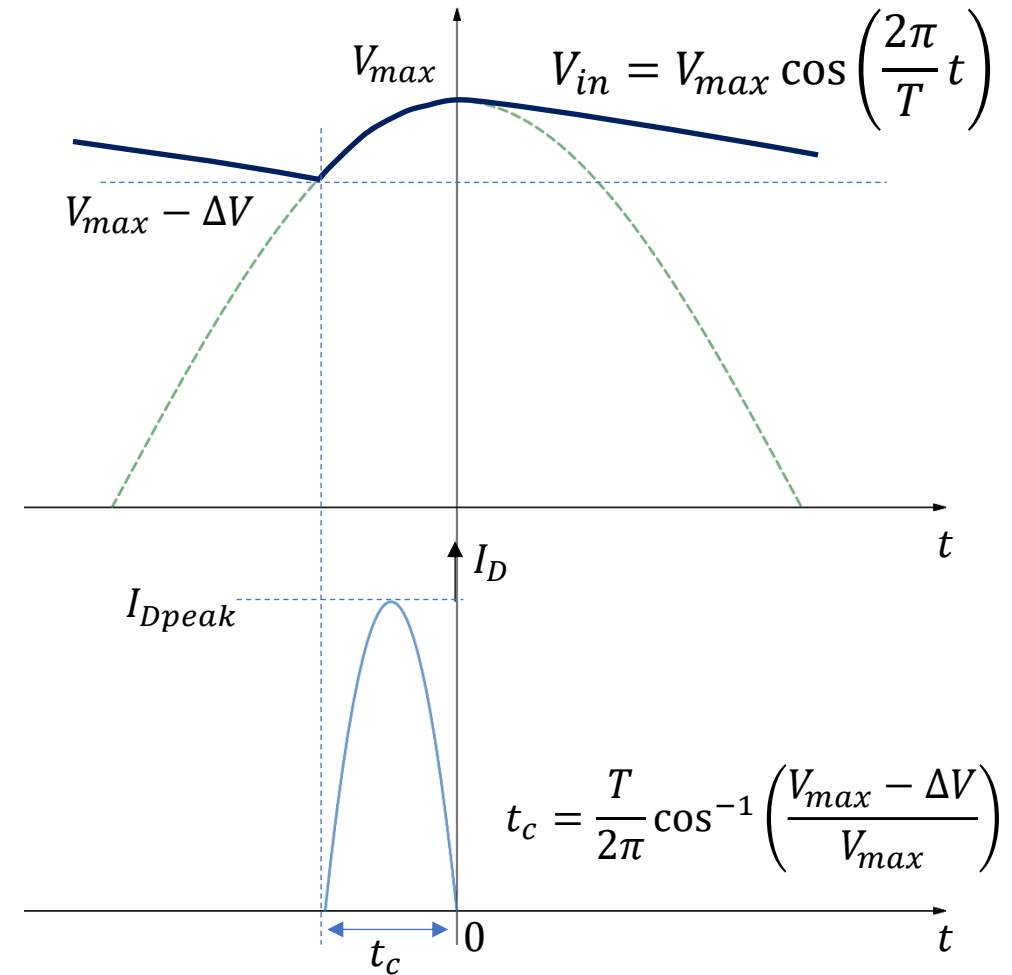
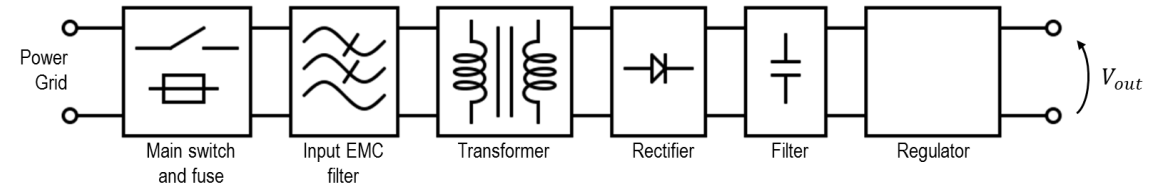
- $\frac{t_c I_{Dpeak}}{2} = \frac{T}{2} I_{out} \Rightarrow I_{Dpeak} = I_{out} \frac{T}{t_c}$

- The higher the conduction time  $t_c$ , the lower the current peak  $I_{Dpeak}$  in the diode

- The higher the residual ripple  $\Delta V$ , the lower the current peak  $I_{Dpeak}$  in the diode

- In the half wave rectifier  $t_c$  is higher, but not twice the conduction time in the full wave rectifier

- The current peak in the half wave rectifier is higher than in the full wave rectifier

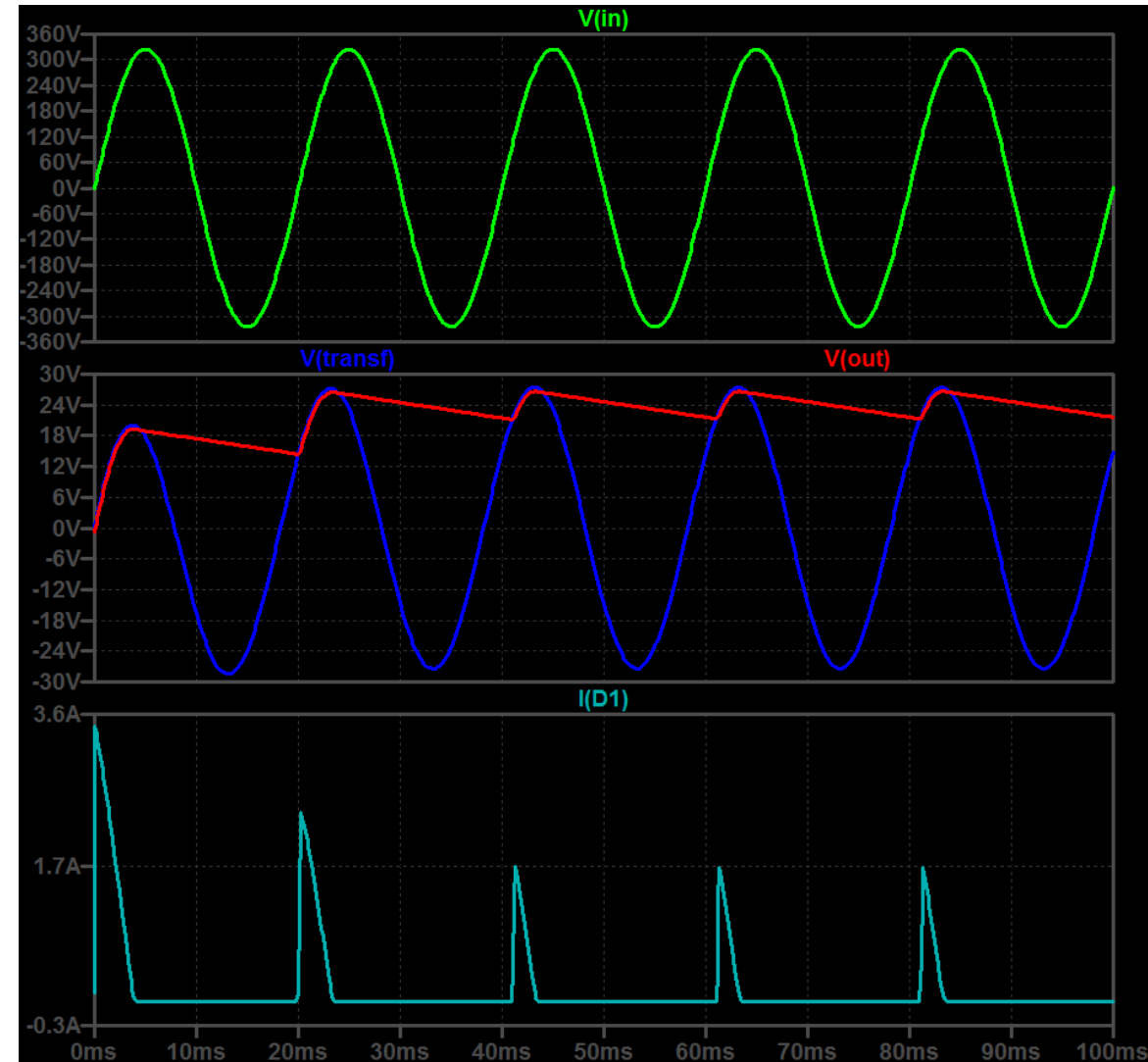
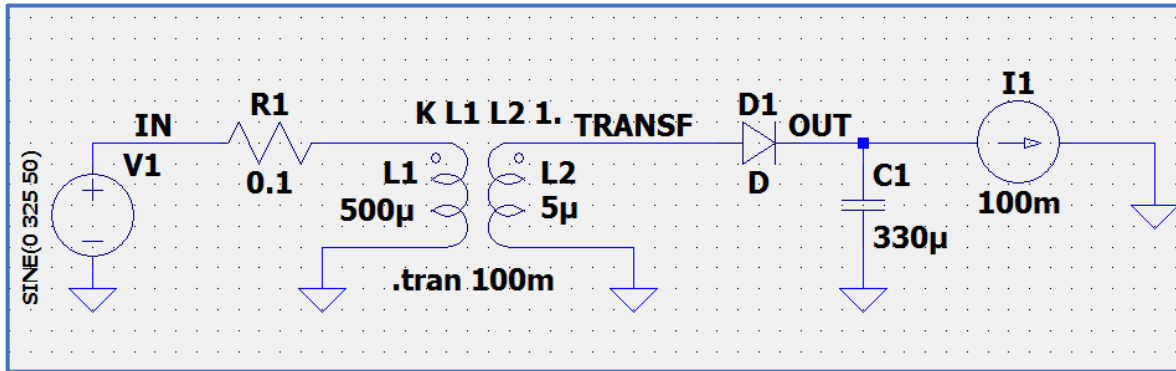


# Power supply

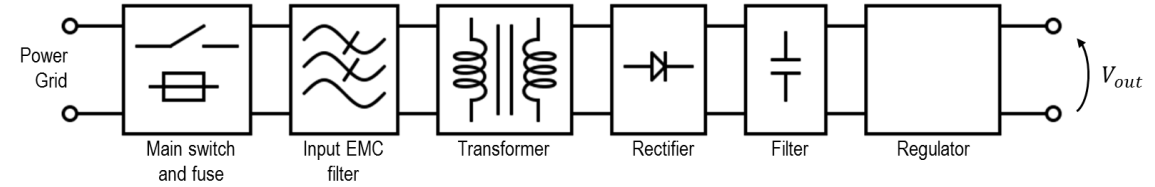
- Filter

- Example

- $V_{in} = 230 V_{RMS} @ f = 50 Hz, T = 20 ms$
- Transformer 10: 1,  $C = 330 \mu F, I_{out} = 100 mA$
- $V_{max} = 27 V, \Delta V = 5.5 V$
- $t_c = \frac{T}{2\pi} \cos^{-1} \left( \frac{V_{max} - \Delta V}{V_{max}} \right) = 2.1 ms$
- $I_{Dpeak} = 2I_{out} \frac{T}{t_c} = 1.9 A$



# Power supply

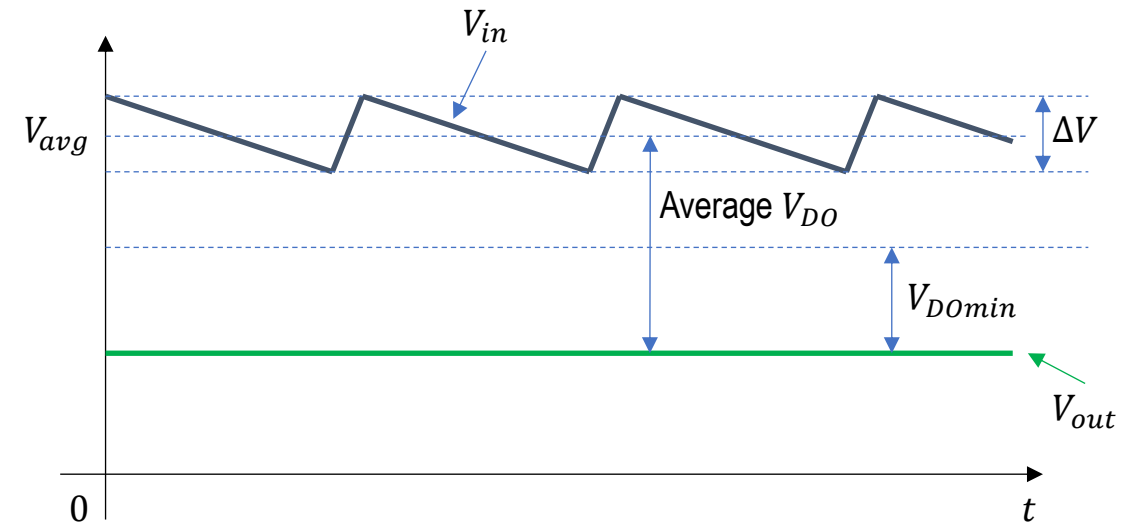
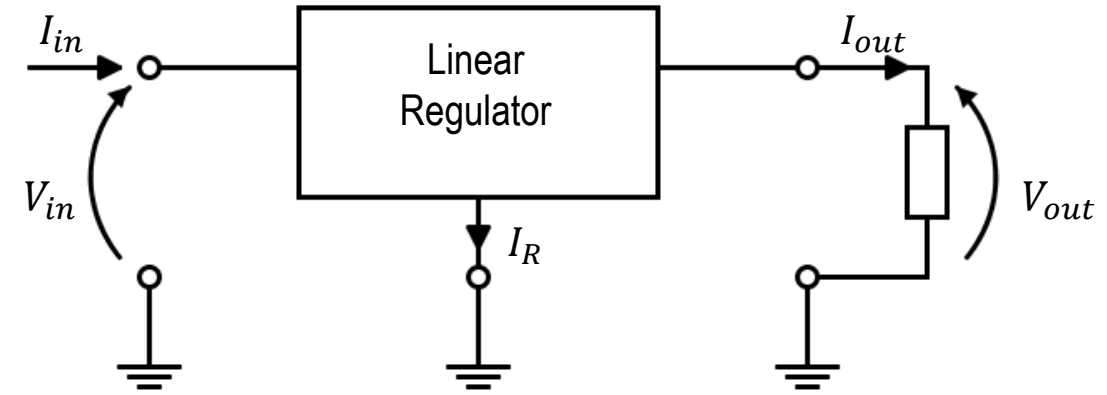


- Summary

- After the filter, there is an average constant output voltage  $V_{avg}$  with a residual ripple  $\Delta V$
- Given a certain load, the ripple mostly depends on the filter capacitor
  - $\Delta V = \frac{I_{load}}{C} \cdot \frac{T}{2}$
- In some cases it already satisfies the specifications
  - The higher the capacitor, the lower the ripple
    - But the capacitor is bulkier and more expensive
  - The higher the capacitor, the higher the current peak in the diodes
    - Diodes are more expensive
- In all other cases
  - Allow a higher residual ripple  $\Delta V$  after the filter
    - Smaller capacitor, smaller diodes
  - Use a regulator to satisfy the load requirements

# Linear power supply

- Linear regulator
  - Three terminal device
    - Input voltage  $V_{in}$  is a pseudo-DC
      - Average and residual ripple
    - Output voltage  $V_{out}$  is the regulated output
      - It is always  $V_{out} < V_{in}$
    - Output current  $I_{out}$  is the current required by the load
    - Current  $I_R$  is the current required by the regulator
      - $I_{in} = I_R + I_{out}$
      - Typically  $I_R \ll I_{out}$
  - Drop-out voltage
    - $V_{DO} = V_{in} - V_{out}$
    - To work correctly, it should be  $V_{DO} \geq V_{DOmin}$ 
      - $V_{DOmin}$  is given in the characteristics of the linear regulator



# Linear power supply

- Linear regulator

- Three terminal device

- Efficiency

- $P_{in} = V_{in} \cdot I_{in}$

- $P_{out} = V_{out} \cdot I_{out}$

- $\eta = \frac{P_{out}}{P_{in}} = \frac{V_{out} \cdot I_{out}}{V_{in} \cdot I_{in}} \simeq \frac{V_{out}}{V_{in}}$

- On average

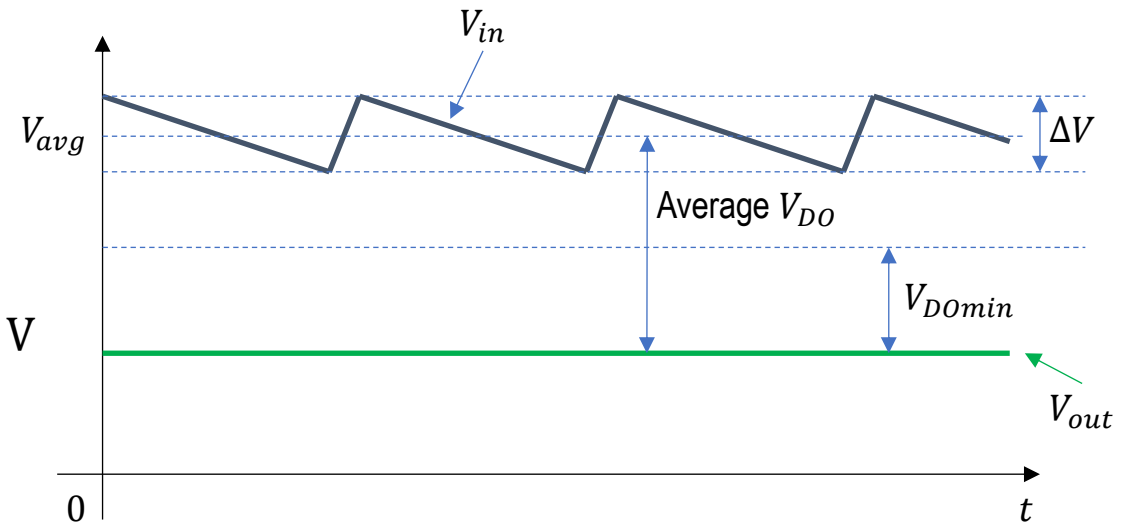
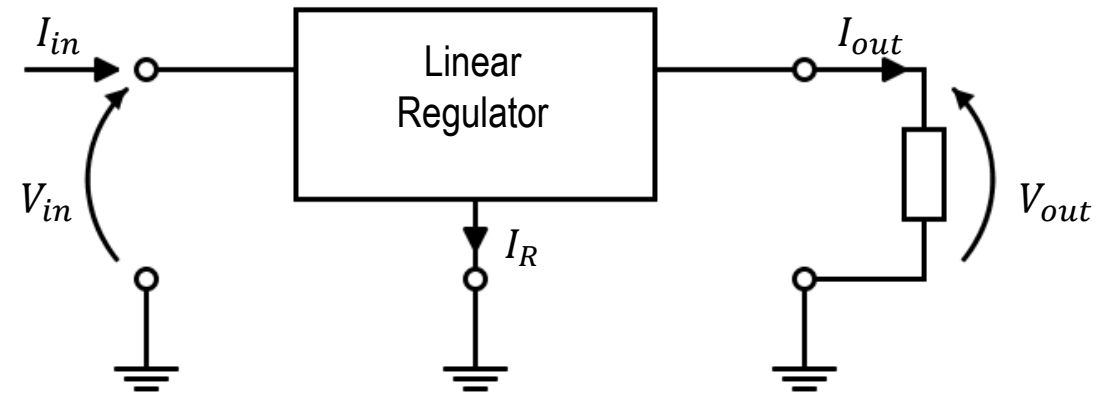
- $\eta \simeq \frac{V_{out}}{V_{in}} = \frac{V_{avg} - V_{DO}}{V_{avg}} = 1 - \frac{V_{DO}}{V_{avg}}$

- $\eta_{max} = 1 - \frac{\frac{\Delta V}{2} + V_{DOmin}}{V_{avg}}$

- Example

- $V_{avg} = 12 \text{ V}, \Delta V = 4 \text{ V}, V_{DOmin} = 3 \text{ V}, V_{out} = 5 \text{ V}$

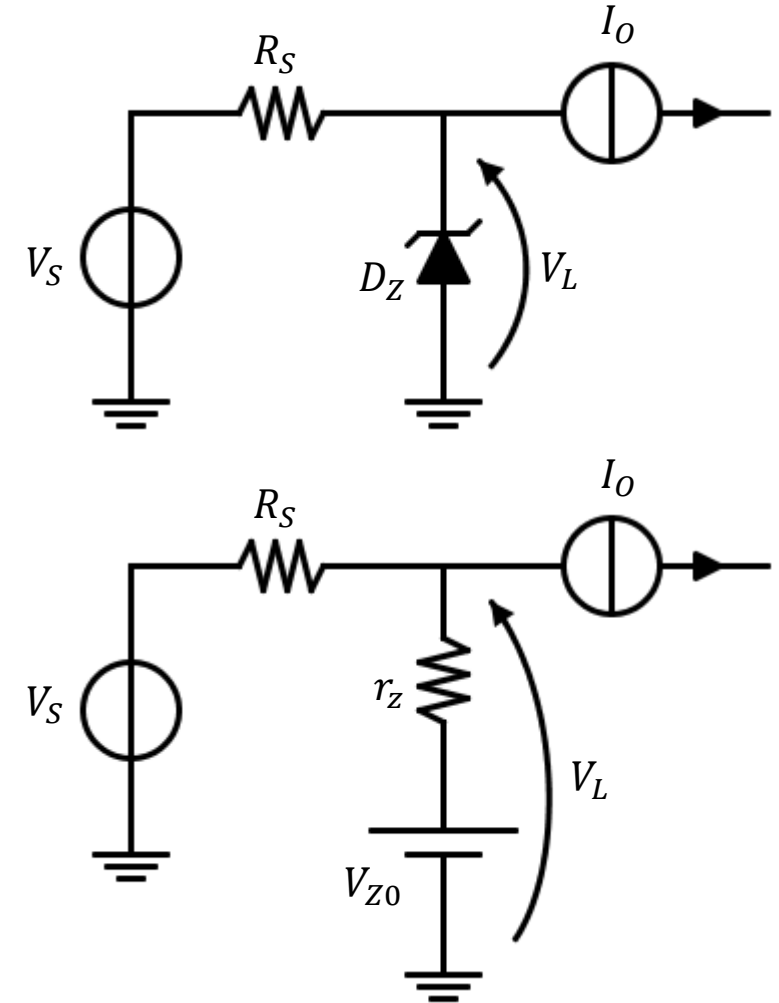
- $\eta_{max} = 58\%, \eta = \frac{5}{12} = 41.7\%$



# Linear power supply

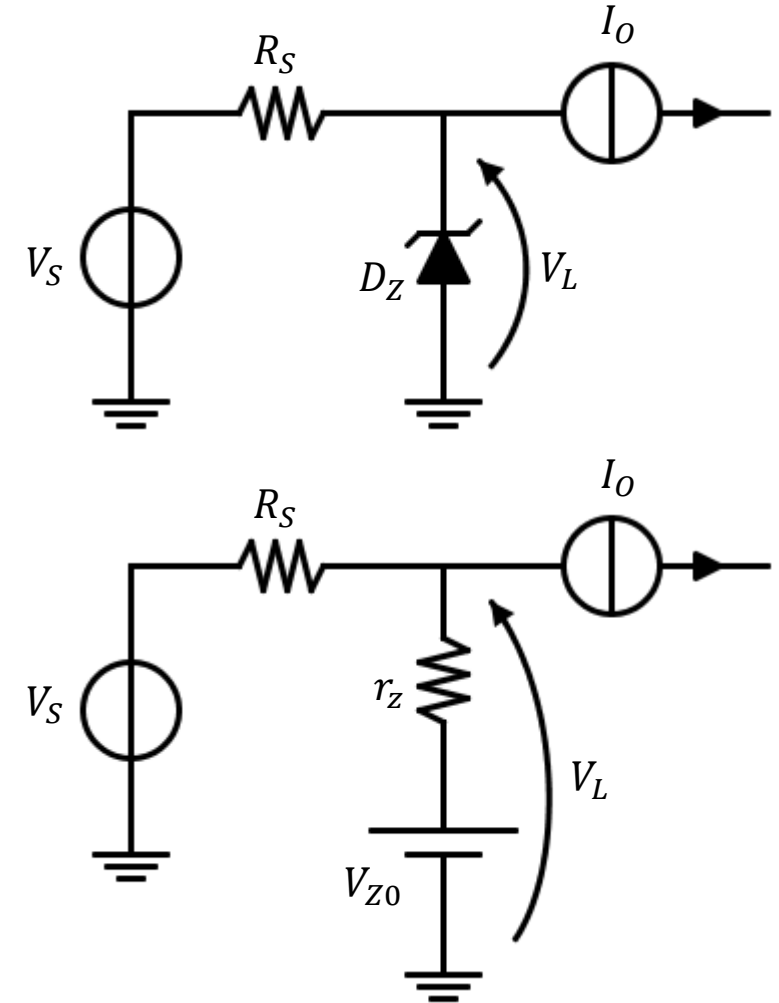
- Zener linear regulator
  - Input voltage  $V_S$  is the pseudo-DC
  - Load current  $I_O$  from  $I_{Omin}$  to  $I_{Omax}$
  - Resistance  $R_S$  to guarantee
    - Minimum Zener current  $I_{Zmin}$
    - Maximum Zener power dissipation less than  $P_{Zmax}$
    - Required current in the load
- Output voltage  $V_L$ 
  - Using the Millman theorem

$$V_L = \frac{\frac{V_S}{R_S} + \frac{V_{Z0}}{r_z} - I_O}{\frac{1}{R_S} + \frac{1}{r_z}}$$



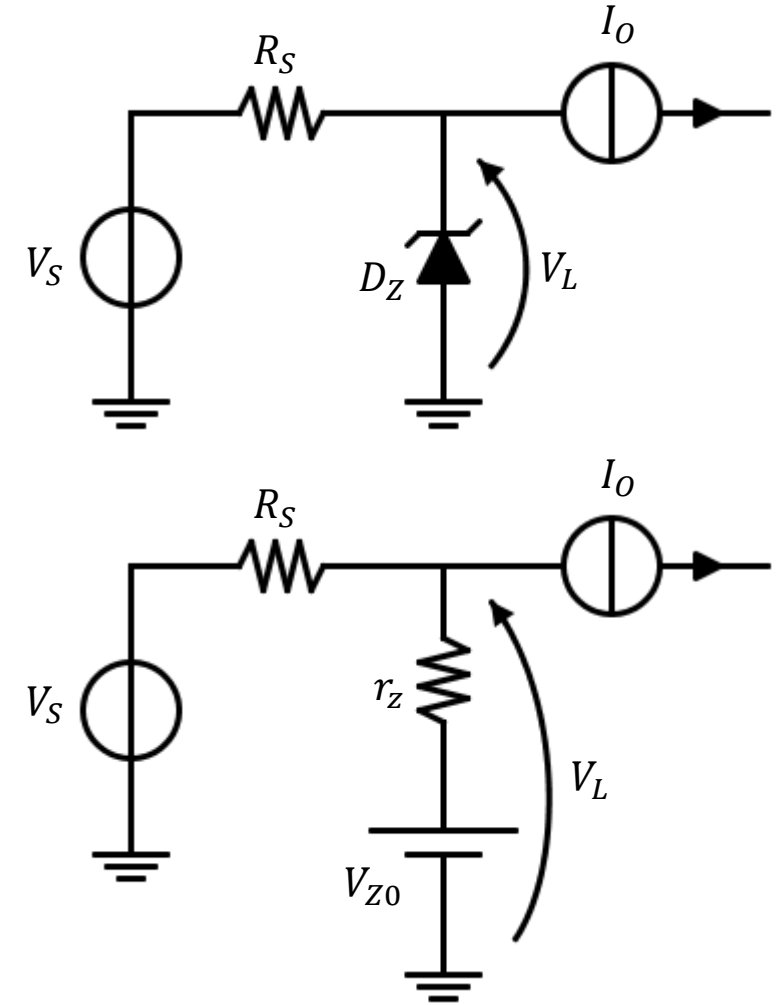
# Linear power supply

- Zener linear regulator
  - Minimum Zener current  $I_{Zmin}$ 
    - Worst case when  $V_S$  is minimum
      - $V_{Smin} = V_{avg} - \frac{\Delta V}{2}$
    - Worst case when  $I_O$  is maximum
      - $I_O = I_{Omax}$
  - It should be
    - $I_Z = \frac{V_L' - V_{Z0}}{r_z} > I_{Zmin}$
    - $V_L' = \frac{\frac{V_{Smin}}{R_S} + \frac{V_{Z0}}{r_z} - I_{Omax}}{\frac{1}{R_S} + \frac{1}{r_z}}$
  - Solving for  $R_S$  gives an upper bound
    - $R_S < R_{Smax} = \frac{V_{Smin} - V_{Z0} - r_z I_{Zmin}}{I_{Omax} + I_{Zmin}}$



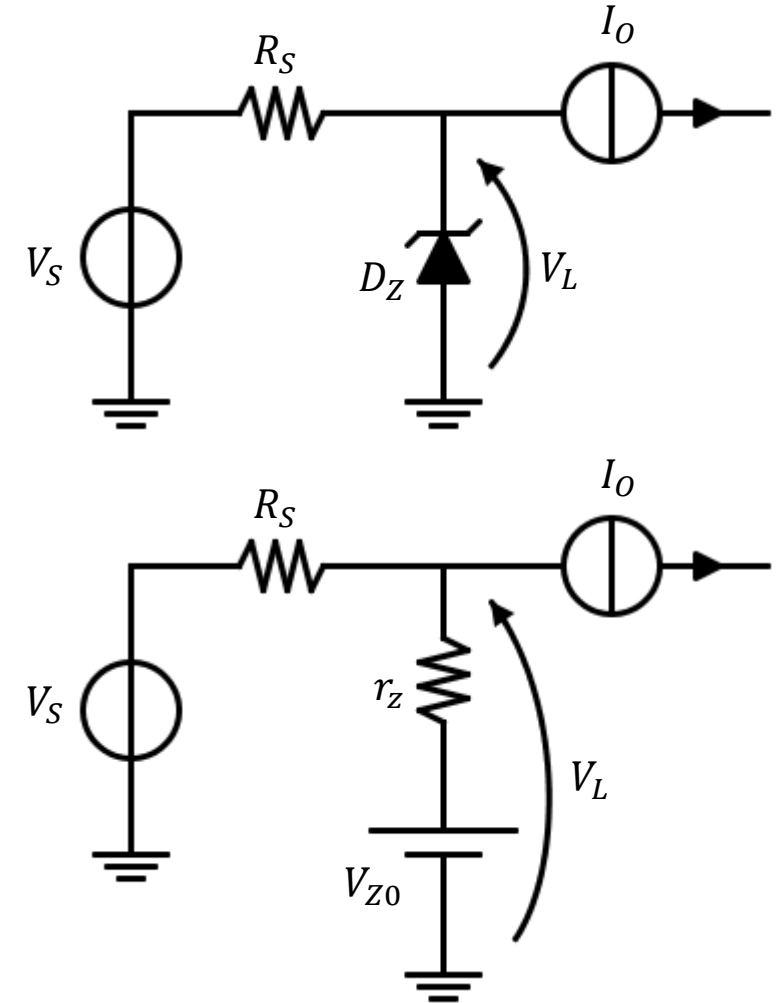
# Linear power supply

- Zener linear regulator
  - Maximum Zener power  $P_{Zmax}$ 
    - Worst case when  $V_S$  is maximum
      - $V_{Smax} = V_{avg} + \frac{\Delta V}{2}$
    - Worst case when  $I_O$  is minimum
      - $I_O = I_{Omin}$  or even worse  $I_O = 0$  (load disconnected)
  - It should be
    - $I_Z V_L'' = \frac{V_L'' - V_{Z0}}{r_z} V_L'' < P_{Zmax}$
    - $V_L'' = \frac{\frac{V_{Smax} + V_{Z0}}{R_S} + \frac{V_{Z0}}{r_z}}{\frac{1}{R_S} + \frac{1}{r_z}}$
  - Solving for  $R_S$  gives a lower bound
    - $R_S > R_{Smin}$



# Linear power supply

- Zener linear regulator
  - If  $R_{Smin} < R_{Smax}$ 
    - A valid range for  $R_S$  exists
  - Else
    - A better circuit is needed
  - Performance parameters
    - Load regulation
      - $S_I = \frac{\Delta V_L}{\Delta I_O} = \frac{R_S r_z}{R_S + r_z} = R_S \parallel r_z$
      - The quality of the Zener determines this parameter
        - The lower  $r_z$  the better
    - Line regulation
      - $S_V = \frac{\Delta V_L}{\Delta V} = \frac{r_z}{R_S + r_z}$ 
        - The lower  $r_z$  the better
        - The higher  $R_S$  the better

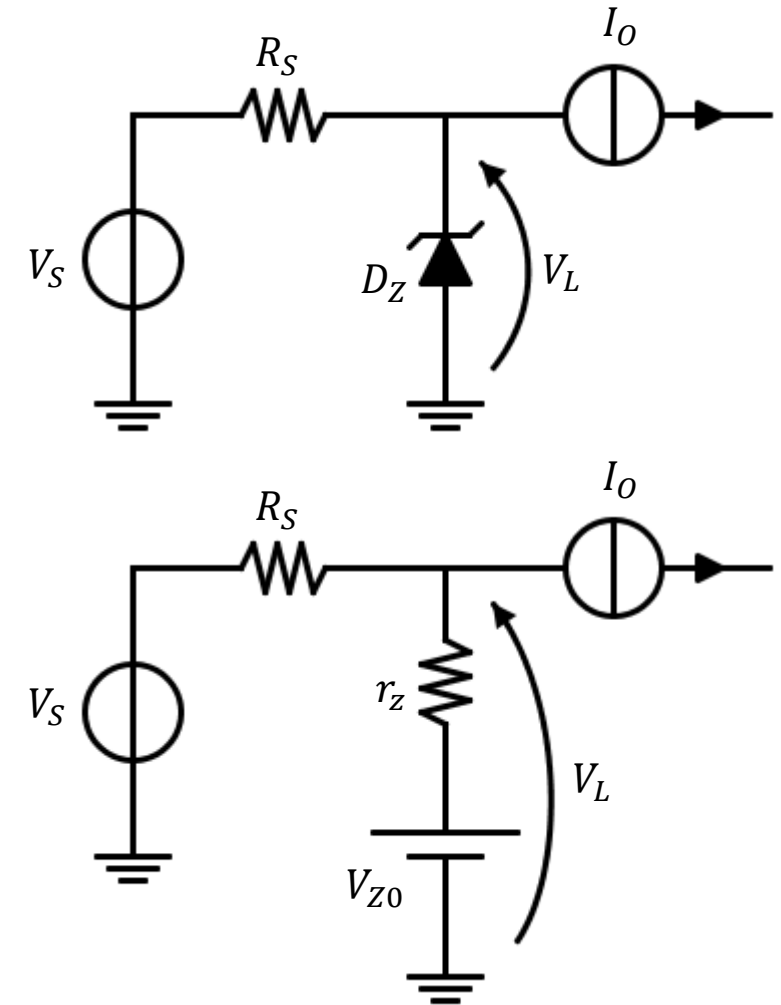


# Linear power supply

- Zener linear regulator

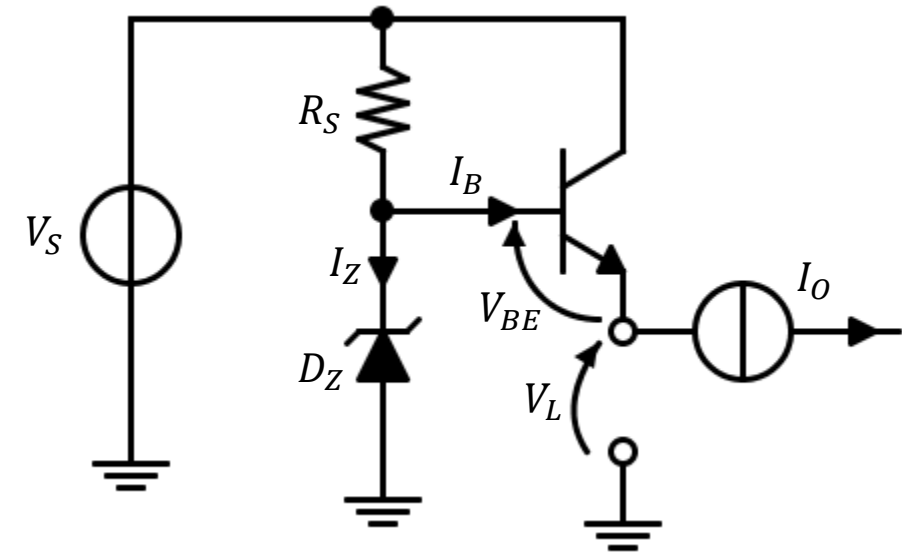
- Example

- $V_{Smin} = 10\text{ V}$ ,  $V_{Smax} = 15\text{ V}$ ,  $\Delta V = 5\text{ V}$
    - $r_z = 12\ \Omega$ ,  $V_{Z0} = 5\text{ V}$ ,  $I_{Zmin} = 5\text{ mA}$ ,  $P_{Zmax} = 2\text{ W}$
    - $I_{Omin} = 20\text{ mA}$ ,  $I_{Omax} = 80\text{ mA}$
    - $28\ \Omega < R_S < 58\ \Omega$
    - Let's choose  $R_S = 57\ \Omega$ 
      - $V_L = 5.8\text{ V}$
      - $S_I = 10\ \Omega$
      - $S_V = 0.174$
      - $\Delta V_L = S_V \Delta V = 0.87\text{ V}$  due to the input ripple
      - $\Delta V_L = S_I \Delta I_O = 0.6\text{ V}$  due to the output current variations
      - $P_{R_S} = 950\text{ mW}$



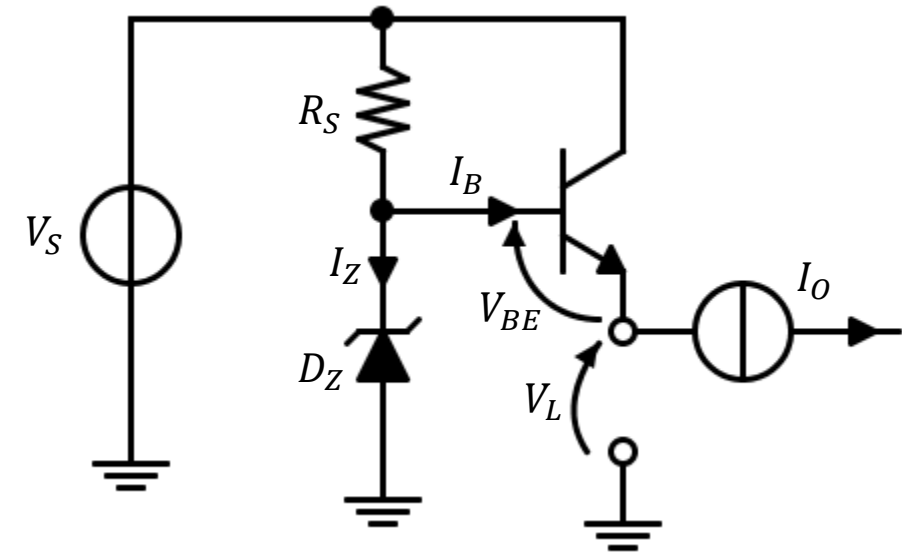
# Linear power supply

- Zener linear regulator with output amplifier
  - A regulator with a simple Zener diode has a problem
    - It needs a high  $R_S$  resistance to get a good line sensitivity  $S_L$
    - But  $R_S$  has an upper bound
      - Mostly due to the maximum output current  $I_{Omax}$
  - Use an output transistor to provide current to the load
    - The rest of the circuit is identical to the Zener regulator
    - Common collector configuration
      - Emitter follower
    - $V_L = V_D - V_{BE} \simeq V_D - 0.7 \text{ V}$
    - Any small signal (ripple) on the base appears unchanged (gain 1) at the output
      - It seems it gives no improvement
    - But the  $R_S$  resistance can now be much higher
      - $R_S < R_{Smax} = \frac{V_{Smin} - V_{Z0} - r_z I_{Zmin}}{\frac{I_{Omax}}{\beta} + I_{Zmin}}$  being  $I_{Bmax} = \frac{I_{Omax}}{\beta}$



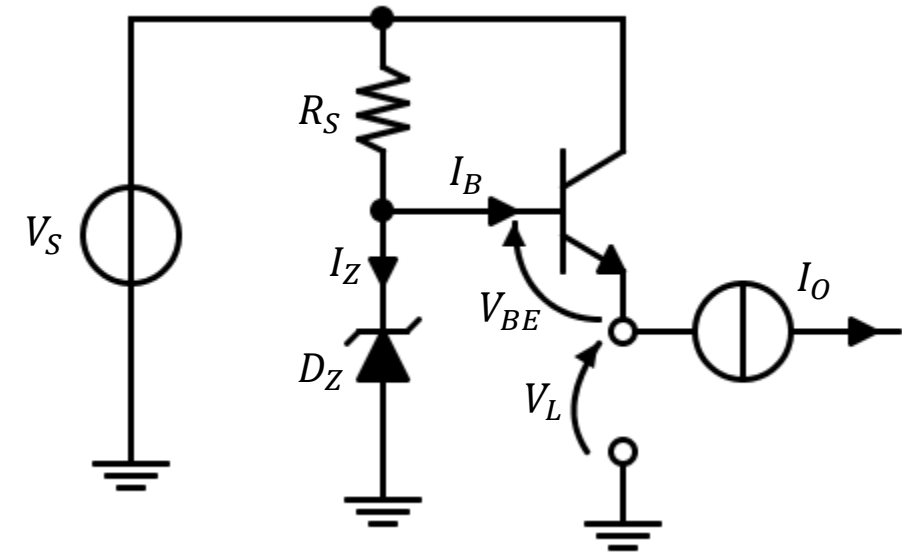
# Linear power supply

- Zener linear regulator with output amplifier
  - Example
    - Same data as before
    - $\beta = 200$
    - $28\ \Omega < R_S < 915\ \Omega$ 
      - The upper bound is now significantly higher
    - Let's choose  $R_S = 820\ \Omega$ 
      - $V_L = 5.1\ \text{V}$ 
        - Decreased by  $0.7\ \text{V}$
      - $S_V = 0.0144$ 
        - It was  $0.174$  before
      - $\Delta V_L = S_V \Delta V = 72\ \text{mV}$  due to the input ripple
        - It was  $870\ \text{mV}$  before
        - Large improvement



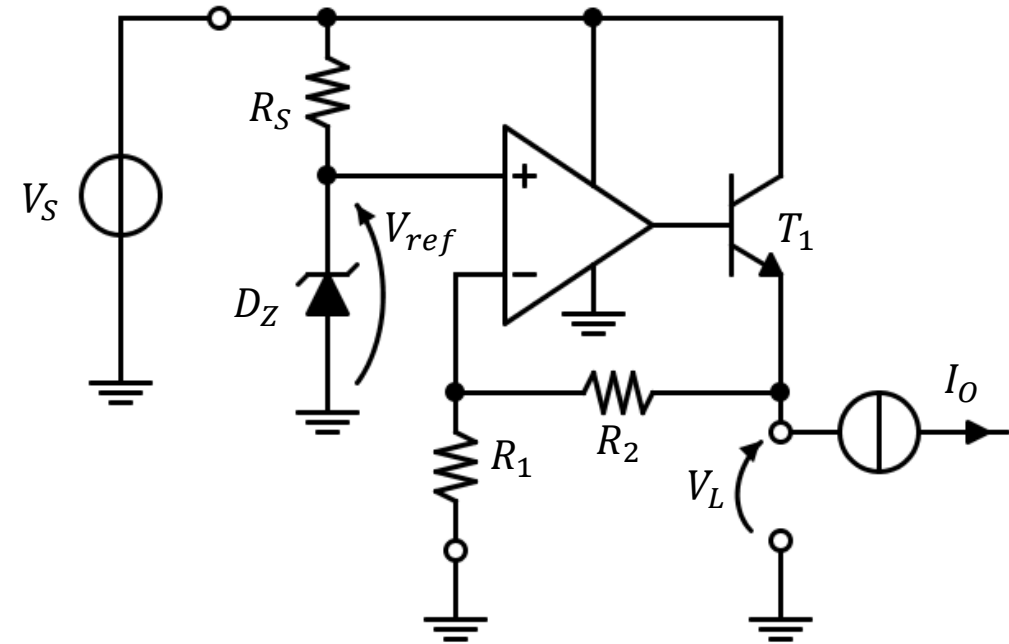
# Linear power supply

- Zener linear regulator with output amplifier
  - The output resistance of the emitter follower is equal to the resistance seen from the base divided by  $\beta$ 
    - The resistance as seen from the base is the output resistance of the simple Zener regulator
  - Example
    - Let's choose  $R_S = 820\ \Omega$ 
      - $S_I = \frac{1}{\beta} \frac{R_S r_z}{R_S + r_z} = 60\ \text{m}\Omega$ 
        - It was  $10\ \Omega$  before
      - $\Delta V_L = S_I \Delta I_O = 3.6\ \text{mV}$  due to the output current variations
        - It was  $600\ \text{mV}$  before
        - Large improvement
  - Drawback
    - It is more expensive



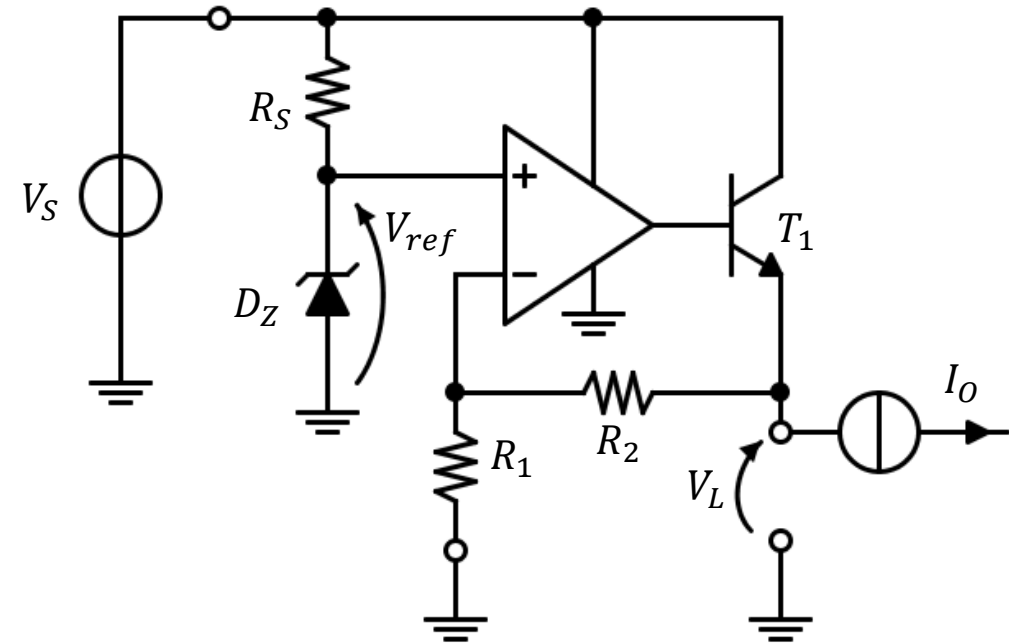
# Linear power supply

- Three terminal regulator
  - Operational amplifier with negative feedback
    - Non-inverting amplifier configuration
  - Output transistor to provide current to the load
    - Negative feedback includes the output transistor
  - $V_L = V_{ref} \left( 1 + \frac{R_2}{R_1} \right)$ 
    - Output voltage always higher than the reference  $V_{ref}$
  - Reference voltage  $V_{ref}$  generated by the Zener diode
    - No current flowing in the opAmp + terminal
    - $R_S < R_{Smax} = \frac{V_{Smin} - V_{Z0} - r_z I_{Zmin}}{I_{Zmin}}$ 
      - Higher upper bound
      - Even better line regulation



# Linear power supply

- Three terminal regulator
  - Line regulation
    - Extremely good
    - It also depends on the Power Supply Rejection Ratio (PSRR) of the operational amplifier
      - Typically in the order of 60 dB
    - Even better if the voltage reference is connected to the output voltage  $V_L$  rather than the input voltage  $V_S$ 
      - Need a bootstrap circuit for power-up
  - Load regulation
    - Extremely good
      - The output resistance of an operation amplifier is very low
      - The output resistance of the emitter follower is very low
      - The negative feedback makes the output resistance even lower

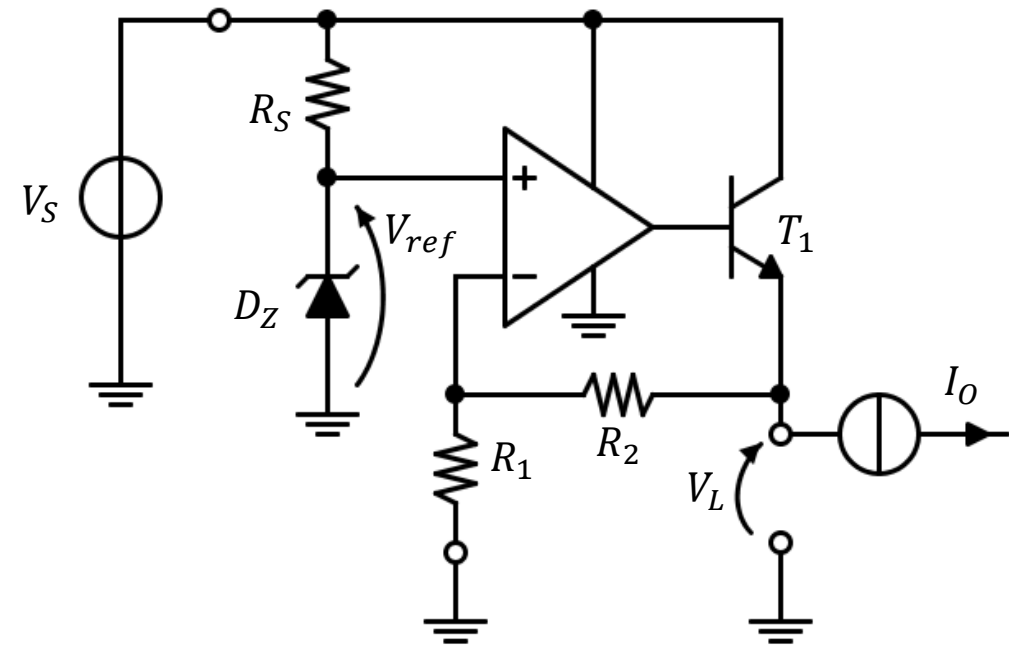


# Linear power supply

- Three terminal regulator

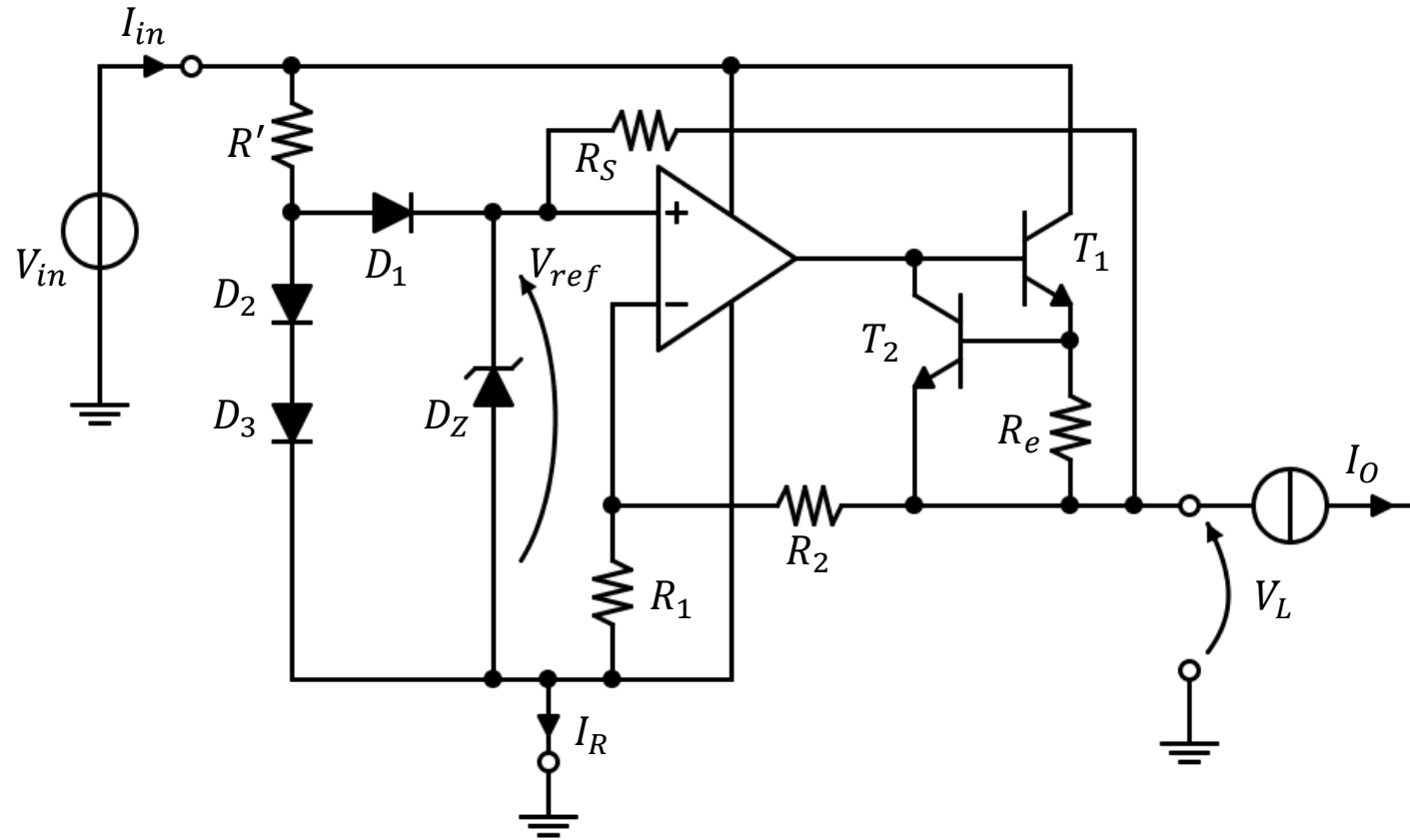
- Problems

- The Zener reference is often too high
      - Use a voltage divider after the reference
    - The Zener reference changes with temperature
      - Use a Band Gap reference voltage generator
    - Input voltage  $V_S$  must always be greater than the output voltage  $V_L$ 
      - By an amount equal to the drop from Voltage Supply to the output of the operational amplifier
        - Normally at least 1 V to remain in linear operation
      - By an additional amount equal to the  $V_{BE}$  of the output transistor
        - Normally around 1 V for power transistors



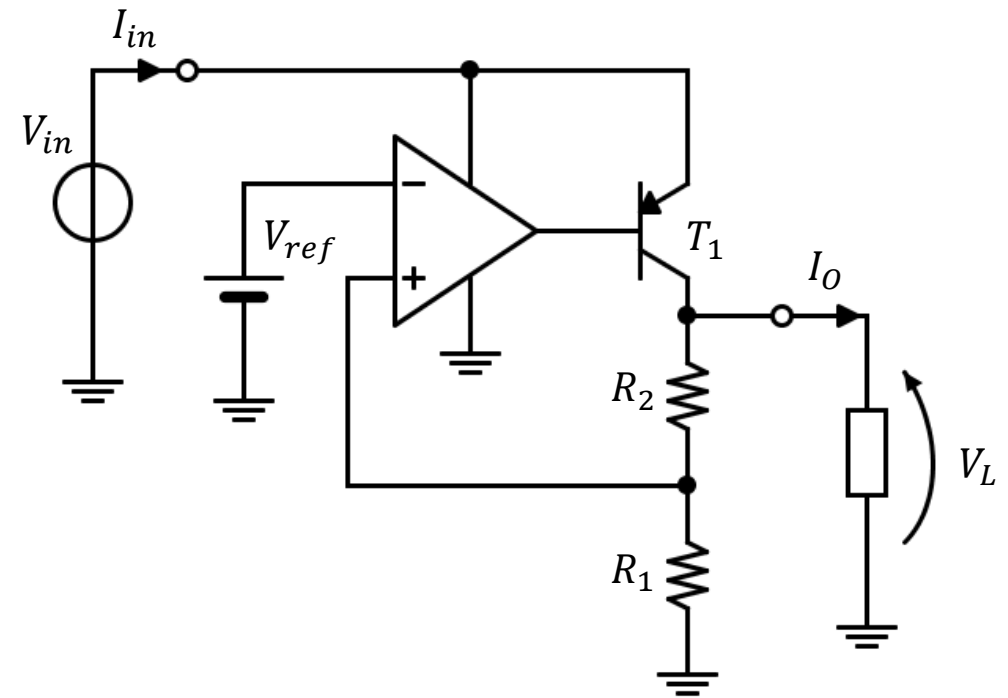
# Linear power supply

- Three terminal regulator
  - Full schematic
    - Excessive output current protection
      - $T_2$  and  $R_e$
      - Choose  $R_e$  so that  $T_2$  starts conducting when the output current is over a threshold
    - Reference connected to output voltage
      - $R_S$  and  $D_Z$
      - Much better line sensitivity
    - Bootstrap circuit
      - $R'$ ,  $D_1$ ,  $D_2$  and  $D_3$



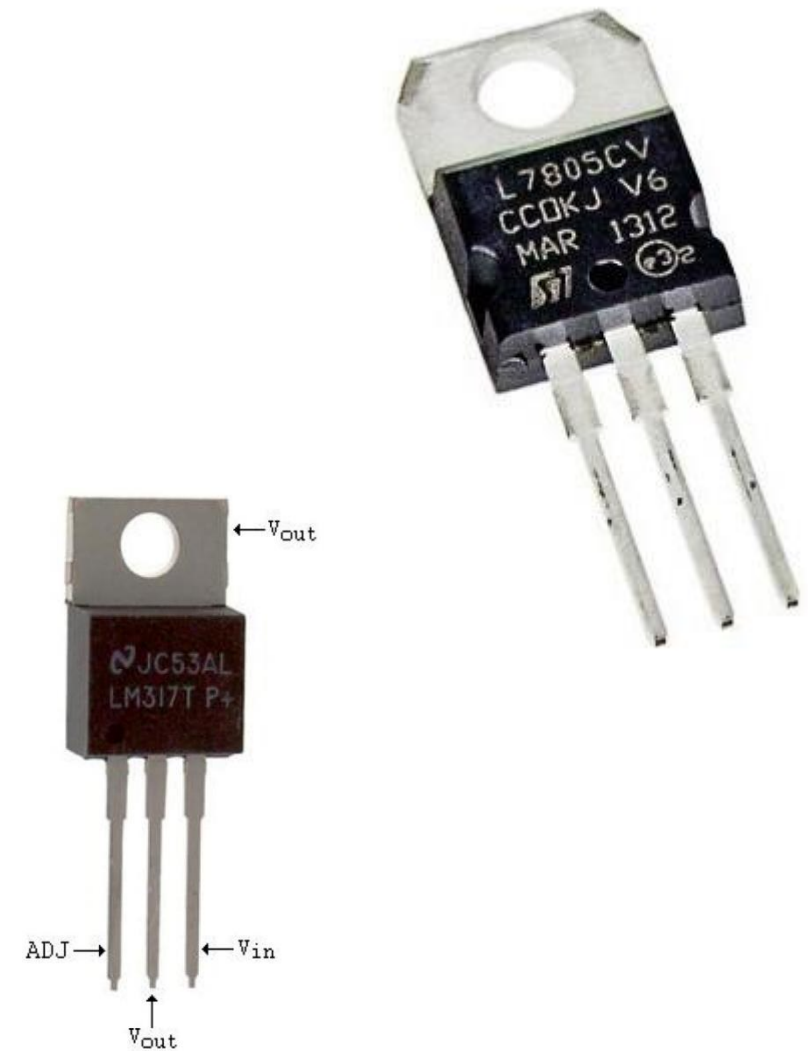
# Linear power supply

- Low Dropout (LDO) regulator
  - Common emitter
    - Using a pnp BJT transistor
  - No longer a voltage follower
    - Amplifier with gain
    - Gain depends on the load
      - Critical for stability
  - Useful for low voltage application
    - Example
      - Generate  $V_{out} = 3.3\text{ V}$
      - Primary source is a lithium polymer battery with average voltage of  $V_{in} = 3.7\text{ V}$
      - A normal linear regulator cannot be used because of the limited margin between input and output voltages



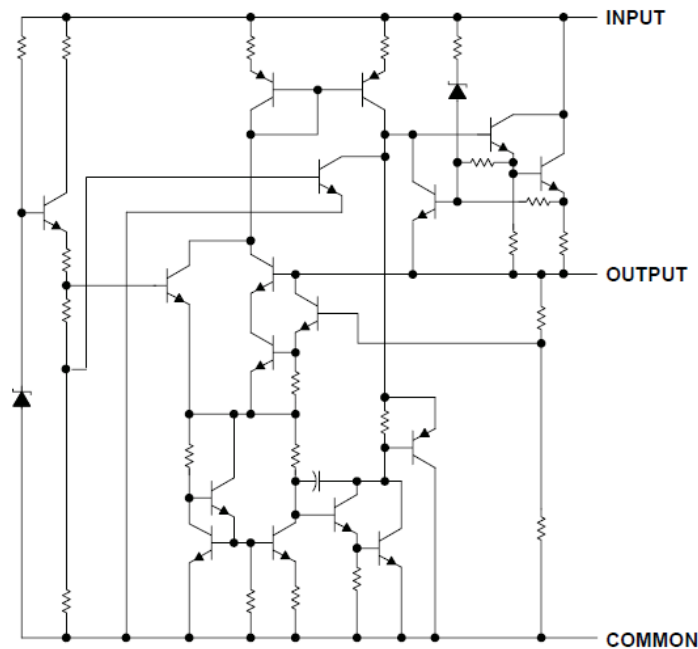
# Linear power supply

- Commercial components
  - Linear regulators
    - 78xx family
      - xx is the output voltage
        - 7805: 5 V output voltage
        - 7812: 12 V output voltage
      - High power package
      - Maximum current depending on the package case
    - 79xx family
      - Negative voltages
    - LM317
      - Variable output
      - Set using external resistances



# Linear power supply

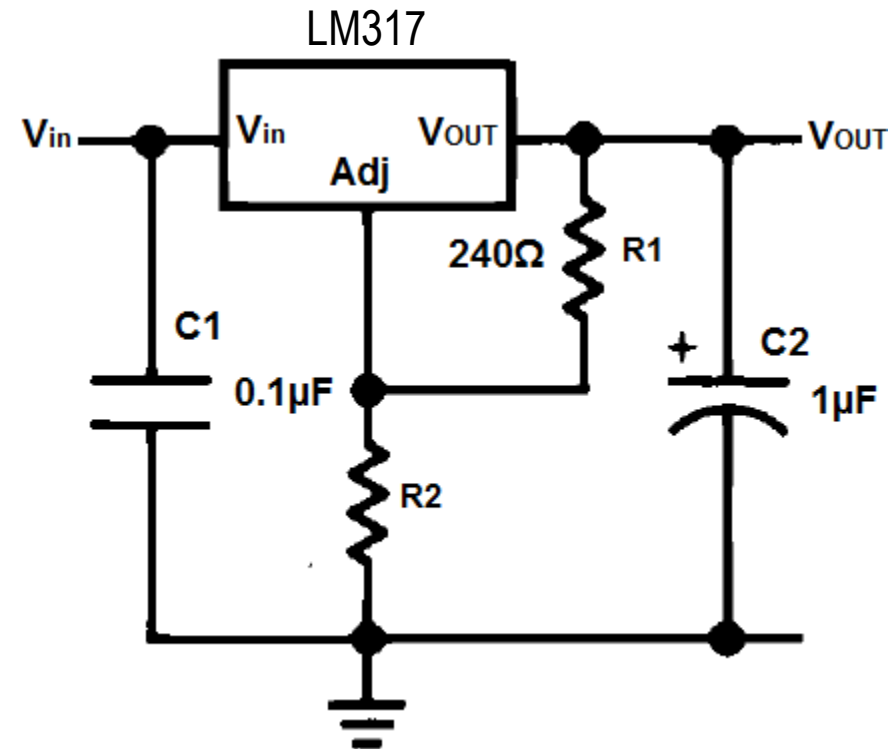
- Commercial components
  - 7805 Datasheet



| PARAMETER                                 | TEST CONDITIONS                                                                                           | $T_J \uparrow$ | $\mu A7805C$ |       |      | UNIT     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------|--------------|-------|------|----------|
|                                           |                                                                                                           |                | MIN          | TYP   | MAX  |          |
| Output voltage                            | $I_O = 5 \text{ mA to } 1 \text{ A},$<br>$P_D \leq 15 \text{ W}$<br>$V_I = 7 \text{ V to } 20 \text{ V},$ | 25°C           | 4.8          | 5     | 5.2  | V        |
|                                           |                                                                                                           | 0°C to 125°C   | 4.75         |       | 5.25 |          |
| Input voltage regulation                  | $V_I = 7 \text{ V to } 25 \text{ V}$                                                                      | 25°C           |              | 3     | 100  | mV       |
|                                           | $V_I = 8 \text{ V to } 12 \text{ V}$                                                                      |                |              | 1     | 50   |          |
| Ripple rejection                          | $V_I = 8 \text{ V to } 18 \text{ V},$<br>$f = 120 \text{ Hz}$                                             | 0°C to 125°C   | 62           | 78    |      | dB       |
| Output voltage regulation                 | $I_O = 5 \text{ mA to } 1.5 \text{ A}$                                                                    | 25°C           |              | 15    | 100  | mV       |
|                                           | $I_O = 250 \text{ mA to } 750 \text{ mA}$                                                                 |                |              | 5     | 50   |          |
| Output resistance                         | $f = 1 \text{ kHz}$                                                                                       | 0°C to 125°C   |              | 0.017 |      | $\Omega$ |
| Temperature coefficient of output voltage | $I_O = 5 \text{ mA}$                                                                                      | 0°C to 125°C   |              | -1.1  |      | mV/°C    |
| Output noise voltage                      | $f = 10 \text{ Hz to } 100 \text{ kHz}$                                                                   | 25°C           |              | 40    |      | $\mu V$  |
| Dropout voltage                           | $I_O = 1 \text{ A}$                                                                                       | 25°C           |              | 2     |      | V        |
| Bias current                              |                                                                                                           | 25°C           |              | 4.2   | 8    | mA       |
| Bias current change                       | $V_I = 7 \text{ V to } 25 \text{ V}$                                                                      | 0°C to 125°C   |              |       | 1.3  | mA       |
|                                           | $I_O = 5 \text{ mA to } 1 \text{ A}$                                                                      |                |              |       | 0.5  |          |
| Short-circuit output current              |                                                                                                           | 25°C           |              | 750   |      | mA       |
| Peak output current                       |                                                                                                           | 25°C           |              | 2.2   |      | A        |

# Linear power supply

- Commercial components
  - LM317 Application example
    - $V_{out} = 1.25 \left(1 + \frac{R_2}{R_1}\right)$
    - If  $R_2$  is a potentiometer
      - Variable power supply
    - Capacitors are suggested for added stability and ripple rejection



# Linear power supply

- Example
  - Load
    - $V_{out} = 5\text{ V} \pm 25\text{ mV}$ , total output maximum ripple is 50 mV,  $I_{out} = 1.5\text{ A}$
  - Input
    - Wall socket  $V_{in} = 230\text{ V} \pm 5\%$  @  $f_{in} = 50\text{ Hz}$
  - Linear regulator
    - $V_{D_{omin}} = 2.5\text{ V}$ , Ripple Rejection  $R_{rej} = 40\text{ dB}$
  - Half wave rectifier
  - Determine
    - Transformer ratio
    - Capacitor value
    - Diode peak current and breakdown voltage
    - There are some degree of freedom

# Linear power supply

- Example

- The voltage at the input of the regulator should be
  - $V_{min} > V_{out} + V_{D0min} = 5 + 2.5 = 7.5 \text{ V}$
- The ripple at the input of the regulator should be
  - $\Delta V < 50 \text{ mV} \cdot 100 = 5 \text{ V}$
- Compute the minimum capacitor value
  - $\Delta V = \frac{I_{out} \cdot T}{C} \Rightarrow C > \frac{I_{out} \cdot T}{\Delta V} = \frac{1.5 \cdot 0.02}{5} = 6 \text{ mF}$
  - We want a high  $\Delta V$  to decrease the peak diode current
    - Choose  $C = 6.7 \text{ mF} \rightarrow \Delta V = 4.48 \text{ V}$
    - Then  $V_{max} > V_{min} + \Delta V > 12 \text{ V}$
- Input voltage considering tolerance
  - $V_{inmin} = 218.5 \text{ V}_{RMS} = 309 \text{ V}_{peak}$
  - $V_{inmax} = 241.5 \text{ V}_{RMS} = 341.5 \text{ V}_{peak}$

# Linear power supply

- Example

- Voltage at the output of the transformer should be

- $V_{ptransf} = V_{max} + V_D > 12 + 1 = 13 \text{ V}$

- Taking some margin, choose

- $V_{max} = 16 \text{ V}$

- In the worst case, when  $V_{in}$  is minimum

- $\frac{N_1}{N_2} = \frac{309}{16} = 19.3$

- We can choose  $\frac{N_1}{N_2} = 20$

- $\rightarrow V_{max} = \frac{309}{20} - 1 = 15.45 - 1 = 14.45 \text{ V}$  ✓

- When  $V_{in}$  is maximum

- $\frac{N_1}{N_2} = 20$

- $\rightarrow V_{max} = \frac{341.5}{20} - 1 = 17.1 - 1 = 16.1 \text{ V}$  ✓

$V_{max} > 12 \text{ V}$

- When  $V_{in}$  is average

- $V_{max} = \frac{325.2}{20} - 1 = 16.3 - 1 = 15.3 \text{ V}$

- $V_{max} > 12 \text{ V}$

- $V_{min} = V_{max} - \Delta V = 15.3 - 4.48 = 10.8 \text{ V}$

- $V_{min} > 7.5 \text{ V}$

- $V_{avg} = \frac{V_{max} + V_{min}}{2} = \frac{15.3 + 10.8}{2} = 13 \text{ V}$

# Linear power supply

- Example

- Conduction time of the diode with  $V_{max} = 14.45 \text{ V}$

- $t_c = \frac{T}{2\pi} \cos^{-1} \left( \frac{V_{max} - \Delta V}{V_{max}} \right) = \frac{0.02}{2\pi} \cos^{-1} \left( \frac{14.45 - 4.48}{14.45} \right) = 2.58 \text{ ms}$

- Conduction time of the diode with  $V_{max} = 16.1 \text{ V}$

- $t_c = \frac{T}{2\pi} \cos^{-1} \left( \frac{V_{max} - \Delta V}{V_{max}} \right) = \frac{0.02}{2\pi} \cos^{-1} \left( \frac{16.1 - 4.48}{16.1} \right) = 2.43 \text{ ms}$

- Peak current in the diode

- $I_{Dpeak} = 2I_{out} \frac{T}{t_c} = 2 \cdot 1.5 \cdot \frac{20}{2.43} = 24.7 \text{ A}$

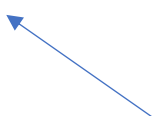
- Choose a diode with at least  $I_{max} = 30 \text{ A}$

- Reverse voltage on the diode

- $V_{BR} > 2 \cdot 17.1 = 34.2 \text{ V}$

- Choose a diode with at least  $V_{BR} = 50 \text{ V}$

Lowest, worst case

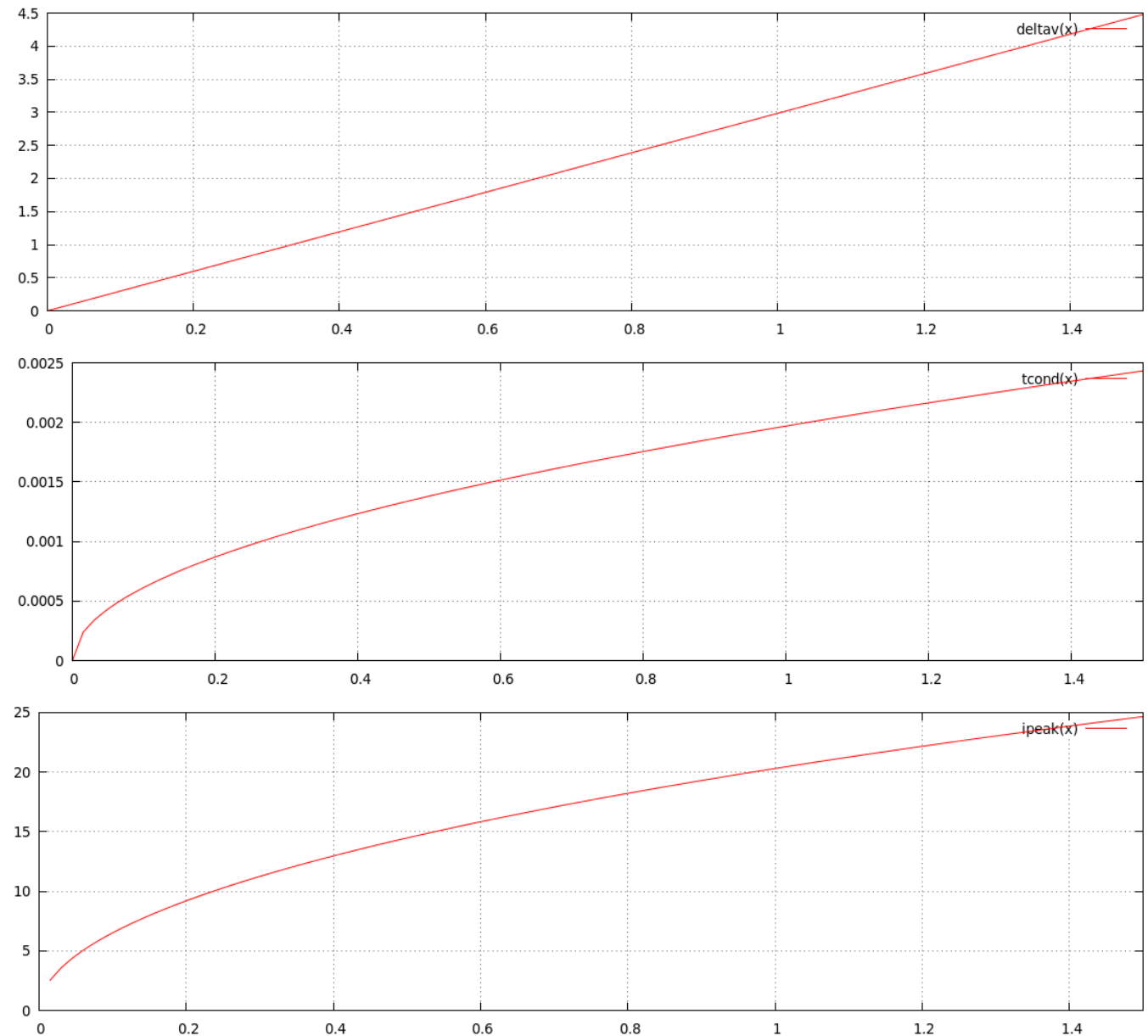


# Linear power supply

- Example
  - Efficiency
    - Assume no power losses in
      - Main switch and fuse
      - EMC filter
      - Transformer
      - Rectifier
      - Filter
    - Compare power at the input and output of the linear regulator
      - Use average values
      - $\eta = \frac{P_{out}}{P_{in\_lin}} \simeq \frac{V_{out}}{V_{avg}} = \frac{5}{13} = 38.5\%$

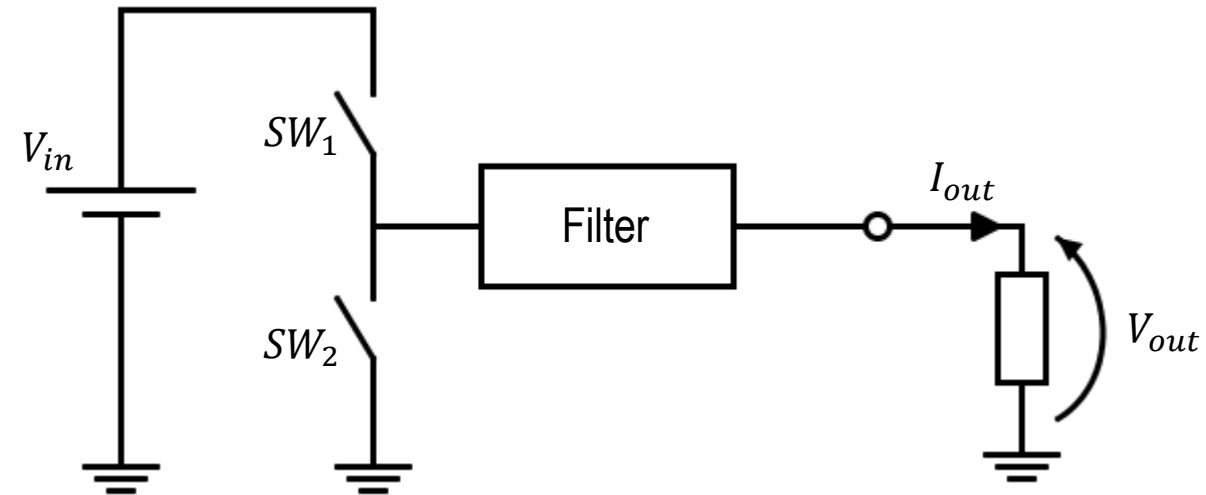
# Linear power supply

- Example
  - What happens if the output current is lower than the maximum 1.5 A?
    - The input ripple to the linear regulator decreases
    - The diode conduction time decreases
      - It would cause an increase of the diode peak current
    - But the output current is lower than the maximum (used for the previous computation)
    - All together, the diode peak current decreases
    - Design at the maximum output current



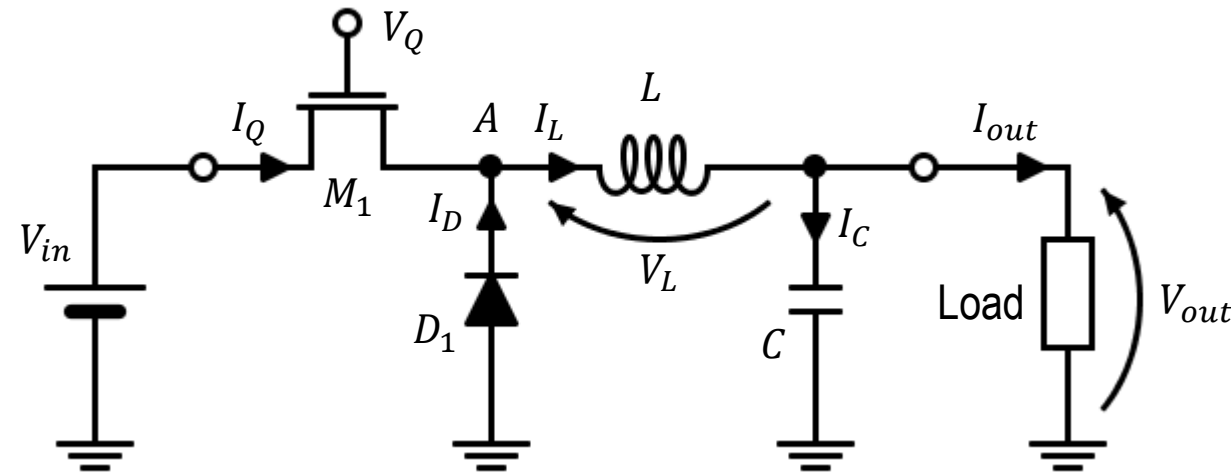
# Switching power supply

- Switching DC-to-DC converter
  - Use a switching architecture to generate a wanted output voltage given a constant input voltage
    - Similar to a PWM technique
    - Switches used to create a square wave
      - With controlled duty cycle  $\delta$
      - With a high switching frequency  $f_{sw}$
      - Switches have a very high efficiency
    - A filter extracts the average value
      - Capacitor and inductor
      - Reactive components
        - No power loss
        - High efficiency
  - Use a feedback network to implement a regulator



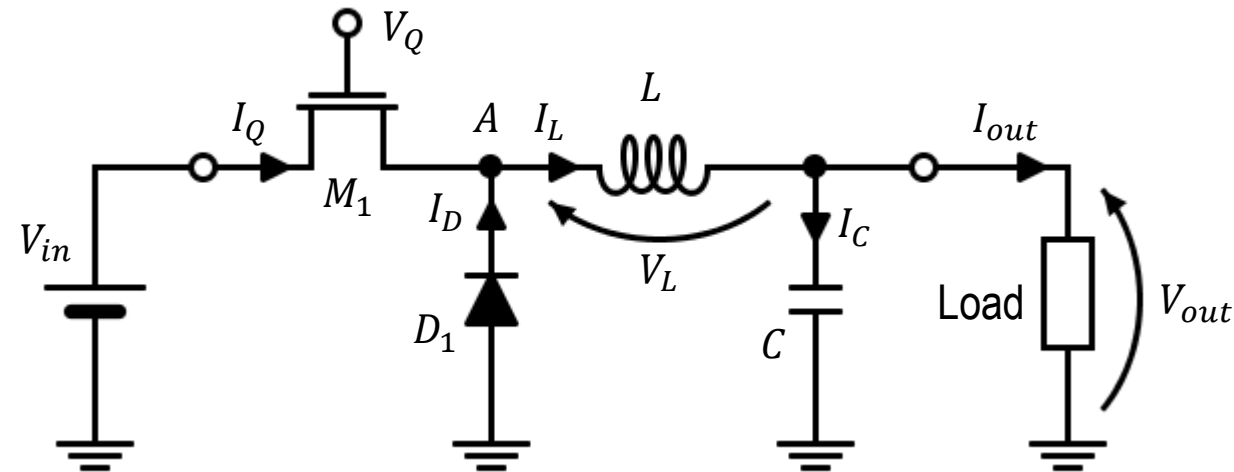
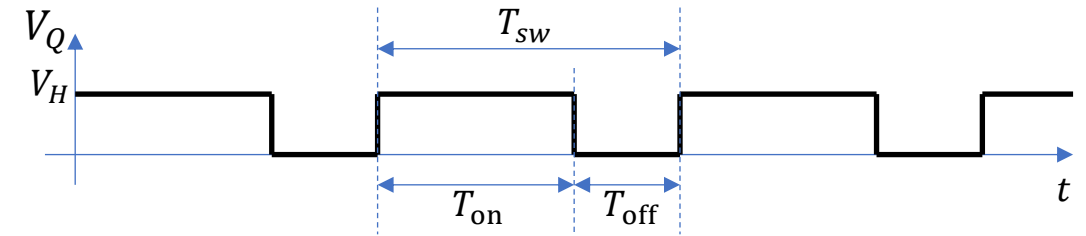
# Switching power supply

- Buck (Step Down) DC-DC converter
  - Produces an output voltage  $V_{out}$  lower than the input voltage  $V_{in}$ 
    - $V_{out} < V_{in}$
  - High side switch  $SW_1$  implemented using transistor  $M_1$
  - Low side switch  $SW_2$  implemented using diode  $D_1$
  - Analyze the circuit at the steady-state
    - Average voltage across an inductor at the steady-state
      - $V_{Lavg} = 0 \text{ V}$
    - Average current flowing in a capacitor at the steady-state
      - $I_{Cavg} = 0 \text{ A}$
    - Assume  $V_{in}$ ,  $V_{out}$  and  $I_{out}$  are constant at the steady-state



# Switching power supply

- Buck (Step Down) DC-DC converter
  - Voltage  $V_Q$  is a periodic square wave
    - Period  $T_{sw}$ , frequency  $f_{sw} = \frac{1}{T_{sw}}$
    - Alternates between a positive voltage  $V_H$  and ground
      - Voltage  $V_H$  high enough to switch on  $M_1$ 
        - The transistor becomes a short circuit
      - When  $V_Q$  is at ground,  $M_1$  is in cut-off
        - The transistor becomes an open circuit
  - Duty cycle  $\delta$ 
    - Time at  $V_H$  is  $T_{on}$
    - Time at ground is  $T_{off}$
    - $T_{sw} = T_{on} + T_{off}$
    - $\delta = \frac{T_{on}}{T_{sw}}$

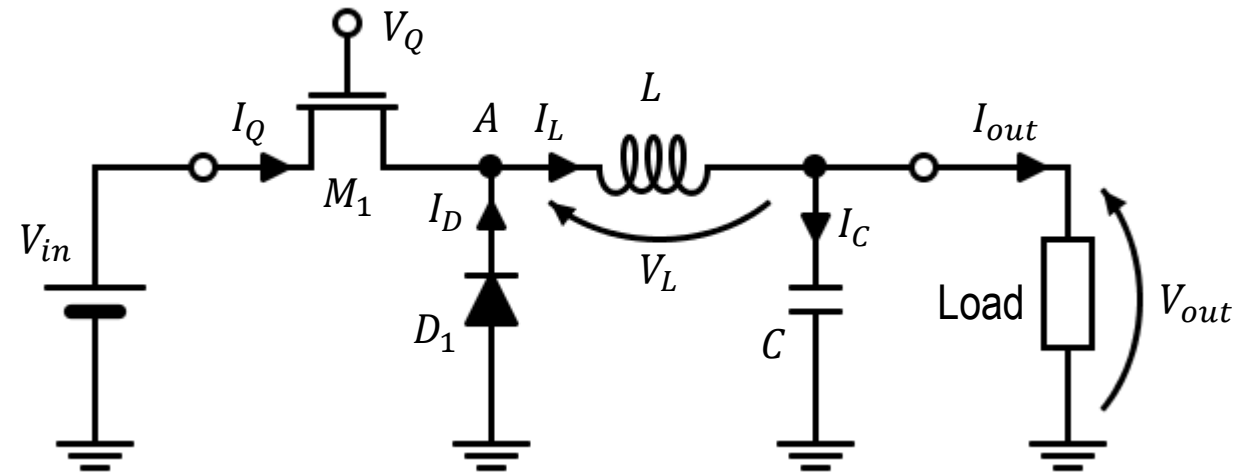


# Switching power supply

- Buck (Step Down) DC-DC converter

- During  $T_{on}$

- Transistor  $M_1$  is conducting
      - High side switch is closed
    - $V_A = V_{in}$
    - Diode  $D_1$  is reversed biased
      - It is an open circuit
      - The low side switch is open
    - Voltage across the inductor
      - $V_L = V_{in} - V_{out}$
      - It is assumed to be constant

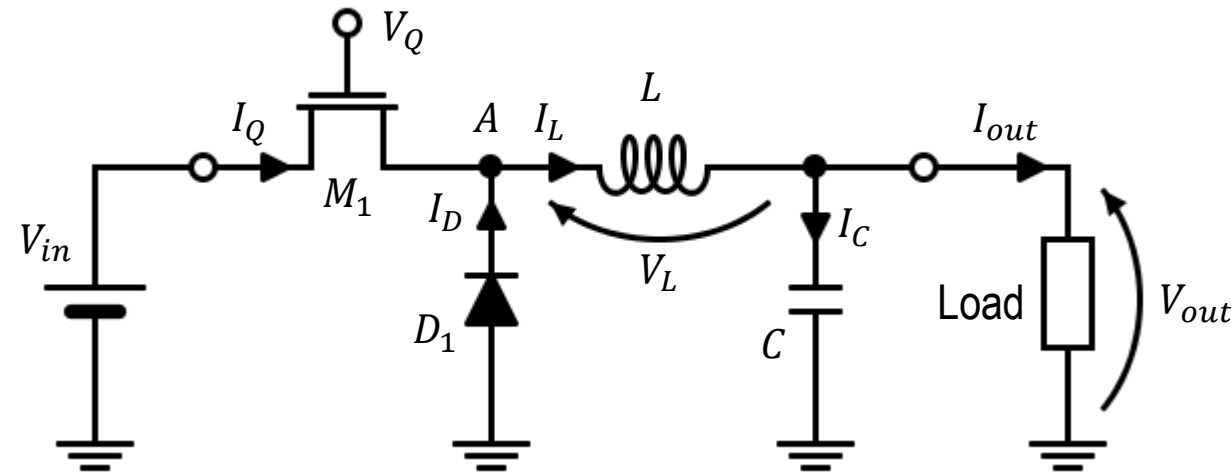


# Switching power supply

- Buck (Step Down) DC-DC converter
  - During  $T_{on}$ 
    - Current flowing in the inductor
      - It changes linearly
      - $\Delta I_{Lon} = \frac{V_L T_{on}}{L} = \frac{1}{L} (V_{in} - V_{out}) T_{on}$
      - Absolute value of  $I_L$  not yet determined
        - Varies between  $I_{Lmin}$  and  $I_{Lmax}$
        - $\Delta I_{Lon} = I_{Lmax} - I_{Lmin}$
    - Current flowing in the transistor  $M_1$ 
      - $I_Q = I_L$
    - Current flowing in the diode  $D_1$ 
      - $I_D = 0$

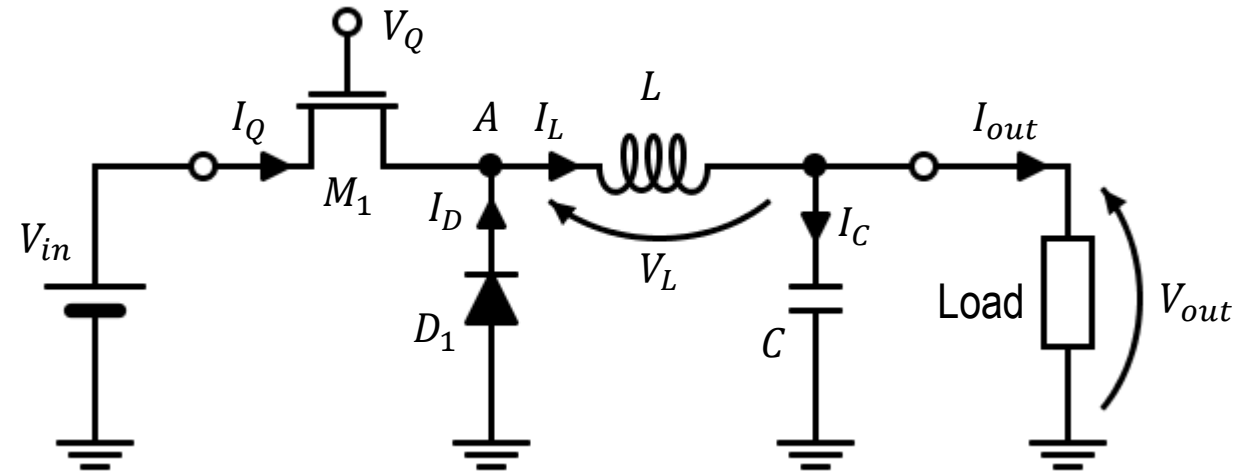
$$I_L(t) = \frac{1}{L} \int V_L(t) dt$$

$$I_{Lon}(t) = \frac{1}{L} \int_0^t (V_{in} - V_{out}) dt = \frac{(V_{in} - V_{out})}{L} t + c$$



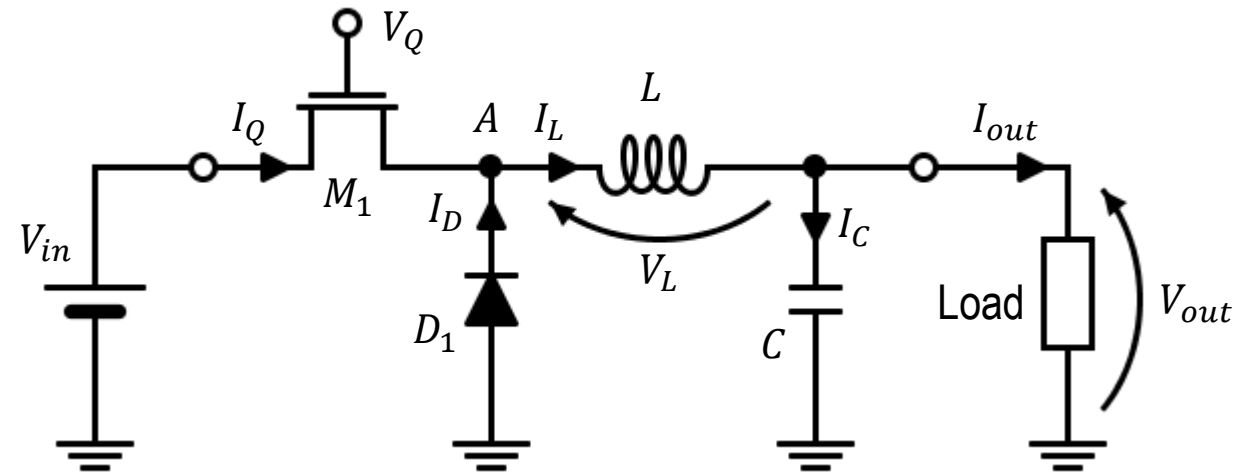
# Switching power supply

- Buck (Step Down) DC-DC converter
  - During  $T_{off}$ 
    - Transistor  $M_1$  is in cut-off
      - High side switch is open
    - Diode  $D_1$  is forced in conduction
      - The low side switch is closed
    - $V_A = 0\text{ V}$ 
      - Neglect voltage drop across the diode
    - Voltage across the inductor
      - $V_L = 0 - V_{out} = -V_{out}$
      - It is assumed to be constant



# Switching power supply

- Buck (Step Down) DC-DC converter
  - During  $T_{\text{off}}$ 
    - Current flowing in the inductor
      - It changes linearly
      - $\Delta I_{L\text{off}} = \frac{V_L T_{\text{off}}}{L} = -\frac{1}{L} V_{\text{out}} T_{\text{off}}$
      - Absolute value of  $I_L$  not yet determined
        - Between  $I_{L\text{max}}$  and  $I_{L\text{min}}$
        - $\Delta I_{L\text{off}} = I_{L\text{min}} - I_{L\text{max}}$
    - Current flowing in the transistor  $M_1$ 
      - $I_Q = 0$
    - Current flowing in the diode  $D_1$ 
      - $I_D = I_L$



# Switching power supply

- Buck (Step Down) DC-DC converter

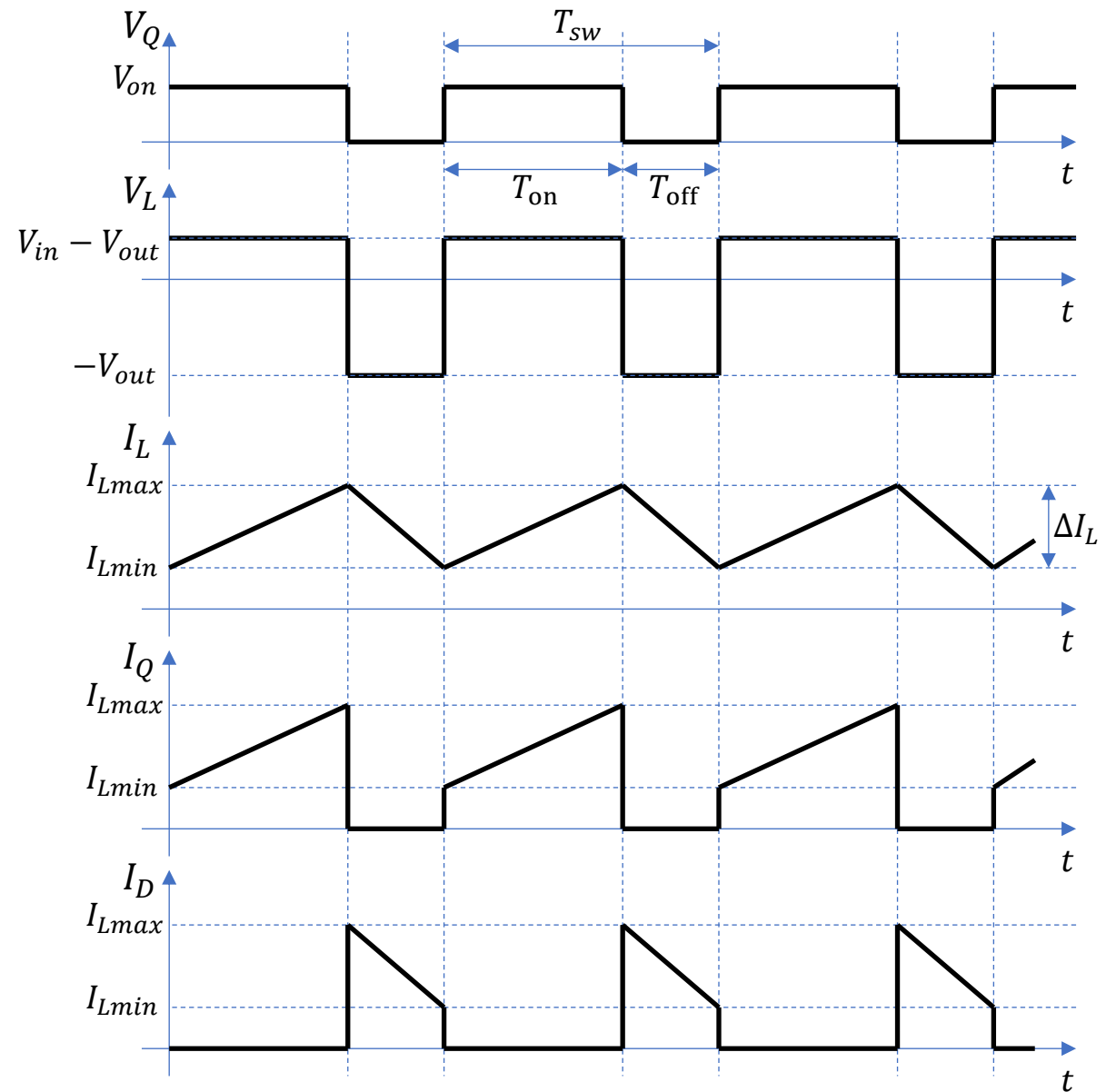
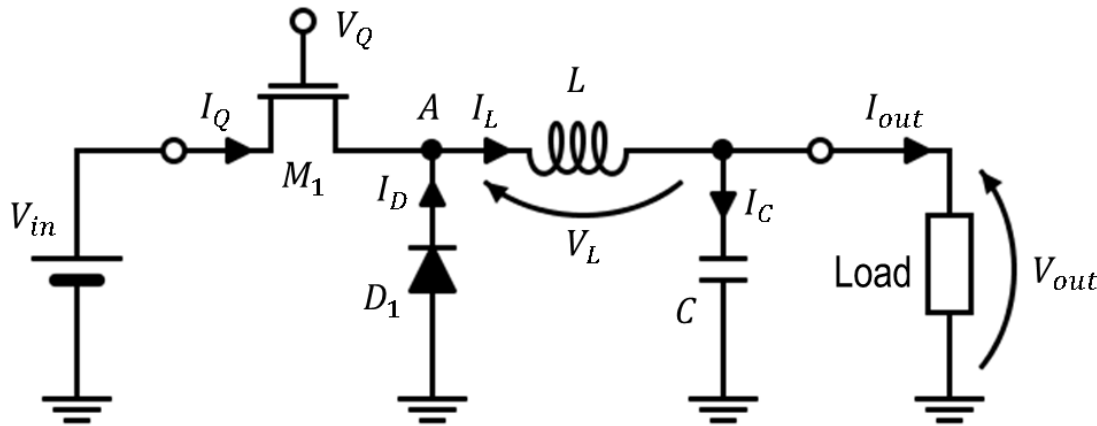
- Waveforms

- Assumptions

- $V_{in} > 0, V_{out} > 0, V_{out} < V_{in}$
      - $I_L > 0 \quad \forall t$

- $V_{Lavg} = 0$  at the steady state

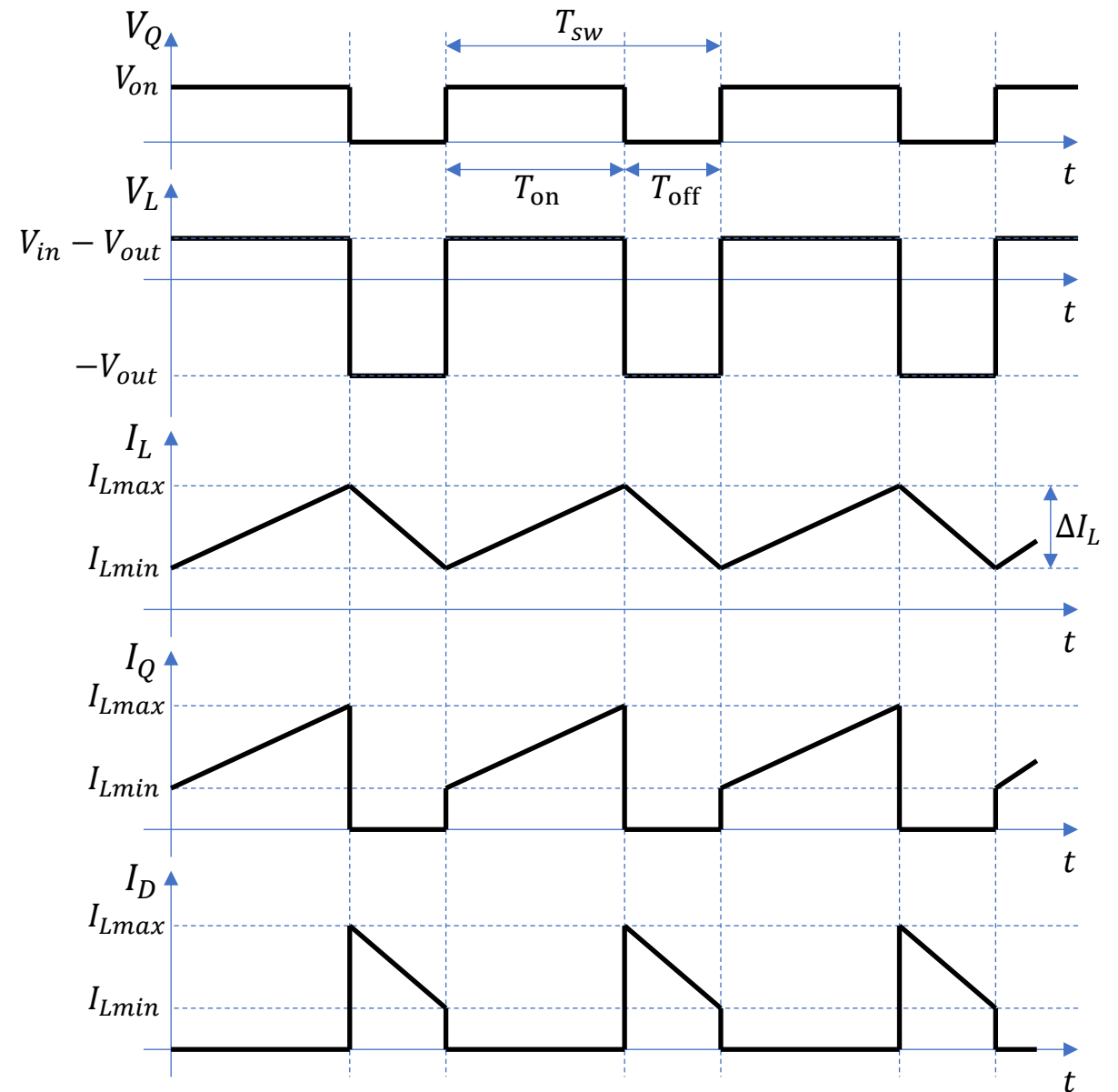
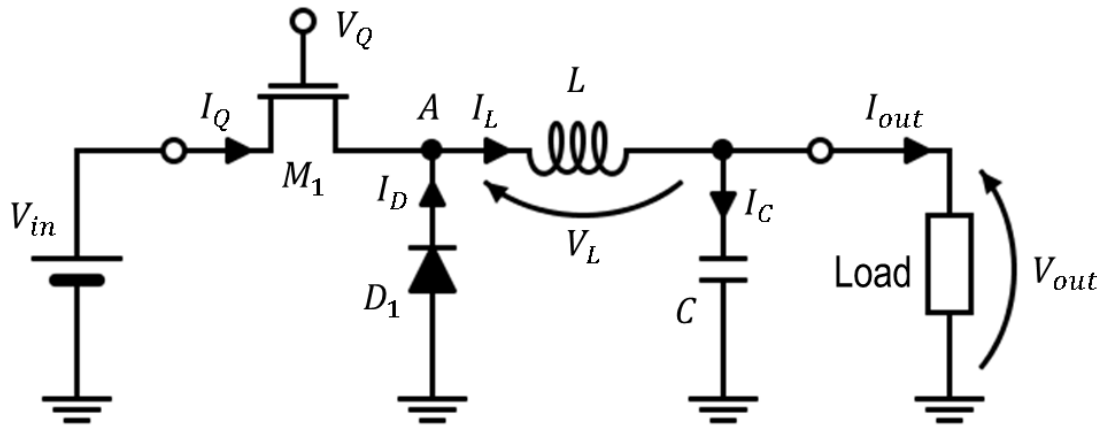
- $I_L$  (integral of  $V_L$ ) does not diverge



# Switching power supply

- Buck (Step Down) DC-DC converter

- $\Delta I_{L\text{on}} + \Delta I_{L\text{off}} = 0$ 
  - $\frac{1}{L}(V_{in} - V_{out})T_{\text{on}} - \frac{1}{L}V_{out}T_{\text{off}} = 0$
  - $\rightarrow \frac{V_{out}}{V_{in}} = \frac{T_{\text{on}}}{T_{\text{sw}}} = \delta$
  - $\rightarrow V_{out} = \delta V_{in}$
  - $V_{out}$  is always lower than  $V_{in}$
- Result valid at the steady-state only



# Switching power supply

- Buck (Step Down) DC-DC converter

- Current in the inductor

- The average current in the load and the average current in the inductor should be identical
      - Because the average current in the capacitor is 0

- Assume  $I_{Lmin} > 0$

- Continuous Current Mode (CCM)

- Write a system of two equations

$$\begin{cases} I_{Lavg} = \frac{I_{Lmax} + I_{Lmin}}{2} = I_{out} \\ \Delta I_L = I_{Lmax} - I_{Lmin} = \frac{1}{L} (V_{in} - V_{out}) T_{on} \end{cases}$$

$$\begin{cases} I_{Lmax} = I_{out} + \frac{(V_{in} - V_{out}) T_{on}}{2L} = I_{out} + \frac{\Delta I_L}{2} \\ I_{Lmin} = I_{out} - \frac{(V_{in} - V_{out}) T_{on}}{2L} = I_{out} - \frac{\Delta I_L}{2} \end{cases}$$

- Solving the system gives  $I_{Lmax}$  and  $I_{Lmin}$

- Currents in the transistor and in the diode

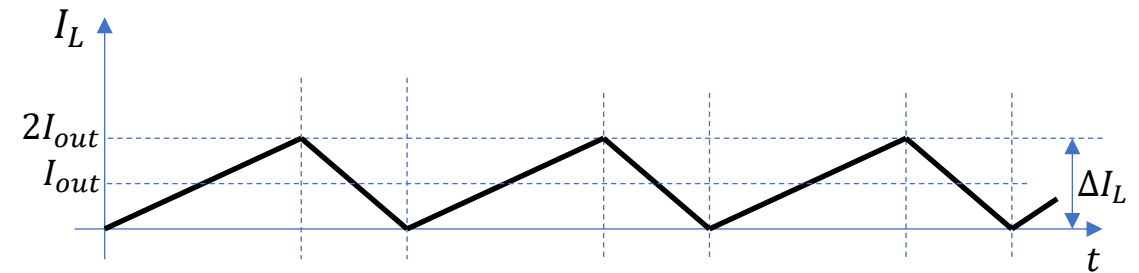
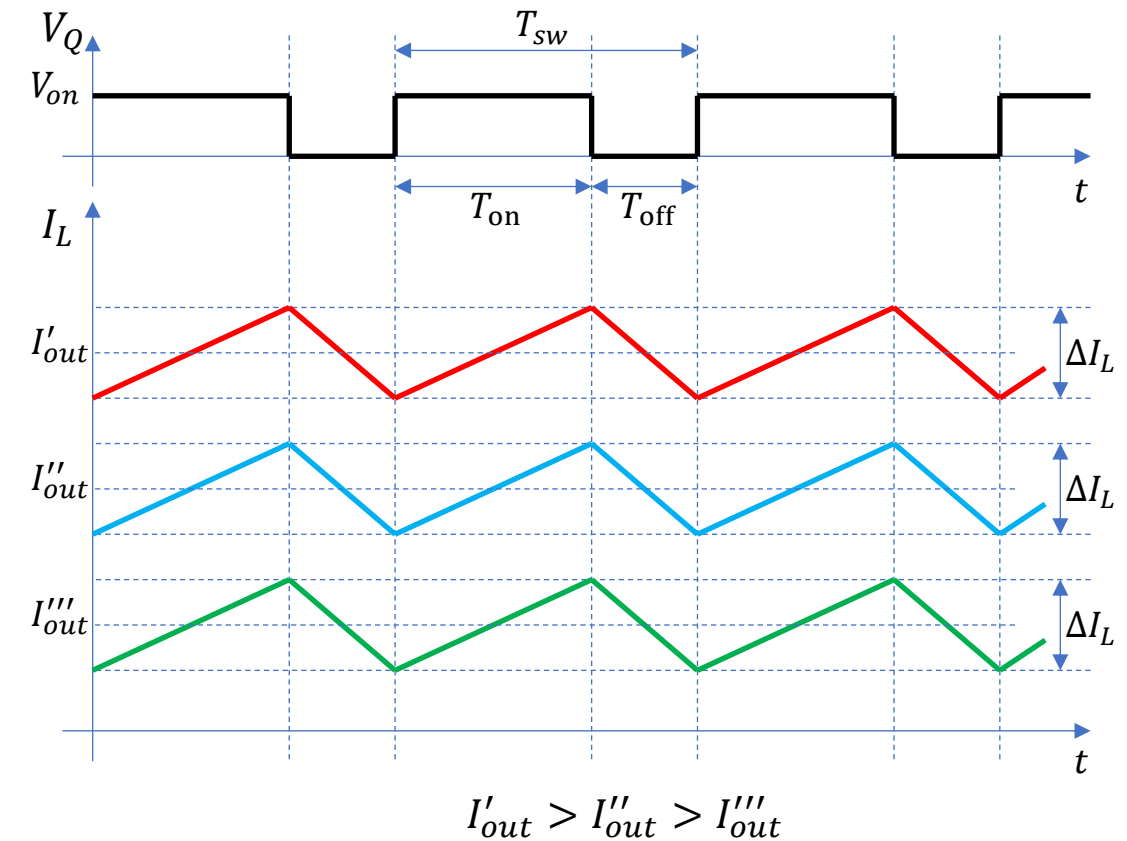
- Identical to the current in the inductor when they conduct

# Switching power supply

- Buck (Step Down) DC-DC converter
  - Current from the voltage source
    - A similar result as in the previous case can be obtained considering the output current of the voltage source
      - $I_{in} = I_Q$
    - Since the converter is loss-less (at least in a first approximation), on average it is
      - $P_{in} = P_{out} \Rightarrow V_{in}I_{inavg} = V_{out}I_{out} \Rightarrow I_{inavg} = \delta I_{out}$
      - $I_{inavg} = \frac{1}{T_{sw}} \left( I_{Lmin} \cdot T_{on} + \frac{1}{2} (I_{Lmax} - I_{Lmin}) T_{on} \right) = \delta \frac{I_{Lmax} + I_{Lmin}}{2}$
    - Substituting  $I_{inavg}$  in the previous expression
      - $\delta V_{in} \frac{I_{Lmax} + I_{Lmin}}{2} = V_{out} I_{out} \Rightarrow \frac{I_{Lmax} + I_{Lmin}}{2} = I_{out}$
    - Same expression as obtained before
    - Add the equation for  $\Delta I_L$  and solve the system for  $I_{Lmax}$  and  $I_{Lmin}$
  - Output current of the voltage source is pulsated
    - Not good for the voltage generator

# Switching power supply

- Buck (Step Down) DC-DC converter
  - Current rating for the inductor
    - When  $I_{out}$  changes,  $I_L$  translates up or down
    - Inductor current rating
      - Evaluate  $I_{Lmax}$  with the maximum  $I_{out}$
    - Or compute  $I_{Lrms}$ 
      - $I_{Lrms}^2 = I_{Lavg}^2 + \frac{1}{12} (\Delta I_L)^2$
      - Worst case contribution of  $\Delta I_L$  when  $I_{Lmin} = 0$ 
        - $\rightarrow I_{Lavg} = I_{out}, \Delta I_L = 2I_{out}$
        - $I_{Lrms}^2 = I_{out}^2 + \frac{1}{3} I_{out}^2$
        - $I_{Lrms} = \frac{2}{\sqrt{3}} I_{out}$
  - Current ratings for the transistor and the diode
    - Use the same rating as for the inductor



# Switching power supply

- Buck (Step Down) DC-DC converter

- Current rating for the inductor

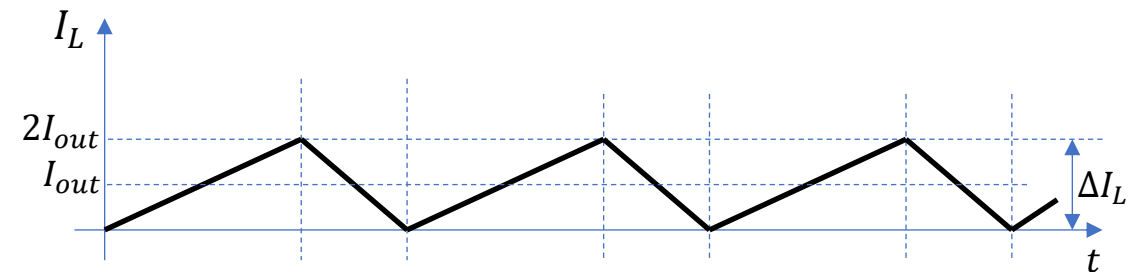
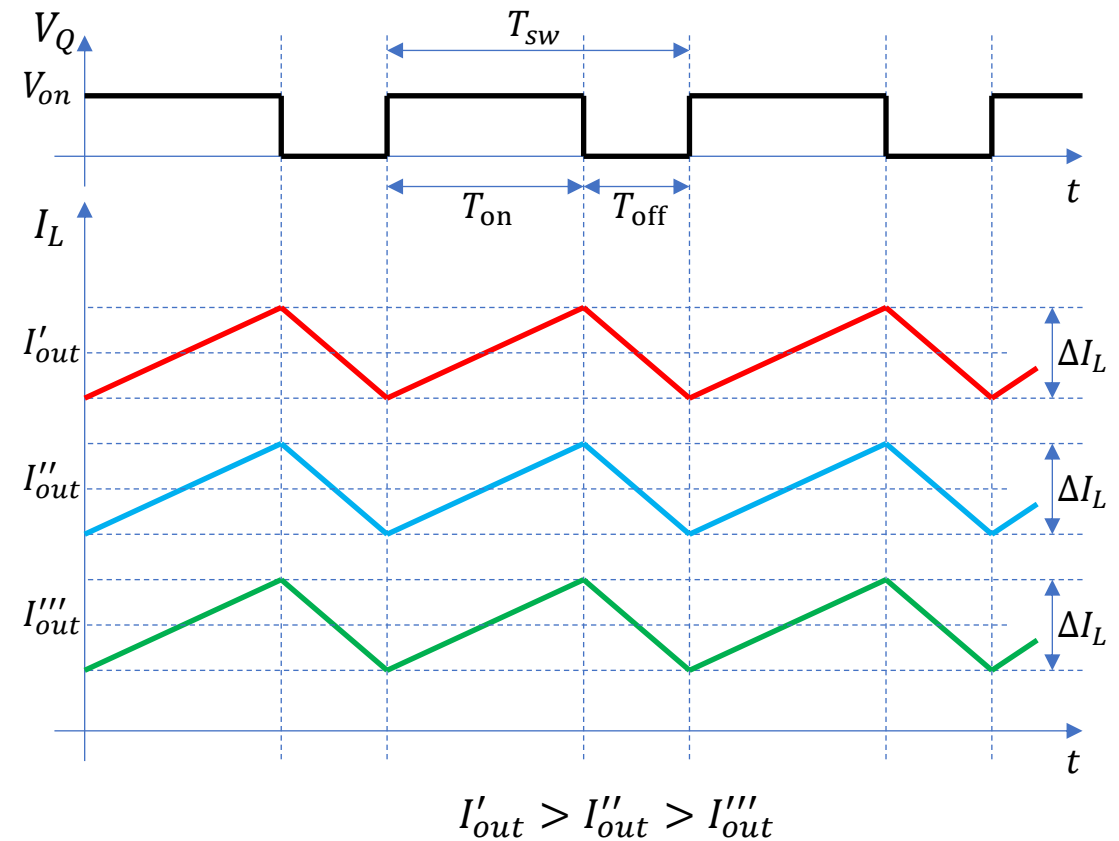
- The RMS (Root Mean Square) current rating  $I_{Lrms}$  is less conservative than the maximum current  $I_{Lmax}$
- Equivalent constant current that produces the same power dissipation of the alternating current
- It is a good measure especially when the switching frequency is high, as it normally happens

- RMS calculation

- Square root of the average of the squares over a period

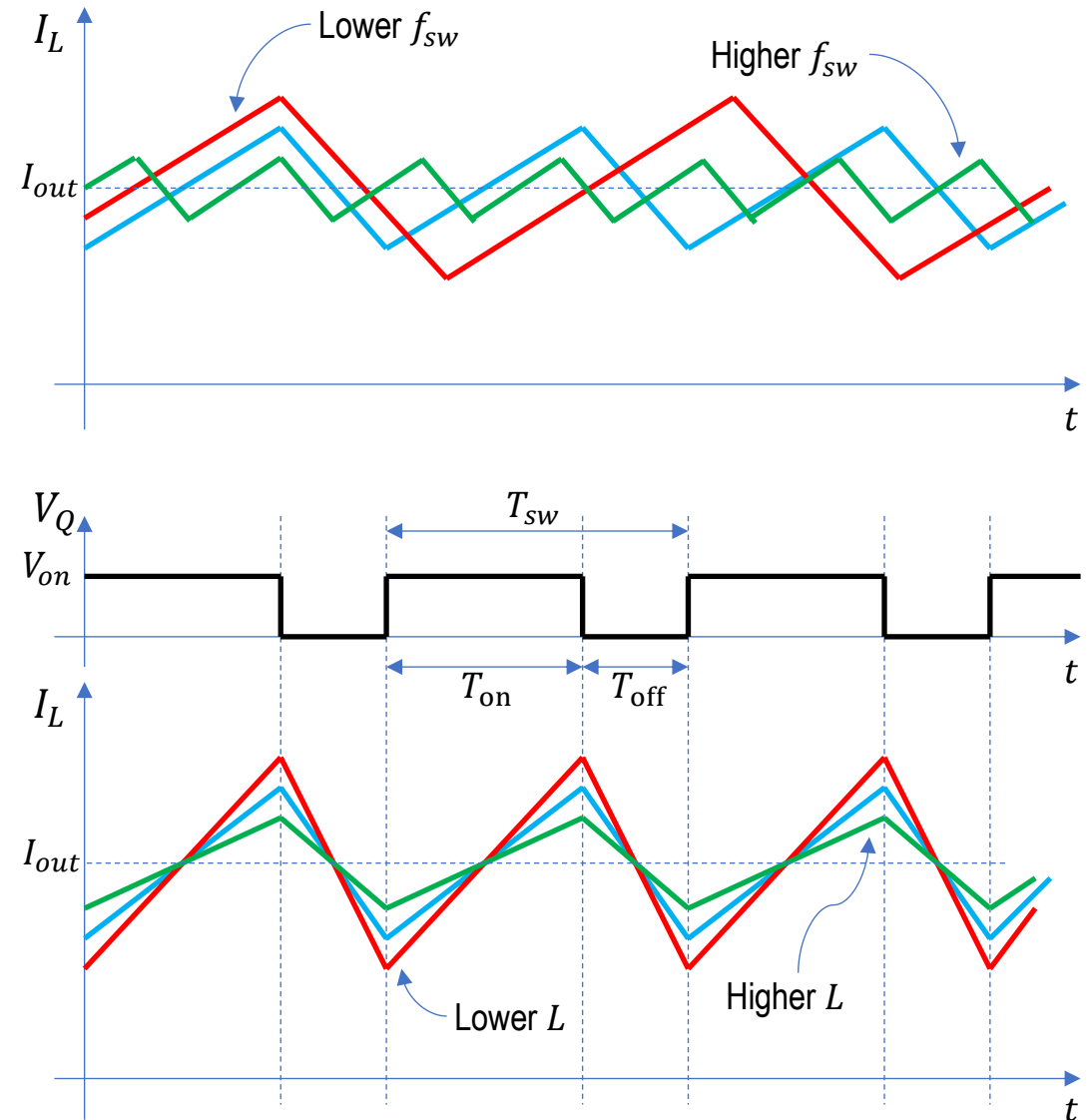
$$I_{Lrms}^2 = \frac{1}{T_{sw}} \int_0^{T_{sw}} I_L^2(t) dt = I_{Lavg}^2 + \frac{1}{3} \left( \frac{\Delta I_L}{2} \right)^2 =$$

$$= I_{Lavg}^2 + \frac{\Delta I_L^2}{12}$$



# Switching power supply

- Buck (Step Down) DC-DC converter
  - Current in the inductor
    - Effect of changing the frequency  $f_{sw}$ 
      - The slopes of  $I_L$  do not change
      - A lower frequency makes  $\Delta I_L$  larger
      - A higher frequency makes  $\Delta I_L$  smaller
    - Effect of changing the value  $L$  of the inductor
      - The slopes of  $I_L$  change
      - A lower  $L$  makes  $\Delta I_L$  larger
      - A higher  $L$  makes  $\Delta I_L$  smaller



# Switching power supply

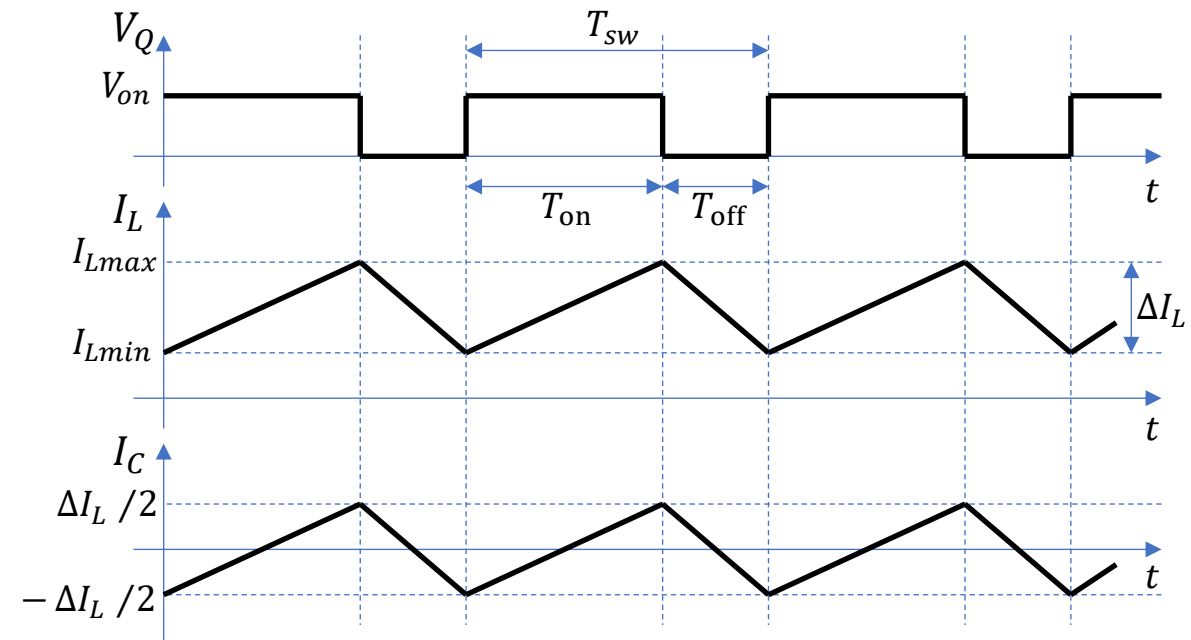
- Buck (Step Down) DC-DC converter

- Current in the capacitor

- $I_C = I_L - I_{out}$
    - Average is 0

- Current rating for the capacitor

- Maximum is  $\frac{\Delta I_L}{2}$
    - Or compute  $I_{Crms}$ 
      - $I_{Crms}^2 = \frac{1}{12} (\Delta I_L)^2$
      - Same worst case as for the inductor,  $\Delta I_L = 2I_{out}$ 
        - $I_{Crms}^2 = \frac{1}{3} I_{out}^2$
        - $I_{Crms} = \frac{1}{\sqrt{3}} I_{out}$



# Switching power supply

- Buck (Step Down) DC-DC converter

- Voltage on the capacitor

- The current in the capacitor is zero only on average

- It changes during a period

- Voltage is the integral of the current

- $V_C = \frac{1}{C} \int I_C dt$

- When current is positive, the voltage increases

- When current is negative, the voltage decreases

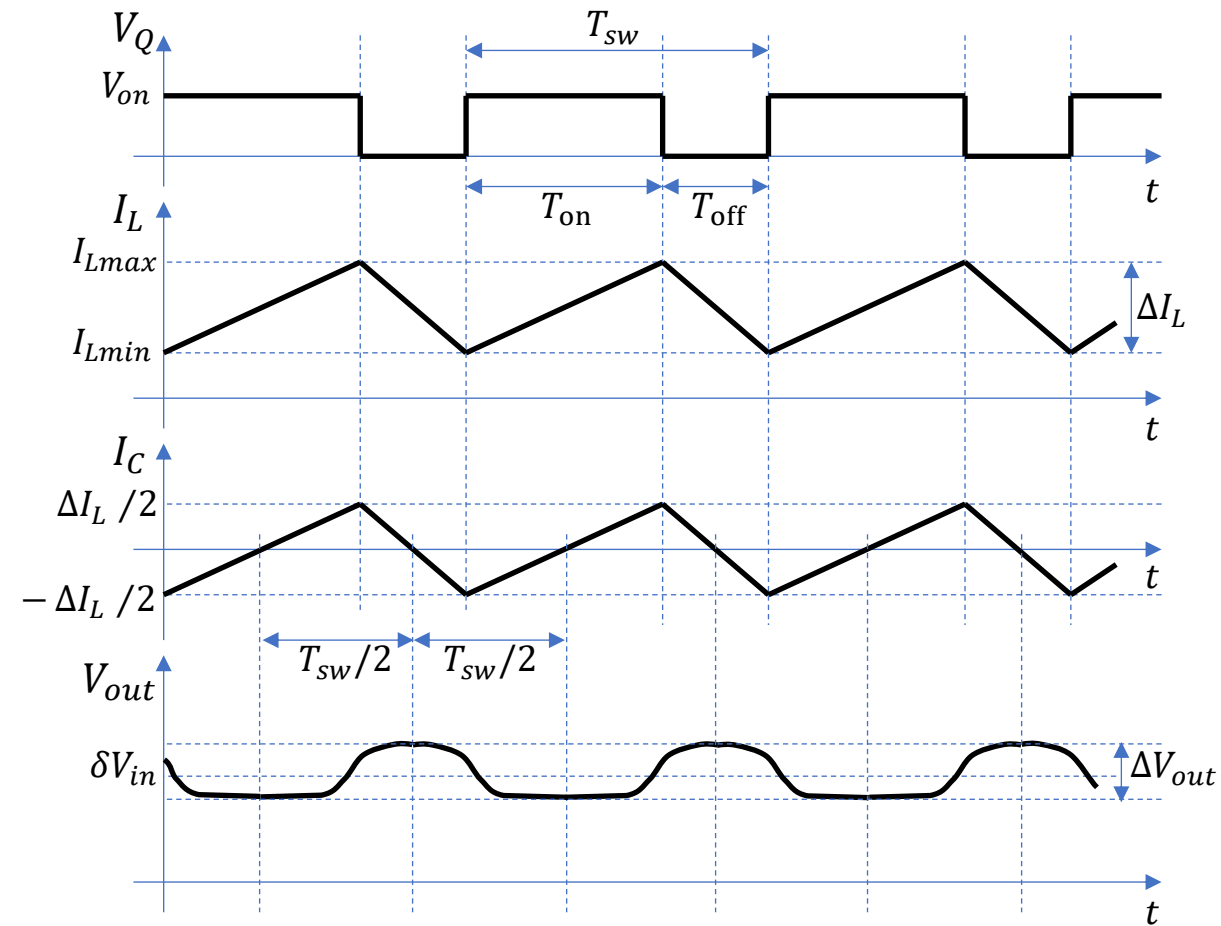
- It causes an output voltage ripple  $\Delta V_{out}$

- Voltage ripple on the capacitor

- $\Delta V_{out} = \frac{1}{C} \frac{1}{2} \frac{T_{sw}}{2} \frac{\Delta I_L}{2} = \frac{\Delta I_L}{8Cf_{sw}}$

- In the worst case, as before  $\Delta I_L = 2I_{out}$

- $\Delta V_{out} = \frac{1}{C} \frac{1}{2} \frac{T_{sw}}{2} \frac{2I_{out}}{2} = \frac{T_{sw}I_{out}}{4C} = \frac{I_{out}}{4Cf_{sw}}$



# Switching power supply

- Buck (Step Down) DC-DC converter
  - Voltage rating for the transistor
    - When it is OFF
      - $V_{DS} = V_{in}$
      - It should be  $BV_{CE0} > V_{in}$ 
        - Take some margin
  - Voltage rating for the diode
    - When it is OFF
      - $V_D = V_{in}$
      - It should be  $V_{BR} > V_{in}$ 
        - Take some margin

# Switching power supply

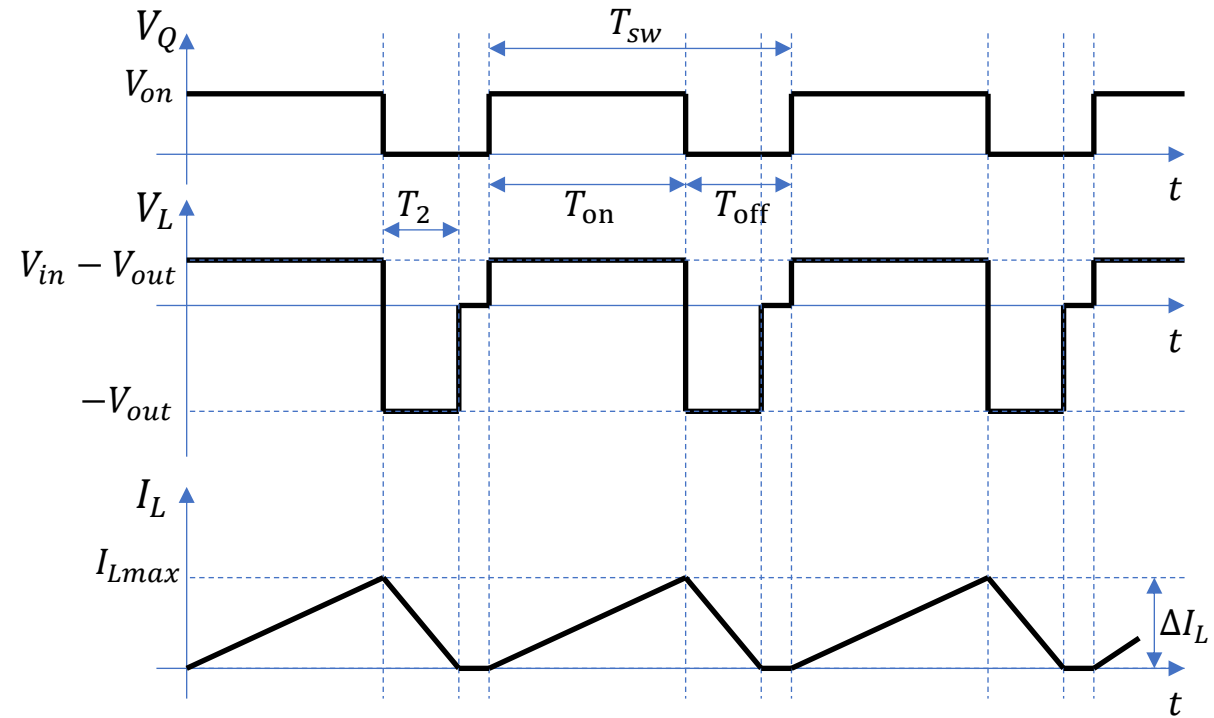
- Buck (Step Down) DC-DC converter
  - Line regulation
    - The output voltage depends on the input voltage
      - When the input changes, the output also changes
  - $S_V = \frac{\Delta V_{out}}{\Delta V_{in}} = \delta$
  - Not good
    - The circuit is a DC-DC converter
    - It is not a regulator
    - Need a negative feedback regulator circuit to control the duty cycle  $\delta$  when the input changes
      - If  $V_{in}$  increases, decrease  $\delta$
      - If  $V_{in}$  decreases, increase  $\delta$

# Switching power supply

- Buck (Step Down) DC-DC converter
  - Load regulation
    - The expression of the output voltage does not depend on the output current  $I_{out}$
    - $S_I = 0$ 
      - The output resistance is zero
    - Excellent result!
    - But it is not strictly true
      - Parasitic resistances were not considered in the expression
      - The analysis is at the steady-state
    - When the output current changes
      - The output voltage goes through a transient
        - During the transient  $V_{out} \neq \delta V_{in}$
      - When the transient terminates, the output voltage returns to the correct value
    - Result is only valid in Continuous Current Mode (CCM)

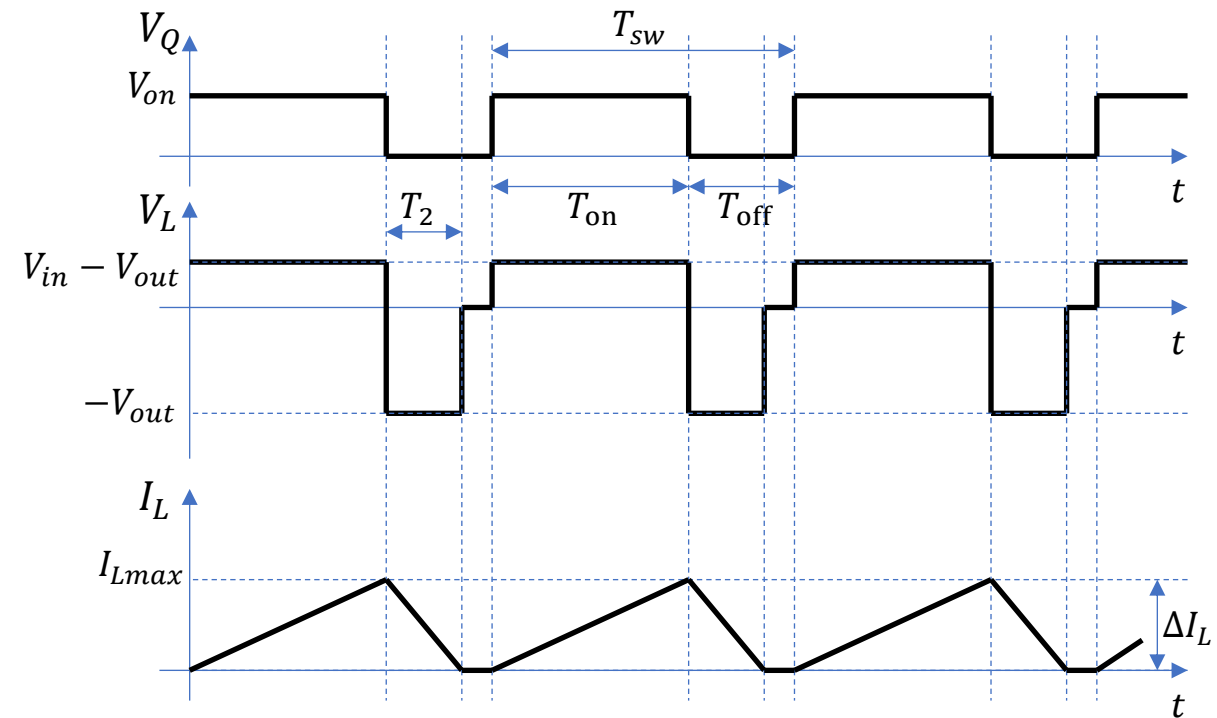
# Switching power supply

- Buck (Step Down) DC-DC converter
  - Discontinuous Current Mode (DCM)
    - The current in the inductor reaches the value 0
      - $I_{Lmin} = 0$
    - The current change during  $T_{on}$  remains the same
      - $\Delta I_{Lon} = \frac{V_L T_{on}}{L} = \frac{1}{L} (V_{in} - V_{out}) T_{on}$
      - $I_{Lmax} = \Delta I_{Lon} = \frac{1}{L} (V_{in} - V_{out}) T_{on}$
    - The current goes to 0 during  $T_2$ 
      - $I_{Lmax} = \frac{1}{L} V_{out} T_2 \Rightarrow T_2 = \frac{V_{in} - V_{out}}{V_{out}} T_{on}$
    - The average value of current  $I_L$  is still equal to the output current  $I_{out}$ 
      - $\frac{1}{2} (T_{on} + T_2) I_{Lmax} = I_{out} T_{sw}$



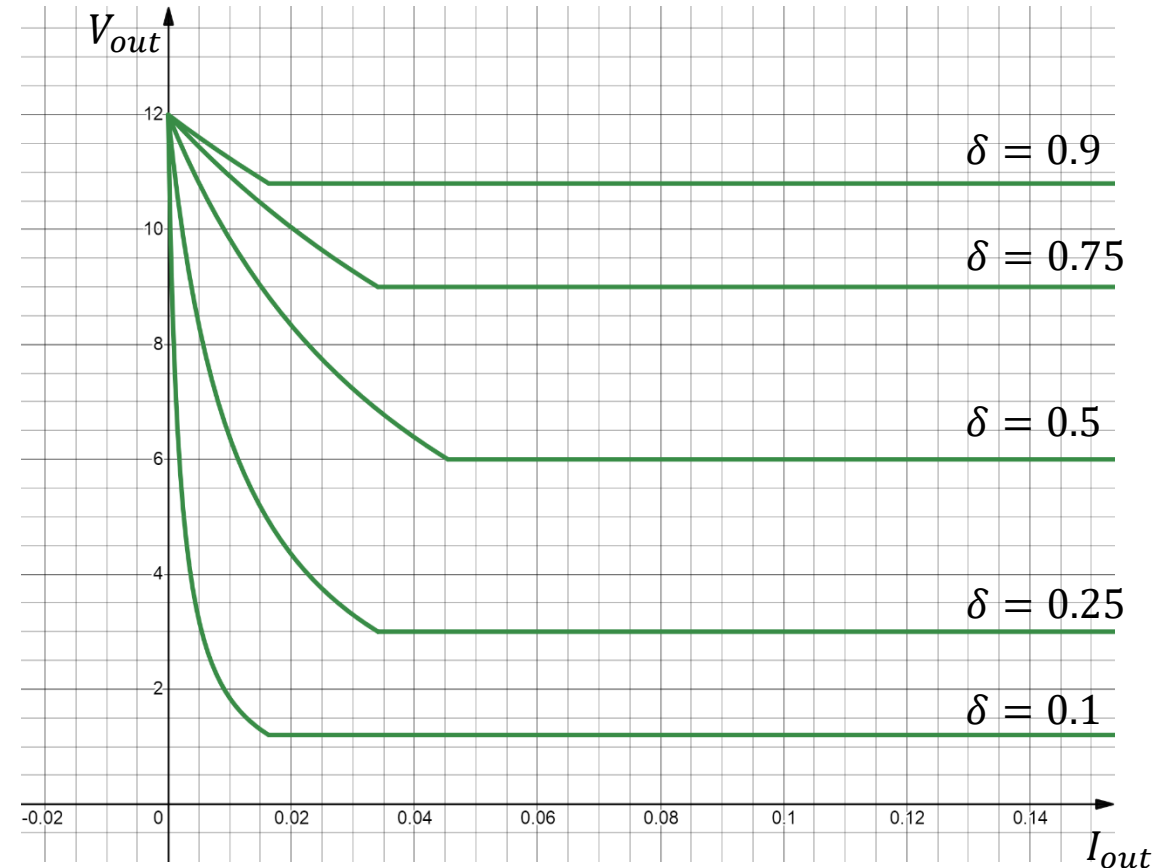
# Switching power supply

- Buck (Step Down) DC-DC converter
  - Discontinuous Current Mode (DCM)
    - Substitute expressions for  $I_{Lmax}$  and  $T_2$
    - Solve for  $V_{out}$ 
      - $$V_{out} = \frac{V_{in}}{\frac{2LI_{out}}{\delta^2 T_{sw} V_{in}} + 1}$$
    - The output voltage now depends also on
      - The inductor value  $L$
      - The switching period  $T_{sw}$
      - The output current  $I_{out}$ 
        - The output resistance is not zero any more
        - $S_I > 0$



# Switching power supply

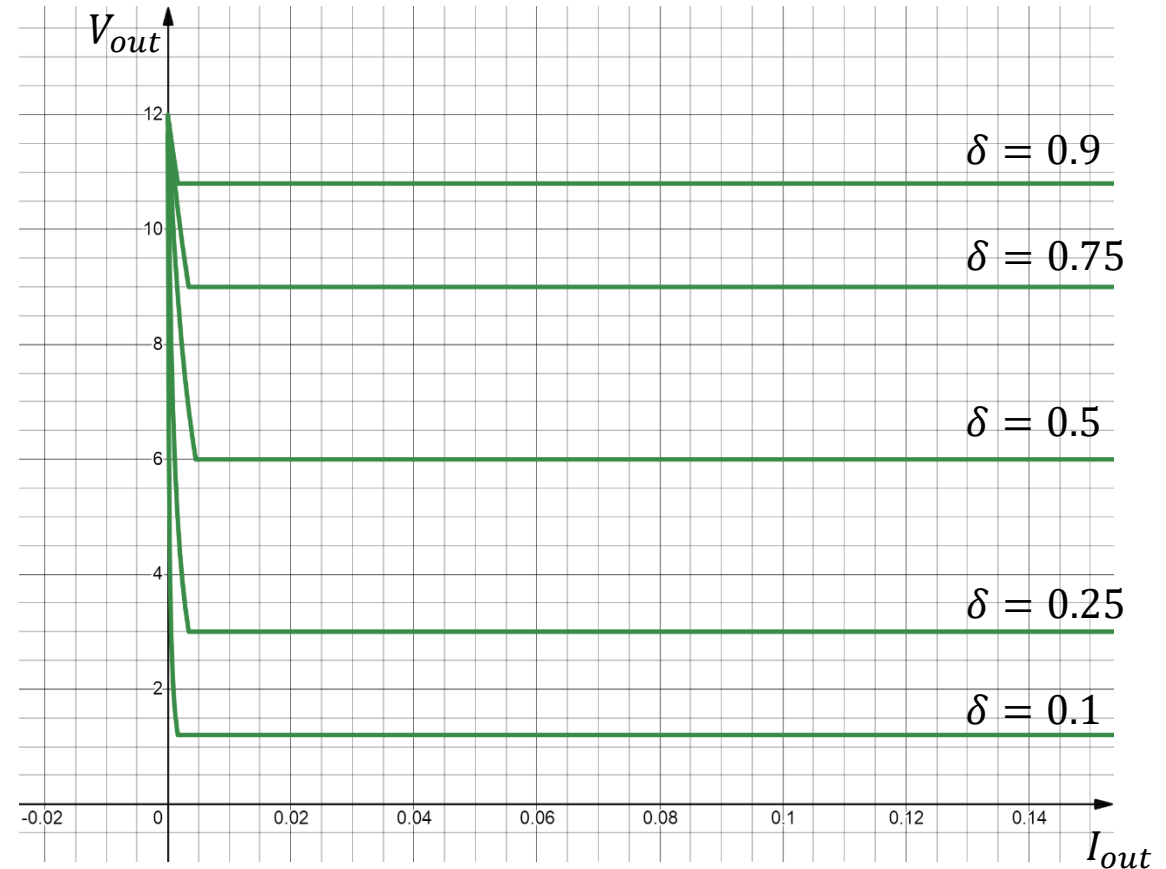
- Buck (Step Down) DC-DC converter
  - Discontinuous Current Mode (DCM)
    - A small output current causes the circuit to work in DCM
      - $I_{DCM} = \frac{\Delta I_{L on}}{2} = \frac{(1-\delta)V_{out}}{2Lf_{sw}}$
      - If  $I_{out} > I_{DCM}$  then the circuit is in CCM
      - If  $I_{out} < I_{DCM}$  then the circuit is in DCM
  - Without a regulator
    - The output voltage  $V_{out}$  increases for low output current  $I_{out}$
    - If the output current  $I_{out} = 0$  the output voltage  $V_{out} = V_{in}$ , regardless of the duty cycle  $\delta$



$$\begin{aligned} V_{in} &= 12 \text{ V} \\ f_{sw} &= 100 \text{ kHz} \\ L &= 330 \text{ } \mu\text{H} \end{aligned}$$

# Switching power supply

- Buck (Step Down) DC-DC converter
  - Discontinuous Current Mode (DCM)
    - A small output current causes the circuit to work in DCM
      - $I_{DCM} = \frac{\Delta I_{L on}}{2} = \frac{(1-\delta)V_{out}}{2Lf_{sw}}$
      - If  $I_{out} > I_{DCM}$  then the circuit is in CCM
      - If  $I_{out} < I_{DCM}$  then the circuit is in DCM
    - Given the minimum output current  $I_{omin}$ , the minimum  $L$  to guarantee CCM is
      - $L > \frac{(1-\delta)V_{out}}{2I_{omin}f_{sw}}$
    - In the worst case  $\delta = 0$ 
      - $L > \frac{V_{out}}{2I_{omin}f_{sw}}$
      - The circuit always works in CCM

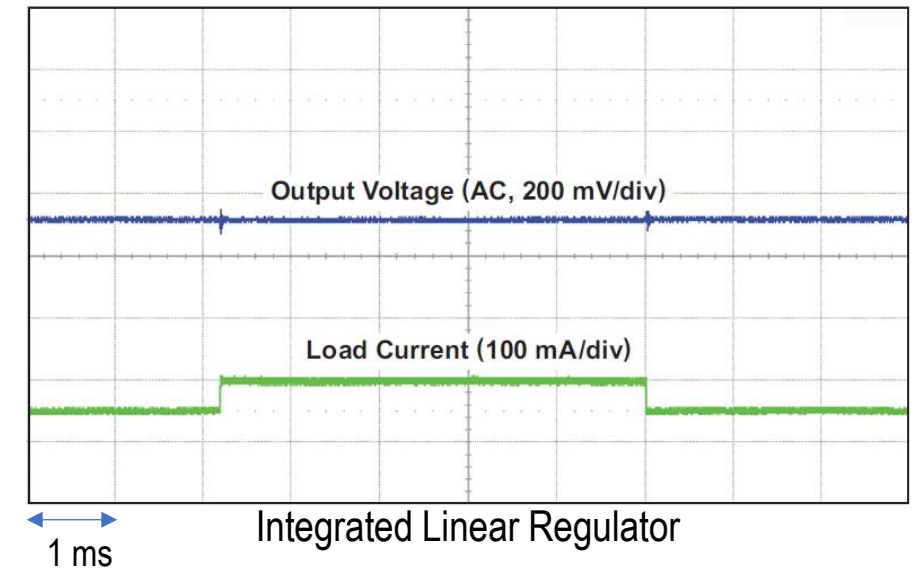
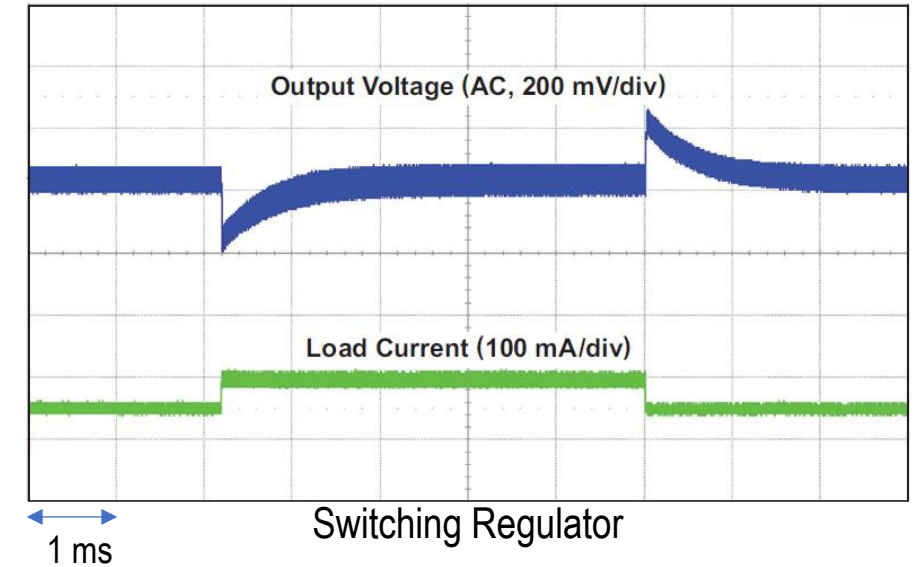


$$\begin{aligned}I_{omin} &= 10 \text{ mA} \\V_{out} &= 6 \text{ V} \\L &> 3 \text{ mH}\end{aligned}$$

$$\begin{aligned}V_{in} &= 12 \text{ V} \\f_{sw} &= 100 \text{ kHz} \\L &= 3,3 \text{ mH}\end{aligned}$$

# Switching power supply

- Buck (Step Down) DC-DC converter
  - Load transient response
    - All previous analyses were at the steady-state
    - Transients can be due to
      - A change of the output current  $I_{out}$
      - A change of the input voltage  $V_{in}$
    - When a regulator is included
      - The output voltage has a short transient
      - It returns to the correct value within a few cycles



R. Nowakowski, R. Taylor, *Linear versus switching regulators in industrial applications with a 24-V bus*, Texas Instruments

# Switching power supply

- Buck (Step Down) DC-DC converter

- Summary

- Obtain any voltage  $0 < V_{out} < V_{in}$ 
      - By controlling the duty cycle  $\delta$
      - With a high efficiency
    - Efficiency is ideally  $\eta = 1$ 
      - Reduced because of static power losses
        - Parasitic resistances of passive components and ON resistances of active components
        - Leakage currents
      - Reduced because of dynamic power losses
        - Turning ON and OFF active components
          - Charge and discharge parasitic capacitances
    - Output noise due to the switching frequency  $f_{sw}$ 
      - Higher frequency, cleaner output
      - But high dynamic power consumption, hence lower efficiency

# Switching power supply

- Buck (Step Down) DC-DC converter

- Summary

- Specifications are  $V_{out}$ ,  $\Delta V_{out}$ ,  $I_{out}$
    - Choose the switching frequency  $f_{sw}$
    - To work in CCM, select an inductor such that
      - $L > \frac{V_{out}}{2I_{out}f_{sw}}$ , where  $I_{out}$  is the minimum output current
    - Select a capacitor to meet the output ripple requirements
      - $\frac{I_{out}}{4Cf_{sw}} < \Delta V_{out} \Rightarrow C > \frac{I_{out}}{4\Delta V_{out}f_{sw}}$
    - Select diode and transistor to meet the switching frequency  $f_{sw}$  timing requirements
    - Select components based on the current and voltage ratings

|         | MOS                          | Diode                        |
|---------|------------------------------|------------------------------|
| Current | $\frac{2}{\sqrt{3}} I_{out}$ | $\frac{2}{\sqrt{3}} I_{out}$ |
| Voltage | $V_{in}$                     | $V_{in}$                     |

|         | Inductor                     | Capacitor                    |
|---------|------------------------------|------------------------------|
| Current | $\frac{2}{\sqrt{3}} I_{out}$ | $\frac{1}{\sqrt{3}} I_{out}$ |
| Voltage | $V_{in}$                     | $V_{out}$                    |

# Switching power supply

- Buck (Step Down) DC-DC converter

- Example

- $L = 330 \mu H, C = 330 \mu F, V_{in} = 12 V, f_{sw} = 100 kHz, \delta = 41.7\%, R_L = 33 \Omega$
    - In the following computation, the worst case scenarios are assuming some of the above parameters may change (i.e. different  $\delta$ , different output current, ...)
    - Compute the switching period  $T_{sw}$  and the duration of the ON and OFF phases
      - $T_{sw} = \frac{1}{f_{sw}} = \frac{1}{100 \cdot 10^3} = 10 \mu s$
      - $T_{on} = \delta T_{sw} = 0.417 \cdot 10 \cdot 10^{-6} = 4.17 \mu s$
      - $T_{off} = (1 - \delta) T_{sw} = 0.583 \cdot 10 \cdot 10^{-6} = 5.83 \mu s$
    - Compute the output voltage  $V_{out}$ , assuming the circuit works in CCM
      - $V_{out} = \delta V_{in} = 0.417 \cdot 12 = 5 V$
    - Compute the output current  $I_{out}$ 
      - $I_{out} = \frac{V_{out}}{R_L} = \frac{5}{33} = 152 mA$

# Switching power supply

- Buck (Step Down) DC-DC converter

- Example

- $L = 330 \mu H, C = 330 \mu F, V_{in} = 12 V, f_{sw} = 100 kHz, \delta = 41.7\%, R_L = 33 \Omega$
- $T_{sw} = 10 \mu s, T_{on} = 4.17 \mu s, T_{off} = 5.83 \mu s, V_{out} = 5 V, I_{out} = 152 mA$
- Compute the inductor current change  $\Delta I_L$  during the ON and OFF phases
  - $\Delta I_{Lon} = \frac{1}{L}(V_{in} - V_{out})T_{on} = \frac{(12-5) \cdot 4.17 \cdot 10^{-6}}{330 \cdot 10^{-6}} = 88.5 mA$
  - $\Delta I_{Loff} = -\frac{1}{L}V_{out}T_{off} = -\frac{5 \cdot 5.83 \cdot 10^{-6}}{330 \cdot 10^{-6}} = -88.3 mA$
  - They should be identical, they differ because of rounding errors  $\rightarrow \Delta I_L = 88.4 mA$
- Compute the minimum and maximum current in the inductor
  - $I_{Lmax} = I_{out} + \frac{\Delta I_L}{2} = 152 \cdot 10^{-3} + \frac{88.4 \cdot 10^{-3}}{2} = 196.2 mA$
  - $I_{Lmin} = I_{out} - \frac{\Delta I_L}{2} = 152 \cdot 10^{-3} - \frac{88.4 \cdot 10^{-3}}{2} = 107.8 mA$
  - The circuit works in Continuous Current Mode (CCM), as initially assumed

# Switching power supply

- Buck (Step Down) DC-DC converter

- Example

- $L = 330 \mu H, C = 330 \mu F, V_{in} = 12 V, f_{sw} = 100 kHz, \delta = 41.7\%, R_L = 33 \Omega$

- $T_{sw} = 10 \mu s, T_{on} = 4.17 \mu s, T_{off} = 5.83 \mu s, V_{out} = 5 V, I_{out} = 152 mA$

- Compute the minimum value of the inductor to keep working in CCM

- $L > \frac{(1-\delta)V_{out}}{2I_{out}f_{sw}} = \frac{0.583 \cdot 5}{2 \cdot 152 \cdot 10^{-3} \cdot 100 \cdot 10^3} = 96 \mu H$

- In the worst case, with  $\delta = 0$ , it is  $L > 165 \mu H$

- Compute the maximum value of the load resistance  $R_L$  to keep working in CCM (with the original  $L$ )

- $I_{DCM} = \frac{V_{out}}{R_{Lmax}} = \frac{(1-\delta)V_{out}}{2Lf_{sw}} = \frac{0.583 \cdot 5}{2 \cdot 330 \cdot 10^{-6} \cdot 100 \cdot 10^3} = 44.2 mA$

- $R_{Lmax} = \frac{2Lf_{sw}}{1-\delta} = \frac{2 \cdot 330 \cdot 10^{-6} \cdot 100 \cdot 10^3}{0.583} = 113 \Omega$

- In the worst case, with  $\delta = 0$ , it is:

- $I_{DCM} = 75.8 mA$

- $R_{Lmax} = 66 \Omega$

# Switching power supply

- Buck (Step Down) DC-DC converter

- Example

- $L = 330 \mu H, C = 330 \mu F, V_{in} = 12 V, f_{sw} = 100 kHz, \delta = 41.7\%, R_L = 33 \Omega$

- $T_{sw} = 10 \mu s, T_{on} = 4.17 \mu s, T_{off} = 5.83 \mu s, V_{out} = 5 V, I_{out} = 152 mA$

- Compute the output ripple

- $\Delta V_{out} = \frac{\Delta I_L}{8Cf_{sw}} = \frac{88.4 \cdot 10^{-3}}{8 \cdot 330 \cdot 10^{-6} \cdot 100 \cdot 10^3} = 335 \mu V$

- Compute the minimum capacitor to have an output ripple  $\Delta V_{out} < 10 mV$

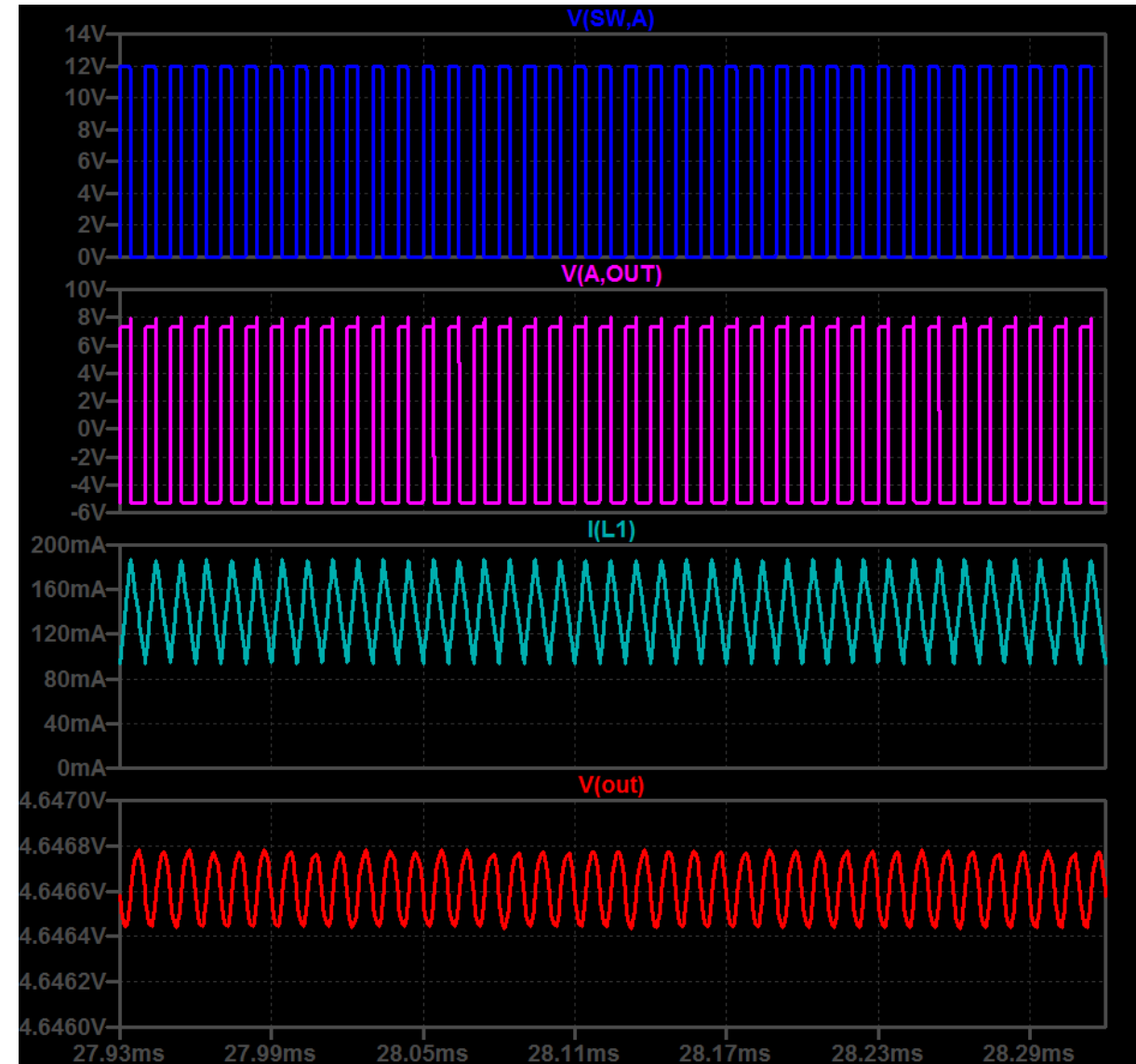
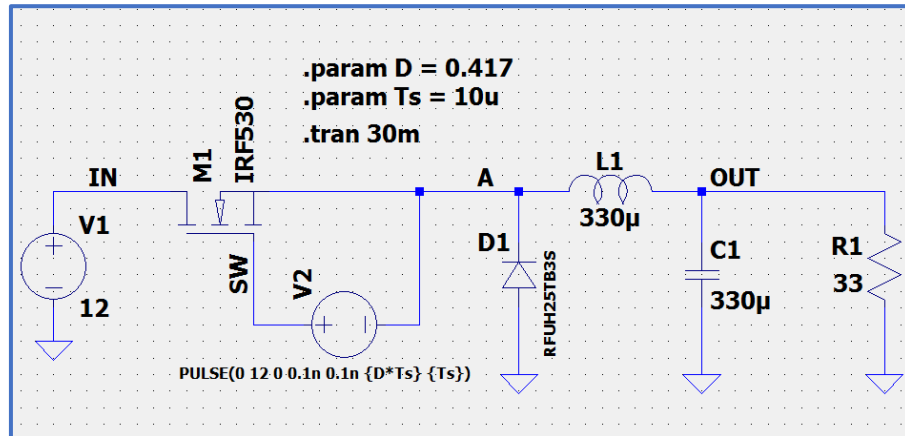
- $C_{min} = \frac{\Delta I_L}{8\Delta V_{out}f_{sw}} = \frac{88.4 \cdot 10^{-3}}{8 \cdot 10 \cdot 10^{-3} \cdot 100 \cdot 10^3} = 11 \mu F$

- In the worst case  $\Delta I_L = 2I_{out}$

- $C_{min} = \frac{I_{out}}{4\Delta V_{out}f_{sw}} = \frac{152 \cdot 10^{-3}}{4 \cdot 10 \cdot 10^{-3} \cdot 100 \cdot 10^3} = 38 \mu F$

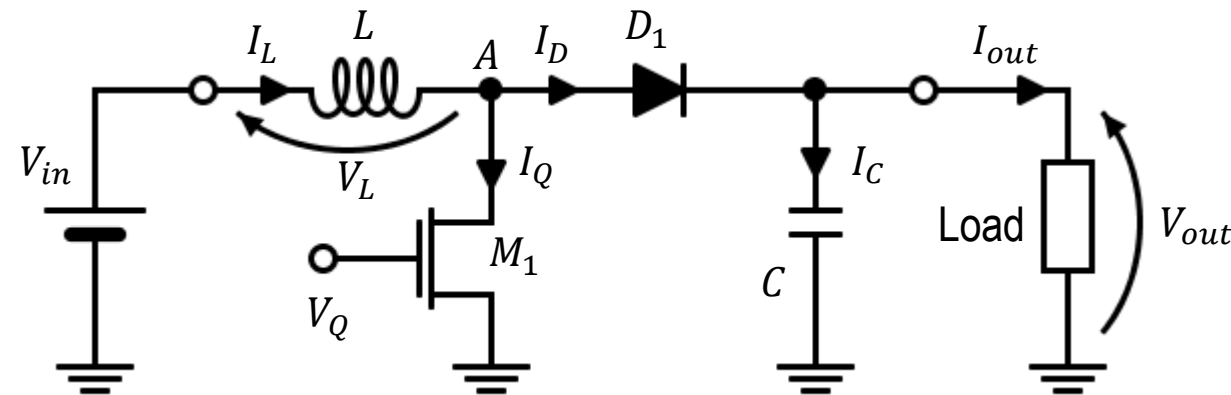
# Switching power supply

- Buck (Step Down) DC-DC converter
  - Example
    - $L = 330 \mu H$ ,  $C = 330 \mu F$ ,  $V_{in} = 12 V$
    - $f_{sw} = 100 kHz$ ,  $\delta = 41.7\%$ ,  $R_L = 33 \Omega$
    - $T_{sw} = 10 \mu s$ ,  $T_{on} = 4.17 \mu s$ ,  $T_{off} = 5.83 \mu s$
    - $V_{out} = 5 V$ ,  $I_{out} = 152 mA$ ,  $\Delta I_L = 88.5 mA$
    - $I_{Lmax} = 196.2 mA$ ,  $I_{Lmin} = 107.8 mA$
    - $\Delta V_{out} = 335 \mu V$



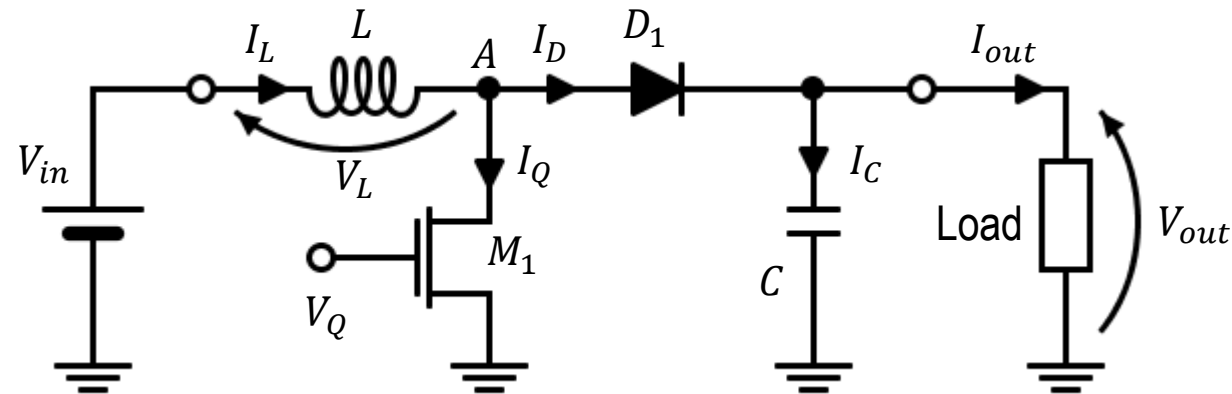
# Switching power supply

- Boost (Step Up) DC-DC converter
  - Produces an output voltage  $V_{out}$  higher than the input voltage  $V_{in}$ 
    - $V_{out} > V_{in}$
  - Similar to a Buck converter
    - Same components
      - Transistor, Diode, Inductor, Capacitor
    - Different topology
- Analyze the circuit at the steady-state
  - When switch is closed
  - When switch is open



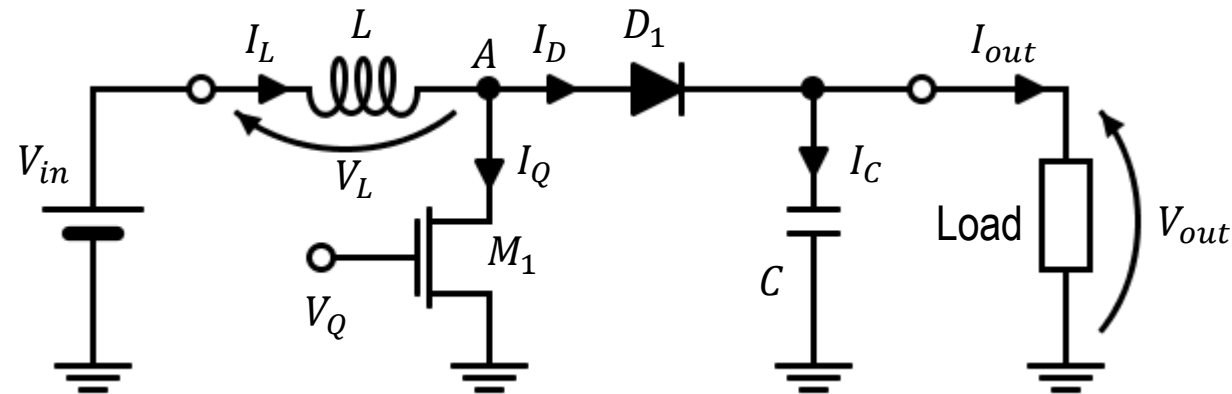
# Switching power supply

- Boost (Step Up) DC-DC converter
  - During  $T_{on}$ 
    - Transistor  $M_1$  is conducting
      - $V_A = 0$  V
    - Diode  $D_1$  is reversed biased
    - Voltage across the inductor
      - $V_L = V_{in}$
    - Current flowing in the inductor
      - It changes linearly
      - $\Delta I_{Lon} = \frac{V_L T_{on}}{L} = \frac{1}{L} V_{in} T_{on}$
    - Current flowing in the transistor  $M_1$ 
      - $I_Q = I_L$
    - Current flowing in the diode  $D_1$ 
      - $I_D = 0$



# Switching power supply

- Boost (Step Up) DC-DC converter
  - During  $T_{\text{off}}$ 
    - Transistor  $M_1$  is in cut-off
    - Diode  $D_1$  is forced in conduction
      - $V_A = V_{\text{out}}$
    - Voltage across the inductor
      - $V_L = V_{\text{in}} - V_{\text{out}}$
    - Current flowing in the inductor
      - It changes linearly
      - $\Delta I_{L\text{off}} = \frac{V_L T_{\text{off}}}{L} = \frac{1}{L} (V_{\text{in}} - V_{\text{out}}) T_{\text{off}}$
    - Current flowing in the transistor  $M_1$ 
      - $I_Q = 0$
    - Current flowing in the diode  $D_1$ 
      - $I_D = I_L$

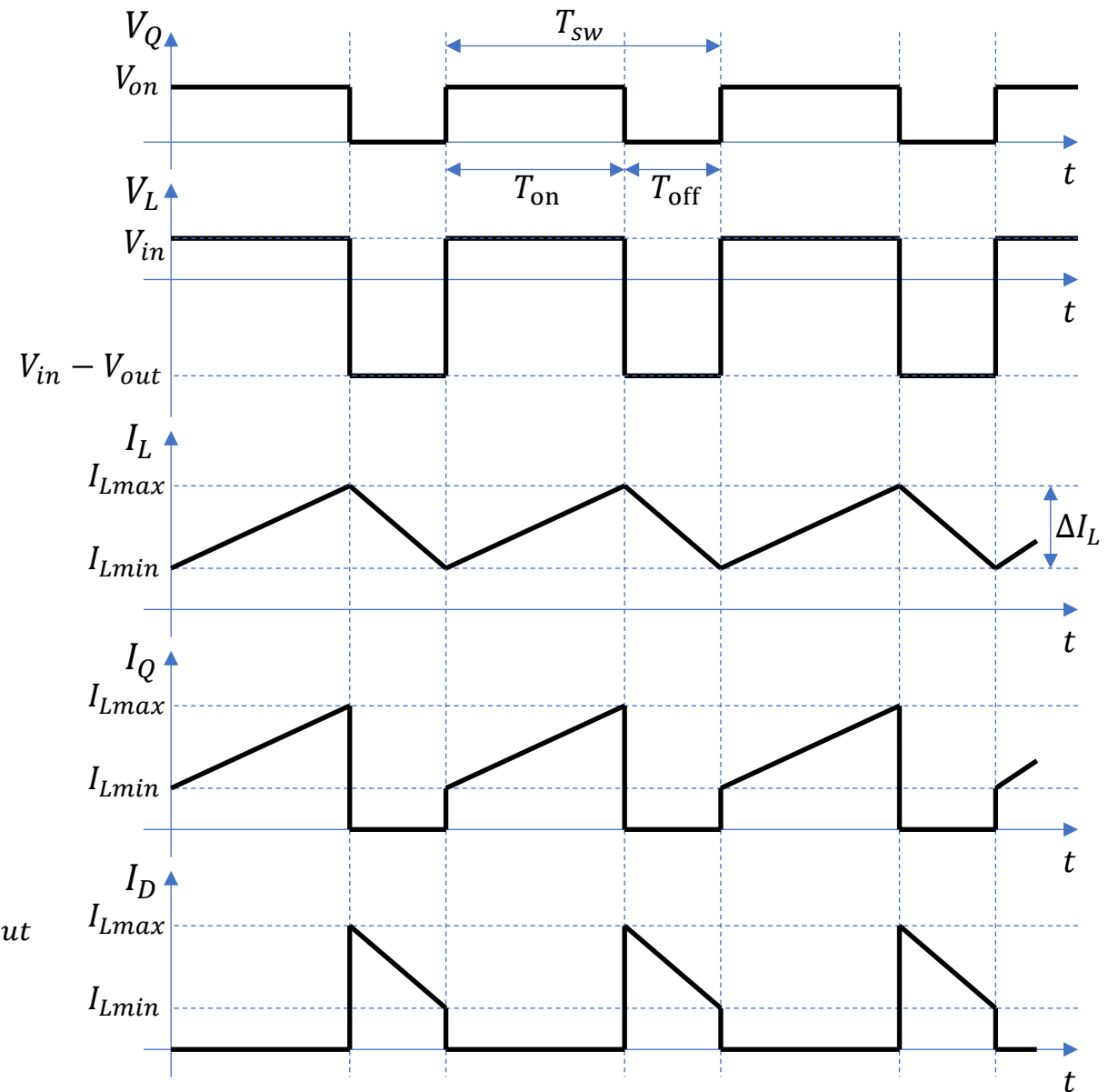
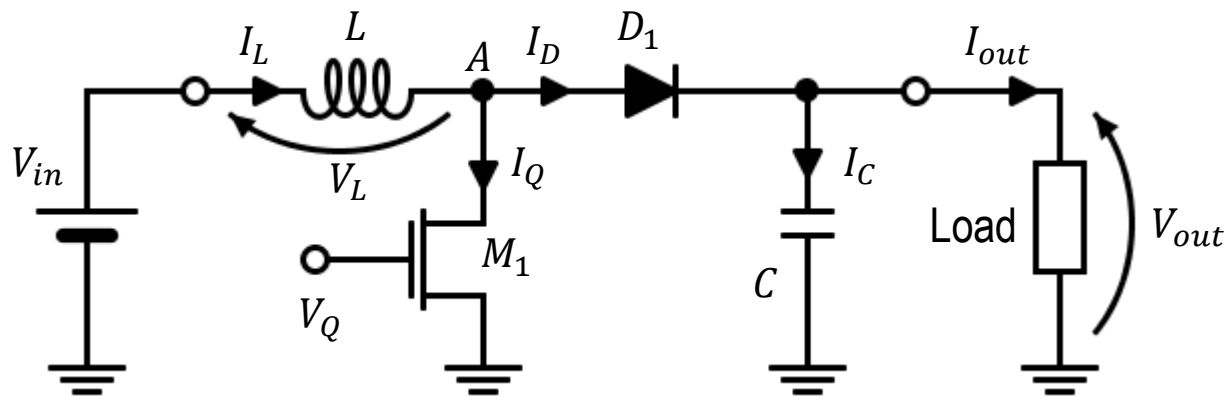


# Switching power supply

- Boost (Step Up) DC-DC converter

- Assumptions

- $V_{in} > 0, V_{out} > 0, V_{out} > V_{in}$
- $\Delta I_{L\text{on}} + \Delta I_{L\text{off}} = 0$
- $\frac{1}{L} V_{out} T_{\text{on}} + \frac{1}{L} (V_{in} - V_{out}) T_{\text{off}} = 0$
- $\rightarrow \frac{V_{out}}{V_{in}} = \frac{T_{\text{sw}}}{T_{\text{sw}} - T_{\text{on}}} = \frac{1}{1 - \delta}, V_{out} = \frac{1}{1 - \delta} V_{in}$
- $V_{out}$  is always higher than  $V_{in}$



# Switching power supply

- Boost (Step Up) DC-DC converter

- Current in the inductor

- The average current in the load and the average current in the diode should be identical
      - Because the average current in the capacitor is 0

- Assume  $I_{Lmin} > 0$

- Continuous Current Mode (CCM)

- Write a system of two equations

$$\left\{ \begin{array}{l} I_{Lmin}T_{off} + \frac{1}{2}(I_{Lmax} - I_{Lmin})T_{off} = I_{out}T_{sw} \\ \Delta I_L = I_{Lmax} - I_{Lmin} = \frac{1}{L}V_{in}T_{on} \end{array} \right. \quad \left\{ \begin{array}{l} I_{Lmax} = \frac{1}{1-\delta}I_{out} + \frac{V_{in}T_{on}}{2L} = \frac{1}{1-\delta}I_{out} + \frac{\Delta I_L}{2} \\ I_{Lmin} = \frac{1}{1-\delta}I_{out} - \frac{V_{in}T_{on}}{2L} = \frac{1}{1-\delta}I_{out} - \frac{\Delta I_L}{2} \end{array} \right.$$

- Solving the system gives  $I_{Lmax}$  and  $I_{Lmin}$

- Currents in the transistor and in the diode

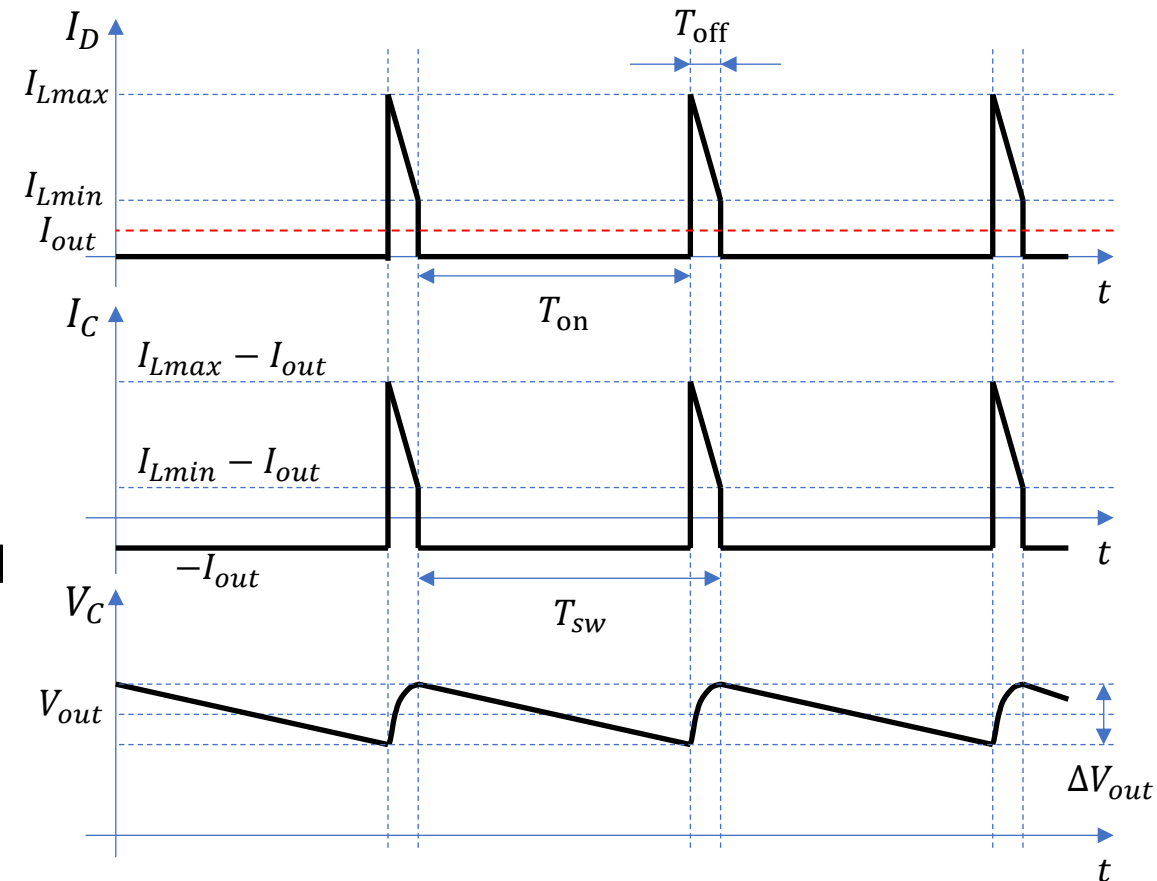
- Identical to the current in the inductor when they conduct

# Switching power supply

- Boost (Step Up) DC-DC converter
  - Current from the voltage source
    - Output current of the voltage source
      - $I_{in} = I_L$
    - Since the converter is loss-less (at least in a first approximation), on average it is
      - $P_{in} = P_{out} \Rightarrow V_{in}I_{inavg} = V_{out}I_{out}$
      - $I_{inavg} = I_{Lavg} = \frac{V_{out}}{V_{in}}I_{out} = \frac{1}{1-\delta}I_{out}$ 
        - Same as first equation of the system in the previous slide
        - Even if they don't look identical, they are really the same
  - Output current of the voltage source has a triangular shape
    - Much better from the point of view of the voltage generator
  - The current of the output stage of the converter is pulsated
    - Noise is injected in the load

# Switching power supply

- Boost (Step Up) DC-DC converter
  - Current rating for the inductor
    - Evaluate  $I_{Lmax}$
    - Or compute  $I_{Lrms}$ 
      - Now  $I_{Lavg} = I_{inavg}$ 
        - $I_{Lrms} = \frac{2}{\sqrt{3}} I_{inavg} = \frac{2}{\sqrt{3}} \frac{1}{1-\delta} I_{out}$
  - Current ratings for the transistor and the diode
    - Use the same rating as for the inductor
  - Current rating for the capacitor
    - Worst case when it provides  $I_{out}$  for most of the period
      - High duty cycle  $\delta$  (close to 1)
      - $I_{Crms} = -I_{out}$
      - $\Delta V_{out} = \frac{I_{out}}{C f_{sw}} = \frac{I_{out} T_{sw}}{C}$

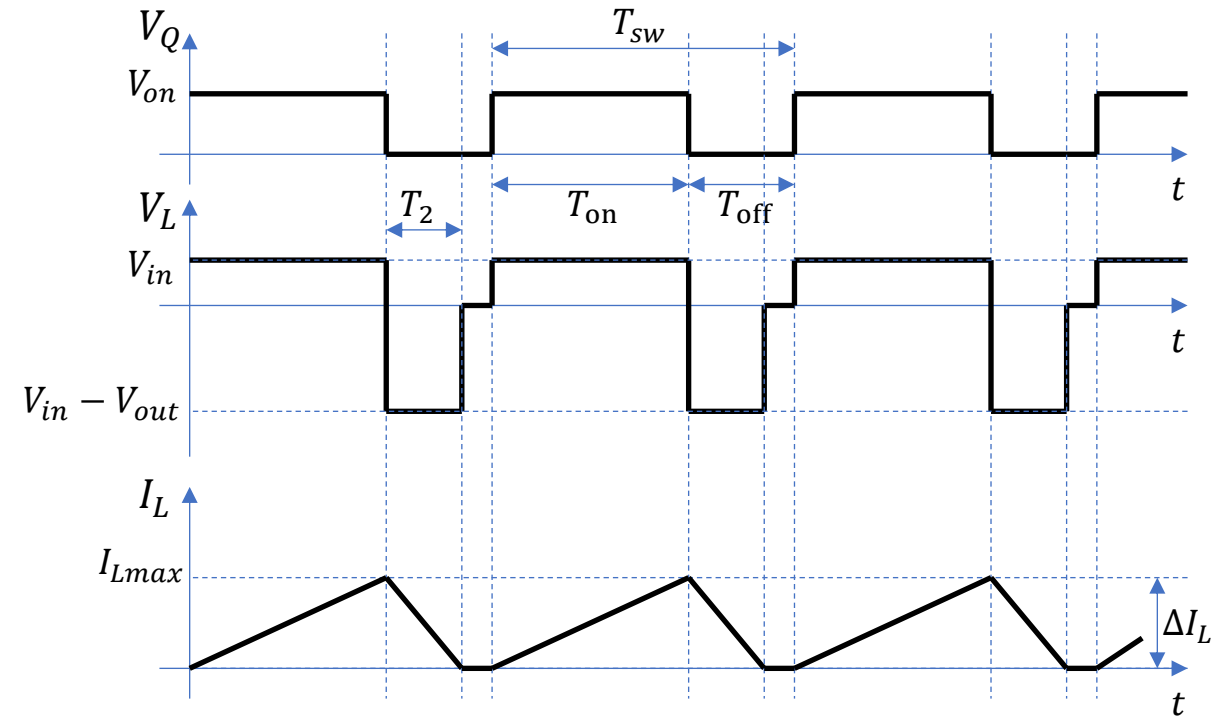


# Switching power supply

- Boost (Step Up) DC-DC converter
  - Line regulation
    - The output voltage depends on the input voltage
      - When the input changes, the output also changes
    - $S_V = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{1}{1-\delta}$
    - Even worse than in the case of the Buck converter
      - Need a regulator
  - Load regulation
    - The expression of the output voltage does not depend on the output current  $I_{out}$
    - $S_I = 0 \Omega$
  - Continuous Current Mode (CCM)
    - Guaranteed when
      - $L > \frac{V_{in}}{2I_{in}f_{sw}}$ , where  $I_{in}$  is the minimum

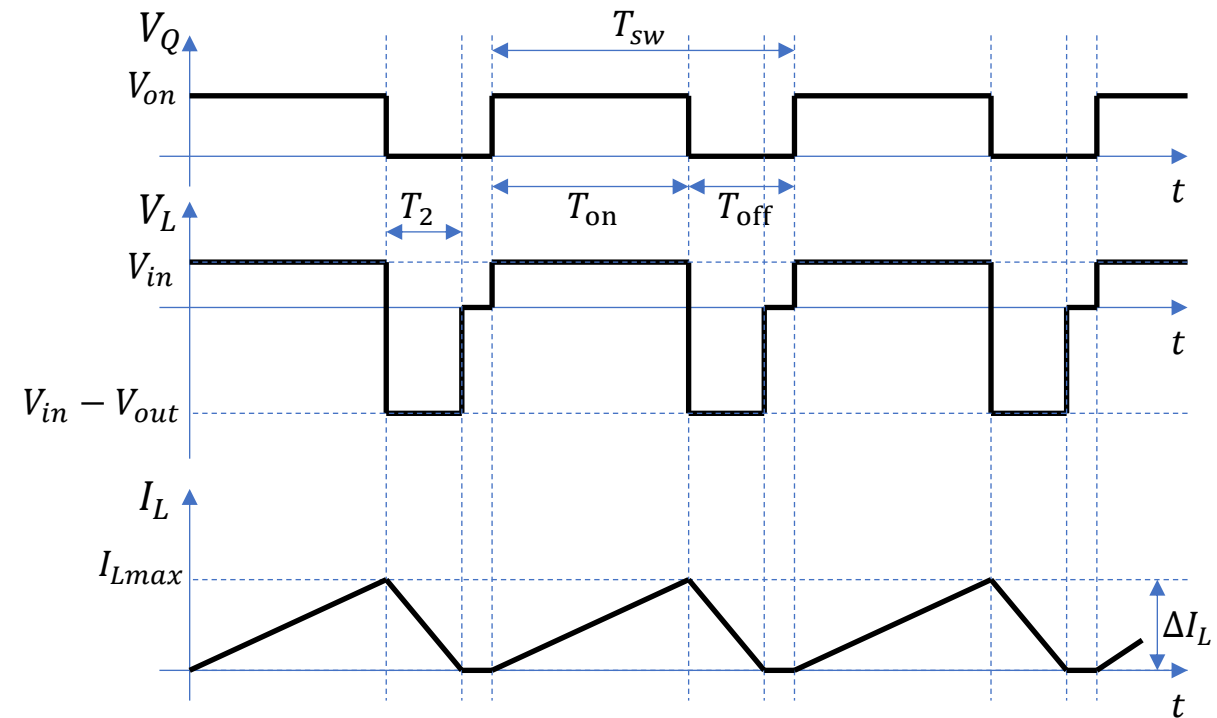
# Switching power supply

- Boost (Step Up) DC-DC converter
  - Discontinuous Current Mode (DCM)
    - The current in the inductor reaches the value 0
      - $I_{Lmin} = 0$
    - The current change during  $T_{on}$  remains the same
      - $\Delta I_{Lon} = \frac{V_L T_{on}}{L} = \frac{1}{L} V_{in} T_{on}$
      - $I_{Lmax} = \Delta I_{Lon} = \frac{1}{L} V_{in} T_{on}$
    - The current goes to 0 during  $T_2$ 
      - $I_{Lmax} = -\frac{1}{L} (V_{in} - V_{out}) T_2 \Rightarrow T_2 = -\frac{V_{in}}{V_{in} - V_{out}} T_{on}$
    - The average value of current  $I_L$  is still equal to  $\frac{V_{out}}{V_{in}} I_{out}$ 
      - $\frac{1}{2} (T_{on} + T_2) I_{Lmax} = \frac{V_{out}}{V_{in}} I_{out} T_{sw}$
    - Or the average current in the diode is still equal to  $I_{out}$ 
      - $\frac{1}{2} T_2 I_{Lmax} = I_{out} T_{sw}$



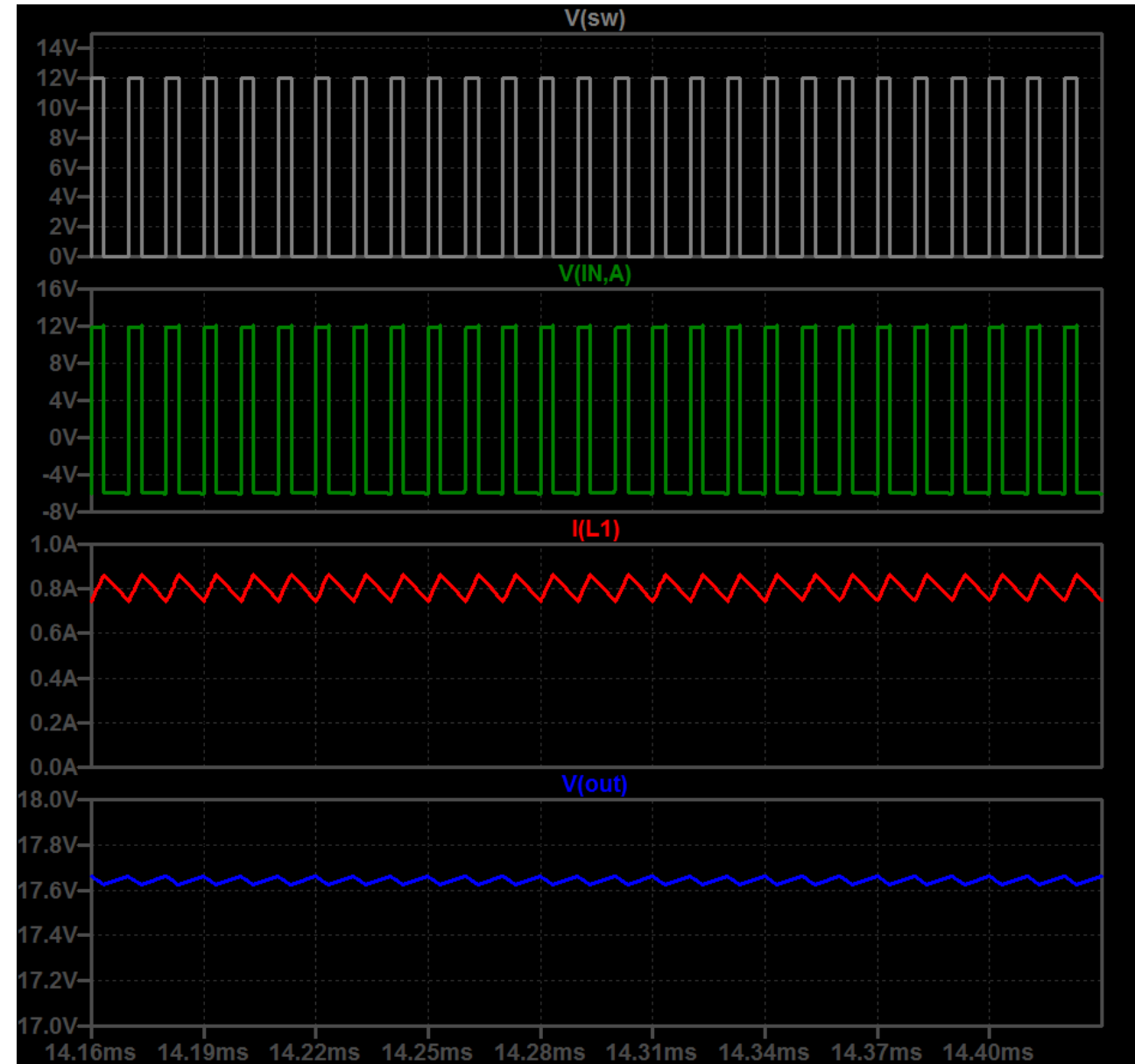
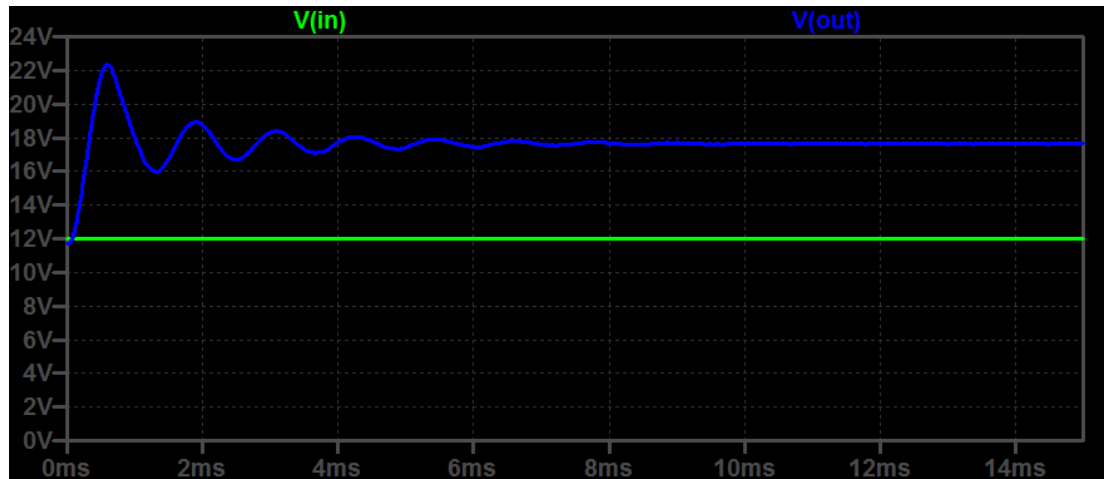
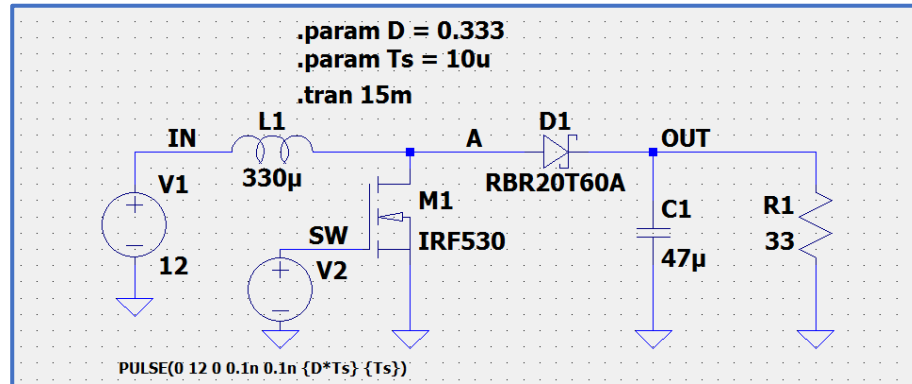
# Switching power supply

- Boost (Step Up) DC-DC converter
  - Discontinuous Current Mode (DCM)
    - Substitute expressions for  $I_{Lmax}$  and  $T_2$
    - Solve for  $V_{out}$ 
      - $V_{out} = V_{in} \left( \frac{\delta^2 T_{sw} V_{in}}{2LI_{out}} + 1 \right)$
    - The output voltage now depends also on
      - The inductor value  $L$
      - The switching period  $T_{sw}$
      - The output current  $I_{out}$ 
        - The output resistance is not zero any more
        - $S_I > 0$



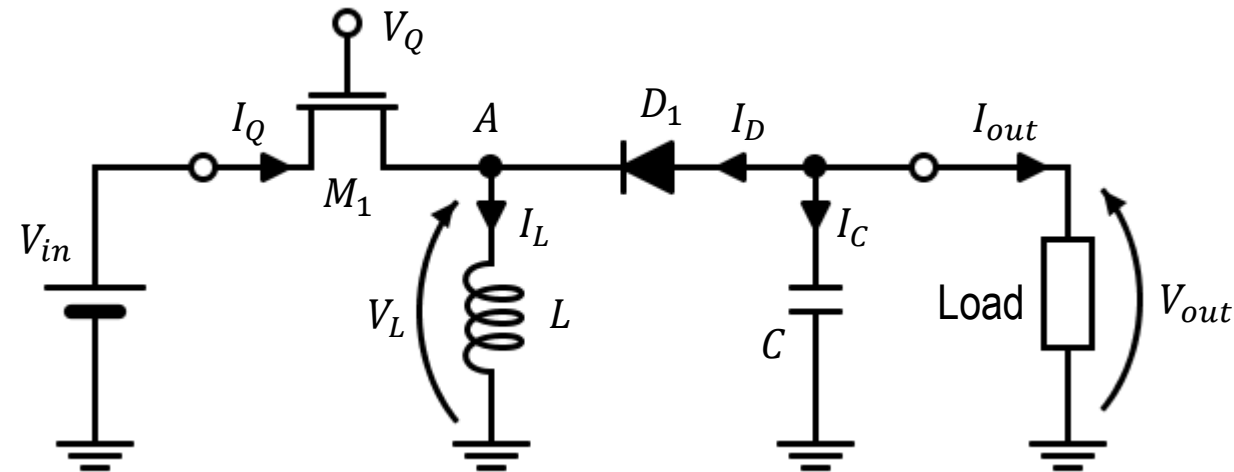
# Switching power supply

- Boost (Step Up) DC-DC converter
  - Example



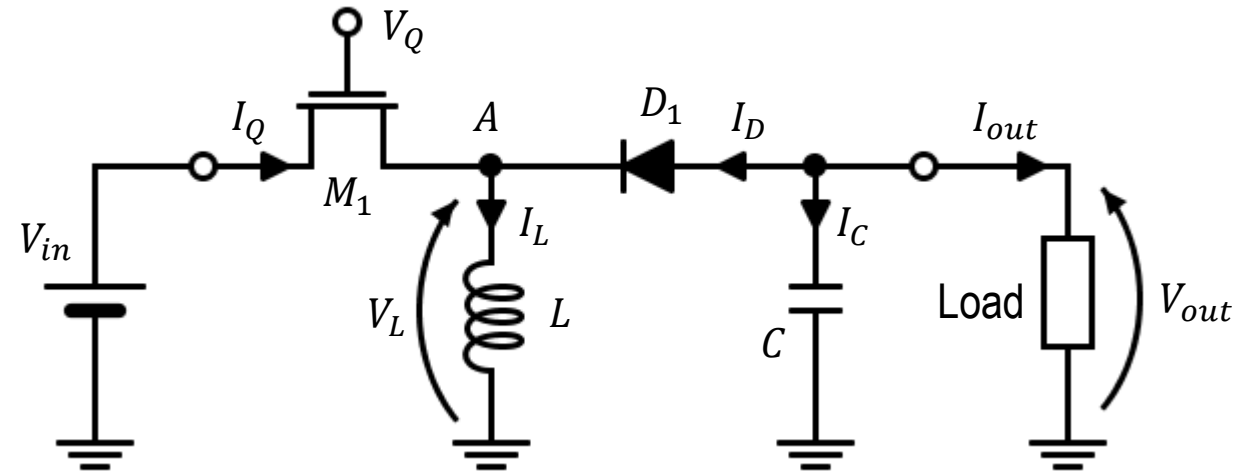
# Switching power supply

- Buck-Boost DC-DC converter
  - Produces a negative output voltage  $V_{out}$  of any value with a positive input voltage  $V_{in}$ 
    - $V_{out} < 0$
  - Similar to a Buck or a Boost converter
    - Same components
      - Transistor, Diode, Inductor, Capacitor
    - Different topology
  - Analyze the circuit at the steady-state
    - When switch is closed
    - When switch is open



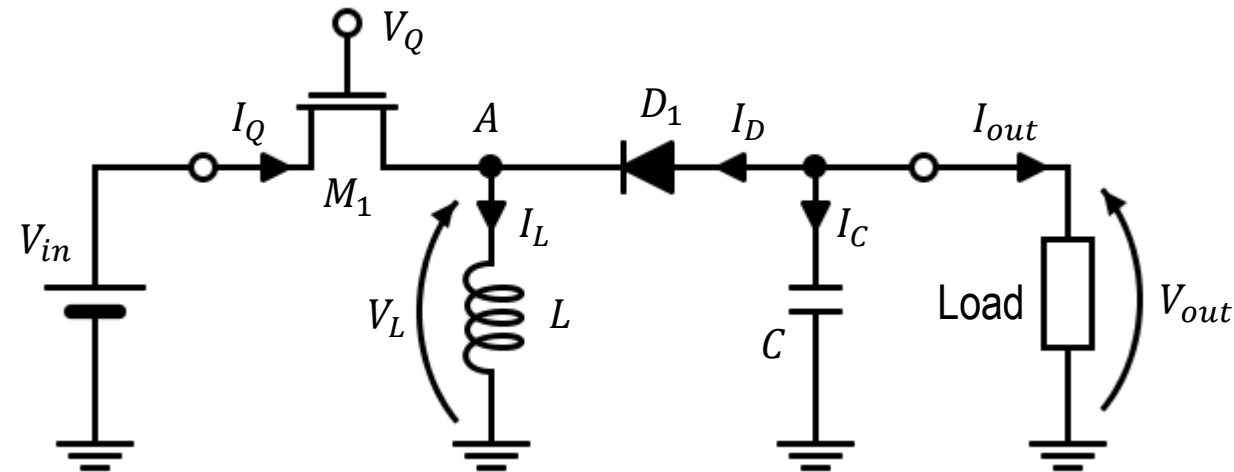
# Switching power supply

- Buck-Boost DC-DC converter
  - During  $T_{on}$ 
    - Transistor  $M_1$  is conducting
      - $V_A = V_{in}$
    - Diode  $D_1$  is reverse biased
    - Voltage across the inductor
      - $V_L = V_{in}$
    - Current flowing in the inductor
      - It changes linearly
      - $\Delta I_{Lon} = \frac{V_L T_{on}}{L} = \frac{1}{L} V_{in} T_{on}$
    - Current flowing in the transistor  $M_1$ 
      - $I_Q = I_L$
    - Current flowing in the diode  $D_1$ 
      - $I_D = 0$



# Switching power supply

- Buck-Boost DC-DC converter
  - During  $T_{\text{off}}$ 
    - Transistor  $M_1$  is in cut-off
    - Diode  $D_1$  is forced in conduction
      - $V_A = V_{\text{out}}$
    - Voltage across the inductor
      - $V_L = V_{\text{out}}$
    - Current flowing in the inductor
      - It changes linearly
      - $\Delta I_{L\text{off}} = \frac{V_L T_{\text{off}}}{L} = \frac{1}{L} V_{\text{out}} T_{\text{off}}$
    - Current flowing in the transistor  $M_1$ 
      - $I_Q = 0$
    - Current flowing in the diode  $D_1$ 
      - $I_D = I_L$



# Switching power supply

- Buck-Boost DC-DC converter

- Assumptions

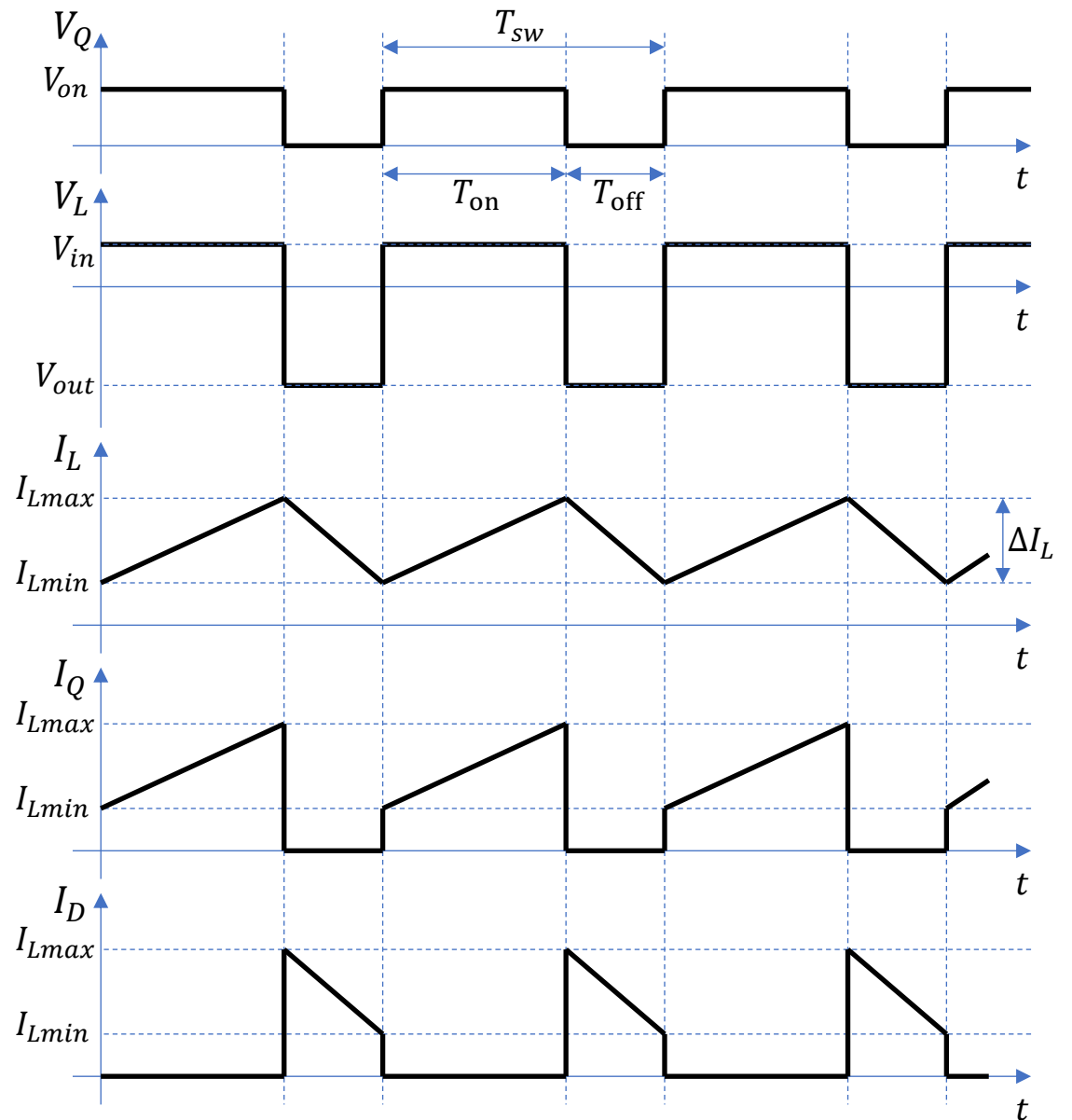
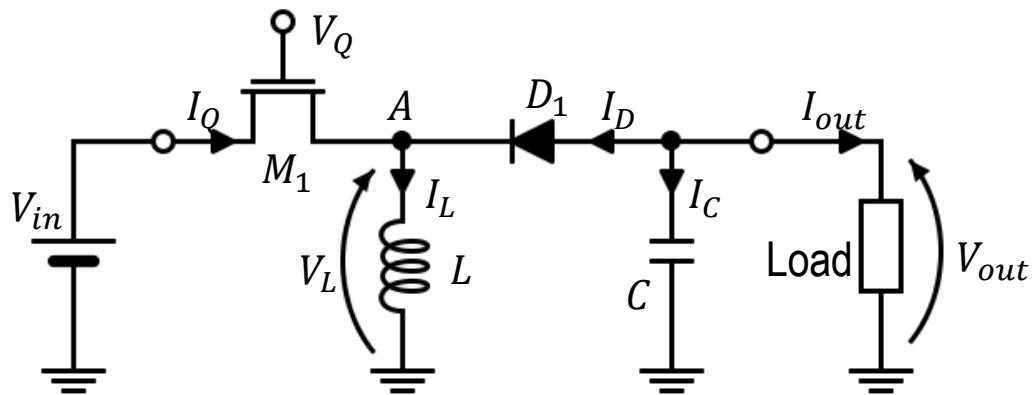
- $V_{in} > 0, V_{out} < 0$

- $\Delta I_{L\text{on}} + \Delta I_{L\text{off}} = 0$

- $\frac{1}{L} V_{out} T_{\text{on}} + \frac{1}{L} V_{out} T_{\text{off}} = 0$

- $\rightarrow \frac{V_{out}}{V_{in}} = -\frac{T_{\text{on}}}{T_{\text{off}}} = -\frac{\delta}{1-\delta}, V_{out} = -\frac{\delta}{1-\delta} V_{in}$

- $V_{out}$  is always negative,  $-\infty < V_{out} < 0$



# Switching power supply

- Buck-Boost DC-DC converter

- Current in the inductor

- The average current in the load and the average current in the diode should be identical in absolute value
      - They have the opposite sign ( $I_{out}$  is negative)
      - Because the average current in the capacitor is 0

- Assume  $I_{Lmin} > 0$

- Continuous Current Mode (CCM)

- Write a system of two equations

$$\left\{ \begin{array}{l} I_{Lmin}T_{off} + \frac{1}{2}(I_{Lmax} - I_{Lmin})T_{off} = -I_{out}T_{sw} \\ \Delta I_L = I_{Lmax} - I_{Lmin} = \frac{1}{L}V_{in}T_{on} \end{array} \right. \quad \left\{ \begin{array}{l} I_{Lmax} = -\frac{1}{1-\delta}I_{out} + \frac{V_{in}T_{on}}{2L} = -\frac{1}{1-\delta}I_{out} + \frac{\Delta I_L}{2} \\ I_{Lmin} = -\frac{1}{1-\delta}I_{out} - \frac{V_{in}T_{on}}{2L} = -\frac{1}{1-\delta}I_{out} - \frac{\Delta I_L}{2} \end{array} \right.$$

- Solving the system gives  $I_{Lmax}$  and  $I_{Lmin}$

- Currents in the transistor and in the diode

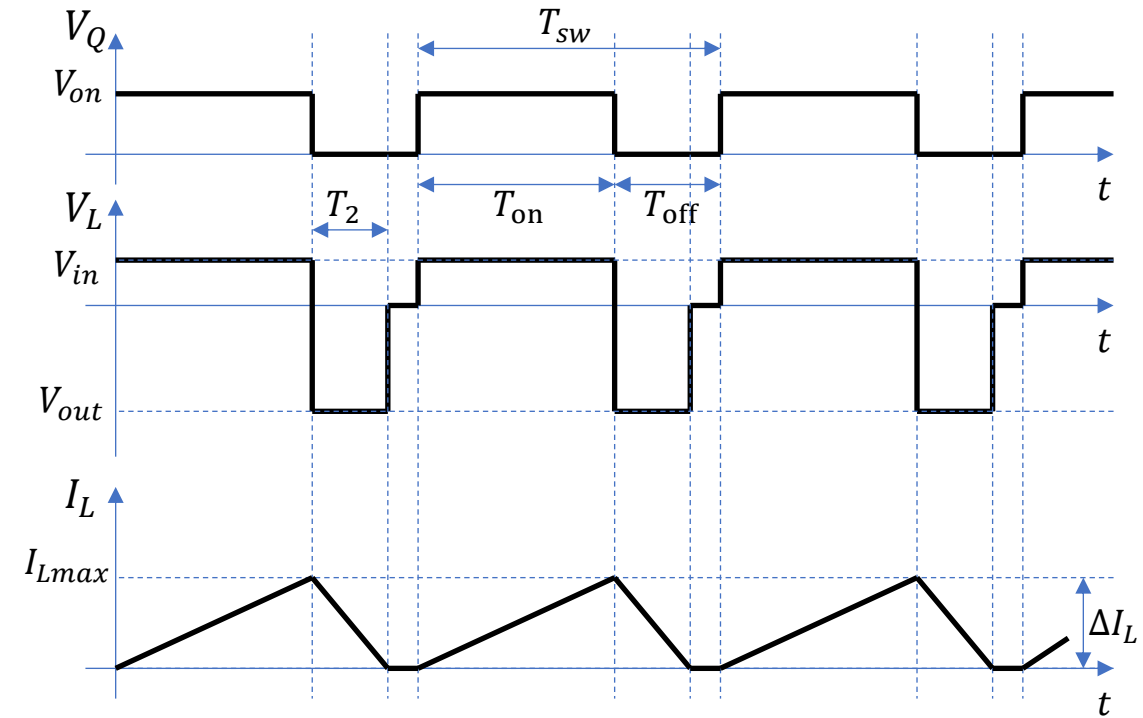
- Identical to the current in the inductor when they conduct

# Switching power supply

- Buck-Boost DC-DC converter
  - Current in the inductor, transistor and diode
    - Similar to the Boost DC-DC converter
      - But now the output current  $I_{out}$  has the opposite sign
  - Output current of the voltage source is pulsated
    - Causes problems to the voltage generator
  - The current of the output stage of the converter is pulsated
    - Noise is injected in the load
  - If duty cycle  $\delta$  is very high
    - The voltage source is almost shorted to ground through the transistor and the inductor
    - Avoid having  $\delta = 1$  to prevent damages

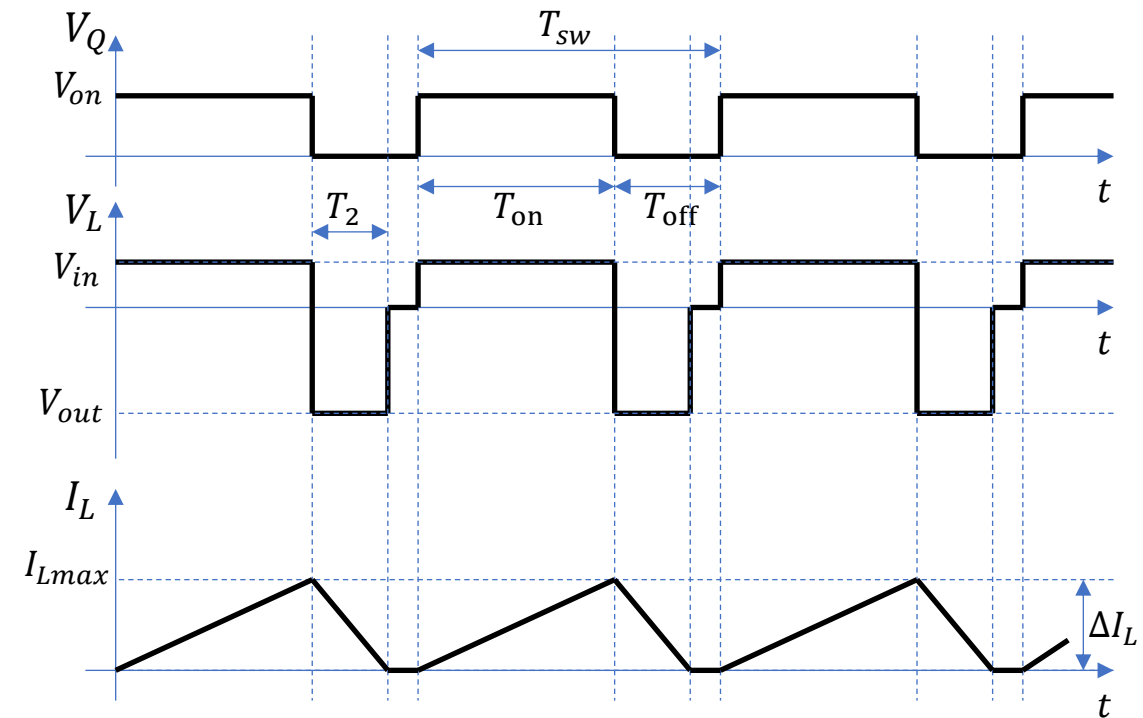
# Switching power supply

- Buck-Boost DC-DC converter
  - Discontinuous Current Mode (DCM)
    - The current in the inductor reaches the value 0
      - $I_{Lmin} = 0$
    - The current change during  $T_{on}$  remains the same
      - $\Delta I_{Lon} = \frac{V_L T_{on}}{L} = \frac{1}{L} V_{in} T_{on}$
      - $I_{Lmax} = \Delta I_{Lon} = \frac{1}{L} V_{in} T_{on}$
    - The current goes to 0 during  $T_2$ 
      - $I_{Lmax} = -\frac{1}{L} V_{out} T_2 \Rightarrow T_2 = -\frac{V_{in}}{V_{out}} T_{on}$
    - The average current in the diode is still equal to  $I_{out}$ 
      - $\frac{1}{2} T_2 I_{Lmax} = I_{out} T_{sw}$



# Switching power supply

- Buck-Boost DC-DC converter
  - Discontinuous Current Mode (DCM)
    - Substitute expressions for  $I_{Lmax}$  and  $T_2$
    - Solve for  $V_{out}$ 
      - $V_{out} = -V_{in} \frac{\delta^2 T_{sw} V_{in}}{2LI_{out}}$
    - The output voltage now depends also on
      - The inductor value  $L$
      - The switching period  $T_{sw}$
      - The output current  $I_{out}$ 
        - The output resistance is not zero any more
        - $S_I > 0$



# Switching power supply

- Flyback DC-DC converter

- Produces a positive output voltage  $V_{out}$  of any value with a positive input voltage  $V_{in}$

- $V_{out} > 0$

- Almost identical to a Buck-Boost converter

- Inductor is now a transformer
  - The secondary is connected on the negative side
  - The diode is turned the other way around

- Analyze the circuit at the steady-state

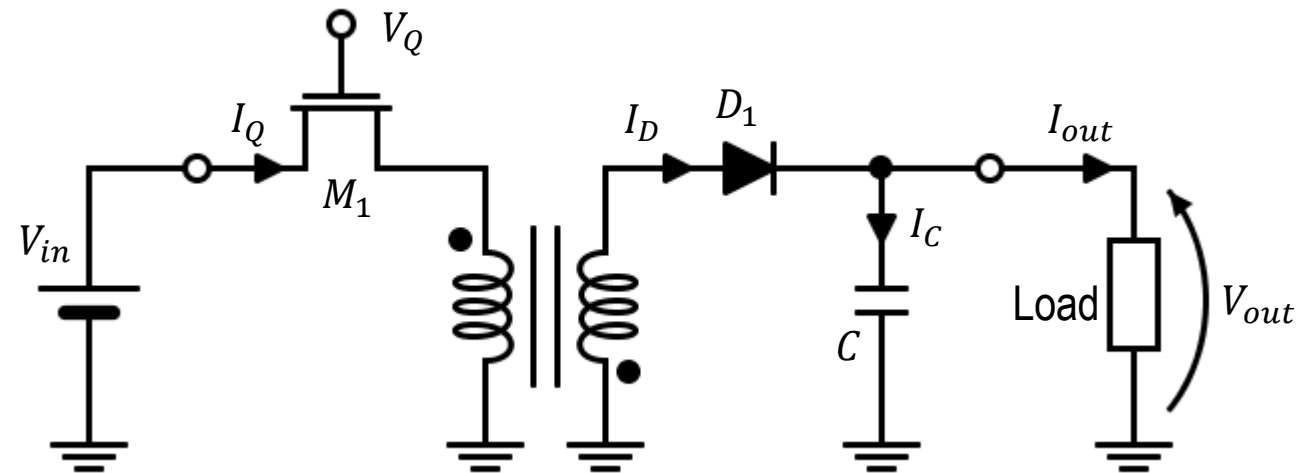
- Same as the Buck-Boost converter

- But positive

- $V_{out} = \frac{\delta}{1-\delta} V_{in}$

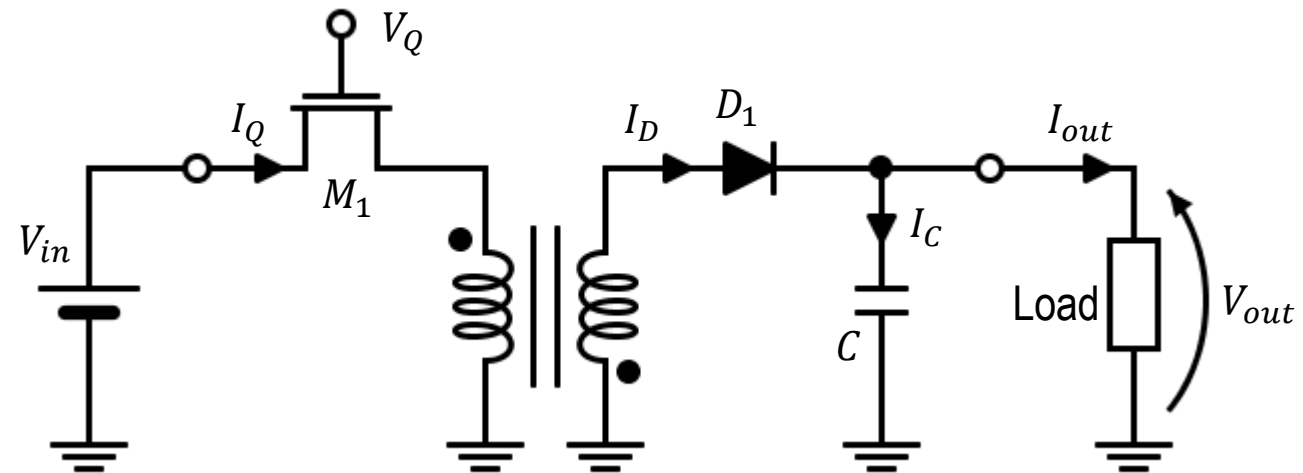
- If the transformer has different turns in the two windings

- Take the ratio into account



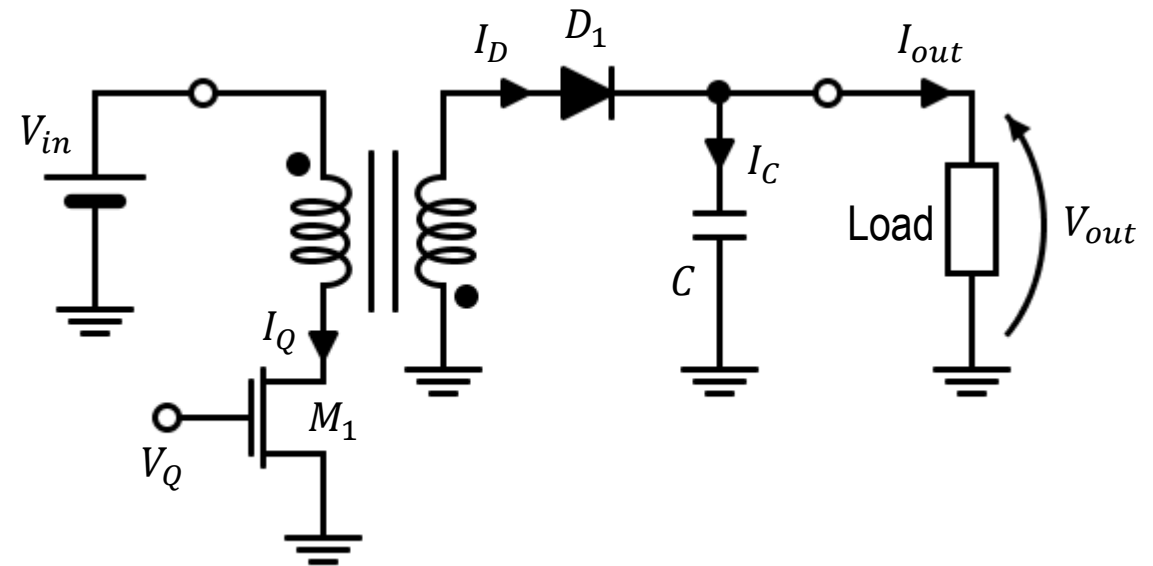
# Switching power supply

- Flyback DC-DC converter
  - The transformer can be used in place of the one in the generic block diagram of a PSU
    - But it works at  $f_{sw}$  rather than at  $f = 50$  Hz
    - Transformer is much smaller, lighter, cheaper
    - The transformer provides galvanic isolation between input and output
      - Good for safety reasons
  - The transformer can have multiple secondaries
    - With different number of turns
    - Generate multiple outputs with different voltages



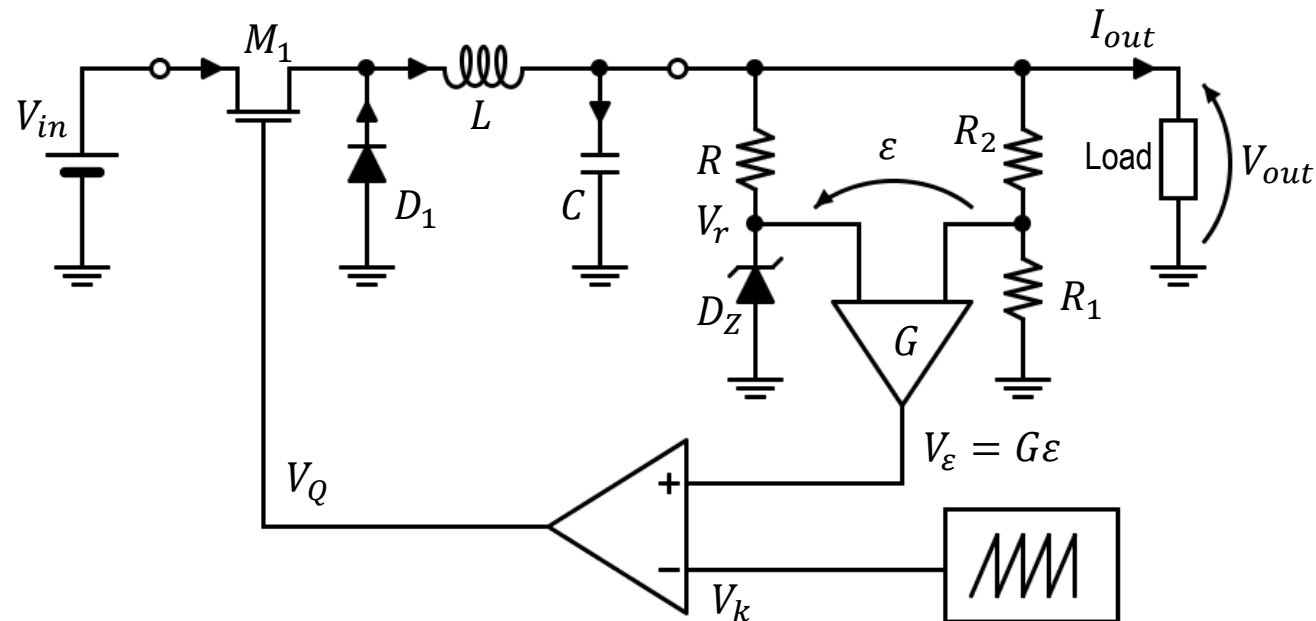
# Switching power supply

- Flyback DC-DC converter
  - The transistor in the Flyback can be moved to a better position
    - Low side driving
    - The Source is at ground
    - Easier to control
      - Compared to a floating switch
  - The circuit behavior does not change
    - When transistor  $M_1$  is OFF there is no current flowing in the primary of the transformer
    - When transistor  $M_1$  is ON, the bottom terminal of the transformer is grounded



# Switching power supply

- Regulator circuit
  - Reference voltage  $V_r$ 
    - Obtained using a Zener diode supplied by the DC-DC converter itself
  - Portion of the output  $\beta V_{out}$  compared to the reference  $V_r$ 
    - $\beta = \frac{R_1}{R_1 + R_2}$
  - Result  $\varepsilon = V_r - \beta V_{out}$  is amplified by gain  $G$ 
    - $\varepsilon$  is the error
    - $V_\varepsilon = G\varepsilon$
  - Control the duty cycle  $\delta$  using  $V_\varepsilon$ 
    - $\delta = HV_\varepsilon$
    - Use a modulator



# Switching power supply

- Regulator circuit

- Modulator

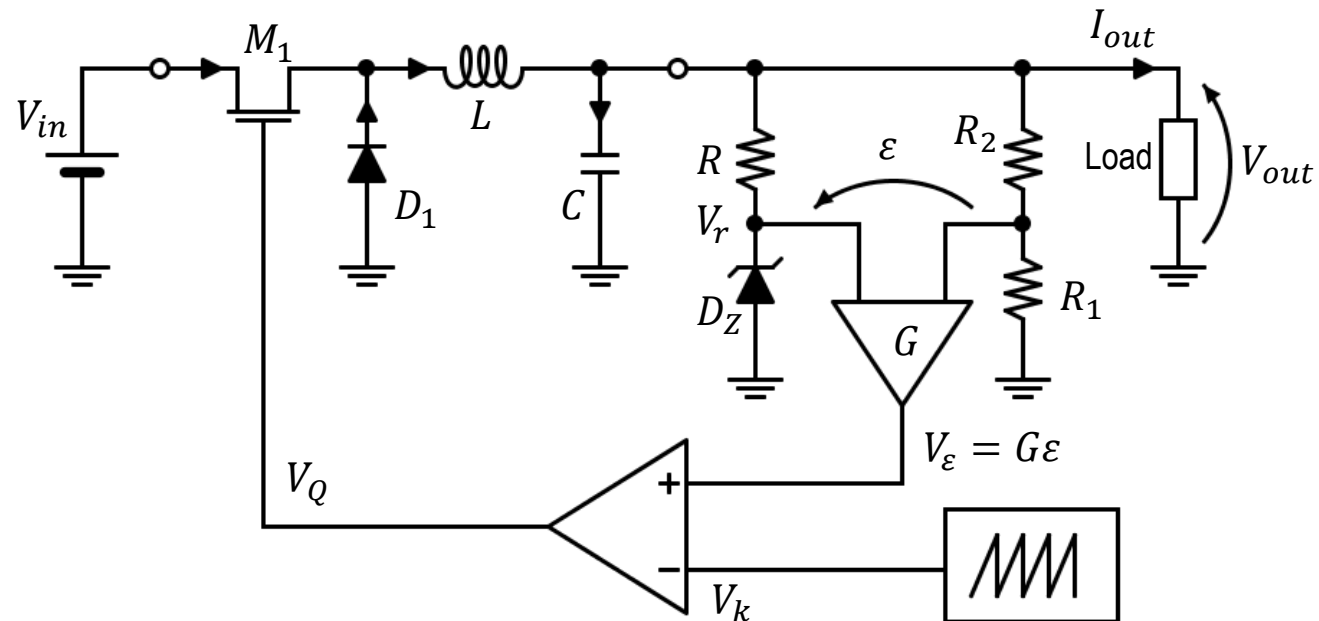
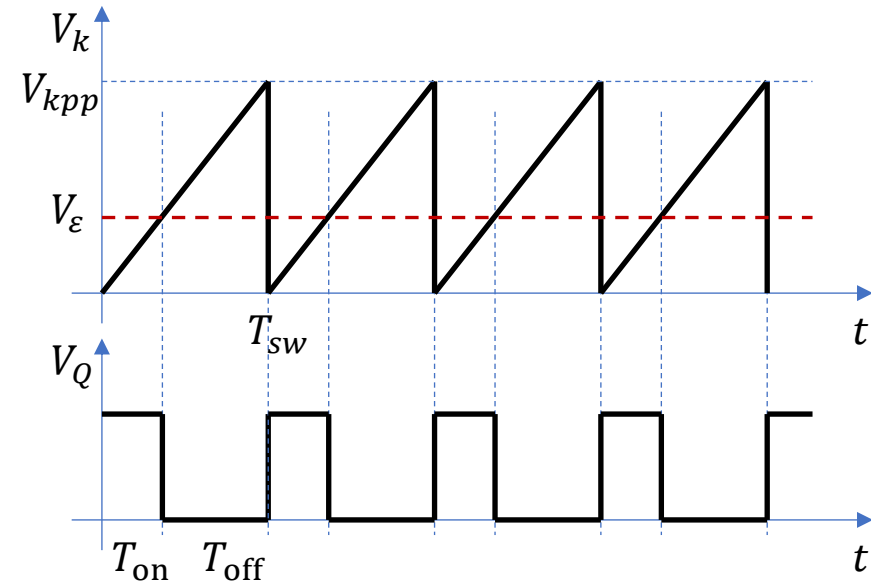
- Saw-tooth waveform  $V_k$ 
  - Frequency  $f_{sw}$
  - Peak-to-peak voltage  $V_{kpp}$

- Comparator

- $V_Q = 1$  when  $V_\varepsilon > V_k$
- $V_Q = 0$  when  $V_\varepsilon < V_k$

- It is

- $T_{on} : T_{sw} = V_\varepsilon : V_{kpp}$
- $\delta = \frac{V_\varepsilon}{V_{kpp}}, \quad H = \frac{1}{V_{kpp}}$



# Switching power supply

- Regulator circuit

- Output voltage  $V_{out}$

- $V_{out} = \delta V_{in} = \frac{V_{\varepsilon}}{V_{kpp}} V_{in} = \frac{G\varepsilon}{V_{kpp}} V_{in}$

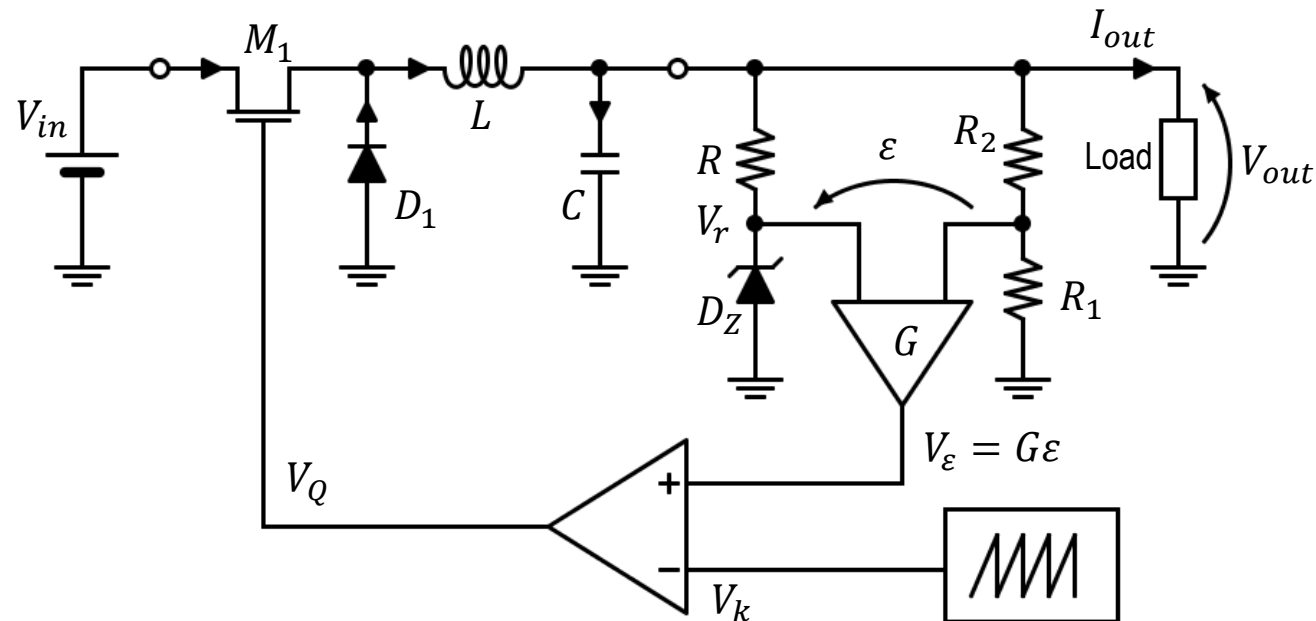
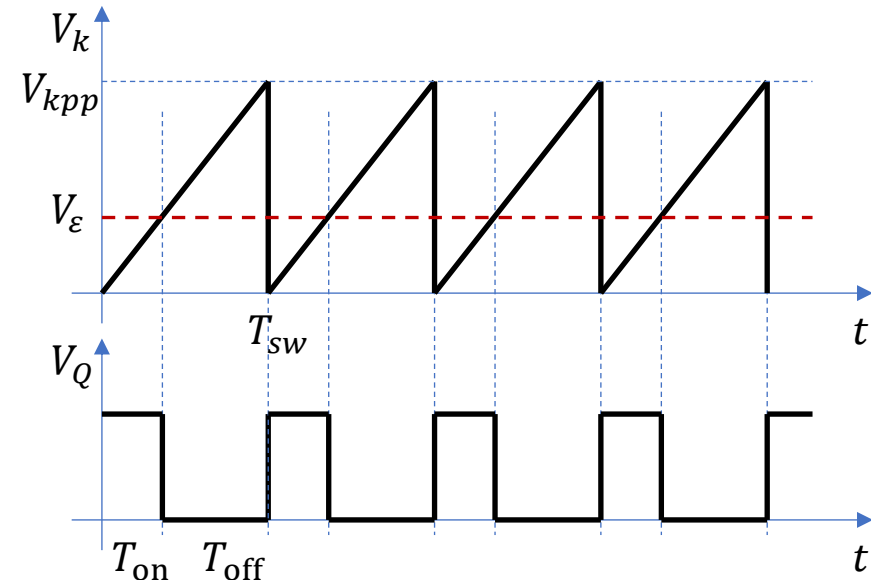
- $V_{out} = \frac{G}{V_{kpp}} (V_r - \beta V_{out}) V_{in}$

- $V_{out} = \frac{G}{V_{kpp}} \left( V_r - \frac{R_1}{R_1 + R_2} V_{out} \right) V_{in}$

- Solve for  $V_{out}$

- $V_{out} = \frac{1}{\beta} \cdot \frac{A\beta}{1+A\beta} V_r$  with  $A = \frac{GV_{in}}{V_{kpp}}$

$$\lim_{A\beta \rightarrow \infty} V_{out} = \frac{1}{\beta} V_r = \left( 1 + \frac{R_2}{R_1} \right) V_r$$



# Switching power supply

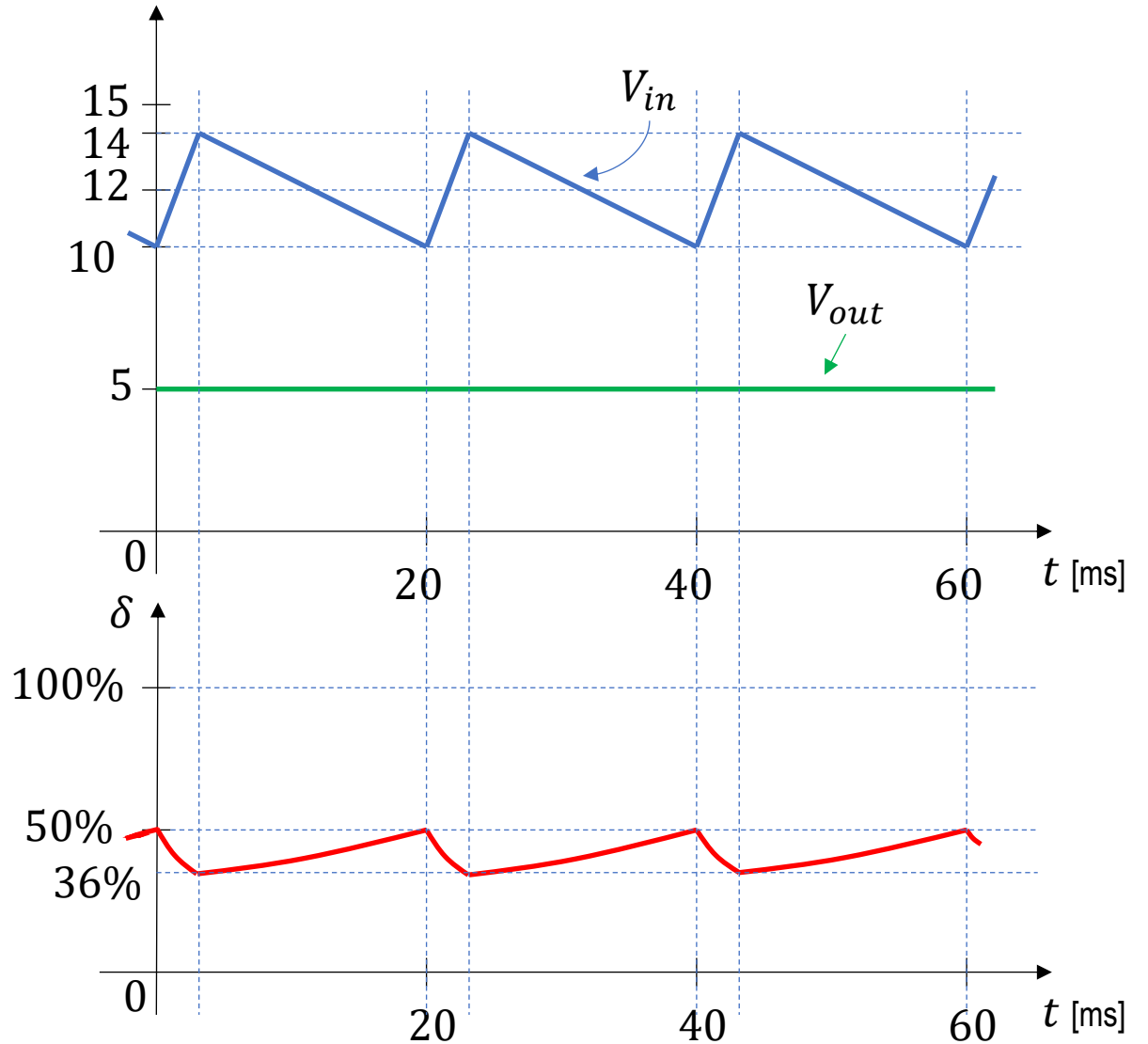
- Regulator circuit

- Example

- $V_{in} = 12\text{ V} \pm 2\text{ V}$  (pseudo DC)
- $V_r = 3,33\text{ V}$ ,  $R_1 = 68\ \Omega$ ,  $R_2 = 33\ \Omega$
- $V_{out} = 5\text{ V}$

| Vin [V] | Duty Cycle % |     |     |     |     |     |     |     |     |     |     |     |
|---------|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|         | 32           | 34  | 36  | 38  | 40  | 42  | 44  | 46  | 48  | 50  | 52  | 54  |
| 14,0    | 4,5          | 4,8 | 5,0 | 5,3 | 5,6 | 5,9 | 6,2 | 6,4 | 6,7 | 7,0 | 7,3 | 7,6 |
| 13,5    | 4,3          | 4,6 | 4,9 | 5,1 | 5,4 | 5,7 | 5,9 | 6,2 | 6,5 | 6,8 | 7,0 | 7,3 |
| 13,0    | 4,2          | 4,4 | 4,7 | 4,9 | 5,2 | 5,5 | 5,7 | 6,0 | 6,2 | 6,5 | 6,8 | 7,0 |
| 12,5    | 4,0          | 4,3 | 4,5 | 4,8 | 5,0 | 5,3 | 5,5 | 5,8 | 6,0 | 6,3 | 6,5 | 6,8 |
| 12,0    | 3,8          | 4,1 | 4,3 | 4,6 | 4,8 | 5,0 | 5,3 | 5,5 | 5,8 | 6,0 | 6,2 | 6,5 |
| 11,5    | 3,7          | 3,9 | 4,1 | 4,4 | 4,6 | 4,8 | 5,1 | 5,3 | 5,5 | 5,8 | 6,0 | 6,2 |
| 11,0    | 3,5          | 3,7 | 4,0 | 4,2 | 4,4 | 4,6 | 4,8 | 5,1 | 5,3 | 5,5 | 5,7 | 5,9 |
| 10,5    | 3,4          | 3,6 | 3,8 | 4,0 | 4,2 | 4,4 | 4,6 | 4,8 | 5,0 | 5,3 | 5,5 | 5,7 |
| 10,0    | 3,2          | 3,4 | 3,6 | 3,8 | 4,0 | 4,2 | 4,4 | 4,6 | 4,8 | 5,0 | 5,2 | 5,4 |

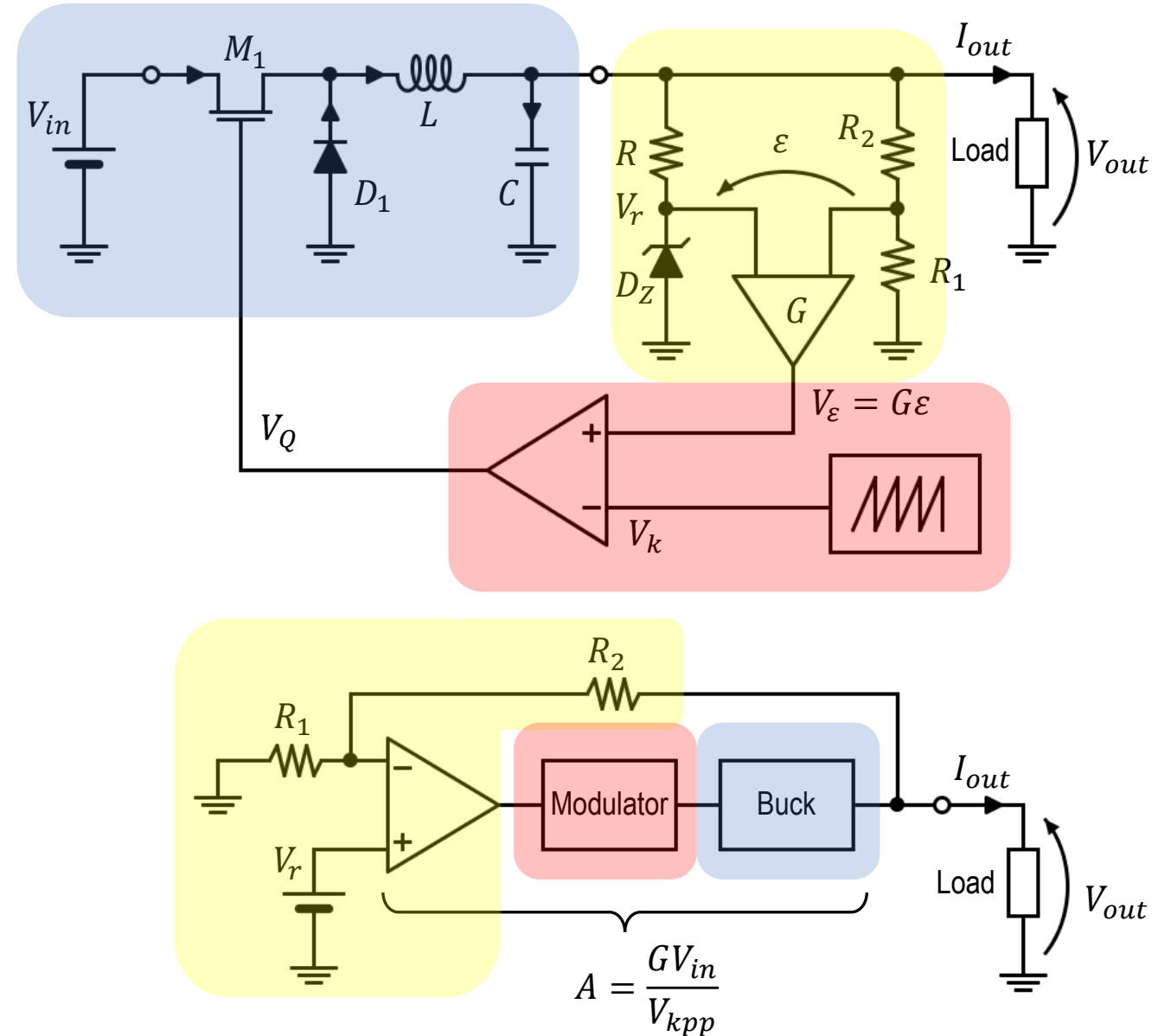
Output  $V_{out}$  as a function of  $V_{in}$  and  $\delta$



# Switching power supply

- Regulator circuit

- Output voltage  $V_{out}$ 
  - $V_{out} = \left(1 + \frac{R_2}{R_1}\right) V_r$
  - Similar to a non-inverting amplifier
    - Negative feedback gives  $\varepsilon \rightarrow 0$
  - It works only as long as  $0 < V_{out} < V_{in}$ 
    - Otherwise the Buck DC-DC converter fails
    - With a Boost it must be  $V_{out} > V_{in}$
    - With a Buck-Boost it must be  $V_{out} < 0$
- The negative feedback may include an opto-coupler
  - When using the flyback configuration
  - To maintain the galvanic isolation between input and output



# Thank you very much

See you at the exam

*“Che sia trenta oppur diciotto  
si festeggia, questo è il motto”  
(Gordon)*